

Lecture Notes Accompanying

STAT2003/STAT7003

Mathematical Probability

Semester 1, 2026

by the

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PREFACE

These notes form a comprehensive, second-year introduction to probability. The Semester 1, 2026 STAT2003/STAT7003 course follows these notes closely.

The notes evolved from previous notes created by Emeritus Prof. Dirk Kroese who has written multiple books about probability, simulation, statistics, machine learning, data science, and related fields. The current notes are edited by A/Prof. Yoni Nazarathy and Dr Kazutoshi (Kasu) Yamazaki.

In addition to the content in these notes, any material discussed in the lectures and given in the tutorial problem sets is *fair game* for the exam. A few selected sections or subsections will be omitted from the exam material, as announced towards the end of the semester. The distribution formula table in the appendix at the end of these notes is supplied with the exam.

To completely master STAT2003/STAT7003 it is important that you

1. visit the lectures, where the lecturing staff provides many extra examples;
2. do the tutorial exercises and the exercises in the appendix, which are there to help you with the “technical” side of things; you will learn here to apply the concepts learned at the lectures,
3. carry out random experiments on the computer, in the simulation project. This will give you a better intuition about how randomness works.
4. read these notes and make sure you can reproduce the examples and derivations (proofs).

All of these will be essential if you wish to understand probability beyond “filling in the formulas”.

Structure of the Course

The course is comprised of 10 units, with each unit having an associated chapter in these notes.

Chapter 1 - Random Experiments, Probability Spaces, and Simulation: In this chapter we introduce the notion of random experiments and probability. With this we present a simplified mathematical description of the probability triple and study simple examples. The probability axioms and resulting properties are studied. We also see how random experiments can be simulated on a computer, a topic sometimes called Monte-Carlo.

Chapter 2 - Counting, Inclusion-Exclusion, and Independence: We continue the study of basic probability by reviewing basics of combinatorics, namely **counting techniques**. These approaches are needed when considering **equiprobable probability spaces**, and related cases. We also study the probability of a union of an arbitrary finite number of events via the so-called **inclusion-exclusion formula**. Finally we define the notion of **independent events**.

Chapter 3 - Discrete Random Variables, their Distributions, and Moments: In this chapter we consider the general notion of random variables, focusing on discrete valued random variables. With this we study their distributions, and moment descriptors, including expectation, mean, and variance. In particular, we study families of random variables associated with independent Bernoulli (coin-flip) trials. This includes the binomial, geometric, and negative Binomial distributions. We also visit the Poisson distribution and the hypergeometric distribution.

Chapter 4 - Conditional Probability and Markov Chains: In this chapter we introduce basics of conditional probability. With this we visit the law of total probability and Bayes' theorem. We also revisit some aspects of discrete random variables from the previous chapter, now through the lens of conditional probability. We also visit the product rule and then introduce the concept of Markov chains. These stochastic processes are a sequence of discrete random variables that exhibit the Markov property. In their elementary study, we see how matrix multiplication is inherently related to their probability laws, and we visit basic concepts such as their limiting distribution at an introductory exploratory level.

Chapter 5 - Continuous Random Variables and their Distributions: In this chapter we introduce continuous distributions, their expectation, moments, and properties. We focus on families of continuous distributions including the Normal distribution, exponential, uniform, Gamma, Beta, Pareto, Cauchy, and a few others.

Chapter 6 - More Aspects of Random Variables and Distributions: In this chapter we explore several additional aspects of random variables and their distributions. We develop techniques for finding the distribution of a transformed random variable, and see how relationships between families of probability distributions arise. We then consider the inverse probability transform, a universal method for simulating random variables. We establish the Markov and Chebyshev inequalities, which bound tail probabilities using only moments. Finally, we encounter random variables whose moments are infinite or even undefined — the Pareto and Cauchy distributions.

Chapter 7 - Joint Distributions: In previous chapters we studied individual random variables — their distributions, moments, and transformations. In this chapter we consider situations where two

or more random variables are observed simultaneously. We develop the theory of joint distributions, marginal and conditional distributions, covariance and correlation, and the multivariate normal and multinomial distributions. We then introduce conditional expectation and the tower property (law of total expectation), and apply these tools to random sums and the law of total variance.

Chapter 8 - More Aspects of Joint Distributions: In this chapter we study transformations of vector random variables — their densities via the Jacobian, linear maps and their effects on mean and covariance, and the Cholesky decomposition for generating multivariate normal samples. We then derive distributions of sums, products, ratios, and order statistics of independent random variables, with convolutions as a central tool.

Chapter 9 - Moment Generating Functions, Related Transforms, and Examples: In this chapter we introduce probability generating functions and moment generating functions — transforms that encode a distribution and turn convolutions into simple algebra. We develop their properties, show how to extract probabilities and moments, and apply them to sums of independent random variables and a branching process model.

Chapter 10 - Random Samples and Limit Theorems: In this chapter we study statistics computed from random samples — the sample mean and sample variance, their distributions, and their unbiasedness. We then establish the laws of large numbers and the central limit theorem, which together describe the behaviour of averages and sums for large samples.

Code Examples in These Notes

This is a course about *probability*, not about computer programming. The emphasis throughout is on understanding probability concepts, distributions, derivations, and problem solving. Nevertheless, using a computer language as an aid for understanding probabilistic ideas is important, and for this reason we include code examples in many of the sections. Some topics, particularly Monte Carlo simulation and random number generation, directly require code to illustrate and explore.

All code examples use Python, but the ideas are what matter, not the specific language. Other languages such as Julia or R would have worked just as well, if not better in some cases. You are encouraged to copy the code snippets, run them, and modify them to explore. Experimenting with the code (changing parameters, increasing sample sizes, trying different distributions) is one of the best ways to build intuition about randomness and probability.

Large language models (LLMs) such as ChatGPT and Claude are also useful tools for generating, explaining, and debugging code. To support this, we include an *LLM prompt* at the bottom of every code snippet: a short natural-language description that, if given to an LLM, would produce code similar to the snippet. You can use these prompts as a starting point and iterate with an LLM to explore variations.

Regarding assessment: students are not required to write code for the exam. However, you *are* expected to understand basic Python code in the context of random number generation and simulation. This does not mean memorising the details of specific libraries or function names (such details would be provided in an exam setting if needed). Rather, it means understanding the *principles*: what it means to generate a uniform random number, how a simulation loop works, how sample means are computed, and so on.

RANDOM EXPERIMENTS, PROBABILITY SPACES, AND SIMULATION

In this chapter we introduce the notion of random experiments and probability. With this we present a simplified mathematical description of the probability triple and study simple examples. The probability axioms and resulting properties are studied. We also see how random experiments can be simulated on a computer, a topic sometimes called Monte-Carlo.

1.1 Random Experiments, Reality, and Simulation

The basic notion in probability is that of a **random experiment**: an experiment whose outcome cannot be determined in advance, but is nevertheless still subject to analysis.

Examples of random experiments are:

1. tossing a die,
2. measuring the amount of rainfall in Brisbane in January,
3. counting the number of calls arriving at a telephone exchange during a fixed time period,
4. selecting a random sample of fifty people and observing the number of left-handers,
5. choosing at random ten people and measuring their height.

These examples illustrate random processes that occur in the real world. However, each can also be simulated on a computer to study their behaviors without having to physically repeat each experiment multiple times.

Example (Coin Tossing). The most *fundamental* stochastic experiment is the experiment where a coin is tossed a number of times, say n times. Indeed, much of probability theory can be based

on this simple experiment, as we shall see in subsequent chapters. To better understand how this experiment behaves, we can carry it out in reality by physically tossing a coin, or simulate it on a digital computer using Python. This duality allows us to both observe and model the behavior of coin tosses more efficiently and repeatedly.

The following simple Python program simulates a sequence of 100 tosses with a fair coin (that is, heads and tails are equally likely), and plots the results in a bar chart.

Simulating 100 fair coin tosses

LLM Prompt > Write Python code that simulates 100 fair coin tosses (as 0s and 1s) using uniform random numbers and displays the results as a bar chart.

Here $\mathbf{x}$ is a vector with 1s and 0s, indicating Heads and Tails, say. Running this code produces the bar chart in Figure 1.1: blue bars indicate “Heads” (=1) and green bars indicate “Tails” (=0).

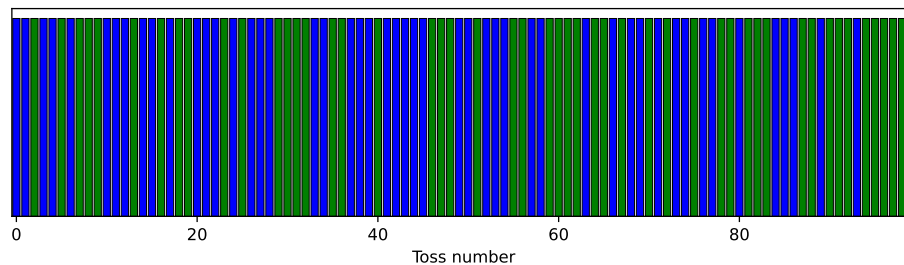


Figure 1.1: Simulated 100 fair coin tosses with seed 2026. Blue bars are “Heads” (=1), green bars are “Tails” (=0).

Notice the line `np.random.seed(2026)` in the code. This sets the *random seed*, which determines the starting point for the pseudorandom number generator. Every time you run the code with the same seed, you get exactly the same sequence of “random” numbers, and hence the same sequence of coin tosses. This is useful for reproducibility. We say more about pseudorandom number generation in Section 1.4.

What happens if we change the seed? The following code is identical except that the seed is set to 2027 instead of 2026.

Simulating 100 fair coin tosses (different seed)

LLM Prompt > Write Python code that simulates 100 fair coin tosses with a different random seed and displays the results as a bar chart.

The result is shown in Figure 1.2. The pattern of Heads and Tails is completely different from Figure 1.1. Setting two different seeds is analogous to tossing the 100 coins on two separate oc-

casions: each run produces a fresh, independent-looking sequence. If we do not specify a seed at all, Python uses whatever value happens to be in the system (based on the current time or operating system entropy), so each run automatically gives a different result.

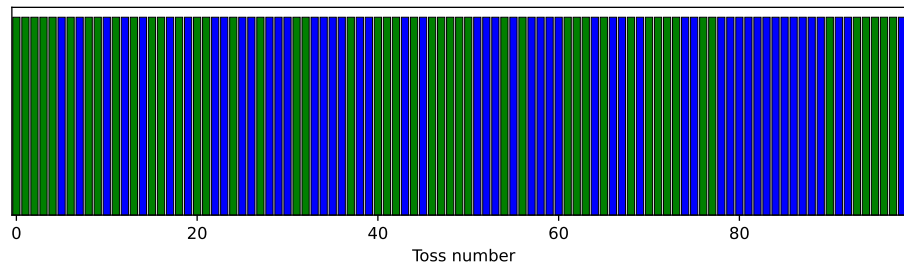


Figure 1.2: Simulated 100 fair coin tosses with seed 2027. The pattern differs from Figure 1.1, just as two separate physical experiments would differ.

We can also plot the running average of “Heads” against the number of tosses. The following code carries out three separate experiments (with seeds 2026, 2027, and 2028) and plots the running average for each.

```
Running average of heads over  $n$  tosses
```

LLM Prompt > *Write Python code that runs three coin-toss experiments (with different seeds), computes the running average of heads for each, and plots all three trajectories on one figure with a dashed line at $1/2$.*

The result is depicted in Figure 1.3. Notice that the average number of Heads seems to converge to $\frac{1}{2}$, but there is a lot of random fluctuation, especially for small n .

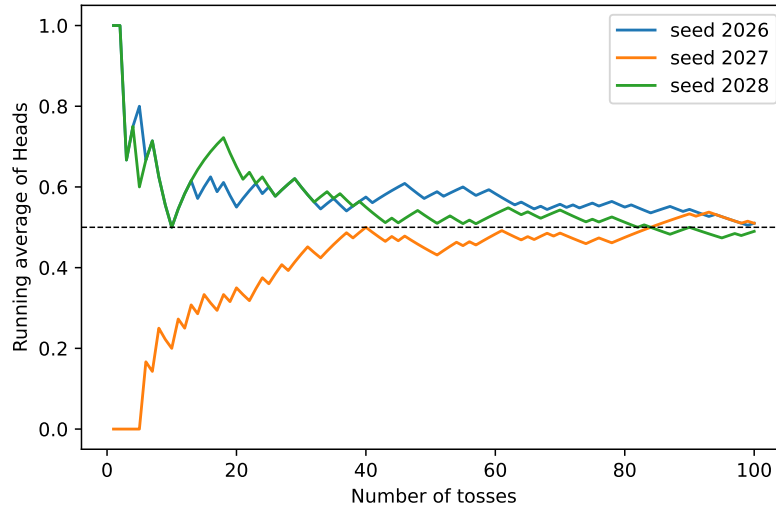


Figure 1.3: The running average of heads in n tosses for three experiments, where $n = 1, \dots, 100$. The dashed line is at $\frac{1}{2}$.

Example (Control Chart). Control charts, see Figure 1.4, are frequently used in manufacturing as a method for *quality control*. In reality, each hour the average output of the process is measured — for example, the average weight of 10 bags of sugar — to assess if the process is still “in control”, i.e., if the machine still puts on average the correct amount of sugar in the bags. Simulations can also be used to predict behaviors of production processes under different configurations and identify potential issues before they occur on the shop floor. When the process $> \text{Upper Control Limit}$ or $< \text{Lower Control Limit}$, an alarm is raised to indicate that the process is out of control, for instance, because the machine either puts too much or not enough sugar in the bags. The question is how to set the control limits, since the random process naturally fluctuates around its “center” or “target” line.

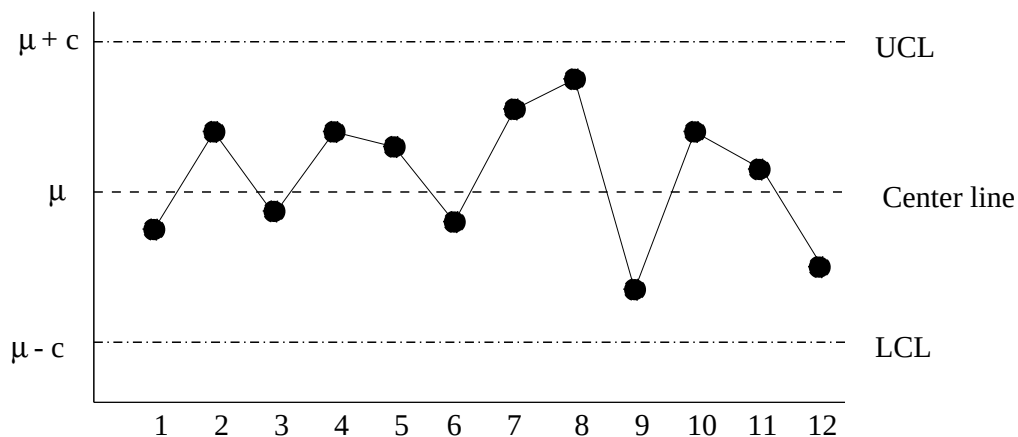


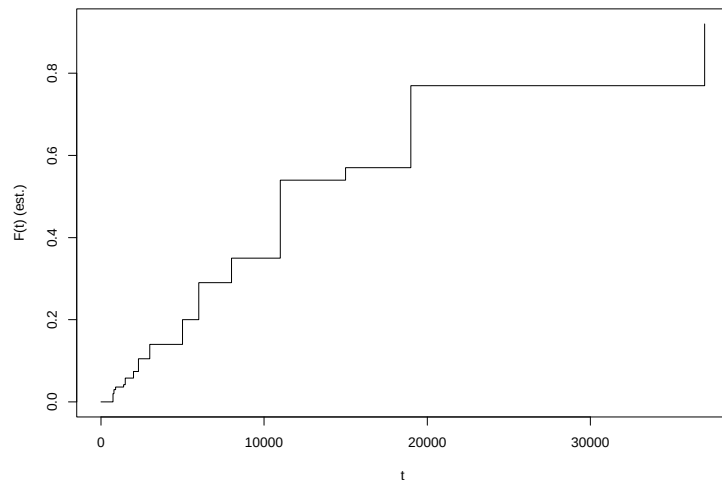
Figure 1.4: Control Chart

Example (Machine Lifetime). Suppose 1000 identical components are monitored for failure in a real-world scenario, up to 50,000 hours. The outcome of such a random experiment is typically

summarized via the cumulative lifetime table and plot, as given in Figure 1.5. The simulation of similar scenarios can help in designing more reliable systems by predicting failures under different stress conditions. Here $\widehat{F}(t)$ denotes the proportion of components that have failed at time t . One question is how $\widehat{F}(t)$ can be modeled via a continuous function F , representing the lifetime distribution of a typical component.

t (h)	failed	$\widehat{F}(t)$
0	0	0.000
750	22	0.022
800	30	0.030
900	36	0.036
1400	42	0.042
1500	58	0.058
2000	74	0.074
2300	105	0.105
3000	140	0.140
5000	200	0.200
6000	290	0.290
8000	350	0.350
11000	540	0.540
15000	570	0.570
19000	770	0.770
37000	920	0.920

(a)



(b)

Figure 1.5: Cumulative lifetime data for 1000 monitored components. (a) Number of failures and empirical cdf $\widehat{F}(t)$ at selected times. (b) Step-function plot of $\widehat{F}(t)$ against time t (hours).

Example (Aeroplane). A 4-engine aeroplane is able to fly on just one engine on each wing. In reality, this can be considered by observing the function of the engines over specified periods to assess reliability. Simulations are pivotal for risk assessment and to calculate the probabilities involved without endangering lives or the actual aircraft. All engines are unreliable.

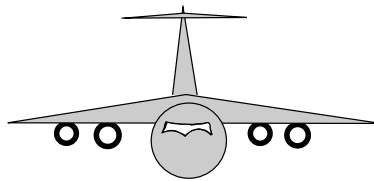


Figure 1.6: An aeroplane with 4 engines, each subject to failure

Number the engines: 1,2 (left wing) and 3,4 (right wing). Observe which engine works properly during a specified period of time. There are $2^4 = 16$ possible outcomes of the experiment. Which outcomes lead to “system failure”? Moreover, if the $\mathbb{P}$ of failure within some time period is known for each of the engines, what is the $\mathbb{P}$ of failure for the entire system? Again this can be viewed as a random experiment.

Example (Reality: Waiting Times at a Bus Stop). Consider a real-world situation where we observe the waiting times (in minutes) for a bus at a particular stop over several days. Assume we record the following data: 5, 8, 4, 7, 6, 5, 9, 4, 8, 6.

In general, given data $x_1, \dots, x_n$, the *sample mean* and *sample standard deviation* are

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}. \quad (1.1)$$

For the waiting-time data above ($n = 10$), we get $\bar{x} = 6.2$ and $s = \sqrt{46/9} \approx 1.75$.

The sample mean gives us an average waiting time, providing an expectation for future waits. The sample standard deviation reflects the variability around the average, capturing the randomness inherent to the process. As more data are collected, these summaries tend to stabilize, better representing the underlying process of bus arrivals. In this course we do not deal much with collected data, but rather with the probabilistic models that describe such situations.

Example (Simulation: Rolling a Dice). Consider simulating the throw of a standard six-sided die 1,000 times using Python to observe the distribution of outcomes.

Simulating 1000 dice rolls

LLM Prompt > Write Python code that simulates 1000 rolls of a fair six-sided die and computes the sample mean and standard deviation of the outcomes.

This simulation generates an array `rolls` of random integers between 1 and 6. With seed 2026, the sample mean is $\bar{x} = 3.527$ and the sample standard deviation is $s = 1.7425$. A theoretical model for a fair die (as we will see in Chapter 3) gives a mean of $\mu = 3.5$ and a standard deviation of $\sigma = \sqrt{35/12} \approx 1.7078$. The simulated values are close to the theoretical ones, but not exactly equal — this is the nature of randomness.

Figure 1.7 shows two more pictures of randomness. Part (a) is a computer-generated “plant”, which looks remarkably like a real plant. Part (b) is real data depicting the number of bytes transmitted over some communications link. An interesting feature is that the data can be shown to exhibit “fractal” behavior, that is, if the data is aggregated into smaller or larger time intervals, a similar picture will appear.

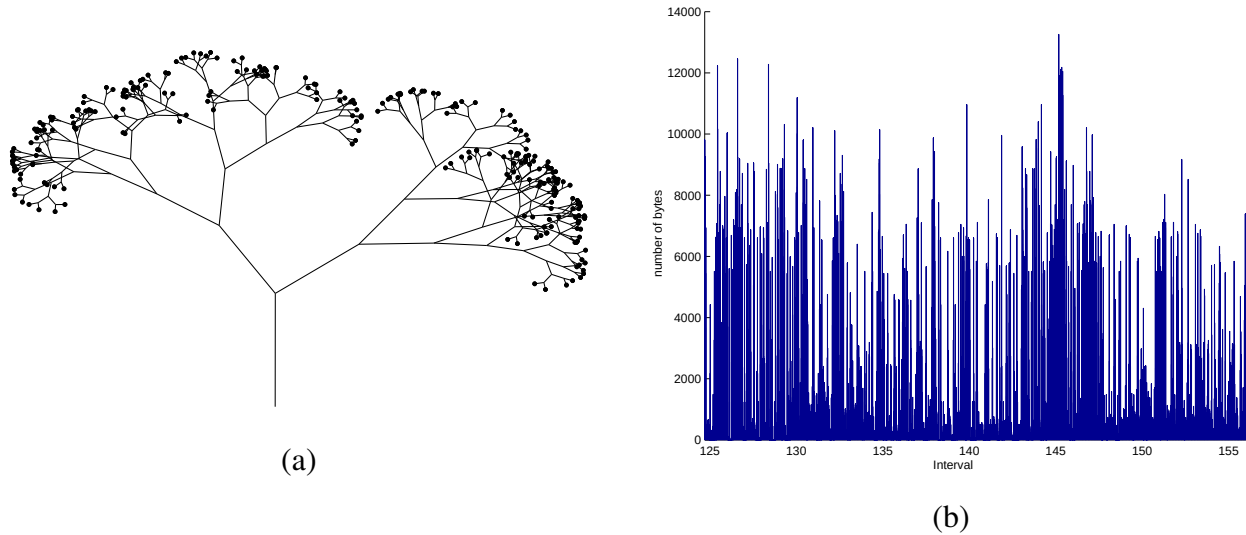


Figure 1.7: Two pictures of randomness. (a) A computer-generated plant produced by a random branching process. (b) Telecommunications traffic data showing the number of bytes transmitted per time interval.

We wish to describe these experiments via appropriate mathematical models. These models consist of three building blocks: a *sample space*, a set of *events*, and a *probability*. We will now describe each of these objects.

1.2 Events and Sets

This section reviews the set theory concepts that underpin probability and introduces the language of events.

Quick Review of Set Theory

A **set** S is a collection of distinct objects called **elements**; we write $x \in S$ to mean x belongs to S . The **empty set** $\emptyset$ contains no elements, and the **universal set** Ω contains all elements under consideration. A set A is a **subset** of B (written $A \subseteq B$) if every element of A also belongs to B . The basic set operations are:

- **Union:** $A \cup B = \{x \in \Omega : x \in A \text{ or } x \in B\}$ — the set of elements belonging to A , or B , or both.
- **Intersection:** $A \cap B = \{x \in \Omega : x \in A \text{ and } x \in B\}$ — the set of elements belonging to both A and B .
- **Complement:** $A^c = \{x \in \Omega : x \notin A\}$ — the set of elements in Ω that do not belong to A .
- **Set difference:** $A \setminus B = \{x \in \Omega : x \in A \text{ and } x \notin B\}$ — the set of elements in A that do not belong to B . Note that $A \setminus B = A \cap B^c$.

These operations can be visualized using a **Venn diagram**, as shown in Figure 1.8.

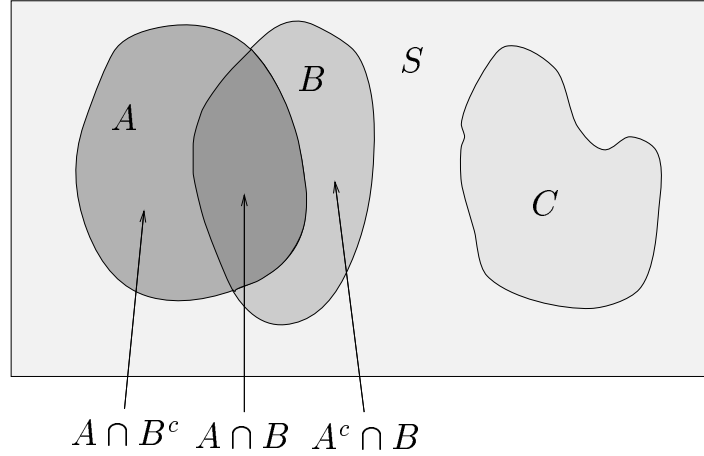


Figure 1.8: A Venn diagram illustrating sets A , B , C within a universal set Ω

These operations extend to indexed collections $\{A_i\}_{i \in I}$:

$$\bigcup_{i \in I} A_i = \{x \in \Omega : x \in A_i \text{ for some } i \in I\}, \quad \bigcap_{i \in I} A_i = \{x \in \Omega : x \in A_i \text{ for all } i \in I\}. \quad (1.2)$$

For a finite family $A_1, \dots, A_N$ we write $\bigcup_{i=1}^N A_i$ and $\bigcap_{i=1}^N A_i$; for an infinite family $A_1, A_2, \dots$ we write $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$.

Claim 1: De Morgan's Laws — For any two sets A and B ,

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c. \quad (1.3)$$

More generally, for any indexed collection $\{A_i\}_{i \in I}$,

$$\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c, \quad \left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c. \quad (1.4)$$

Proof: We prove the first identity for two sets; the second follows by replacing each set with its complement. An element x belongs to $(A \cup B)^c$ if and only if $x \notin A \cup B$, i.e., $x \notin A$ and $x \notin B$, i.e., $x \in A^c \cap B^c$. The general case follows by the same argument: $x \notin \bigcup_{i \in I} A_i$ if and only if $x \notin A_i$ for every $i \in I$, i.e., $x \in \bigcap_{i \in I} A_i^c$. $\square$

The Power Set

The **power set** of a set S , denoted $\mathcal{P}(S)$, is the set of *all* subsets of S (including $\emptyset$ and S itself). If S has n elements, then $\mathcal{P}(S)$ has 2^n elements.

Example (Power set of $\{1, 2, 3\}$). If $S = \{1, 2, 3\}$, the power set is

$$\mathcal{P}(S) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}, \quad (1.5)$$

which has $2^3 = 8$ elements.

Events

In probability, the **sample space** Ω plays the role of the universal set: it contains all possible outcomes of an experiment. **Events** are subsets of Ω , denoted by capital letters $A, B, C, \dots$. An event A **occurs** if the experimental outcome belongs to A . For example:

- The event that the sum of two dice is 10 or more: $A = \{(4, 6), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)\}$.
- The event that a machine lives less than 1000 days: $A = [0, 1000)$.

Set operations translate directly: $A \cup B$ means “ A **or** B (or both) occur”, $A \cap B$ means “ A **and** B both occur”, and A^c means “ A does **not** occur”. For indexed families, $\bigcup_{i \in I} A_i$ is the event that at least one A_i occurs, and $\bigcap_{i \in I} A_i$ that all A_i occur. Two events are **disjoint** (or **mutually exclusive**) if $A \cap B = \emptyset$. If $A \subseteq B$, we say that A **implies** B : whenever A occurs, the outcome belongs to A and hence also to B , so B must occur as well.

Example (Coin Tossing). Suppose a coin is tossed three times, recording each head (H) and tail (T). The sample space is:

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\} . \quad (1.6)$$

The event A that the third toss is heads is:

$$A = \{HHH, HTH, THH, TTH\} . \quad (1.7)$$

Example (Casting Two Dice). Suppose we cast two dice. The sample space is:

$$\Omega = \{(1, 1), (1, 2), \dots, (1, 6), (2, 1), \dots, (6, 6)\} . \quad (1.8)$$

Let $A = \{(6, 1), \dots, (6, 6)\}$ be the event that the first die is 6, and let $B = \{(1, 6), \dots, (6, 6)\}$ be the event that the second die is 6. Then,

$$A \cap B = \{(6, 1), \dots, (6, 6)\} \cap \{(1, 6), \dots, (6, 6)\} = \{(6, 6)\} \quad (1.9)$$

is the event that both dice show 6.

Example (System Reliability). In Figure 1.9, three systems are depicted with three unreliable components. The **series** system works **iff** all components work; the **parallel** system works **iff** at least one works; and the **2-out-of-3** system works **iff** at least 2 out of 3 work.

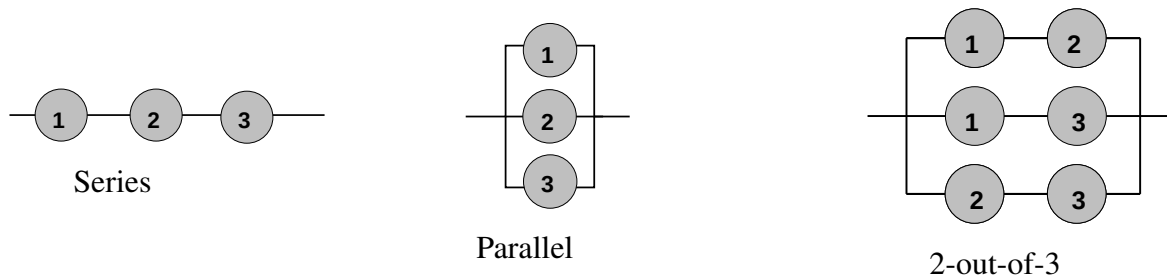


Figure 1.9: Three unreliable systems

Let A_i be the event that the i th component functions, $i = 1, 2, 3$; and D_a, D_b, D_c the events that the series, parallel, and 2-out-of-3 systems function, respectively. Then:

$$D_a = A_1 \cap A_2 \cap A_3, \quad D_b = A_1 \cup A_2 \cup A_3, \quad D_c = (A_1 \cap A_2) \cup (A_1 \cap A_3) \cup (A_2 \cap A_3). \quad (1.10)$$

1.3 The Probability Triple, Axioms, and Properties

In Section 1.2 we described sample spaces and events using the language of set theory. We now formalize what it means to assign *probabilities* to events. Ideas of probability emerged as early as the 16th and 17th centuries through the work of mathematicians such as Cardano, Pascal, and Fermat, yet the field was only placed on rigorous mathematical foundations in the 20th century, when A.N. Kolmogorov formulated his axioms using the language of *measure theory*.

The central object is the **probability triple** $(\Omega, \mathcal{F}, \mathbb{P})$. Its three components are:

- Ω , the **sample space** — the set of all possible outcomes (introduced in Section 1.1),
- $\mathcal{F}$, the **collection of events** — the subsets of Ω to which we assign probabilities,
- $\mathbb{P}$, the **probability measure** — a function that assigns a probability to each event in $\mathcal{F}$.

For the purposes of this course, we can conceptually think of $\mathcal{F}$ as the power set $\mathcal{P}(\Omega)$ — the collection of all subsets of Ω . Yet in a more advanced course, we may take $\mathcal{F}$ to be a *Borel sigma-algebra* — something you learn about in STAT3004 and/or a measure theory course. This subtlety is discussed briefly in the subsection on sigma-algebras at the end of this section.

Definition : Probability Triple — A **probability triple** (or probability space) is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where:

- Ω is the sample space,
- $\mathcal{F}$ is a collection of subsets of Ω (the events),
- $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a probability measure satisfying Kolmogorov's axioms below.

That is, $\mathbb{P}$ is a function that operates on subsets (events) of Ω and returns a real value between 0 and 1. The higher the value, the more probability we attribute to the event. The probability measure $\mathbb{P}$ must satisfy certain fundamental rules, known as **Kolmogorov's axioms**, which ensure that probability behaves in a consistent and mathematically rigorous way.

Definition : Kolmogorov's Axioms — A function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a **probability measure** if it satisfies:

1. **Non-negativity:** $\mathbb{P}(A) \geq 0$ for every event $A \in \mathcal{F}$.
2. **Normalization:** $\mathbb{P}(\Omega) = 1$.

3. **Finite additivity:** If $A_1, \dots, A_n$ are finitely many disjoint events in $\mathcal{F}$, then

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mathbb{P}(A_i). \quad (1.11)$$

4. **Countable additivity** (σ -additivity): If $\{A_i\}_{i=1}^\infty$ is a countable collection of disjoint events in $\mathcal{F}$, then

$$\mathbb{P}\left(\bigcup_{i=1}^\infty A_i\right) = \sum_{i=1}^\infty \mathbb{P}(A_i). \quad (1.12)$$

Note that countable additivity implies finite additivity (set $A_i = \emptyset$ for $i > n$), so axiom 4 is the stronger statement. However, finite additivity is listed separately because it is the version used most often in practice.

From these four axioms alone, a rich collection of properties can be derived. The following claim lists the most fundamental ones.

Claim 2: Properties of Probability — Let A and B be events. Then,

1. $\mathbb{P}(\emptyset) = 0$.
2. **Monotonicity:** $A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$.
3. $\mathbb{P}(A) \leq 1$.
4. **Complement Rule:** $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
5. **Inclusion-Exclusion:** $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Proof:

1. Since $\Omega = \Omega \cup \emptyset$, we have by the sum rule $\mathbb{P}(\Omega) = \mathbb{P}(\Omega) + \mathbb{P}(\emptyset)$, and therefore, by the second axiom, $1 = 1 + \mathbb{P}(\emptyset)$, from which it follows that $\mathbb{P}(\emptyset) = 0$.
2. If $A \subseteq B$, then $B = A \cup (B \cap A^c)$, where A and $B \cap A^c$ are disjoint. Hence, by the sum rule, $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \cap A^c)$, which is (by the first axiom) greater than or equal to $\mathbb{P}(A)$.
3. This follows directly from property 2 and axiom 2, since $A \subseteq \Omega$.
4. $\Omega = A \cup A^c$, where A and A^c are disjoint. Hence, by the sum rule and axiom 2: $1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c)$, and thus $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
5. Write $A \cup B$ as the disjoint union of A and $B \cap A^c$. Then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \cap A^c)$. Also, $B = (A \cap B) \cup (B \cap A^c)$, so that $\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(B \cap A^c)$. Combining these two equations gives $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

□

Example (Rolling a Die). Suppose a fair six-sided die is rolled. Let $A = \{2, 4, 6\}$ (even) and $B = \{1, 2, 3\}$ (at most 3). Then $\mathbb{P}(A) = \mathbb{P}(B) = \frac{1}{2}$. By the complement rule, $\mathbb{P}(A^c) = 1 - \frac{1}{2} = \frac{1}{2}$. By

inclusion-exclusion, since $A \cap B = \{2\}$,

$$\mathbb{P}(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{6} = \frac{5}{6}. \quad (1.13)$$

Sigma-Algebras

We said that $\mathcal{F}$ is “the collection of events” to which we assign probabilities. But what qualifies as a valid $\mathcal{F}$? The answer is that $\mathcal{F}$ must be a *sigma-algebra*.

Definition : Sigma-Algebra — A collection $\mathcal{F}$ of subsets of Ω is called a **sigma-algebra** (or σ -algebra) if:

1. $\Omega \in \mathcal{F}$.
2. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ (closed under complementation).
3. If $\{A_i\}_{i=1}^{\infty}$ is a sequence of sets in $\mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ (closed under countable unions).

It follows that a sigma-algebra is also closed under countable intersections. Indeed, if $\{A_i\}_{i=1}^{\infty}$ is a sequence of sets in $\mathcal{F}$, then each $A_i^c \in \mathcal{F}$ by condition 2, so $\bigcup_{i=1}^{\infty} A_i^c \in \mathcal{F}$ by condition 3, and therefore $\bigcap_{i=1}^{\infty} A_i = \left(\bigcup_{i=1}^{\infty} A_i^c\right)^c \in \mathcal{F}$ by condition 2 again, where the equality is De Morgan’s law.

Example (Sigma-algebras on $\Omega = \{1, 2, 3\}$). Consider $\Omega = \{1, 2, 3\}$. The power set

$$\mathcal{P}(\Omega) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\} \quad (1.14)$$

is a sigma-algebra (all 8 subsets are included, so any complement or union of members is again a member).

A smaller collection can also qualify. Consider $\widetilde{\mathcal{F}} = \{\emptyset, \{1\}, \{2, 3\}, \{1, 2, 3\}\}$. We verify the three conditions:

1. $\Omega = \{1, 2, 3\} \in \widetilde{\mathcal{F}}$. ✓
2. Complements: $\{1\}^c = \{2, 3\} \in \widetilde{\mathcal{F}}$, $\{2, 3\}^c = \{1\} \in \widetilde{\mathcal{F}}$, $\emptyset^c = \{1, 2, 3\} \in \widetilde{\mathcal{F}}$, and $\{1, 2, 3\}^c = \emptyset \in \widetilde{\mathcal{F}}$. ✓
3. Unions: every union of members of $\widetilde{\mathcal{F}}$ is again in $\widetilde{\mathcal{F}}$ (for instance, $\{1\} \cup \{2, 3\} = \{1, 2, 3\} \in \widetilde{\mathcal{F}}$). ✓

Both $\mathcal{P}(\Omega)$ and $\widetilde{\mathcal{F}}$ are valid choices for the event collection in a probability triple on this specific Ω . If we use the power set $\mathcal{P}(\Omega)$, then every subset of Ω is an event and we can speak about its probability. But if we use the smaller $\widetilde{\mathcal{F}}$, then the subsets $\{2\}$, $\{3\}$, $\{1, 2\}$, and $\{1, 3\}$ are not in $\widetilde{\mathcal{F}}$ and are therefore not events — we cannot assign probabilities to them.

Why sigma-algebras? For finite (or countable) sample spaces, the power set $\mathcal{P}(\Omega)$ always works as $\mathcal{F}$, and there is no issue. The subtlety arises for uncountable sample spaces such as $\Omega = \mathbb{R}$.

In that setting, it can be shown that *no* probability measure satisfying Kolmogorov's axioms can be consistently defined on *all* subsets of Ω (i.e., the full power set). To guarantee that a valid probability measure exists, we must restrict to a smaller, well-behaved collection of subsets — a sigma-algebra. This is the central insight of the measure-theoretic formulation of probability.

The formal definition of a sigma-algebra is *not* used further in this course. For our purposes, $\mathcal{F}$ can be conceptually thought of as the power set.

Continuity of Probability

Continuity of probability is a slightly more advanced property that connects limits of sequences of events to limits of their probabilities. It is a direct consequence of countable additivity.

Claim 3: Continuity of Probability —

1. If $\{A_n\}_{n=1}^{\infty}$ is an increasing sequence of events ($A_1 \subseteq A_2 \subseteq \dots$), then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n) . \quad (1.15)$$

2. If $\{B_n\}_{n=1}^{\infty}$ is a decreasing sequence of events ($B_1 \supseteq B_2 \supseteq \dots$), then

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} B_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(B_n) . \quad (1.16)$$

Proof: For part 1, define $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The C_n are disjoint and $\bigcup_{n=1}^{\infty} C_n = \bigcup_{n=1}^{\infty} A_n$. By countable additivity,

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(C_n) = \lim_{N \rightarrow \infty} \sum_{n=1}^N \mathbb{P}(C_n) = \lim_{N \rightarrow \infty} \mathbb{P}(A_N) . \quad (1.17)$$

For part 2, let $D_n = B_n^c$, which is an increasing sequence. By part 1,

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} B_n\right) = 1 - \mathbb{P}\left(\bigcup_{n=1}^{\infty} D_n\right) = 1 - \lim_{n \rightarrow \infty} \mathbb{P}(D_n) = \lim_{n \rightarrow \infty} \mathbb{P}(B_n) . \quad (1.18)$$

□

Example (An event with probability 1 that is not Ω). Let $\Omega = [0, 1]$ with uniform probability, so $\mathbb{P}([a, b]) = b - a$ for $0 \leq a < b \leq 1$. Define $A_n = [0, 1 - 1/n)$ for $n = 1, 2, \dots$. Then $A_1 \subseteq A_2 \subseteq \dots$ and $\mathbb{P}(A_n) = 1 - 1/n$. By continuity of probability,

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) = 1 . \quad (1.19)$$

Yet $\bigcup_{n=1}^{\infty} A_n = [0, 1) \neq \Omega$, illustrating that an event can have probability 1 without being the entire sample space.

Continuity of probability is not used further in this course, but it illustrates the power of the countable additivity axiom and is important in more advanced probability.

1.4 More on Simulation

In Section 1.1 we saw how to simulate simple experiments such as coin tosses and dice rolls. We now develop the ideas behind computer-based random number generation more carefully.

The Uniform Distribution on $[0, 1]$

In Section 1.3 we introduced the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$. Consider the specific case where the sample space is the unit interval, $\Omega = [0, 1]$, and the probability of any sub-interval $[a, b] \subseteq [0, 1]$ is simply its length:

$$\mathbb{P}([a, b]) = b - a . \quad (1.20)$$

This is called the **uniform distribution on $[0, 1]$** , written $\mathcal{U}(0, 1)$. Intuitively, it describes the experiment of “picking a point at random” from the interval $[0, 1]$, with no part of the interval favoured over any other. Continuous probability distributions will be treated formally in Chapter 5; for now, this intuitive description is sufficient.

The $\mathcal{U}(0, 1)$ distribution is the fundamental building block of computer-based simulation. A computer can generate numbers that behave as if they were drawn independently from this probability space. Given such samples, we can estimate the probability of any event $A \subseteq [0, 1]$: generate a large number of samples, check how many fall in A , and compute the proportion. By the law of large numbers (Chapter 10), this proportion converges to $\mathbb{P}(A)$ as the number of samples grows. This idea — using random sampling to estimate quantities that may be difficult to compute analytically — is the essence of **Monte Carlo simulation**, named after the famed Monaco casino. Applications span cryptography, financial modeling, physical sciences, and engineering.

Pseudorandom Sequence Algorithms

But how does a computer, a deterministic machine, produce numbers that “look random”? Early simulation relied on manual techniques (coin flipping, dice rolling) or physical devices (noise diodes). Modern approaches use deterministic algorithms called *pseudorandom number generators* that are fast, reproducible, and pass statistical tests for randomness. The most classical family is the *linear congruential generator*, which produces a sequence via

$$X_{i+1} = a X_i + c \pmod{m} , \quad (1.21)$$

where X_0 is the *seed*, and a, c, m are positive integers called the *multiplier*, *increment*, and *modulus*. The modulo- m operation takes the remainder after dividing by m , so each $X_i \in \{0, 1, \dots, m-1\}$. The *pseudorandom numbers*

$$U_i = \frac{X_i}{m} , \quad (1.22)$$

then approximate independent $\mathcal{U}(0, 1)$ random variables.

Example (Linear Congruential Generator). Let $a = 3$, $c = 2$, $m = 7$, and $X_0 = 1$. Applying (1.21) repeatedly:

$$\begin{array}{c|ccccccccc} i & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & \dots \\ \hline X_i & 1 & 5 & 3 & 4 & 0 & 2 & 1 & 5 & \dots \\ U_i & 0.14 & 0.71 & 0.43 & 0.57 & 0.00 & 0.29 & 0.14 & 0.71 & \dots \end{array} \quad (1.23)$$

The sequence has period 6 (then repeats). The choice of a , c , m strongly affects quality; number-theoretic results guide good choices. In practice, m is chosen as a large prime fitting the computer word size (e.g. $m = 2^{31} - 1$ for 32-bit systems).

Linear congruential generators are now considered outdated for serious simulation work, due to short periods and known statistical deficiencies (for example, consecutive outputs can exhibit lattice structure in higher dimensions). Modern pseudorandom number generators include the *Mersenne Twister* (period $2^{19937} - 1$, used by default in many languages including Python's `random` module), the *Permuted Congruential Generator* (PCG, used by NumPy since version 1.17), and the *xoshiro/xoroshiro* family. These algorithms offer vastly longer periods, better statistical properties, and faster performance. Their design and analysis are beyond the scope of this course.

Most languages provide a built-in pseudorandom generator. In Python, the `rand` function from `numpy.random` produces $\mathcal{U}(0, 1)$ samples.

Generating Uniform Random Variables in Python

Generating a single uniform random number

LLM Prompt > Show how to generate and print a single $U(0, 1)$ random number in Python using NumPy.

```
[0.88171397]
```

Generating a vector of uniform random numbers

LLM Prompt > Show how to generate and print a vector of 5 independent $U(0, 1)$ random numbers in Python.

```
[0.86880489 0.37527345 0.90842683 0.21716382 0.71134196]
```

Setting and resetting the random seed

LLM Prompt > Demonstrate in Python how setting the random seed to a fixed value and then resetting it produces the same sequence of random numbers.

```
[0.19151945 0.62210877 0.43772774 0.78535858 0.77997581]
```



```
[0.27259261 0.27646426 0.80187218 0.95813935 0.87593263]
[0.19151945 0.62210877 0.43772774 0.78535858 0.77997581]
```

Generating Bernoulli Random Variables

A *Bernoulli random variable* (formally defined in Chapter 3) takes the value 1 with probability p and 0 with probability $1 - p$. Since `rand()` produces a $\mathcal{U}(0, 1)$ value, the comparison `rand() <= p` returns True (which Python treats as 1) with probability p . We can use this to simulate coin flips and other binary outcomes.

Generating Bernoulli random variables with $p = 0.3$

LLM Prompt > Write Python code that generates 20 Bernoulli random variables with $p = 0.3$ by comparing uniform random numbers to p , and counts how many are 1.

```
[0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0]
Number of 1's: 5 out of 20
```

COUNTING, INCLUSION-EXCLUSION, AND INDEPENDENCE

We continue the study of basic probability by reviewing basics of combinatorics, namely **counting techniques**. These approaches are needed when considering **equiprobable probability spaces**, and related cases. We also study the probability of a union of an arbitrary finite number of events via the so-called **inclusion-exclusion formula**. Finally we define the notion of **independent events**.

2.1 Equiprobable Probability Spaces

For a set S , we write $|S|$ for the **cardinality** of the set, which counts the number of elements in the set. When the set is finite, this is a non-negative integer. For example, $|\emptyset| = 0$ and $|\{a, b, c\}| = 3$.

The most basic way one can think of probability—but certainly not the only way—is to assume that each outcome in the sample space Ω is equally likely to occur. Such spaces are called **equiprobable** probability spaces. When $|\Omega|$ is finite and all outcomes are equally likely, the probability of an event A is given by:

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} \quad (2.1)$$

where $|A|$ denotes the number of outcomes in A and $|\Omega|$ the total number of outcomes in the sample space. This simple formula reduces computing probabilities to a *counting* problem: determine $|A|$ and $|\Omega|$.

Example (Rolling a Fair Die). Consider the experiment of rolling a fair six-sided die. The sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let A be the event of rolling an even number, so $A = \{2, 4, 6\}$. Then

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{3}{6} = \frac{1}{2}.$$

Example (Flipping Two Coins). Suppose we flip two fair coins. The sample space is

$$\Omega = \{HH, HT, TH, TT\},$$

so $|\Omega| = 4$. Let A be the event that at least one coin shows heads. Then $A = \{HH, HT, TH\}$ and

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{3}{4}.$$

These examples had small sample spaces where counting $|A|$ and $|\Omega|$ was straightforward — we could simply list all outcomes. But what happens when the sample space is large? The next example shows that even a natural equiprobable model can lead to counting challenges that require systematic techniques.

Example (The Birthday Problem). Suppose there are n people in a room. Assuming that each person's birthday is equally likely to fall on any of the 365 days of the year (ignoring leap years), what is the probability that at least two people share a birthday?

The sample space Ω consists of all possible birthday assignments for the n people. Since each person can have any of 365 birthdays, $|\Omega| = 365^n$.

Let A be the event that at least two people share a birthday. It is easier to compute $\mathbb{P}(A^c)$, the probability that *all* n birthdays are different. For all birthdays to be distinct, the first person can have any of 365 days, the second any of the remaining 364, and so on:

$$|A^c| = 365 \cdot 364 \cdots (365 - n + 1) = \frac{365!}{(365 - n)!}.$$

The quantity $\frac{365!}{(365-n)!}$ is a *permutation count*, which we study in detail in Section 2.2. One can also arrive at this result using conditional probability and the product rule (discussed in Chapter 4): the probability that all birthdays differ is $\frac{365}{365} \cdot \frac{364}{365} \cdots \frac{365-n+1}{365}$. Either way,

$$\mathbb{P}(A) = 1 - \mathbb{P}(A^c) = 1 - \frac{365!}{(365 - n)! \cdot 365^n}.$$

Perhaps surprisingly, for $n = 23$ we already get $\mathbb{P}(A) \approx 0.507$ — in a group of just 23 people, there is a better than even chance that two share a birthday! With $n = 50$ people the probability exceeds 0.97.

The following Python code computes $\mathbb{P}(A)$ as a function of n and plots the result (Figure 2.1).

Birthday problem: probability of a shared birthday

LLM Prompt > Write Python code that computes and plots the probability that at least two out of n people share a birthday, for $n = 1, \dots, 70$. Highlight the point $n = 23$ where the probability first exceeds 0.5.

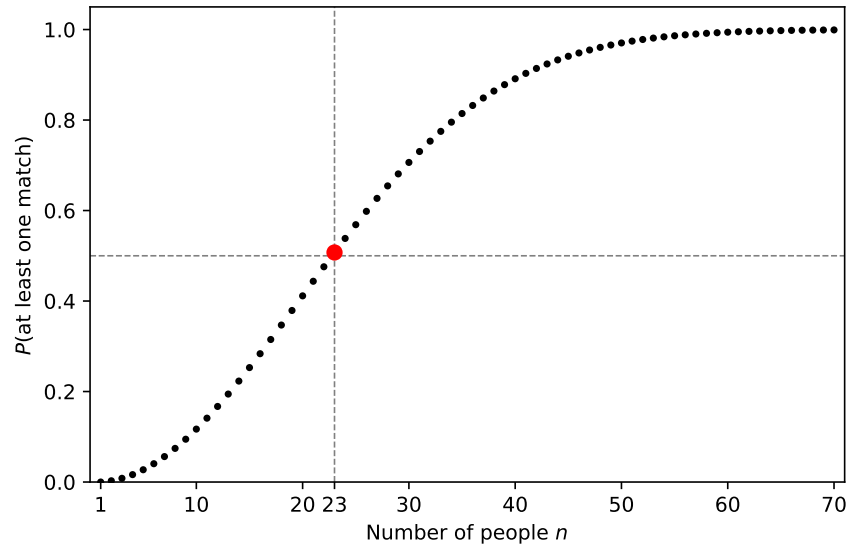


Figure 2.1: Probability that at least two out of n people share a birthday, as a function of n . The dashed lines mark $n = 23$, where the probability first exceeds 0.5.

The birthday problem illustrates that even in an equiprobable probability space, computing $|A|$ and $|\Omega|$ can be non-trivial when the sample space is large.

Uniform Continuous Sampling

The equilikely principle above applies when the sample space Ω is finite. However, a similar idea works when Ω is a continuous region — a line segment, a region in the plane, a volume in three-dimensional space, and so on — provided there is a well-defined notion of “size” (length, area, volume, etc.). We will return to continuous distributions and random variables in Chapter 3 and beyond; for now, we simply note that one can sample a point “uniformly at random” from a region Ω , meaning that every position in Ω is equally likely.

In this setting, the probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ is specified as follows:

- Ω is the region from which we sample (e.g., a disk, a rectangle, an interval).
- $\mathcal{F}$ is a suitable collection of subsets of Ω (for our purposes, any “reasonable” geometric region is an event).
- $\mathbb{P}$ assigns probabilities via the ratio of sizes:

$$\mathbb{P}(A) = \frac{\text{Size}(A)}{\text{Size}(\Omega)},$$

where $\text{Size}(\cdot)$ denotes length, area, or volume as appropriate.

This is the continuous analogue of $\mathbb{P}(A) = |A|/|\Omega|$: cardinality is replaced by a geometric measure of size.

Example (Uniform Point in an Annulus). Suppose we select a point uniformly at random from the annulus (ring-shaped region)

$$\Omega = \{(x, y) : 1 \leq x^2 + y^2 \leq 4\},$$

that is, the region between the circles of radius 1 and 2 centred at the origin. What is the probability that the point lies in the upper half-plane ($y > 0$)?

Let $A = \{(x, y) \in \Omega : y > 0\}$ be the upper half of the annulus. By symmetry, the upper half has exactly half the area of Ω . We can verify: the area of the full annulus is $\pi \cdot 2^2 - \pi \cdot 1^2 = 3\pi$, and the upper half has area $3\pi/2$. Hence,

$$\mathbb{P}(A) = \frac{\text{Area}(A)}{\text{Area}(\Omega)} = \frac{3\pi/2}{3\pi} = \frac{1}{2}.$$

As a follow-up: what is the probability that the point is closer to the outer circle than to the inner circle? The point (x, y) is at distance $r = \sqrt{x^2 + y^2}$ from the origin, and it is closer to the outer circle (radius 2) than to the inner circle (radius 1) when $2 - r < r - 1$, i.e., $r > 3/2$. So we need $B = \{(x, y) \in \Omega : x^2 + y^2 > 9/4\}$. The area of B is $\pi \cdot 2^2 - \pi \cdot (3/2)^2 = 4\pi - 9\pi/4 = 7\pi/4$, giving

$$\mathbb{P}(B) = \frac{7\pi/4}{3\pi} = \frac{7}{12}.$$

In the next section we develop systematic counting methods — permutations and combinations — that allow us to tackle problems with large finite sample spaces efficiently.

2.2 Counting Techniques and Combinatorics

Counting is not always easy. Let us first look at some examples:

1. A multiple choice form has 20 questions; each question has 3 choices. In how many possible ways can the exam be completed? **Answer: 3,486,784,401.**
2. Consider a horse race with 8 horses. How many ways are there to gamble on the placings (1st, 2nd, 3rd). **Answer: 336.**
3. Jessica has a collection of 20 CDs¹, she wants to take 3 of them to work. How many possibilities does she have? **Answer: 1140.**
4. How many different throws are possible with 3 identically looking dice? Here a throw is considered a result on the three dice, e.g., (1, 1, 5), (2, 3, 6), (4, 4, 4), etc. **Answer: 56.**

Solving counting problems requires experience and *practice*, and some remain exceedingly hard. Many, however, can be cast into a simple framework.

¹A CD (Compact Disc) was a shiny, round plastic disc used to store music. People would collect hundreds of them in enormous shelving units, only to later rip them all to MP3s and wonder why they kept the shelves.

Factorials and Binomial Coefficients

The **factorial** of a non-negative integer n is

$$n! = n \times (n-1) \times \cdots \times 2 \times 1, \quad 0! = 1. \quad (2.2)$$

Equivalently, $n! = n \cdot (n-1)!$ for $n \geq 1$. The factorial $n!$ counts the number of ways to arrange n distinct objects in a row.

The **binomial coefficient** $\binom{n}{k}$ (read “ n choose k ”) is defined for integers $0 \leq k \leq n$ by

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}. \quad (2.3)$$

Basic properties: $\binom{n}{0} = \binom{n}{n} = 1$; symmetry: $\binom{n}{k} = \binom{n}{n-k}$; Pascal’s rule: $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$; binomial theorem: $(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$; setting $a = b = 1$: $\sum_{k=0}^n \binom{n}{k} = 2^n$.

The Four Cases of Elementary Counting

Fortunately, many counting problems can be cast into the simple framework of drawing balls from an urn (Figure 2.2a). Consider an urn with n different balls, numbered $1, \dots, n$ from which k balls are drawn. This can be done in a number of different ways. First, the balls can be drawn one-by-one, or one could draw all the k balls at the same time. In the first case the **order** in which the balls are drawn can be noted, in the second case that is not possible. In the latter case we can (and will) still assume the balls are drawn one-by-one, but that the order is not noted.

Second, once a ball is drawn, it can either be put back into the urn (after the number is recorded), or left out. This is called, respectively, drawing with and without **replacement**. All in all there are 4 possible experiments: (ordered, with replacement), (ordered, without replacement), (unordered, without replacement) and (unordered, with replacement). An equivalent viewpoint is to distribute k balls into n labelled bins (Figure 2.2b).

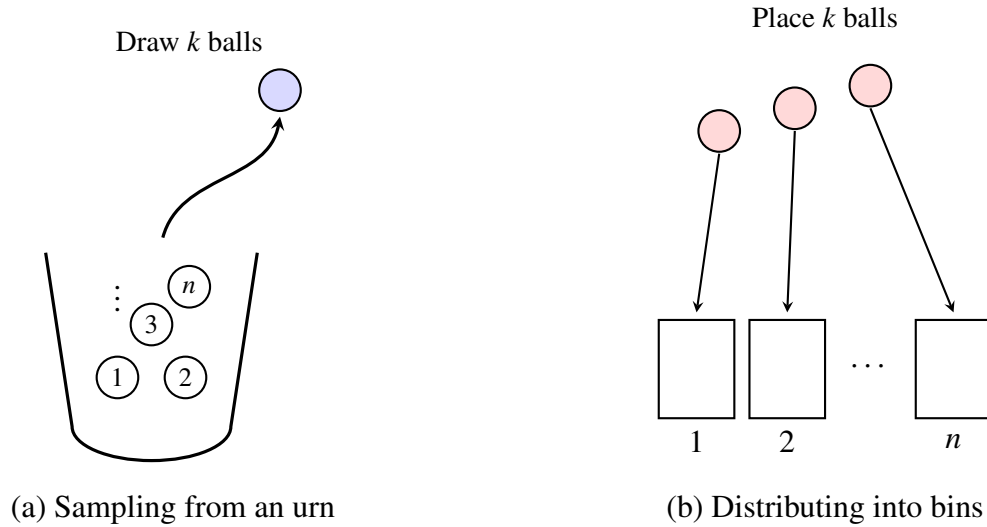


Figure 2.2: Two equivalent views of counting problems. (a) Draw k balls from an urn containing n distinguishable balls; the choices of replacement and ordering yield four counting cases. (b) Equivalently, place k balls into n labelled bins; the choices of ball distinguishability and bin capacity yield the same four cases.

The art is to recognise a seemingly unrelated counting problem as one of these four urn problems. For the 4 examples above we have the following

1. Example 1 (multiple choice) can be viewed as drawing $k = 20$ balls from an urn containing $n = 3$ balls, noting the order, and with replacement. Equivalently, place $k = 20$ different-colour balls into $n = 3$ bins with no limit per bin.
2. Example 2 (horse race) is equivalent to drawing $k = 3$ balls from an urn containing $n = 8$ balls, noting the order, and without replacement. Equivalently, place $k = 3$ different-colour balls into $n = 8$ bins with at most one ball per bin.
3. Example 3 (CDs) is a case of taking $k = 3$ balls from an urn containing $n = 20$ balls, not noting the order, and without replacement. Equivalently, place $k = 3$ same-colour balls into $n = 20$ bins with at most one ball per bin.
4. Example 4 (dice) is a case of drawing $k = 3$ balls from an urn containing $n = 6$ balls, not noting the order, and with replacement. Equivalently, place $k = 3$ same-colour balls into $n = 6$ bins with no limit per bin.

Before we proceed it is important to introduce a notation that reflects whether the outcomes/arrangements are ordered or not. In particular, we denote ordered arrangements by *vectors*, e.g., $(1, 2, 3) \neq (3, 2, 1)$, and unordered arrangements by *sets*, e.g., $\{1, 2, 3\} = \{3, 2, 1\}$. Each case can be viewed from two equivalent perspectives: the **sampling from an urn** view and the **balls-into-bins** view (distributing balls into labelled bins). The table below shows both; bold labels give the sampling perspective and parenthetical annotations give the balls-into-bins perspective. Different-colour balls correspond to ordered sampling (each ball carries its draw number), same-colour to

unordered; at most one ball per bin corresponds to sampling without replacement, no limit to with replacement.

	With Replacement (no limit per bin)	Without Replacement (≤ 1 ball per bin)
Ordered (different-colour balls)	Case I: Sequences n^k	Case II: Permutations $\frac{n!}{(n-k)!}$
Unordered (same-colour balls)	Case IV: Multisets $\binom{n+k-1}{k}$	Case III: Combinations $\binom{n}{k}$

Each case is treated in its own subsection below. We illustrate each with $n = 4$ and $k = 3$, then state the general formula.

Case I: Sequences (Ordered, With Replacement)

We draw a ball, note its number, put it back, and repeat k times. With $n = 4, k = 3$, outcomes such as $(1, 1, 1), (4, 1, 2), (2, 3, 2), \dots$ are possible. There are $k = 3$ positions to fill, each from $\{1, 2, 3, 4\}$, giving $4 \times 4 \times 4 = 4^3 = 64$ outcomes. This is illustrated in Figure 2.3.

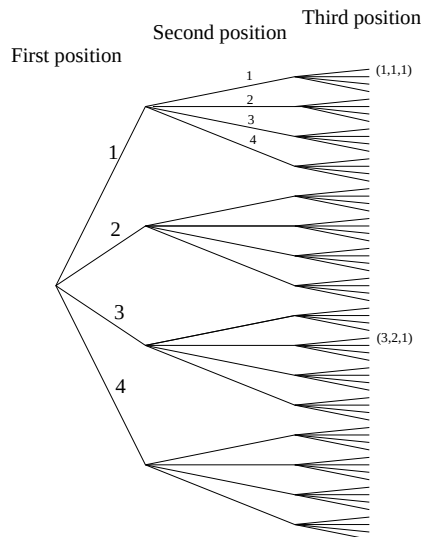


Figure 2.3: Combinatorial tree with duplication

In general, the number of ordered arrangements of k items chosen from $\{1, \dots, n\}$ with replacement is n^k .

Case II: Permutations (Ordered, Without Replacement)

Now balls are not replaced. With $n = 4, k = 3$, outcomes such as $(1, 4, 2), (3, 2, 1), \dots$ are possible, but $(1, 1, 2)$ is not. Such an ordered arrangement is called a **permutation** of size k from $\{1, \dots, n\}$.

The first position has 4 choices, the second 3, the third 2, giving $4 \times 3 \times 2 = 24$ permutations, as shown in Figure 2.4.

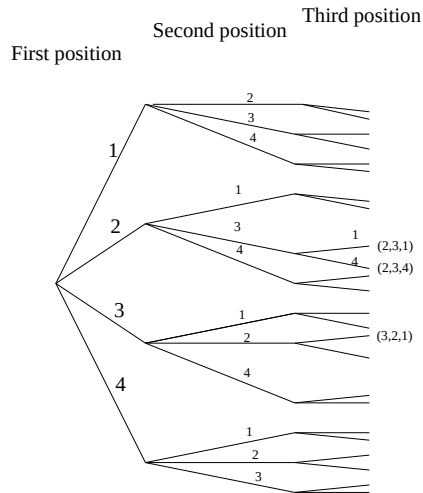


Figure 2.4: Combinatorial tree without duplication

In general, the number of permutations of size k from $\{1, \dots, n\}$ is

$$n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!} . \quad (2.4)$$

When $k = n$, this gives $n!$, the number of ways to arrange all n items.

Case III: Combinations (Unordered, Without Replacement)

We draw k items without replacement but do not record the order (equivalently, grab k balls at once). With $n = 4, k = 3$, outcomes are sets such as $\{1, 2, 4\}, \{1, 2, 3\}$, etc. Each such set corresponds to $k! = 6$ permutations (e.g., $\{1, 2, 4\}$ gives $(1, 2, 4), (1, 4, 2), (2, 1, 4), (2, 4, 1), (4, 1, 2), (4, 2, 1)$). An unordered arrangement without replication is called a **combination** of size k from $\{1, \dots, n\}$. Since there are $\frac{n!}{(n-k)!}$ permutations in total, the number of combinations is $\frac{n!}{(n-k)!} \cdot \frac{1}{k!} = \binom{n}{k}$. For $n = 4, k = 3$: $24/6 = 4$.

In general, the number of combinations of size k from $\{1, \dots, n\}$ is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} . \quad (2.5)$$

Remark on notation. Some textbooks write nP_k for the number of permutations $\frac{n!}{(n-k)!}$ and nC_k for the binomial coefficient $\binom{n}{k}$. We will not use this notation.

Case IV: Multisets (Unordered, With Replacement)

With $n = 4, k = 3$, outcomes are multisets such as $\{3, 3, 4\}, \{1, 2, 4\}, \{2, 2, 2\}$, etc. To count these, represent each outcome by a vector $(x_1, \dots, x_n)$ recording how many times each element appears: e.g., $\{3, 3, 4\} \leftrightarrow (0, 0, 2, 1)$ and $\{1, 2, 4\} \leftrightarrow (1, 1, 0, 1)$. Equivalently, place $k = 3$ identical balls into $n = 4$ bins. To count the arrangements, distribute $n - 1 = 3$ separators and $k = 3$ balls over $n - 1 + k = 6$ positions, as in Figure 2.5.

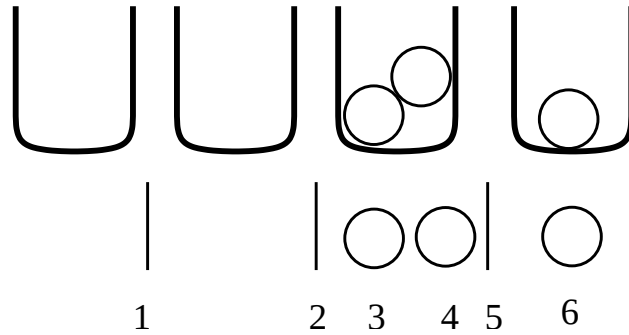


Figure 2.5: Distributing $k = 3$ balls over $n = 4$ bins using $n - 1 = 3$ separators and $k = 3$ balls in $n - 1 + k = 6$ positions.

Each arrangement of balls and separators corresponds to a unique multiset: the balls between the i -th and $(i + 1)$ -th separator (or boundary) indicate how many times element i was chosen. For example, the arrangement shown in Figure 2.5 corresponds to the multiset $\{3, 3, 4\}$. The number of ways to choose k positions for the balls out of $n - 1 + k$ positions is $\binom{n+k-1}{k}$.

In general, the number of multisets of size k from $\{1, \dots, n\}$ is $\binom{n+k-1}{k}$.

Returning to our original four problems, we can now solve them easily:

1. The total number of ways the exam can be completed is $3^{20} = 3,486,784,401$.
2. The number of placings is $\frac{8!}{5!} = 8 \times 7 \times 6 = 336$.
3. The number of possible combinations of CDs is $\binom{20}{3} = 1140$.
4. The number of different throws with three dice is $\binom{6+3-1}{3} = \binom{8}{3} = 56$.

The Size of the Power Set

Recall from Chapter 1 that the **power set** $\mathcal{P}(S)$ of a set S is the collection of all subsets of S . If S has n elements, then $|\mathcal{P}(S)| = 2^n$. To see why, note that forming a subset of $S = \{s_1, \dots, s_n\}$ amounts to making n independent yes/no decisions — for each element, either include it or not.

There are 2 choices per element and n elements, giving 2^n subsets in total. This is the reason 2^S is sometimes used as notation for the power set.

The binomial coefficient $\binom{n}{k}$ counts the number of subsets of size k from a set of n elements. Since every subset has some size $k \in \{0, 1, \dots, n\}$, the total number of subsets is

$$\sum_{k=0}^n \binom{n}{k} = 2^n, \quad (2.6)$$

which is exactly the identity obtained by setting $a = b = 1$ in the binomial theorem. In other words, the binomial theorem $(1 + 1)^n = \sum_{k=0}^n \binom{n}{k}$ is a counting statement: the left side counts all subsets by the multiplication principle, while the right side counts them grouped by size.

More examples

Here are some more examples. Not all problems can be directly related to the 4 problems above. Some require additional reasoning. However, the counting principles remain the same.

1. In how many ways can the numbers $1, \dots, 5$ be arranged, such as 13524, 25134, etc?

Answer: $5! = 120$.

2. How many different arrangements are there of the numbers $1, 2, \dots, 7$, such that the first 3 numbers are 1, 2, 3 (in any order) and the last 4 numbers are 4, 5, 6, 7 (in any order)?

Answer: $3! \times 4!$.

3. How many different arrangements are there of the word “arrange”, such as “aarrnge”, “ar-rngea”, etc?

Answer: Convert this into a ball drawing problem with 7 balls, numbered $1, \dots, 7$. Balls 1 and 2 correspond to ‘a’, balls 3 and 4 to ‘r’, ball 5 to ‘n’, ball 6 to ‘g’ and ball 7 to ‘e’. The total number of permutations of the numbers is $7!$. However, since, for example, $(1, 2, 3, 4, 5, 6, 7)$ is identical to $(2, 1, 3, 4, 5, 6, 7)$ (when substituting the letters back), we must divide $7!$ by $2! \times 2!$ to account for the 4 ways the two ‘a’s and ‘r’s can be arranged. So the answer is $7!/4 = 1260$.

4. An urn has 1000 balls, labelled 000, 001, $\dots$, 999. How many balls are there that have all numbers in ascending order (for example 047 and 489, but not 033 or 321)?

Answer: There are $10 \times 9 \times 8 = 720$ balls with different numbers. Each triple of numbers can be arranged in $3! = 6$ ways, and only one of these is in ascending order. So the total number of balls in ascending order is $720/6 = 120$.

5. In a group of 20 people each person has a different birthday. How many different arrangements of these birthdays are there (assuming each year has 365 days)?

Answer: $\frac{365!}{345!}$.

Back to Equiprobable Probability Spaces

Once we've learned how to count, we can apply the equilikely principle to calculate probabilities. Here are some examples:

1. What is the probability that out of a group of 40 people all have different birthdays?

Answer: Choosing the birthdays is like choosing 40 balls with replacement from an urn containing the balls $1, \dots, 365$. Thus, our sample space Ω consists of vectors of length 40, whose components are chosen from $\{1, \dots, 365\}$. There are $|\Omega| = 365^{40}$ such vectors possible, and all are *equally likely*. Let A be the event that all 40 people have different birthdays. Then, $|A| = 365!/325!$. It follows that $\mathbb{P}(A) = |A|/|\Omega| \approx 0.109$, so not very big!

2. What is the probability that in 10 tosses with a fair coin we get exactly 5 Heads and 5 Tails?

Answer: Here Ω consists of vectors of length 10 consisting of 1s (Heads) and 0s (Tails), so there are 2^{10} of them, and all are *equally likely*. Let A be the event of exactly 5 heads. We must count how many binary vectors there are with exactly 5 1s. This is equivalent to determining in how many ways the positions of the 5 1s can be chosen out of 10 positions, that is, $\binom{10}{5}$. Consequently, $\mathbb{P}(A) = \binom{10}{5}/2^{10} = 252/1024 \approx 0.25$.

3. We draw at random 13 cards from a full deck of cards. What is the probability that we draw 4 Hearts and 3 Diamonds?

Answer: Give the cards a number from 1 to 52. Suppose 1–13 is Hearts, 14–26 is Diamonds, etc. Ω consists of unordered sets of size 13, without repetition, e.g., $\{1, 2, \dots, 13\}$. There are $|\Omega| = \binom{52}{13}$ of these sets, and they are all equally likely. Let A be the event of 4 Hearts and 3 Diamonds. To form A we have to choose 4 Hearts out of 13, and 3 Diamonds out of 13, followed by 6 cards out of 26 Spades and Clubs. Thus, $|A| = \binom{13}{4} \times \binom{13}{3} \times \binom{26}{6}$. So that $\mathbb{P}(A) = |A|/|\Omega| \approx 0.074$.

2.3 Inclusion-Exclusion Principle

The inclusion-exclusion principle is a fundamental concept in probability and combinatorics, used to calculate the probability of the union of several events. It accounts for the fact that when we simply add the probabilities of individual events, we may be overcounting the outcomes that occur in the intersection of these events. In a combinatorics course (e.g. MATH1061/MATH1081 at UQ), this principle is typically stated in terms of the cardinality of sets. Here we state it in terms of probability. Note that it applies to *any* probability space, not just equiprobable ones.

Two or Three Events

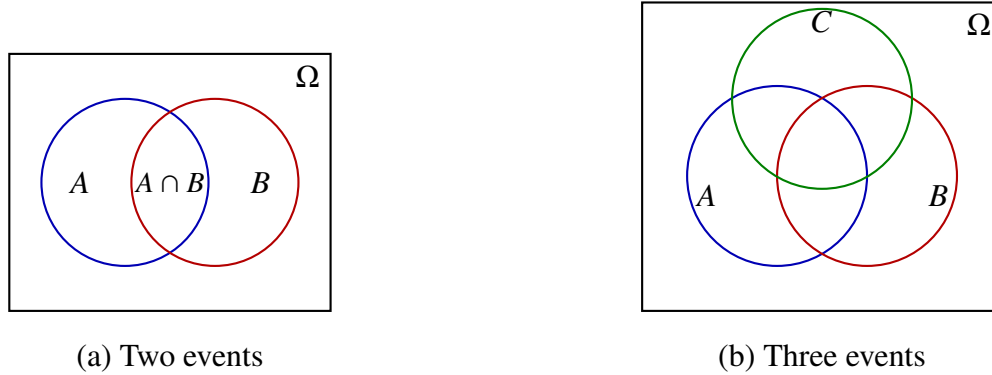


Figure 2.6: Venn diagrams for the inclusion-exclusion principle. (a) Two events A and B . (b) Three events A , B , and C .

For two events A and B (Figure 2.6a), the principle states:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \quad (2.7)$$

In words: to find the probability that at least one of A or B occurs, we add their individual probabilities, then subtract the probability of their overlap, which was counted twice.

For three events A , B , and C (Figure 2.6b), the principle extends to:

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C). \quad (2.8)$$

In words: we first add the three individual probabilities, then subtract the three pairwise overlaps (each of which was double-counted), then add back the triple overlap (which was subtracted too many times).

Note that in both cases, if the events are **disjoint** (i.e., all intersections have probability zero), the formulas reduce to simply adding the individual probabilities. This is just Axiom 3 (finite additivity) from Chapter 1.

The General Case: n Events

For n events $A_1, A_2, \dots, A_n$, the inclusion-exclusion principle states:

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}). \quad (2.9)$$

The subscript $1 \leq i_1 < i_2 < \dots < i_k \leq n$ under the inner sum is one way to denote summation over all $\binom{n}{k}$ subsets of size k from $\{1, \dots, n\}$.

For $n = 2$, the sum has terms with $k = 1$ and $k = 2$: the $k = 1$ terms give $\mathbb{P}(A_1) + \mathbb{P}(A_2)$ and the single $k = 2$ term gives $-\mathbb{P}(A_1 \cap A_2)$, recovering the two-event formula above.

For $n = 3$, the $k = 1$ terms give the three individual probabilities, the $\binom{3}{2} = 3$ terms with $k = 2$ give the three pairwise intersections (with a minus sign), and the single $k = 3$ term gives $+\mathbb{P}(A_1 \cap A_2 \cap A_3)$, recovering the three-event formula above.

Claim 4: Inclusion-Exclusion Principle — For any n events $A_1, \dots, A_n$, the inclusion-exclusion formula (2.9) holds.

Proof: We proceed by induction on n .

Base case ($n = 1$): The formula reduces to $\mathbb{P}(A_1) = \mathbb{P}(A_1)$, which holds trivially.

Inductive step: Assume (2.9) holds for any collection of n events. We wish to show it holds for $n + 1$ events $A_1, \dots, A_n, A_{n+1}$. First, group the union as two pieces:

$$\bigcup_{i=1}^{n+1} A_i = \left(\bigcup_{i=1}^n A_i \right) \cup A_{n+1}. \quad (2.10)$$

Applying the two-event inclusion-exclusion formula to these two pieces gives

$$\mathbb{P}\left(\bigcup_{i=1}^{n+1} A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) + \mathbb{P}(A_{n+1}) - \mathbb{P}\left(\left(\bigcup_{i=1}^n A_i\right) \cap A_{n+1}\right). \quad (2.11)$$

We now handle each term on the right:

- The first term $\mathbb{P}(\bigcup_{i=1}^n A_i)$ is a union of n events, so the inductive hypothesis applies directly, expanding it via (2.9).
- The second term $\mathbb{P}(A_{n+1})$ is already a single probability.
- For the third term, use the distributive law of intersection over union:

$$\left(\bigcup_{i=1}^n A_i\right) \cap A_{n+1} = \bigcup_{i=1}^n (A_i \cap A_{n+1}). \quad (2.12)$$

This is again a union of n events (namely $A_1 \cap A_{n+1}, A_2 \cap A_{n+1}, \dots, A_n \cap A_{n+1}$), so the inductive hypothesis applies once more.

Substituting the inductive expansions of the first and third terms, we obtain

$$\mathbb{P}\left(\bigcup_{i=1}^{n+1} A_i\right) = \underbrace{\sum_{k=1}^n (-1)^{k+1} \sum_{\substack{S \subseteq \{1, \dots, n\} \\ |S|=k}} \mathbb{P}\left(\bigcap_{i \in S} A_i\right)}_{\text{inductive hypothesis on } A_1, \dots, A_n} + \mathbb{P}(A_{n+1}) \quad (2.13)$$

$$- \underbrace{\sum_{k=1}^n (-1)^{k+1} \sum_{\substack{S \subseteq \{1, \dots, n\} \\ |S|=k}} \mathbb{P}\left(\bigcap_{i \in S} A_i \cap A_{n+1}\right)}_{\text{inductive hypothesis on } A_1 \cap A_{n+1}, \dots, A_n \cap A_{n+1}}. \quad (2.14)$$

To verify that this equals (2.9) for $n + 1$ events, we must check that every subset $T \subseteq \{1, \dots, n + 1\}$ of size j contributes $(-1)^{j+1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_j})$. There are two cases:

1. *Subsets not containing $n + 1$:* If $T \subseteq \{1, \dots, n\}$ with $|T| = j$, then $\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_j})$ appears in the first sum with $k = j$ and sign $(-1)^{j+1}$. ✓

2. *Subsets containing $n + 1$* : Write $T = S \cup \{n + 1\}$ where $S \subseteq \{1, \dots, n\}$ has $|S| = j - 1$.

- If $j = 1$ (so $S = \emptyset$ and $T = \{n + 1\}$), the term $\mathbb{P}(A_{n+1})$ appears as the second term above with sign $+1 = (-1)^{1+1}$. ✓
- If $j \geq 2$ (so $|S| = j - 1 \geq 1$), the term $\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_{j-1}} \cap A_{n+1})$ appears in the third sum with $k = j - 1$. Its sign is $-(-1)^{(j-1)+1} = (-1)^{j+1}$. ✓

In every case the sign is $(-1)^{j+1}$, so combining all terms gives exactly (2.9) for $n + 1$ events. $\square$

Example (Bags in the Locker Room). Suppose n students each place their bag in a pile in a dark locker room. They then each pick up a bag uniformly at random (without replacement), resulting in a random permutation of the n bags. What is the probability that *no one* picks up their own bag?

The sample space consists of all $n!$ permutations, each equally likely. As a warm-up, note that there is exactly one permutation in which *everyone* has their correct bag (the identity permutation), so the probability that everyone is correct is simply $1/n!$.

The complementary question — that *no one* is correct — is harder, and this is where inclusion-exclusion comes in. Let A_i be the event that student i picks up their own bag. We want

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i^c\right) = 1 - \mathbb{P}\left(\bigcup_{i=1}^n A_i\right), \quad (2.15)$$

i.e., the probability that none of the events $A_1, \dots, A_n$ occurs. We compute $\mathbb{P}(\bigcup_{i=1}^n A_i)$ using inclusion-exclusion (2.9). The key observation is that for any subset of k students $\{i_1, \dots, i_k\}$, the event $A_{i_1} \cap \dots \cap A_{i_k}$ (all k of them have their own bag) requires the remaining $n - k$ bags to be distributed among the remaining $n - k$ students, giving $(n - k)!$ favourable permutations out of $n!$:

$$\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) = \frac{(n - k)!}{n!}. \quad (2.16)$$

Since this depends only on k (not on which k students), and there are $\binom{n}{k}$ such subsets, the inclusion-exclusion formula gives

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \frac{(n - k)!}{n!} = \sum_{k=1}^n (-1)^{k+1} \frac{1}{k!}. \quad (2.17)$$

Therefore, the probability that no one has their own bag is

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i^c\right) = 1 - \sum_{k=1}^n (-1)^{k+1} \frac{1}{k!} = \sum_{k=0}^n \frac{(-1)^k}{k!}. \quad (2.18)$$

Recalling the Taylor series $e^{-1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$, we see that as $n \rightarrow \infty$,

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i^c\right) \rightarrow e^{-1} \approx 0.368. \quad (2.19)$$

The following code illustrates this result for $n = 3, \dots, 7$, plotting the exact formula, the asymptotic value $1/e$, and a Monte Carlo estimate from 10^6 simulated permutations.

Probability that no one has their own bag: exact, asymptotic, and Monte Carlo

LLM Prompt > Suppose n students each place their bag in a pile and then each pick up a bag uniformly at random. For each $n = 3, 4, \dots, 7$: (1) compute the exact probability that no student gets their own bag, $\sum_{k=0}^n (-1)^k / k!$, and (2) estimate it by simulating 10^6 random permutations and counting the fraction where no element is in its original position. Plot the exact and simulated probabilities on the same axes, and include a horizontal line at the asymptotic value $1/e$.

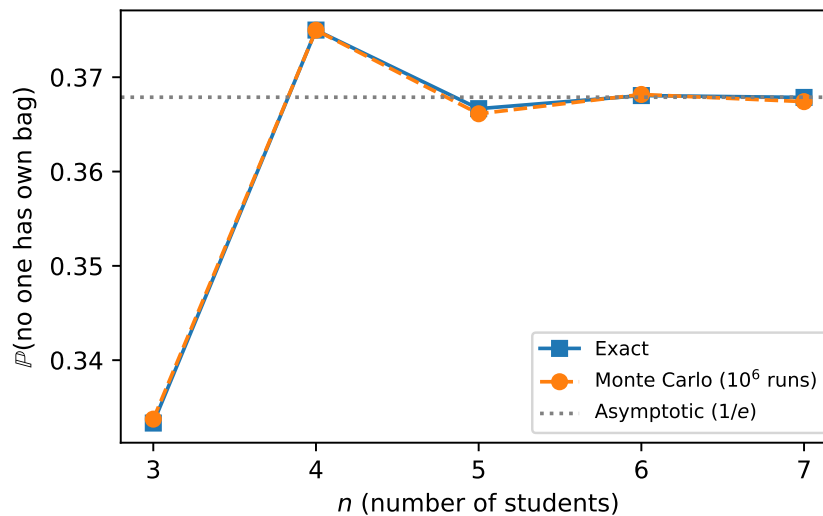


Figure 2.7: Probability that no one has their own bag for $n = 3, \dots, 7$: exact formula, Monte Carlo estimate (10^6 runs), and asymptotic value $1/e \approx 0.368$.

2.4 Independence of Events

Independence is a very important concept in probability and statistics. Informally, two events are independent if the occurrence of one has no bearing on the likelihood of the other. The formal definition requires only the multiplication rule:

$$A, B \text{ independent} \Leftrightarrow \mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B)$$

We will revisit independence using the language of conditional probability in Chapter 4, and in the context of random variables starting in Chapter 3. We can extend the definition to arbitrarily many events:

Definition : Independent Events — The events $A_1, A_2, \dots$, are said to be **(mutually) independent** if for any k and any choice of distinct indices $i_1, \dots, i_k$,

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = \mathbb{P}(A_{i_1}) \mathbb{P}(A_{i_2}) \dots \mathbb{P}(A_{i_k}). \quad (2.20)$$

Remark (Independence in Practice). Independence enters in two ways. First, when *constructing* a probability model — for example, modelling repeated trials as independent. Second, as a *derived property* that must be proved — for example, it can be shown that the sample mean and sample variance from a normal sample are independent (see Chapter 10), a result that is far from obvious and requires careful analysis.

Example (A coin and a die). A fair coin and a fair six-sided die are thrown together. Since the two objects do not physically affect each other, we expect the coin outcome and the die outcome to be independent. We construct the probability space in two ways and show they agree.

Construction 1 (Equiprobable space). The sample space lists all $2 \times 6 = 12$ outcomes:

$$\Omega = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\} . \quad (2.21)$$

Since both the coin and the die are fair, every outcome is equally likely, so $\mathbb{P}(\{\omega\}) = 1/12$ for each $\omega \in \Omega$. Now let $A = \{H1, H2, H3, H4, H5, H6\}$ be the event that the coin shows heads, and $B = \{H5, H6, T5, T6\}$ the event that the die shows 5 or more. Then

$$\mathbb{P}(A) = \frac{6}{12} = \frac{1}{2}, \quad \mathbb{P}(B) = \frac{4}{12} = \frac{1}{3}, \quad \mathbb{P}(A \cap B) = \mathbb{P}(\{H5, H6\}) = \frac{2}{12} = \frac{1}{6} . \quad (2.22)$$

Since $\mathbb{P}(A \cap B) = \frac{1}{6} = \frac{1}{2} \cdot \frac{1}{3} = \mathbb{P}(A)\mathbb{P}(B)$, the events A and B are independent. One can verify that the same holds for any event defined purely in terms of the coin and any event defined purely in terms of the die.

Construction 2 (Independence by design). Alternatively, we can build the model from the start using independence. Define the coin probability space with $\Omega_1 = \{H, T\}$ and $\mathbb{P}_1(\{H\}) = \mathbb{P}_1(\{T\}) = 1/2$, and the die probability space with $\Omega_2 = \{1, 2, 3, 4, 5, 6\}$ and $\mathbb{P}_2(\{k\}) = 1/6$ for each k . The joint sample space is the Cartesian product $\Omega = \Omega_1 \times \Omega_2$, and we *define* the joint probability measure by the product rule:

$$\mathbb{P}(\{(c, d)\}) = \mathbb{P}_1(\{c\}) \cdot \mathbb{P}_2(\{d\}) , \quad c \in \Omega_1, d \in \Omega_2 . \quad (2.23)$$

For example, $\mathbb{P}(\{(H, 4)\}) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$. This gives each of the 12 outcomes probability $1/12$, exactly as in Construction 1. Independence of coin events and die events is guaranteed by construction: for any $A_1 \subseteq \Omega_1$ and $A_2 \subseteq \Omega_2$,

$$\mathbb{P}(A_1 \times A_2) = \mathbb{P}_1(A_1) \cdot \mathbb{P}_2(A_2) . \quad (2.24)$$

The two constructions yield the same probability space. Construction 1 starts from an equiprobable space and *proves* independence. Construction 2 starts from separate experiments and *builds* independence into the model via the product rule. The second approach generalises naturally to combining any number of independent experiments.

Independence versus Disjointness

A common misconception is to confuse *independent* events with *disjoint* (mutually exclusive) events. In fact, for events with positive probability, the two notions are *opposites*.

If A and B are disjoint, then $A \cap B = \emptyset$, so $\mathbb{P}(A \cap B) = 0$. But if both $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$, then $\mathbb{P}(A)\mathbb{P}(B) > 0$. Therefore

$$\mathbb{P}(A \cap B) = 0 \neq \mathbb{P}(A)\mathbb{P}(B) > 0, \quad (2.25)$$

which means A and B are *not* independent. In other words, two events with positive probability cannot be both disjoint and independent.

Intuitively, disjoint events are as far from independent as possible: knowing that one occurred tells us the other *certainly did not occur*. Independent events with positive probability, on the other hand, always *overlap* — they must satisfy $\mathbb{P}(A \cap B) > 0$.

Example (Disjoint but Not Independent). Roll a fair die. Let $A = \{1, 2\}$ (roll is at most 2) and $B = \{5, 6\}$ (roll is at least 5). These events are disjoint: $A \cap B = \emptyset$. We have $\mathbb{P}(A) = 1/3$ and $\mathbb{P}(B) = 1/3$, but $\mathbb{P}(A \cap B) = 0 \neq 1/9 = \mathbb{P}(A)\mathbb{P}(B)$. If we know the roll is 5 or 6, there is zero chance it is 1 or 2. The events are not independent.

Pairwise Independence Does Not Imply Mutual Independence

The definition of mutual independence requires the multiplication rule to hold for *every* subcollection of events, not just for pairs. The following example shows that pairwise independence alone is not enough.

Example (Pairwise but Not Mutually Independent). Toss two fair coins. The sample space is $\Omega = \{HH, HT, TH, TT\}$, each with probability $1/4$. Define three events:

$$A = \{\text{first coin is H}\} = \{HH, HT\}, \quad (2.26)$$

$$B = \{\text{second coin is H}\} = \{HH, TH\}, \quad (2.27)$$

$$C = \{\text{both coins show the same face}\} = \{HH, TT\}. \quad (2.28)$$

Each event has probability $1/2$. The pairwise intersections are:

$$A \cap B = \{HH\}, \quad A \cap C = \{HH\}, \quad B \cap C = \{HH\}, \quad (2.29)$$

each with probability $1/4 = \frac{1}{2} \cdot \frac{1}{2}$. So every pair satisfies the multiplication rule: the events are *pairwise independent*.

However, $A \cap B \cap C = \{HH\}$ has probability $1/4$, while $\mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$. Since $1/4 \neq 1/8$, the three events are *not* mutually independent. Intuitively, knowing that A and B both occurred (first coin H, second coin H) tells us with certainty that C occurred (both coins match), so C is not independent of the pair $\{A, B\}$.

DISCRETE RANDOM VARIABLES, THEIR DISTRIBUTIONS, AND MOMENTS

In this chapter we consider the general notion of random variables, focusing on discrete valued random variables. With this we study their distributions, and moment descriptors, including expectation, mean, and variance. In particular, we study families of random variables associated with independent Bernoulli (coin-flip) trials. This includes the binomial, geometric, and negative Binomial distributions. We also visit the Poisson distribution and the hypergeometric distribution.

3.1 Discrete Random Variables and Their Distributions

A **random variable** is a function from a sample space to the real numbers.

Definition : Random Variable — A **random variable** is a (measurable) function $X : \Omega \rightarrow \mathbb{R}$. That is, X assigns a real number $X(\omega)$ to each outcome $\omega \in \Omega$.

Measurability is a technical condition ensuring that the sets below are events (elements of the σ -algebra); we will not dwell on it in this course.

Although random variables are mathematically *functions*, it is often helpful to think of a random variable as a quantity whose value will be revealed once we carry out the experiment. We usually denote random variables with capital letters from the end of the alphabet, e.g. X, Y, Z .

Since X is a function on Ω , any statement about the values of X defines an event — a subset of Ω . For a set $C \subseteq \mathbb{R}$, we write

$$\{X \in C\} := \{\omega \in \Omega : X(\omega) \in C\}. \quad (3.1)$$

This is the set of all outcomes ω for which the value $X(\omega)$ falls in C . Probabilities involving X are therefore ordinary probabilities of events:

$$\mathbb{P}(X \in C) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in C\}). \quad (3.2)$$

Common special cases include $\{X = x\} = \{\omega : X(\omega) = x\}$, $\{X \leq x\} = \{\omega : X(\omega) \leq x\}$, and $\{a < X \leq b\} = \{\omega : a < X(\omega) \leq b\}$. These abbreviations are standard and we use them freely throughout.

Example (Sum of Two Dice). Suppose we toss two fair dice and note their sum. If we observe each die separately, the sample space is $\Omega = \{(i, j) : i, j \in \{1, \dots, 6\}\}$, with $|\Omega| = 36$ equally likely outcomes. The function X defined by $X(i, j) = i + j$ is a random variable mapping each outcome to the corresponding sum, as depicted in Figure 3.1.

All outcomes in the circled set are mapped to 8. This set is precisely the event $\{X = 8\} = \{\omega \in \Omega : X(\omega) = 8\}$. Since it contains 5 outcomes, and all outcomes in Ω are equally likely,

$$\mathbb{P}(X = 8) = \mathbb{P}(\{\omega \in \Omega : X(\omega) = 8\}) = \frac{5}{36}. \quad (3.3)$$

More generally, one can verify that

$$\mathbb{P}(X = x) = \frac{6 - |7 - x|}{36}, \quad x = 2, 3, \dots, 12. \quad (3.4)$$

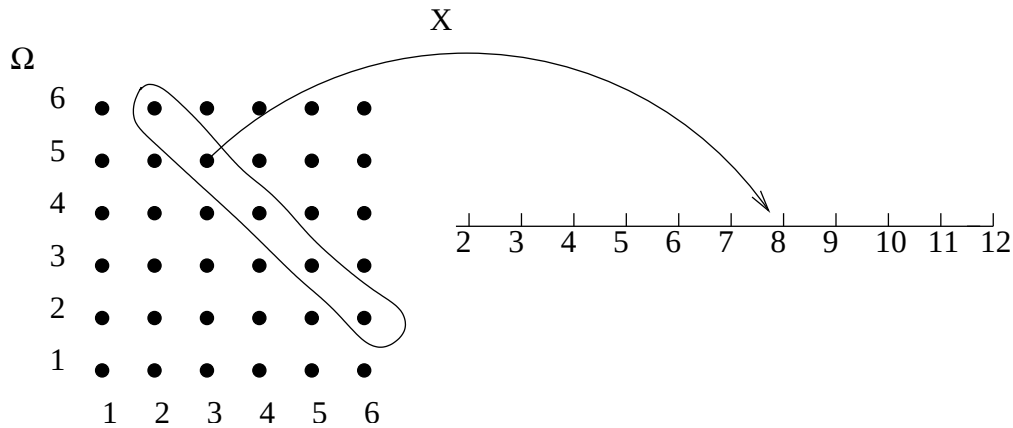


Figure 3.1: A random variable representing the sum of two dice. The function X maps each outcome $(i, j) \in \Omega$ to the sum $i + j$.

We distinguish between two broad types of random variables:

Discrete random variables can only take *isolated* values — their range is a finite or countably infinite subset of $\mathbb{R}$.

Continuous random variables can take values in an *interval* (or more generally a union of intervals), and their distribution is described by a probability density function (see Chapter 5).

The description of continuous random variables here is informal; the formal definition via probability density functions is given in Chapter 5. Some random variables are neither purely discrete nor purely continuous (for example, mixtures of the two types), though we will mostly not encounter these in this course.

In this chapter we focus on discrete random variables. Continuous random variables are treated in Chapter 5. Along the way, we also introduce general concepts — such as the cumulative distribution function, expectation, and variance — that apply to both discrete and continuous random variables.

In this course our random variables take values in $\mathbb{R}$ (or a subset of it); from Chapter 7 onwards we also consider *random vectors* taking values in $\mathbb{R}^n$. However, in more advanced settings — particularly in machine learning — one often considers probability distributions over very complicated spaces. For example, a generative model for images defines a probability distribution over the space of all possible colour images. The core ideas of probability mass functions, densities, and expectations carry over to those settings, but the mathematical framework requires additional machinery beyond the scope of this course.

The Distribution of a Random Variable

The *distribution* of a random variable is a somewhat fluid term: it refers to any complete description of the probabilities of events associated with the random variable. There are several common ways to specify a distribution:

- The **probability mass function** (pmf) — for discrete random variables (this chapter).
- The **probability density function** (pdf) — for continuous random variables (Chapter 5).
- The **cumulative distribution function** (cdf) — works for both discrete and continuous random variables, and is defined below.

There are also many other ways to characterise a distribution, including the hazard rate function, probability generating functions, moment generating functions, and characteristic functions (see Chapter 9), among others. All of these are equivalent in the sense that knowing any one of them (under mild conditions) determines the full distribution.

Example (Sum of Two Dice — Continued). Returning to the random variable $X = i + j$ from Example 3.1, its pmf is

$$f(x) = \mathbb{P}(X = x) = \frac{6 - |7 - x|}{36}, \quad x = 2, 3, \dots, 12. \quad (3.5)$$

Figure 3.2 shows three common ways to display a pmf graphically. The thin-bar (stem) plot emphasises that X takes only isolated values, while the thick-bar chart is the more conventional style

seen in most textbooks. The impulse plot (c) uses upward-pointing arrows to represent each probability; this notation is common in electrical engineering and signal processing, where the discrete pmf is written as a sum of Dirac delta functions, $f(x) = \sum_k p_k \delta(x - x_k)$, effectively treating the discrete case within a continuous framework. All three convey the same information. The Dirac delta formalism is outside the scope of this course; we mention it only so that students encountering it in other subjects can recognise the connection.

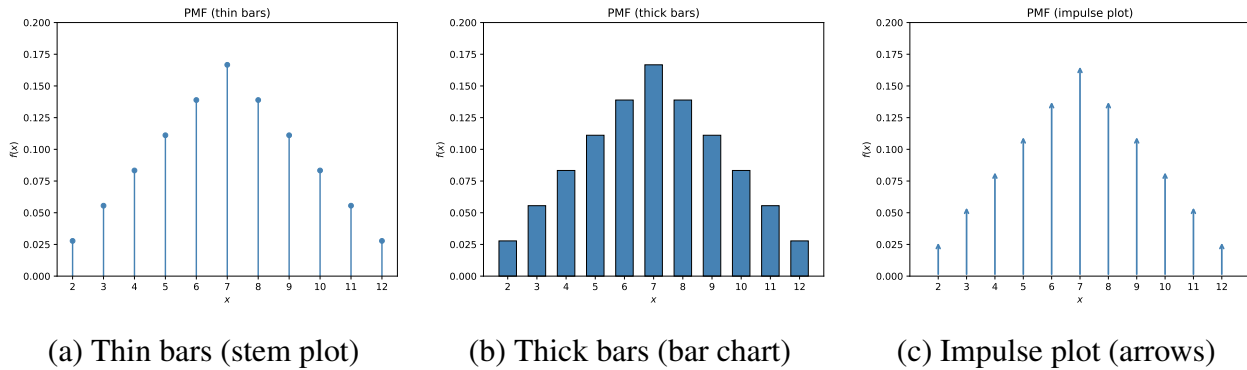


Figure 3.2: Three styles of PMF plot for $X = \text{sum of two dice}$ (Example 3.1).

Probability Mass Function

Let's now study the pmf and its properties.

Definition : Probability Mass Function — Let X be a discrete random variable with possible values $x_1, x_2, \dots$. The **probability mass function** (pmf) of X is the function f defined by

$$f(x) = \mathbb{P}(X = x), \quad \text{for each } x \in \{x_1, x_2, \dots\}. \quad (3.6)$$

We sometimes write f_X instead of f to stress that the pmf refers to the random variable X .

Formally, we can treat f as a function $f: \mathbb{R} \rightarrow [0, 1]$ by setting $f(x) = 0$ for $x \notin \{x_1, x_2, \dots\}$. The set of values where $f(x) > 0$ is called the **support** of the distribution.

Note that some authors call $f(x)$ a “density”; in this course we reserve the term *density* (or *probability density function*) exclusively for continuous distributions, introduced in Chapter 5. Also, some texts write $p(x)$ for the probability mass function; we use $f(x)$ throughout for both the discrete and continuous cases.

The pmf satisfies the following two properties:

1. $f(x) \geq 0$ for all x ,
2. $\sum_x f(x) = 1$ (the sum is over all values in the support of X).

If we know the pmf, we can calculate any probability involving X :

$$\mathbb{P}(X \in B) = \sum_{x \in B} f(x) \quad (3.7)$$

for any subset B of the range of X .

Whenever we have a function that satisfies the properties of the pmf, it means there is a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and a random variable X , where the distribution of X implied by the probability measure $\mathbb{P}$ is described by the pmf.

Cumulative Distribution Function

The cdf is often not considered as the first choice for the description of the distribution of a random variable, yet it has the advantage that it is not limited only to discrete random variables. Note that in probability some people just call it the “distribution function” without “cumulative”.

Definition : Cumulative Distribution Function — The **cumulative distribution function** (cdf) of a random variable X is the function $F : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F(x) := \mathbb{P}(X \leq x), \quad x \in \mathbb{R}. \quad (3.8)$$

Note that the cdf is defined for **all** real values of x , regardless of what the support of X is.

Claim 5: Basic Properties of the CDF — For any random variable X with cdf F ,

1. $0 \leq F(x) \leq 1$ for all x ,
2. F is non-decreasing: $x \leq y \Rightarrow F(x) \leq F(y)$,
3. $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.

Proof:

1. Since $F(x) = \mathbb{P}(X \leq x)$ is a probability, it lies in $[0, 1]$.
2. If $x \leq y$, then $\{X \leq x\} \subseteq \{X \leq y\}$, so $F(x) = \mathbb{P}(X \leq x) \leq \mathbb{P}(X \leq y) = F(y)$ by Claim 2(2).
3. As $x \rightarrow -\infty$, the events $\{X \leq x\}$ decrease to $\emptyset$, so $F(x) \rightarrow \mathbb{P}(\emptyset) = 0$ by continuity of probability (Claim 3). As $x \rightarrow \infty$, the events $\{X \leq x\}$ increase to Ω , so $F(x) \rightarrow \mathbb{P}(\Omega) = 1$.

□

Claim 6: Probabilities of Half-Open Intervals — For any random variable X with cdf F , and any $a < b$,

$$\mathbb{P}(a < X \leq b) = F(b) - F(a). \quad (3.9)$$

Proof: The event $\{X \leq b\}$ can be written as the disjoint union $\{X \leq a\} \cup \{a < X \leq b\}$. By additivity,

$$\mathbb{P}(X \leq b) = \mathbb{P}(X \leq a) + \mathbb{P}(a < X \leq b), \quad (3.10)$$

so $\mathbb{P}(a < X \leq b) = F(b) - F(a)$. $\square$

As cdfs will reappear in Chapter 5 with continuous random variables, there it will **not** matter whether the inequalities are $<$ or $\leq$, but here in the discrete case it does.

Figure 3.3 shows the graph of a cdf of a random variable that has both discrete and continuous components. Notice the jumps (corresponding to point masses), the smooth sections (corresponding to continuous parts), the right-continuity (closed dots on the right side of each jump), and the limits at $\pm\infty$.

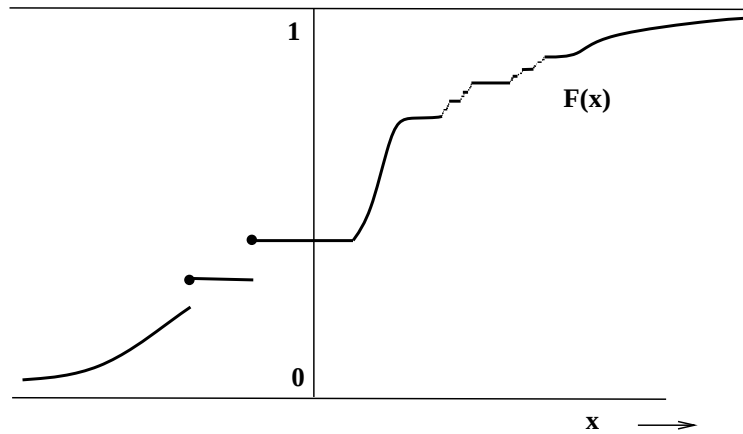


Figure 3.3: A cumulative distribution function. The jumps correspond to values with positive probability; between jumps the function is constant (for a discrete random variable) or continuously increasing (for a continuous one).

For discrete random variables, F is a *step function* with jumps at the support points. Note that the proof of the following claim is not required as part of the material.

Claim 7: Right-Continuity of the CDF — For any random variable X with cdf F , the function F is right-continuous:

$$\lim_{h \downarrow 0} F(x + h) = F(x) \quad \text{for all } x \in \mathbb{R}. \quad (3.11)$$

Proof: Let $h_1 > h_2 > \dots > 0$ be any sequence with $h_n \downarrow 0$. Define the events $A_n = \{X \leq x + h_n\}$. Since $h_1 > h_2 > \dots$, we have $A_1 \supseteq A_2 \supseteq \dots$, i.e. the A_n form a decreasing sequence of events.

Their intersection is

$$\bigcap_{n=1}^{\infty} A_n = \bigcap_{n=1}^{\infty} \{X \leq x + h_n\} = \{X \leq x\}, \quad (3.12)$$

since a real number satisfies $X(\omega) \leq x + h_n$ for every n if and only if $X(\omega) \leq x$. By the continuity of probability for decreasing sequences (Claim 3),

$$\lim_{n \rightarrow \infty} F(x + h_n) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = \mathbb{P}(X \leq x) = F(x). \quad (3.13)$$

Since this holds for every sequence $h_n \downarrow 0$, we conclude $\lim_{h \downarrow 0} F(x + h) = F(x)$. $\square$

A related function is the **complementary cumulative distribution function** (ccdf), defined by $\bar{F}(x) = 1 - F(x) = \mathbb{P}(X > x)$. It is also called the **survival function**, especially in reliability theory and survival analysis.

PMF $\leftrightarrow$ CDF Conversions for Discrete Distributions

For discrete random variables, given one of the pmf or cdf we can always recover the other (see Figure 3.4 below for an illustration). Hence both the cdf and pmf are alternative ways to describe the distribution of the random variable.

PMF to CDF. For a discrete random variable,

$$F(x) = \sum_{k \leq x} f(k). \quad (3.14)$$

Note that the summation is restricted to the support points of f that are smaller or equal to x .

CDF to PMF. The pmf at a point x equals the *jump size* of F at x :

$$f(x) = F(x) - F(x^-), \quad (3.15)$$

where $F(x^-) = \lim_{h \downarrow 0} F(x - h)$ denotes the left-hand limit of F at x . That is, $F(x^-)$ is the value of the cdf just before the point x , and $F(x)$ is the value at x itself. The difference is exactly the size of the jump at x , which equals $\mathbb{P}(X = x) = f(x)$. If x is not a support point, there is no jump, and $f(x) = 0$.

Example (A Three-Valued Random Variable). Suppose a discrete random variable X takes values 0, 1, and 2 with probabilities

$$\mathbb{P}(X = 0) = 0.3, \quad \mathbb{P}(X = 1) = 0.5, \quad \mathbb{P}(X = 2) = 0.2. \quad (3.16)$$

The pmf is summarised in the following table:

x	0	1	2	Σ
$f(x)$	0.3	0.5	0.2	1

The cdf is built by accumulating the pmf values from left to right:

$$F(x) = \begin{cases} 0 & x < 0, \\ 0.3 & 0 \leq x < 1, \\ 0.8 & 1 \leq x < 2, \\ 1 & x \geq 2. \end{cases} \quad (3.17)$$

As an illustration, we compute

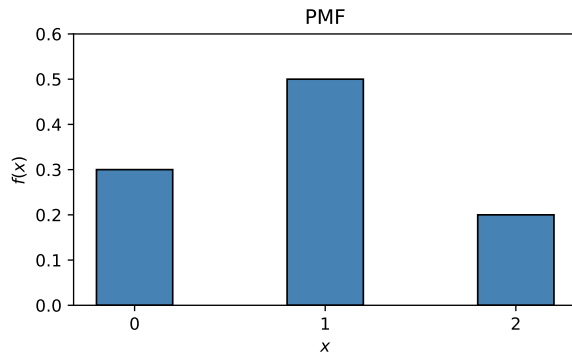
$$\mathbb{P}(0 < X \leq 2) = F(2) - F(0) = 1 - 0.3 = 0.7, \quad (3.18)$$

which we can verify from the pmf: $\mathbb{P}(X = 1) + \mathbb{P}(X = 2) = 0.5 + 0.2 = 0.7$.

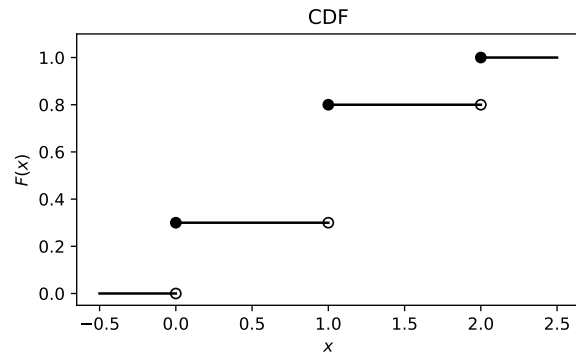
The following Python code generates bar and step plots of the pmf and cdf (Figure 3.4).

Plotting the PMF and CDF of a discrete random variable

LLM Prompt > Write Python code that plots the PMF (as a bar chart) and CDF (as a step function with open and closed dots at each jump) for a discrete random variable taking values 0, 1, 2 with probabilities 0.3, 0.5, 0.2.



(a) Probability mass function



(b) Cumulative distribution function

Figure 3.4: PMF and CDF of X in Example 3.1.

Distributions on Infinite Support

A discrete random variable need not have finitely many possible values. In the examples below, the support is the entire set of positive integers $\{1, 2, 3, \dots\}$. For the pmf to be valid, the series $\sum_{i=1}^{\infty} f(i)$ must converge to 1.

Example (A pmf proportional to $1/i^2$). Suppose we wish to define a random variable X on $\{1, 2, 3, \dots\}$ with probabilities proportional to $1/i^2$, that is, $f(i) = c/i^2$ for some normalising constant

$c > 0$. For this to be a valid pmf we need

$$1 = \sum_{i=1}^{\infty} \frac{c}{i^2} = c \sum_{i=1}^{\infty} \frac{1}{i^2}. \quad (3.19)$$

The series $\sum_{i=1}^{\infty} 1/i^2$ is known to converge. Its exact value, first established by Euler in 1734 (the *Basel problem*), is

$$\sum_{i=1}^{\infty} \frac{1}{i^2} = \frac{\pi^2}{6}. \quad (3.20)$$

Therefore $c = 6/\pi^2 \approx 0.6079$ and the pmf is

$$f(i) = \frac{6}{\pi^2 i^2}, \quad i = 1, 2, 3, \dots \quad (3.21)$$

The following code computes the cdf $F(x) = \sum_{i=1}^{\lfloor x \rfloor} f(i)$ and plots it as a step function (Figure 3.5). Note how F climbs steeply at first and then levels off, approaching 1 slowly.

CDF of the distribution with pmf proportional to $1/i^2$

LLM Prompt > Write Python code that plots the CDF as a step function (with open and closed dots) for the discrete distribution with pmf $f(i) = (6/\pi^2)/i^2$, $i = 1, 2, \dots, 15$.

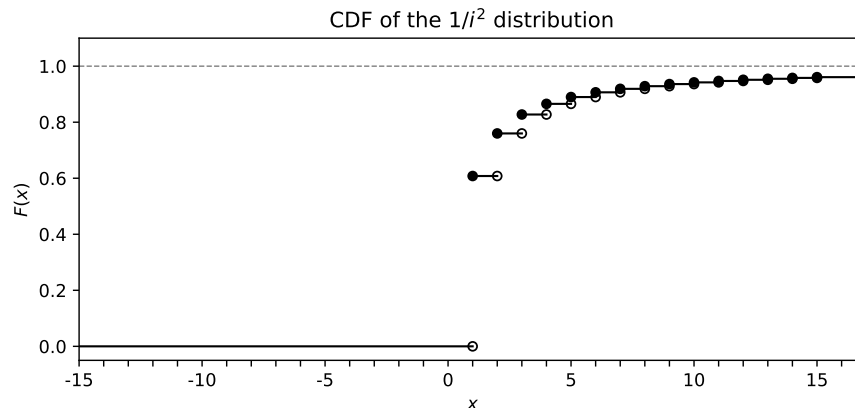


Figure 3.5: Cumulative distribution function of the discrete distribution with pmf $f(i) = 6/(\pi^2 i^2)$. The first value alone carries probability $f(1) = 6/\pi^2 \approx 0.608$.

Example (Why $1/i$ does not work). One might try a similar construction with $f(i) = c/i$ for $i = 1, 2, 3, \dots$. However, the *harmonic series*

$$\sum_{i=1}^{\infty} \frac{1}{i} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots \quad (3.22)$$

diverges to infinity. No finite constant $c > 0$ can make $\sum_{i=1}^{\infty} c/i = 1$, so $f(i) = c/i$ cannot define a valid pmf.

The contrast between $1/i^2$ and $1/i$ illustrates that convergence of $\sum f(i)$ is essential: probabilities proportional to $1/i^2$ yield a valid distribution, while probabilities proportional to $1/i$ do not.

3.2 Moments: Expectation, Mean, and Variance

Now that we can describe the distribution of a discrete random variable via its pmf or cdf, we turn to numerical summaries that capture key features of the distribution. The most fundamental of these are the expectation (a measure of centre) and the variance (a measure of spread).

Expectation (Mean)

The **expectation** (or **mean**) of a random variable is a weighted average of its possible values, weighted by their probabilities. It represents the “centre of mass” of the distribution.

Definition : Expectation (Mean) — The **expectation** (or **mean**) of a discrete random variable X with pmf f is

$$\mathbb{E}[X] = \sum_x x f(x), \quad (3.23)$$

provided the sum converges absolutely ($\sum_x |x| f(x) < \infty$). We also write $\mu = \mathbb{E}[X]$ or $\mu_X = \mathbb{E}[X]$.

The expectation can be thought of as the long-run average: if we repeat the experiment many times and record the values of X , the sample average will converge to $\mathbb{E}[X]$. This is made precise by the Law of Large Numbers (Chapter 10). Even if we only get to observe a single realisation of X , the expectation describes a “typical value”. Note, however, that this typical value may not be one of the possible values of X , as we now illustrate.

Consider the random variable X from Example 3.1, which takes values 0, 1, 2 with probabilities 0.3, 0.5, 0.2. Its mean is

$$\mathbb{E}[X] = 0 \cdot 0.3 + 1 \cdot 0.5 + 2 \cdot 0.2 = 0 + 0.5 + 0.4 = 0.9. \quad (3.24)$$

Example (Tossing a Fair Die). Let X be the outcome of a toss of a fair die, so that $\mathbb{P}(X = k) = 1/6$ for $k = 1, \dots, 6$. Then

$$\mathbb{E}[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \dots + 6 \cdot \frac{1}{6} = \frac{21}{6} = \frac{7}{2}. \quad (3.25)$$

Note that $\mathbb{E}[X] = 3.5$ is not itself a possible outcome of the experiment — the expectation need not be a value that X can actually take.

The expectation as a centre of mass

The expectation has a physical interpretation as a *centre of mass*. Imagine that point masses with weights $p_1, p_2, \dots, p_n$ are placed at positions $x_1, x_2, \dots, x_n$ on the real line (Figure 3.6). The centre

of mass — the point at which the system balances — is

$$\text{centre of mass} = x_1 p_1 + x_2 p_2 + \cdots + x_n p_n, \quad (3.26)$$

which is exactly $\mathbb{E}[X]$ for a discrete random variable X taking values $x_1, \dots, x_n$ with probabilities $p_1, \dots, p_n$.

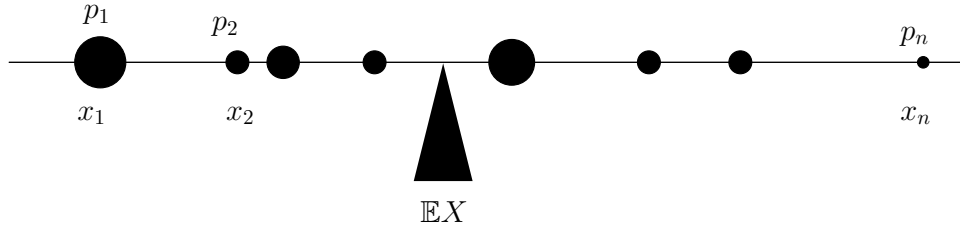


Figure 3.6: The expectation as a centre of mass. Point masses $p_1, \dots, p_n$ are placed at positions $x_1, \dots, x_n$; the balance point is $\mathbb{E}[X]$.

An immediate consequence is that if the pmf is *symmetric* about a point c — that is, $f(c + y) = f(c - y)$ for all y — then $\mathbb{E}[X] = c$ (when the expectation exists).

Expectation of a Function of a Random Variable

We often need to compute the expectation of some function $h(X)$ of a random variable X . One approach is to first derive the pmf of the new random variable $Y = h(X)$ and then compute $\mathbb{E}[Y]$ directly. We consider transformations of random variables in Chapter 6. However, there is a much more convenient shortcut.

Claim 8: Law of the Unconscious Statistician (LOTUS) — If X is a discrete random variable with pmf f and $h : \mathbb{R} \rightarrow \mathbb{R}$ is any function, then

$$\mathbb{E}[h(X)] = \sum_x h(x) f(x), \quad (3.27)$$

provided the sum converges absolutely.

The name reflects the fact that one can compute $\mathbb{E}[h(X)]$ “unconsciously” — without first deriving the distribution of $h(X)$.

For the random variable X from Example 3.1, taking $h(x) = x^2$ gives

$$\mathbb{E}[X^2] = 0^2 \cdot 0.3 + 1^2 \cdot 0.5 + 2^2 \cdot 0.2 = 0 + 0.5 + 0.8 = 1.3. \quad (3.28)$$

Why does LOTUS work? (Informal argument.) Recall that X is a function from the sample space Ω to $\mathbb{R}$, mapping each outcome ω to a value $X(\omega)$. When we define $Y = h(X)$, we are composing two functions: first $\omega \mapsto X(\omega)$, then $X(\omega) \mapsto h(X(\omega))$.

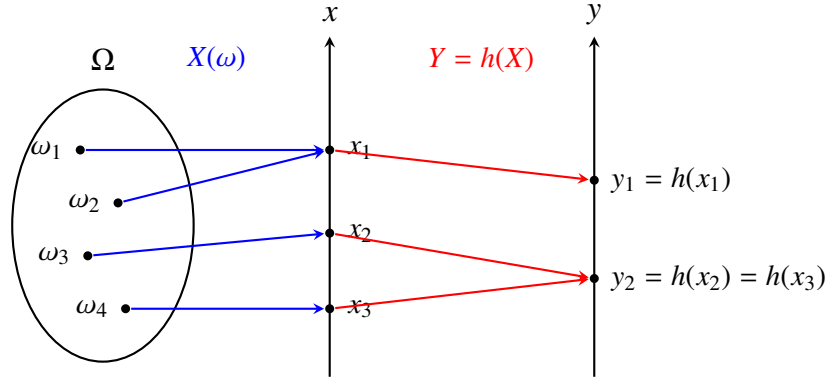


Figure 3.7: The composition $\omega \mapsto X(\omega) \mapsto h(X(\omega))$. Multiple outcomes in Ω may map to the same x , and multiple x -values may map to the same y .

By definition of expectation applied to $Y = h(X)$, we should sum over the distinct values y of Y :

$$\mathbb{E}[Y] = \sum_y y \cdot \mathbb{P}(Y = y). \quad (3.29)$$

But each value y may be produced by several values of x (those with $h(x) = y$), so $\mathbb{P}(Y = y) = \sum_{x: h(x)=y} f(x)$. Substituting and regrouping,

$$\mathbb{E}[Y] = \sum_y y \sum_{x: h(x)=y} f(x) = \sum_x h(x) f(x), \quad (3.30)$$

where the last step simply collects all terms by x instead of by y : each x contributes $h(x)f(x)$ exactly once. This is precisely the LOTUS formula.

Moments

An important family of functions arises by taking $h(x) = x^k$ for non-negative integers k , or $\tilde{h}(x) = (x - \mu)^k$ where $\mu = \mathbb{E}[X]$, as used for central moments.

Definition : Moments — The k -th moment of a random variable X is

$$\mathbb{E}[X^k] = \sum_x x^k f(x), \quad k = 0, 1, 2, \dots \quad (3.31)$$

The k -th central moment is

$$\mathbb{E}[(X - \mu)^k] = \sum_x (x - \mu)^k f(x), \quad \text{where } \mu = \mathbb{E}[X]. \quad (3.32)$$

The zeroth moment is always $\mathbb{E}[X^0] = 1$. The first moment $\mathbb{E}[X]$ is the mean. The second central moment $\mathbb{E}[(X - \mu)^2]$ is the variance, which we define next.

Moments are important for two reasons. First, the most commonly used summary quantities — the mean and the variance — are themselves moments/central-moments. Second, the infinite sequence of moments often characterises the distribution entirely: under mild conditions, if two

distributions share all their moments, they must be the same distribution. Keep in mind, however, that knowing only a few moments does not determine the distribution exactly, but the more moments we know, the more information we have about it.

Variance and Standard Deviation

The mean tells us where the distribution is centred, but says nothing about how spread out the values are. The **variance** measures this spread.

Definition : Variance and Standard Deviation — The **variance** of a random variable X with mean $\mu = \mathbb{E}[X]$ is

$$\mathbb{V}\text{ar}(X) = \mathbb{E}[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x). \quad (3.33)$$

The **standard deviation** is $\text{SD}(X) = \sqrt{\mathbb{V}\text{ar}(X)}$.

The variance is the expected squared deviation from the mean — it is always non-negative. The standard deviation has the same units as X and is often more interpretable.

For the random variable X from Example 3.1 with $\mu = 0.9$,

$$\mathbb{V}\text{ar}(X) = (0 - 0.9)^2 \cdot 0.3 + (1 - 0.9)^2 \cdot 0.5 + (2 - 0.9)^2 \cdot 0.2 = 0.243 + 0.005 + 0.242 = 0.49, \quad (3.34)$$

and $\text{SD}(X) = \sqrt{0.49} = 0.7$.

Properties of Expectation and Variance

The following results describe how the mean and variance behave under a linear transformation $X \mapsto aX + b$.

Claim 9: Expectation under Linear Transformation — For any constants $a, b \in \mathbb{R}$,

$$\mathbb{E}[aX + b] = a \mathbb{E}[X] + b. \quad (3.35)$$

Proof: Using LOTUS,

$$\mathbb{E}[aX + b] = \sum_x (ax + b) f(x) = a \sum_x x f(x) + b \sum_x f(x) \quad (3.36)$$

$$= a \mathbb{E}[X] + b \cdot 1 = a \mathbb{E}[X] + b. \quad (3.37)$$

□

In particular, taking $a = 1$ gives $\mathbb{E}[X + b] = \mathbb{E}[X] + b$ (shifting does not change the mean apart from the shift itself), and taking $b = 0$ gives $\mathbb{E}[aX] = a \mathbb{E}[X]$ (scaling).

Claim 10: Variance under Linear Transformation — For any constants $a, b \in \mathbb{R}$,

$$\mathbb{V}\text{ar}(aX + b) = a^2 \mathbb{V}\text{ar}(X). \quad (3.38)$$

Proof: Let $\mu = \mathbb{E}[X]$. By the linearity of expectation, $\mathbb{E}[aX + b] = a\mu + b$. Therefore,

$$\mathbb{V}\text{ar}(aX + b) = \mathbb{E}[(aX + b - (a\mu + b))^2] \quad (3.39)$$

$$= \mathbb{E}[(a(X - \mu))^2] \quad (3.40)$$

$$= a^2 \mathbb{E}[(X - \mu)^2] = a^2 \mathbb{V}\text{ar}(X). \quad (3.41)$$

□

Notice that the additive constant b does not affect the variance — shifting a distribution does not change its spread. The scaling factor enters squared, so multiplying X by a multiplies the variance by a^2 (and the standard deviation by $|a|$).

Claim 11: Additivity of Expectation for Functions of a Random Variable — If X is a discrete random variable with pmf f , and $g, h : \mathbb{R} \rightarrow \mathbb{R}$ are any functions, then

$$\mathbb{E}[g(X) + h(X)] = \mathbb{E}[g(X)] + \mathbb{E}[h(X)], \quad (3.42)$$

provided both expectations on the right-hand side exist.

Proof: By LOTUS,

$$\mathbb{E}[g(X) + h(X)] = \sum_x (g(x) + h(x))f(x) \quad (3.43)$$

$$= \sum_x g(x)f(x) + \sum_x h(x)f(x) = \mathbb{E}[g(X)] + \mathbb{E}[h(X)], \quad (3.44)$$

where the second step uses the ordinary linearity of summation. □

The following result provides a convenient computational formula for the variance in terms of the first two moments.

Claim 12: Computational Formula for Variance —

$$\mathbb{V}\text{ar}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \quad (3.45)$$

Proof: Let $\mu = \mathbb{E}[X]$. Expanding the square in the definition,

$$\mathbb{V}\text{ar}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \quad (3.46)$$

$$= \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 \quad (3.47)$$

$$= \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - \mu^2. \quad (3.48)$$

In the second line we used the additivity of expectation (Claim 11) to split the expectation across the three terms. In the third line we used $\mathbb{E}[X] = \mu$. □

This formula is often easier to use than the definition, especially when μ is not a “nice” number. We can verify it with the random variable from Example 3.1: we computed $\mathbb{E}[X] = 0.9$ and $\mathbb{E}[X^2] = 1.3$, so

$$\mathbb{V}\text{ar}(X) = 1.3 - 0.9^2 = 1.3 - 0.81 = 0.49, \quad (3.49)$$

which agrees with the direct calculation above.

Useful Summation Formulas

The following identities, which can be proved by induction, will be used repeatedly in the examples below:

$$\sum_{k=1}^N k = \frac{N(N+1)}{2}, \quad (3.50)$$

$$\sum_{k=1}^N k^2 = \frac{N(N+1)(2N+1)}{6}, \quad (3.51)$$

$$\sum_{k=1}^N k^3 = \left[\frac{N(N+1)}{2} \right]^2. \quad (3.52)$$

For example, $\sum_{k=1}^N k$ gives 1, 3, 6, 10, 15 for $N = 1, 2, 3, 4, 5$. Similarly, $\sum_{k=1}^N k^2$ gives 1, 5, 14, 30, 55, and $\sum_{k=1}^N k^3$ gives 1, 9, 36, 100, 225 (which are $1^2, 3^2, 6^2, 10^2, 15^2$ — the squares of the first formula).

Example: Discrete Uniform Distribution on $\{1, \dots, N\}$

Example (Discrete Uniform on $\{1, \dots, N\}$). Suppose X is uniformly distributed on $\{1, 2, \dots, N\}$, so that

$$f(x) = \frac{1}{N}, \quad x = 1, 2, \dots, N. \quad (3.53)$$

We write $X \sim \text{DU}(\{1, \dots, N\})$. Note that the symbol “ $\sim$ ” means “distributed as”. Different texts use different notation (and sometimes different parameterisations) for distributions; for us, DU stands for “Discrete Uniform”.

Mean. Using (3.50),

$$\mathbb{E}[X] = \frac{1}{N} \sum_{k=1}^N k = \frac{1}{N} \cdot \frac{N(N+1)}{2} = \frac{N+1}{2}. \quad (3.54)$$

Second moment. Using (3.51),

$$\mathbb{E}[X^2] = \frac{1}{N} \sum_{k=1}^N k^2 = \frac{(N+1)(2N+1)}{6}. \quad (3.55)$$

Variance. By the computational formula for variance,

$$\mathbb{V}\text{ar}(X) = \frac{(N+1)(2N+1)}{6} - \left(\frac{N+1}{2} \right)^2 = (N+1) \left[\frac{2N+1}{6} - \frac{N+1}{4} \right] = (N+1) \cdot \frac{N-1}{12} = \frac{N^2-1}{12}. \quad (3.56)$$

In summary,

$$X \sim \text{DU}(\{1, \dots, N\}) : \quad \mathbb{E}[X] = \frac{N+1}{2}, \quad \mathbb{V}\text{ar}(X) = \frac{N^2-1}{12}. \quad (3.57)$$

Example: Triangular Distribution on $\{1, \dots, N\}$

Example (Triangular Distribution on $\{1, \dots, N\}$). Suppose X takes values in $\{1, 2, \dots, N\}$ with probabilities proportional to $1, 2, \dots, N$. That is, $f(x) = cx$ for some normalising constant $c > 0$. Since the pmf must sum to one,

$$1 = \sum_{k=1}^N ck = c \cdot \frac{N(N+1)}{2}, \quad \text{so} \quad c = \frac{2}{N(N+1)}. \quad (3.58)$$

Hence

$$f(x) = \frac{2x}{N(N+1)}, \quad x = 1, 2, \dots, N. \quad (3.59)$$

Mean. Using (3.51),

$$\mathbb{E}[X] = \sum_{k=1}^N k \cdot \frac{2k}{N(N+1)} = \frac{2}{N(N+1)} \sum_{k=1}^N k^2 = \frac{2}{N(N+1)} \cdot \frac{N(N+1)(2N+1)}{6} = \frac{2N+1}{3}. \quad (3.60)$$

Second moment. Using (3.52),

$$\mathbb{E}[X^2] = \frac{2}{N(N+1)} \sum_{k=1}^N k^3 = \frac{2}{N(N+1)} \cdot \frac{N^2(N+1)^2}{4} = \frac{N(N+1)}{2}. \quad (3.61)$$

Variance. By the computational formula,

$$\mathbb{V}\text{ar}(X) = \frac{N(N+1)}{2} - \left(\frac{2N+1}{3} \right)^2 = \frac{9N(N+1) - 2(2N+1)^2}{18}. \quad (3.62)$$

Expanding the numerator,

$$9N(N+1) - 2(2N+1)^2 = 9N^2 + 9N - 8N^2 - 8N - 2 = N^2 + N - 2 = (N-1)(N+2). \quad (3.63)$$

Therefore,

$$\mathbb{E}[X] = \frac{2N+1}{3}, \quad \mathbb{V}\text{ar}(X) = \frac{(N-1)(N+2)}{18}. \quad (3.64)$$

Compared with the discrete uniform distribution on $\{1, \dots, N\}$, the triangular distribution has a larger mean (the distribution is skewed towards larger values) and a smaller variance for $N \geq 2$.

Example: Discrete Uniform Distribution on $\{a, \dots, b\}$

Example (Discrete Uniform on $\{a, \dots, b\}$). Let Y be uniformly distributed on $\{a, a+1, \dots, b\}$, where $a \leq b$ are integers. Rather than repeating the summation calculations, we use the linear transformation properties established above.

Let $N = b - a + 1$ (the number of values) and let $X \sim \text{DU}(\{1, \dots, N\})$. Then

$$Y = X + (a - 1) \quad (3.65)$$

is uniformly distributed on $\{a, a+1, \dots, b\}$, since adding $a-1$ shifts $\{1, \dots, N\}$ to $\{a, \dots, b\}$.

Mean. By the linearity of expectation and the result of Example 3.2,

$$\mathbb{E}[Y] = \mathbb{E}[X] + (a - 1) = \frac{N + 1}{2} + a - 1 = \frac{b - a + 2}{2} + a - 1 = \frac{a + b}{2}. \quad (3.66)$$

Variance. Since adding a constant does not change the variance,

$$\mathbb{V}\text{ar}(Y) = \mathbb{V}\text{ar}(X) = \frac{N^2 - 1}{12} = \frac{(b - a + 1)^2 - 1}{12} = \frac{(b - a)(b - a + 2)}{12}. \quad (3.67)$$

In summary,

$$Y \sim \text{DU}(\{a, \dots, b\}) : \quad \mathbb{E}[Y] = \frac{a + b}{2}, \quad \mathbb{V}\text{ar}(Y) = \frac{(b - a)(b - a + 2)}{12}. \quad (3.68)$$

These results are intuitive: the mean is the midpoint of the range, and the variance depends only on the width $b - a$ of the range, not on the location. This example illustrates the power of the linear transformation properties — we obtained the mean and variance of $\text{DU}(\{a, \dots, b\})$ without any new summation calculations.

3.3 Sequences of Random Variables

So far we have considered a single random variable X defined on a sample space Ω . But many situations involve *several* random variables, all defined on the same underlying experiment.

Multiple Random Variables on the Same Sample Space

Since a random variable is simply a function $X : \Omega \rightarrow \mathbb{R}$, nothing prevents us from defining two, three, or more such functions on the same sample space. For instance, if we roll two dice so that $\Omega = \{(i, j) : i, j \in \{1, \dots, 6\}\}$, we might define $X_1(i, j) = i$ (the first die) and $X_2(i, j) = j$ (the second die). Both X_1 and X_2 are random variables on Ω .

More generally, we can consider a **sequence of random variables** $X_1, X_2, \dots, X_n$ (or even an infinite sequence $X_1, X_2, X_3, \dots$), all defined as functions on the same sample space Ω . Each X_i has its own pmf, mean, and variance, but the random variables may also be related to one another — knowing the value of X_1 might give us information about X_2 .

Independence of Random Variables

In Chapter 2 we defined independence of *events*. We now extend this to random variables.

Definition : Independent Random Variables — Two discrete random variables X and Y are **independent** if for all values x and y ,

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y). \quad (3.69)$$

More generally, random variables $X_1, X_2, \dots, X_n$ are **mutually independent** if

$$\mathbb{P}(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = \mathbb{P}(X_1 = x_1)\mathbb{P}(X_2 = x_2) \cdots \mathbb{P}(X_n = x_n) \quad (3.70)$$

for all values $x_1, x_2, \dots, x_n$.

Independence means that knowing the value of any subset of the random variables gives no information about the others. In our two-dice example, X_1 and X_2 are independent: the outcome of the first die tells us nothing about the second.

Remark (Joint distributions). The probability $\mathbb{P}(X = x, Y = y)$ is called the **joint probability mass function** of X and Y . In Chapter 7 we will study joint distributions in detail, providing a richer framework for analysing the relationship between multiple random variables. For now, the definition above is all we need.

IID Sequences

An especially important special case arises when we repeat the same experiment independently many times.

Definition : IID Sequence — A sequence of random variables $X_1, X_2, \dots$ is **independent and identically distributed (iid)** if:

1. the random variables are mutually independent, and
2. they all have the same probability distribution (the same pmf, the same mean, the same variance, etc.).

We write $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} F$ to indicate that each X_i has distribution F and the sequence is iid.

The iid assumption is one of the most common modelling assumptions in probability and statistics. It captures the idea of *repeatable, non-interacting* experiments: each trial is a fresh copy of the same random mechanism, unaffected by what happened before.

Example (IID Dice Throws). Suppose we roll a fair die repeatedly. Let X_i be the outcome of the i -th roll. Then $X_1, X_2, X_3, \dots$ is an iid sequence, with each $X_i \sim \text{DU}(\{1, 2, 3, 4, 5, 6\})$. Knowledge of $X_1, \dots, X_k$ gives no information about X_{k+1} — each roll is independent and has the same uniform distribution.

The following code simulates 15 iid rolls of a fair die.

Simulating an iid sequence of dice throws

LLM Prompt > Write Python code that simulates and prints 15 iid rolls of a fair six-sided die.

IID dice rolls: [2 3 1 6 6 6 5 5 4 6 5 5 3 6 4]

Each roll is independent of all others and has the same distribution, so this is an iid sequence.

Remark (Looking ahead). Many of the distributions in the next two sections — binomial, geometric, negative binomial — arise from counting patterns in an iid sequence of Bernoulli random variables. The iid assumption is what makes these counts tractable.

In Chapter 4 we will encounter *Markov chains*, which are sequences of random variables that are *not* iid: the distribution of X_{n+1} depends on the current value X_n . The contrast between iid sequences and Markov chains is a central theme in probability.

3.4 Bernoulli Trials and Related Distributions

A **Bernoulli trial** is an experiment with exactly two outcomes, conventionally called *success* and *failure*. We denote the probability of success by $p \in [0, 1]$ and the probability of failure by $q = 1 - p$. A sequence of **independent Bernoulli trials** consists of repeated independent performances of such an experiment, each with the same success probability p .

Many important discrete distributions arise from an iid sequence of Bernoulli random variables. Specifically, let $X_1, X_2, X_3, \dots$ be an iid sequence with each $X_i \sim \text{Ber}(p)$ (as defined below). Each remaining distribution in this section — binomial, geometric, and negative binomial — is obtained as a function (mapping) of this iid sequence:

- The **binomial** counts the total number of successes in the first n trials: $S_n = X_1 + \dots + X_n$.
- The **geometric** records the index of the first success: $T = \min\{i \geq 1 : X_i = 1\}$.
- The **negative binomial** records the index of the r -th success: $T_r = \min\{i : X_1 + \dots + X_i = r\}$.

Thus all three are simply different ways of summarising the same underlying iid Bernoulli sequence.

Bernoulli Distribution

The simplest random variable associated with a Bernoulli trial records whether the trial resulted in success or failure.

Definition : Bernoulli Distribution — A random variable X has a **Bernoulli distribution** with parameter $p \in [0, 1]$, written $X \sim \text{Ber}(p)$, if X takes the value 1 (success) with probability p and 0 (failure) with probability $q = 1 - p$. The pmf is

$$f(x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}. \quad (3.71)$$

Or equivalently,

$$f(x) = \begin{cases} 1 - p & \text{if } x = 0, \\ p & \text{if } x = 1. \end{cases} \quad (3.72)$$

Mean.

$$\mathbb{E}[X] = 0 \cdot (1 - p) + 1 \cdot p = p. \quad (3.73)$$

Second moment. Since X takes values 0 and 1 only, $X^2 = X$, so $\mathbb{E}[X^2] = \mathbb{E}[X] = p$.

Variance. By the computational formula for variance,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p). \quad (3.74)$$

In summary,

$$X \sim \text{Ber}(p) : \quad \mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p). \quad (3.75)$$

Note that the variance is maximised at $p = 1/2$ (where $\text{Var}(X) = 1/4$) and equals zero at $p = 0$ and $p = 1$ (where there is no randomness).

The cdf of the Bernoulli distribution is a step function with a single jump at $x = 0$ of size $1 - p$ and a second jump at $x = 1$ of size p (Figure 3.8).

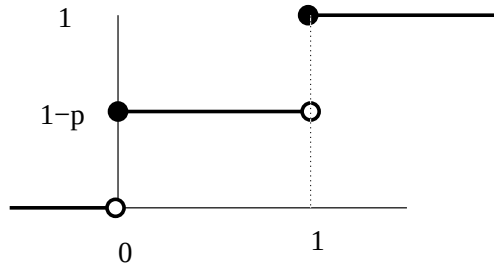


Figure 3.8: The cumulative distribution function of the $\text{Ber}(p)$ distribution.

Monte Carlo simulation of a Bernoulli random variable

A Bernoulli random variable can be simulated by generating a uniform random number U on $(0, 1)$ and setting $X = 1$ if $U < p$ and $X = 0$ otherwise. The following code estimates the mean and variance of $\text{Ber}(p)$ by simulation for a grid of p values and compares them with the theoretical curves.

Monte Carlo estimation of Bernoulli mean and variance

LLM Prompt > Write Python code that estimates the mean and variance of $\text{Ber}(p)$ by Monte Carlo simulation for a grid of p values from 0 to 1, and plots the empirical estimates against the theoretical curves p and $p(1 - p)$ side by side.

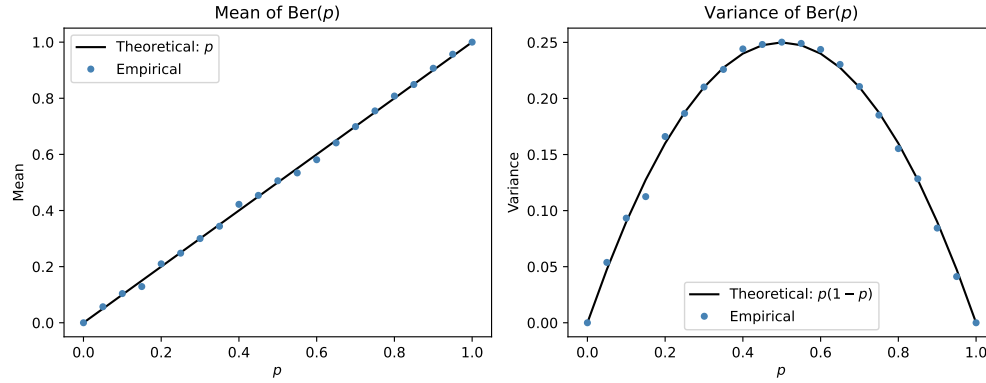


Figure 3.9: Theoretical (solid curve) and empirical (dots, $n = 1000$ samples) mean and variance of $\text{Ber}(p)$ for $p \in [0, 1]$.

Binomial Distribution

The binomial distribution generalises the Bernoulli distribution from a single trial to n independent trials.

Definition : Binomial Distribution — Let X be the number of successes in n independent Bernoulli trials, each with success probability p . Then X has a **binomial distribution** with parameters n and p , written $X \sim \text{Bin}(n, p)$. The pmf is

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n. \quad (3.76)$$

Here $\binom{n}{x}$ counts the number of ways to choose which x of the n trials are successes, p^x is the probability that those x trials are successes, and $(1-p)^{n-x}$ is the probability that the remaining trials are failures. Note that $\text{Bin}(1, p) = \text{Ber}(p)$.

We verify that the pmf sums to one using the binomial theorem:

$$\sum_{x=0}^n \binom{n}{x} p^x (1-p)^{n-x} = (p + (1-p))^n = 1. \quad (3.77)$$

Mean of the binomial (direct calculation)

We compute the mean directly from the definition:

$$\mathbb{E}[X] = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=1}^n k \cdot \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \quad (3.78)$$

$$= \sum_{k=1}^n \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k}. \quad (3.79)$$

Substituting $j = k - 1$ and factoring out np ,

$$\mathbb{E}[X] = np \sum_{j=0}^{n-1} \frac{(n-1)!}{j!(n-1-j)!} p^j (1-p)^{n-1-j} \quad (3.80)$$

$$= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{n-1-j} = np \cdot 1 = np, \quad (3.81)$$

where the inner sum equals 1 by the binomial theorem (it is the sum of $\text{Bin}(n-1, p)$ probabilities).

A similar but more tedious calculation shows that $\mathbb{V}\text{ar}(X) = np(1-p)$. We omit this derivation here; in a later chapter we will obtain both the mean and variance much more easily by writing X as a sum of n independent $\text{Ber}(p)$ random variables and using properties of sums of independent random variables.

In summary,

$$X \sim \text{Bin}(n, p) : \quad \mathbb{E}[X] = np, \quad \mathbb{V}\text{ar}(X) = np(1-p). \quad (3.82)$$

Plots of the binomial pmf

The shape of the binomial pmf depends on p : for small p it is right-skewed, for $p = 0.5$ it is symmetric, and for large p it is left-skewed.

Binomial PMF for different values of p

LLM Prompt > Give me Python code that plots the PMF of a Binomial distribution with $n = 10$ for three values of p (e.g. 0.1, 0.5, 0.9). Plot them side by side.

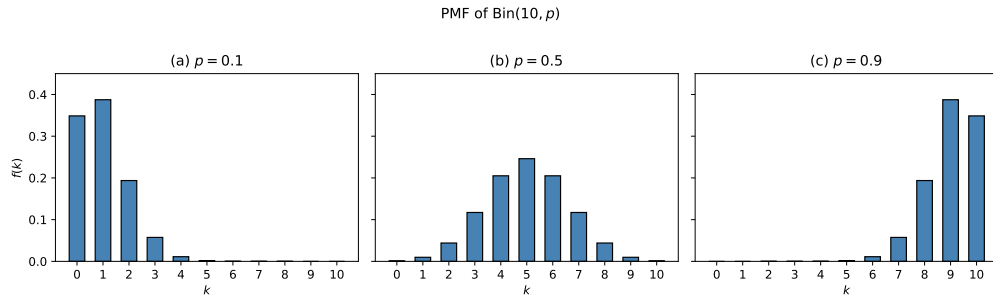


Figure 3.10: Probability mass function of $\text{Bin}(10, p)$ for (a) $p = 0.1$, (b) $p = 0.5$, (c) $p = 0.9$.

Example (Guessing in a Multiple Choice Exam). A multiple choice exam has 20 questions, each with four options (a)–(d), exactly one of which is correct. A student guesses each answer independently and uniformly at random. Let X be the number of correct answers. Since each question is an independent Bernoulli trial with success probability $p = 1/4$, we have $X \sim \text{Bin}(20, 1/4)$.

Mean and variance. The expected number of correct answers is

$$\mathbb{E}[X] = np = 20 \cdot \frac{1}{4} = 5, \quad (3.83)$$

and the variance is

$$\text{Var}(X) = np(1 - p) = 20 \cdot \frac{1}{4} \cdot \frac{3}{4} = 3.75, \quad \text{SD}(X) = \sqrt{3.75} \approx 1.94. \quad (3.84)$$

So by pure guessing, a student expects to get 5 out of 20 correct, with a standard deviation of about 2.

Probability of passing. Suppose a passing mark is 10 or more correct answers. Then

$$\mathbb{P}(X \geq 10) = \sum_{x=10}^{20} \binom{20}{x} \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{20-x} \approx 0.0139. \quad (3.85)$$

The probability of passing by pure guessing is less than 1.4%. We can verify this with the following code.

Computing the probability of passing by guessing

LLM Prompt > Write Python code that computes the probability of getting 10 or more correct answers out of 20 multiple choice questions with 4 options each, by pure guessing. Also compute the mean and variance.

Geometric Distribution

The geometric distribution models the number of trials needed to observe the first success in a sequence of independent Bernoulli trials.

Definition : Geometric Distribution (Counting Trials) — Let X be the number of trials until the first success in a sequence of independent $\text{Ber}(p)$ trials. Then X has a **geometric distribution** with parameter p , written $X \sim \text{Geom}(p)$. The support is $\{1, 2, 3, \dots\}$ and the pmf is

$$f(x) = (1 - p)^{x-1} p, \quad x = 1, 2, 3, \dots \quad (3.86)$$

The event $\{X = k\}$ means the first $k - 1$ trials are failures (each with probability $1 - p$) and the k -th trial is a success (probability p).

We verify that the pmf sums to one using the geometric series formula:

$$\sum_{k=1}^{\infty} (1 - p)^{k-1} p = p \sum_{j=0}^{\infty} (1 - p)^j = p \cdot \frac{1}{1 - (1 - p)} = 1. \quad (3.87)$$

Mean of the geometric distribution

Claim 13: Mean of the Geometric Distribution — If $X \sim \text{Geom}(p)$ with support $\{1, 2, 3, \dots\}$, then

$$\mathbb{E}[X] = \frac{1}{p}. \quad (3.88)$$

Proof: We compute

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k (1 - p)^{k-1} p = p \sum_{k=1}^{\infty} k (1 - p)^{k-1}. \quad (3.89)$$

To evaluate this sum, recall that for $|r| < 1$,

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1 - r}. \quad (3.90)$$

Differentiating both sides with respect to r ,

$$\sum_{k=1}^{\infty} k r^{k-1} = \frac{1}{(1 - r)^2}. \quad (3.91)$$

Setting $r = 1 - p$ (which satisfies $|r| < 1$ for $0 < p \leq 1$),

$$\mathbb{E}[X] = p \cdot \frac{1}{(1 - (1 - p))^2} = p \cdot \frac{1}{p^2} = \frac{1}{p}. \quad (3.92)$$

□

The mean $1/p$ is intuitive: if each trial has a 1-in-5 chance of success ($p = 0.2$), we expect to need 5 trials on average.

In summary,

$$X \sim \text{Geom}(p), X \in \{1, 2, 3, \dots\} : \quad \mathbb{E}[X] = \frac{1}{p}, \quad \mathbb{V}\text{ar}(X) = \frac{1-p}{p^2}. \quad (3.93)$$

The formula for the variance can be obtained with similar computations as the mean, using the second derivative of the geometric series.

Alternative parameterisation: counting failures

Some textbooks define the geometric distribution as the number of *failures before* the first success, so the support is $\{0, 1, 2, \dots\}$. If $X \sim \text{Geom}(p)$ counts trials (as above), then

$$Y = X - 1 \quad (3.94)$$

counts failures before the first success. The pmf of Y is

$$f_Y(x) = (1-p)^x p, \quad x = 0, 1, 2, \dots \quad (3.95)$$

By the linear transformation properties of expectation and variance,

$$\mathbb{E}[Y] = \mathbb{E}[X] - 1 = \frac{1}{p} - 1 = \frac{1-p}{p}, \quad \mathbb{V}\text{ar}(Y) = \mathbb{V}\text{ar}(X) = \frac{1-p}{p^2}. \quad (3.96)$$

The variance is unchanged because subtracting a constant does not affect the spread.

Remark (Which geometric?). Whenever you encounter the geometric distribution, check whether the author counts *trials* (support starting at 1) or *failures* (support starting at 0). The two conventions differ by a shift of 1. In these notes, $\text{Geom}(p)$ counts trials unless stated otherwise.

Negative Binomial Distribution

The geometric distribution counts the number of trials until the *first* success. The negative binomial distribution generalises this to the r -th success.

Definition : Negative Binomial Distribution — Let X be the number of trials until the r -th success in a sequence of independent $\text{Ber}(p)$ trials. Then X has a **negative binomial distribution** with parameters r and p , written $X \sim \text{NegBin}(r, p)$. The support is $\{r, r+1, r+2, \dots\}$ and the pmf is

$$f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, r+2, \dots \quad (3.97)$$

To see why, consider the event $\{X = x\}$: the x -th trial must be a success (and the r -th overall), so among the first $x-1$ trials there are exactly $r-1$ successes. The factor $\binom{x-1}{r-1}$ counts the arrangements of those $r-1$ successes among $x-1$ trials, p^r is the probability of the r successes, and $(1-p)^{x-r}$ is the probability of the $x-r$ failures.

Note that $\text{NegBin}(1, p) = \text{Geom}(p)$.

The mean and variance can be computed in several ways which we do not show here. One approach is to recognise that X is a sum of r independent $\text{Geom}(p)$ random variables (one for each successive success) and use properties of sums of independent random variables. Another approach uses probability generating functions (transforms). The result is intuitive: if on average it takes $1/p$ trials to get one success, it should take r/p trials to get r successes.

In summary,

$$X \sim \text{NegBin}(r, p), X \in \{r, r+1, \dots\} : \quad \mathbb{E}[X] = \frac{r}{p}, \quad \mathbb{V}\text{ar}(X) = \frac{r(1-p)}{p^2}. \quad (3.98)$$

Alternative parameterisation: counting failures

Analogously to the geometric case, we can define $Y = X - r$ as the number of *failures* before the r -th success. Then Y has support $\{0, 1, 2, \dots\}$ and pmf

$$f_Y(x) = \binom{x+r-1}{r-1} p^r (1-p)^x, \quad x = 0, 1, 2, \dots \quad (3.99)$$

By the linear transformation properties,

$$\mathbb{E}[Y] = \mathbb{E}[X] - r = \frac{r}{p} - r = \frac{r(1-p)}{p}, \quad \mathbb{V}\text{ar}(Y) = \mathbb{V}\text{ar}(X) = \frac{r(1-p)}{p^2}. \quad (3.100)$$

Again, the variance is unchanged by the shift. Setting $r = 1$ recovers the two geometric parameterisations.

Simulation of the negative binomial distribution

Example (Simulating the Negative Binomial from First Principles). We can simulate $Y \sim \text{NegBin}(r, p)$ (counting failures) directly from its definition: repeatedly perform independent $\text{Ber}(p)$ trials, counting the number of failures until the r -th success is observed. The following code implements this, draws $n = 1000$ samples with $r = 5$ and $p = 0.4$, and compares the empirical histogram with the theoretical pmf.

Simulating negative binomial random variables from first principles

LLM Prompt > Write Python code that simulates negative binomial random variables (counting failures before the r -th success) from first principles using Bernoulli trials. Use $r = 5$, $p = 0.4$, generate 1000 samples, and compare the empirical histogram with the theoretical PMF using stem plots.

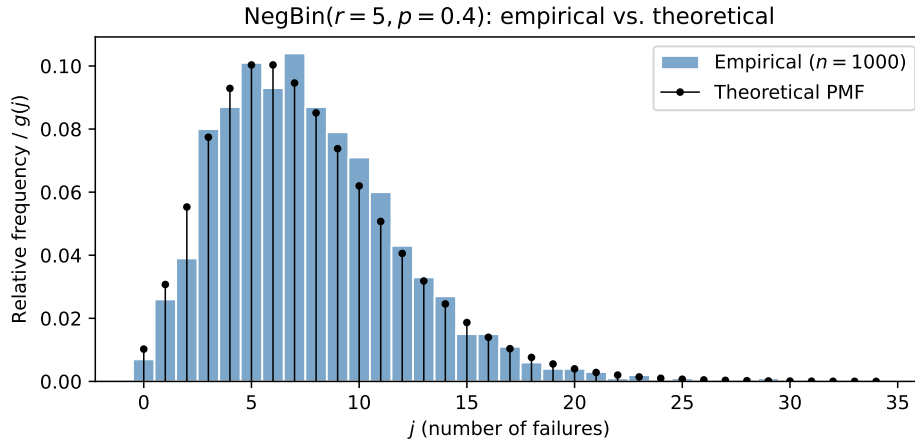


Figure 3.11: Empirical histogram ($n = 1000$ samples) versus the theoretical pmf for $Y \sim \text{NegBin}(5, 0.4)$ counting failures. The stochastic variation in the histogram is visible due to the moderate sample size.

3.5 Poisson and Hypergeometric Distributions

In this section we introduce two further discrete distributions: the Poisson distribution, which models the count of events during a time interval, and the hypergeometric distribution, which models sampling without replacement. Both are closely related to the binomial distribution, as we establish at the end of the section.

Poisson Distribution

The **Poisson distribution** arises naturally as a model for the number of events occurring in a fixed interval of time or space, when events happen independently at a constant average rate. Typical examples include the number of emails received per hour, the number of typos on a page, or the number of radioactive decays per second.

Definition : Poisson Distribution — A random variable X has a **Poisson distribution** with parameter $\lambda > 0$, written $X \sim \text{Poi}(\lambda)$, if

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots \quad (3.101)$$

The parameter λ represents the average rate (or expected count). We verify that the pmf sums to one using the Taylor series for the exponential function:

$$\sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} \cdot e^{\lambda} = 1. \quad (3.102)$$

Mean of the Poisson distribution

Claim 14: Mean of the Poisson Distribution — If $X \sim \text{Poi}(\lambda)$, then $\mathbb{E}[X] = \lambda$.

Proof:

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda} \lambda^k}{k!} = \sum_{k=1}^{\infty} \frac{e^{-\lambda} \lambda^k}{(k-1)!} = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda, \quad (3.103)$$

where we substituted $j = k - 1$. □

A similar calculation yields the variance. We state the result without proof.

In summary,

$$X \sim \text{Poi}(\lambda) : \quad \mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda. \quad (3.104)$$

A distinctive feature of the Poisson distribution is that its mean and variance are equal. This provides a quick diagnostic: if the sample mean and sample variance of count data are roughly equal, a Poisson model may be appropriate.

Plots of the Poisson pmf

Poisson PMF for different values of λ

LLM Prompt > Write Python code that plots the PMF of the Poisson distribution for $\lambda = 1, 5, 15$ as bar charts side by side. Implement the Poisson PMF formula directly without using scipy.

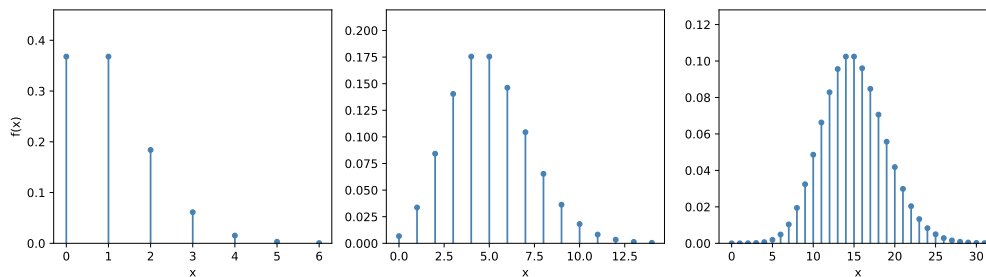


Figure 3.12: Probability mass function of $\text{Poi}(\lambda)$ for (a) $\lambda = 1$, (b) $\lambda = 5$, (c) $\lambda = 15$. As λ increases, the distribution becomes more spread out and more symmetric.

Hypergeometric Distribution

The **hypergeometric distribution** arises when sampling *without replacement* from a finite population of two types. A typical scenario: an urn contains N balls, of which K are red and $N - K$ are blue. We draw n balls without replacement and count the number of red balls drawn.

Definition : Hypergeometric Distribution — Suppose a population of N items contains K items of type I (“successes”) and $N - K$ items of type II (“failures”). If n items are drawn *without replacement*, then X , the number of successes drawn, has a **hypergeometric distribution**, written $X \sim \text{Hyp}(N, K, n)$. The pmf is

$$f(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, \quad x = \max(0, n - N + K), \dots, \min(n, K). \quad (3.105)$$

The numerator counts the number of ways to choose x successes from the K available and $n - x$ failures from the $N - K$ available. The denominator counts the total number of ways to choose n items from N . The situation is summarised in the following table:

	Success	Failure	Total
Selected	x	$n - x$	n
Not Selected	$K - x$	$N - n - K + x$	$N - n$
Total	K	$N - K$	N

The $\binom{K}{x}$ factor counts the ways to fill the top-left cell (choosing x successes from K), the $\binom{N-K}{n-x}$ factor counts the ways to fill the top-right cell (choosing $n - x$ failures from $N - K$), and $\binom{N}{n}$ counts the total number of ways to select the n items.

We verify that the pmf sums to one by a combinatorial argument: the total number of ways to choose n items from N is $\binom{N}{n}$, and partitioning by the number x of successes selected gives

$$\sum_x \binom{K}{x} \binom{N-K}{n-x} = \binom{N}{n}. \quad (3.106)$$

Mean and variance of the hypergeometric distribution

The mean can be derived by a direct (but somewhat involved) combinatorial argument. We state the results.

$$X \sim \text{Hyp}(N, K, n) : \quad \mathbb{E}[X] = n \frac{K}{N}, \quad \mathbb{V}\text{ar}(X) = n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}. \quad (3.107)$$

Setting $p = K/N$ (the proportion of successes in the population), the mean is $\mathbb{E}[X] = np$, which is the same as the binomial mean. The variance is

$$\mathbb{V}\text{ar}(X) = np(1-p) \cdot \frac{N-n}{N-1}. \quad (3.108)$$

The factor $(N-n)/(N-1)$ is called the **finite population correction**. It is always at most 1 (with equality only when $n = 1$), so the hypergeometric variance is smaller than the corresponding binomial variance $np(1-p)$. Intuitively, sampling without replacement reduces variability because the draws are negatively correlated.

Example (Quality Control). A shipment of $N = 50$ components contains $K = 5$ defective items. An inspector draws $n = 10$ components at random without replacement. The number of defective items in the sample is $X \sim \text{Hyp}(50, 5, 10)$.

The expected number of defectives is $\mathbb{E}[X] = 10 \cdot 5/50 = 1$, and the variance is

$$\mathbb{V}\text{ar}(X) = 10 \cdot \frac{5}{50} \cdot \frac{45}{50} \cdot \frac{40}{49} = 0.9 \cdot \frac{40}{49} \approx 0.735. \quad (3.109)$$

For comparison, a binomial model $\text{Bin}(10, 0.1)$ (sampling with replacement) would give $\mathbb{V}\text{ar}(X) = 0.9$, which is larger.

Relations to the Binomial Distribution

Both the Poisson and hypergeometric distributions are intimately connected to the binomial distribution through limiting arguments.

Poisson limit of the binomial

When the number of trials n is large and the success probability p is small, with the product $np \approx \lambda$ held roughly constant, the binomial distribution is well approximated by a Poisson distribution.

Claim 15: Poisson Limit Theorem — Let $X_n \sim \text{Bin}(n, p_n)$ where $p_n = \lambda/n$ for a fixed $\lambda > 0$. Then for every $k = 0, 1, 2, \dots$,

$$\lim_{n \rightarrow \infty} \binom{n}{k} p_n^k (1-p_n)^{n-k} = \frac{e^{-\lambda} \lambda^k}{k!}. \quad (3.110)$$

That is, as $n \rightarrow \infty$, the $\text{Bin}(n, \lambda/n)$ pmf converges pointwise to the $\text{Poi}(\lambda)$ pmf.

Proof: Fix $k \geq 0$ and write $p_n = \lambda/n$. Then

$$\binom{n}{k} p_n^k (1 - p_n)^{n-k} = \frac{n!}{k!(n-k)!} \cdot \frac{\lambda^k}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^{n-k}. \quad (3.111)$$

Rearranging the first two factors,

$$\frac{n!}{(n-k)! n^k} = \frac{n(n-1) \cdots (n-k+1)}{n^k} = 1 \cdot \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) \xrightarrow{n \rightarrow \infty} 1. \quad (3.112)$$

For the remaining factor, since k is fixed,

$$\left(1 - \frac{\lambda}{n}\right)^{n-k} = \left(1 - \frac{\lambda}{n}\right)^n \cdot \left(1 - \frac{\lambda}{n}\right)^{-k} \xrightarrow{n \rightarrow \infty} e^{-\lambda} \cdot 1. \quad (3.113)$$

Combining,

$$\binom{n}{k} p_n^k (1 - p_n)^{n-k} \rightarrow \frac{1}{k!} \cdot \lambda^k \cdot e^{-\lambda} = \frac{e^{-\lambda} \lambda^k}{k!}. \quad (3.114)$$

□

This result is sometimes called the “law of rare events”: when many independent trials each have a small probability of success, the total number of successes is approximately Poisson.

Example (Poisson approximation to the binomial). Suppose a factory produces $n = 1000$ items per day, each independently defective with probability $p = 0.002$. The exact distribution of the number of defectives is $\text{Bin}(1000, 0.002)$, but since n is large and p is small, we can approximate this by $\text{Poi}(\lambda)$ with $\lambda = np = 2$. For instance,

$$\mathbb{P}(X = 0) = \binom{1000}{0} (0.002)^0 (0.998)^{1000} \approx 0.1353, \quad (3.115)$$

while the Poisson approximation gives $e^{-2} \approx 0.1353$ — the agreement is excellent.

Hypergeometric convergence to the binomial

When the population size N is much larger than the sample size n , sampling without replacement behaves almost like sampling with replacement, and the hypergeometric distribution is well approximated by a binomial.

Claim 16: Hypergeometric–Binomial Convergence — Let $X \sim \text{Hyp}(N, K, n)$ with $K/N \rightarrow p \in (0, 1)$ as $N \rightarrow \infty$ (and n fixed). Then for every $k = 0, 1, \dots, n$,

$$\lim_{N \rightarrow \infty} \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}} = \binom{n}{k} p^k (1-p)^{n-k}. \quad (3.116)$$

That is, the $\text{Hyp}(N, K, n)$ pmf converges pointwise to the $\text{Bin}(n, p)$ pmf.

Proof: Write $K = \lfloor pN \rfloor$ so that $K/N \rightarrow p$. For the ratio of binomial coefficients,

$$\frac{\binom{K}{k}}{\binom{N}{k}} = \frac{K(K-1)\cdots(K-k+1)}{N(N-1)\cdots(N-k+1)} = \prod_{j=0}^{k-1} \frac{K-j}{N-j} \xrightarrow{N \rightarrow \infty} p^k, \quad (3.117)$$

since each factor $(K-j)/(N-j) \rightarrow p$. Similarly,

$$\frac{\binom{N-K}{n-k}}{\binom{N-k}{n-k}} = \prod_{j=0}^{n-k-1} \frac{N-K-j}{N-k-j} \xrightarrow{N \rightarrow \infty} (1-p)^{n-k}. \quad (3.118)$$

Combining and using $\binom{N}{n} = \binom{N}{k} \binom{N-k}{n-k} / \binom{n}{k}$, we obtain

$$\frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}} \rightarrow \binom{n}{k} p^k (1-p)^{n-k}. \quad (3.119)$$

□

Intuitively, when the population is vast compared to the sample, removing a few items barely changes the composition of the remaining population, so each draw is nearly independent with the same success probability $p = K/N$. The finite population correction $(N-n)/(N-1) \rightarrow 1$ in this regime, confirming that the variance also converges to $np(1-p)$.

Example (When to use the binomial approximation). A standard rule of thumb is that the binomial approximation to the hypergeometric is adequate when $n/N \leq 0.05$ (the sample is at most 5% of the population). For the quality control example (Example 3.5), $n/N = 10/50 = 0.2$, so the correction matters. But if the shipment had $N = 5000$ components with $K = 500$ defective and we still sampled $n = 10$, then $n/N = 0.002$ and $\text{Bin}(10, 0.1)$ would be an excellent approximation.

3.6 Simulation of Discrete Random Variables

We have met several families of discrete distributions — Bernoulli, binomial, geometric, negative binomial, Poisson, and hypergeometric. A natural question is: how can we generate random samples from a given distribution on a computer? We have already seen some answers for specific cases. Here we develop a general method.

Specific Algorithms for Specific Distributions

For each of the distributions studied so far, there is a natural simulation strategy based on the definition of the distribution:

- **Bernoulli.** Draw $U \sim \mathcal{U}(0, 1)$ and set $X = 1$ if $U < p$, else $X = 0$ (see Section 3.4).
- **Binomial.** Sum n independent Bernoulli trials: $X = \sum_{i=1}^n I_i$ where each $I_i \sim \text{Ber}(p)$.
- **Geometric.** Repeat independent $\text{Ber}(p)$ trials until the first success; count the trial number.

- **Negative Binomial.** Repeat independent $\text{Ber}(p)$ trials until the r -th success; count failures (Example 3.4).
- **Discrete Uniform** on $\{a, a + 1, \dots, b\}$: draw a uniform integer from this range.

For the discrete uniform case, Python provides a direct function:

Simulating a discrete uniform random variable

LLM Prompt > Show how to generate a single draw from a discrete uniform distribution on $\{3, 4, \dots, 8\}$ in Python using NumPy.

There are also specialised algorithms for other distributions (e.g. efficient methods for the Poisson distribution exist, but we will not cover them here). The question is: is there a **single method** that works for *any* discrete distribution?

The Cumulative Probability Method

Suppose X takes values $x_1, x_2, x_3, \dots$ with probabilities $p_1, p_2, p_3, \dots$ (where $\sum_k p_k = 1$). The idea is to partition the interval $[0, 1)$ into consecutive sub-intervals of lengths $p_1, p_2, p_3, \dots$:

$$\underbrace{[0, p_1)}_{x_1} \underbrace{[p_1, p_1 + p_2)}_{x_2} \underbrace{[p_1 + p_2, p_1 + p_2 + p_3)}_{x_3} \cdots \quad (3.120)$$

Now draw $U \sim \mathcal{U}(0, 1)$: whichever sub-interval U falls in determines X . Since U is uniform on $[0, 1)$, the probability that U falls in the k -th sub-interval is exactly p_k , as required.

Concretely, set $X = x_k$ where k is the smallest index such that

$$p_1 + p_2 + \cdots + p_k > U. \quad (3.121)$$

This gives the following algorithm.

Algorithm 3.6.1: Cumulative Probability Method for a Discrete Distribution

1. Generate $U \sim \mathcal{U}(0, 1)$.
 2. Set $S = 0, k = 1$.
 3. While $S \leq U$: set $S = S + p_k, k = k + 1$.
 4. Return $X = x_{k-1}$.
-

This method works whether the support is finite or infinite: if the support is infinite the while loop in Step 3 will always terminate in finitely many steps (with probability 1), since the cumulative sum converges to 1.

Remark (Preview of the inverse-transform method). Algorithm 3.6.1 is a special case of the *inverse-transform method* (also called the *inverse probability transform*), which provides a general technique for simulating from any distribution — discrete or continuous. We will study this method formally in Section 6.2.

Example: Finite Support

Example (Simulating a Loaded Die). Consider a loaded die with probabilities $\mathbf{p} = (0.1, 0.1, 0.1, 0.2, 0.2, 0.3)$ for faces 1–6. Since the support is finite, we can use `np.cumsum` to compute all the cumulative sums at once and then find the first index where the cumulative sum exceeds U :

Simulating a loaded die using the cumulative probability method

LLM Prompt > Write Python code that simulates 10,000 rolls of a loaded die with probabilities (0.1, 0.1, 0.1, 0.2, 0.2, 0.3) using the cumulative probability method, and compares the empirical frequencies with the theoretical probabilities in a grouped bar chart.

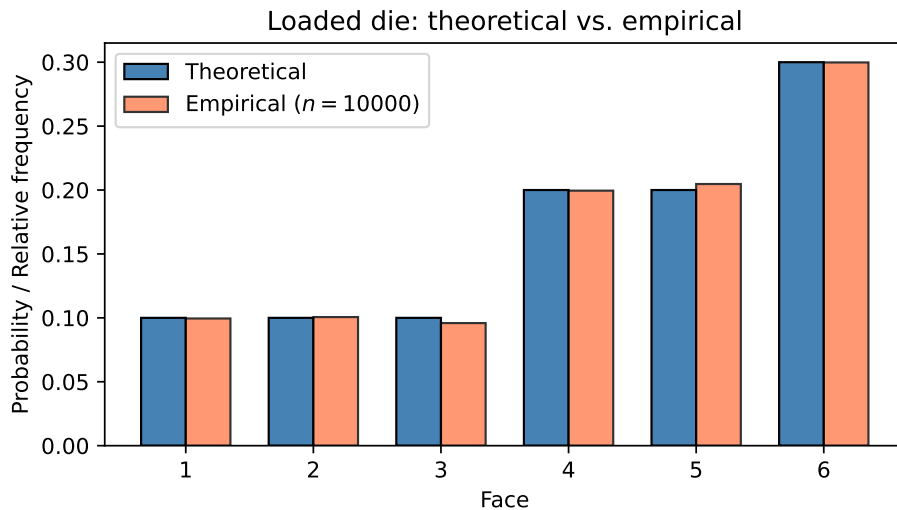


Figure 3.13: Theoretical probabilities and empirical relative frequencies ($n = 10,000$ samples) for a loaded die with $\mathbf{p} = (0.1, 0.1, 0.1, 0.2, 0.2, 0.3)$, generated using the cumulative probability method.

Example: Infinite Support

Example (The Inverse-Square Distribution). Consider a random variable X taking values $1, 2, 3, \dots$ with

$$\mathbb{P}(X = i) = \frac{c}{i^2}, \quad i = 1, 2, 3, \dots \quad (3.122)$$

where $c = 6/\pi^2$, since $\sum_{i=1}^{\infty} 1/i^2 = \pi^2/6$ (the Basel series).

Since the support is infinite, we cannot compute `np.cumsum` of the entire probability vector. Instead, we apply Algorithm 3.6.1 directly: accumulate the cumulative sum one term at a time until it exceeds U , then stop.

Simulating from an infinite-support distribution

LLM Prompt > Write Python code that samples from the infinite-support distribution $P(X = i) = (6/\pi^2)/i^2$ using the cumulative probability method (accumulating one term at a time until exceeding a uniform draw). Generate 50,000 samples and compare with the theoretical PMF for the first 20 values.

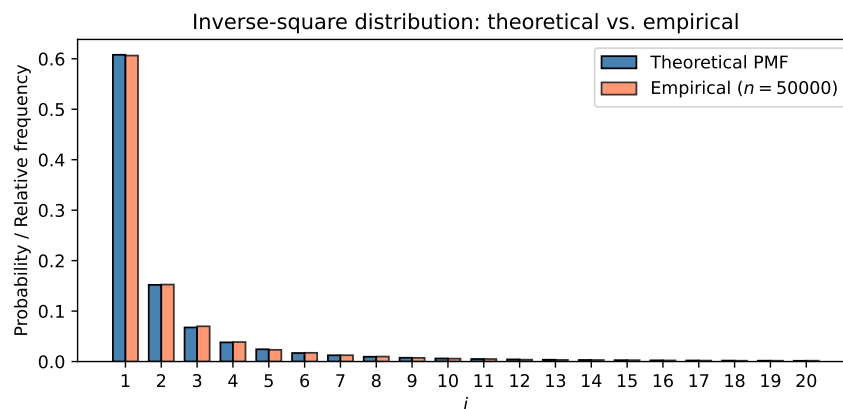


Figure 3.14: Theoretical pmf and empirical relative frequencies ($n = 50,000$ samples) for the inverse-square distribution $\mathbb{P}(X = i) = (6/\pi^2)/i^2$ over the first 20 values, generated using the cumulative probability method.

CONDITIONAL PROBABILITY AND MARKOV CHAINS

In this chapter we introduce basics of conditional probability. With this we visit the law of total probability and Bayes' theorem. We also revisit some aspects of discrete random variables from the previous chapter, now through the lens of conditional probability. We also visit the product rule and then introduce the concept of Markov chains. These stochastic processes are a sequence of discrete random variables that exhibit the Markov property. In their elementary study, we see how matrix multiplication is inherently related to their probability laws, and we visit basic concepts such as their limiting distribution at an introductory exploratory level.

4.1 Conditional Probability

How do probabilities change when we know some event $B \subseteq \Omega$ has occurred? Suppose B has occurred. Thus, we know that the outcome lies in B . Then A will occur if and only if $A \cap B$ occurs, and the relative chance of A occurring is therefore $\mathbb{P}(A \cap B)/\mathbb{P}(B)$.

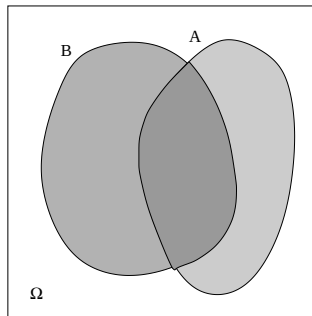


Figure 4.1: What is the conditional probability of A given that B occurs?

This leads to the following definition.

Definition : Conditional Probability — Let B be an event with $\mathbb{P}(B) > 0$. The **conditional probability** of A given B is:

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (4.1)$$

Example (Throwing Two Dice). We throw two dice. Given that the sum of the dice is 10, what is the probability that one 6 is cast?

Let B be the event that the sum is 10,

$$B = \{(4, 6), (5, 5), (6, 4)\}.$$

Let A be the event that one 6 is cast,

$$A = \{(1, 6), \dots, (5, 6), (6, 1), \dots, (6, 5)\}. \quad (4.2)$$

Then, $A \cap B = \{(4, 6), (6, 4)\}$. And, since all elementary events are equally likely, we have

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{2/36}{3/36} = \frac{2}{3}. \quad (4.3)$$

The Probability Space under Conditioning

In Section 1.3 we introduced Kolmogorov's axioms and the properties that every probability measure must satisfy. A natural question is: does the conditional probability $\mathbb{P}(\cdot | B)$, viewed as a function of events, itself define a valid probability measure?

The answer is yes. Provided $\mathbb{P}(B) > 0$, the function $\mathbb{P}(\cdot | B): \mathcal{F} \rightarrow [0, 1]$ defined by $\mathbb{P}(A | B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ satisfies all three of Kolmogorov's axioms:

1. **Non-negativity.** For every event $A \in \mathcal{F}$,

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \geq 0, \quad (4.4)$$

since $\mathbb{P}(A \cap B) \geq 0$ and $\mathbb{P}(B) > 0$.

2. **Normalization.**

$$\mathbb{P}(\Omega | B) = \frac{\mathbb{P}(\Omega \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} = 1. \quad (4.5)$$

3. **Countable additivity.** If $\{A_i\}_{i=1}^{\infty}$ is a sequence of pairwise disjoint events, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i \mid B\right) = \frac{\mathbb{P}\left(\left(\bigcup_{i=1}^{\infty} A_i\right) \cap B\right)}{\mathbb{P}(B)} = \frac{\mathbb{P}\left(\bigcup_{i=1}^{\infty} (A_i \cap B)\right)}{\mathbb{P}(B)} = \frac{\sum_{i=1}^{\infty} \mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \sum_{i=1}^{\infty} \mathbb{P}(A_i | B), \quad (4.6)$$

where the second equality uses the fact that the sets $A_i \cap B$ are themselves pairwise disjoint, together with countable additivity of $\mathbb{P}$.

Since $\mathbb{P}(\cdot | B)$ satisfies the axioms, all properties derived from them in Section 1.3 — the complement rule, monotonicity, inclusion–exclusion, and so on — automatically hold for conditional probabilities as well. In other words, $(\Omega, \mathcal{F}, \mathbb{P}(\cdot | B))$ is a valid probability triple, and we may work with conditional probabilities using exactly the same tools we developed in Chapter 1.

Constructing a Probability Model via Conditional Probability

In the dice example above we started with a fully specified probability space and then *derived* conditional probabilities from it. In practice, models are often built the other way around: we *specify* conditional probabilities as part of the model and use them to determine the joint probability of all events.

Example (Two-Stage Urn). A box contains 3 red and 2 blue balls. We draw one ball at random; then, without replacing it, we draw a second ball. Rather than listing all $5 \times 4 = 20$ ordered pairs and assigning each probability $1/20$, we can describe the experiment by specifying:

- The first-draw probabilities: $\mathbb{P}(R_1) = 3/5$ and $\mathbb{P}(B_1) = 2/5$.
- The conditional second-draw probabilities:

$$\mathbb{P}(R_2 | R_1) = \frac{1}{2}, \quad \mathbb{P}(B_2 | R_1) = \frac{1}{2}, \quad \mathbb{P}(R_2 | B_1) = \frac{3}{4}, \quad \mathbb{P}(B_2 | B_1) = \frac{1}{4}. \quad (4.7)$$

This structure is naturally represented by a probability tree:

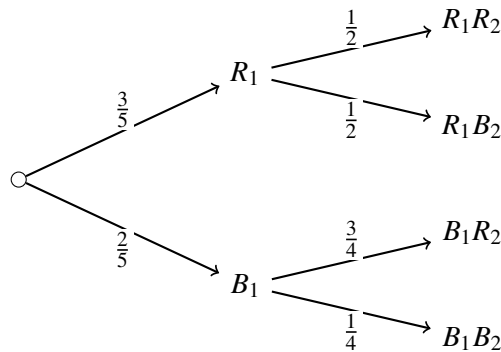


Figure 4.2: Probability tree for the two-stage urn. Edge labels are probabilities; second-stage edges are conditional probabilities given the first draw.

From these we can compute any joint probability. For instance,

$$\mathbb{P}(R_1 \cap B_2) = \mathbb{P}(R_1) \mathbb{P}(B_2 | R_1) = \frac{3}{5} \cdot \frac{1}{2} = \frac{3}{10}, \quad \mathbb{P}(B_1 \cap B_2) = \frac{2}{5} \cdot \frac{1}{4} = \frac{1}{10}. \quad (4.8)$$

The conditional probabilities, together with the first-draw probabilities, fully determine the probability space. This “specify conditionals first” approach is often more natural than enumerating the entire sample space, and it becomes essential for multi-stage experiments and Markov chains (Section 4.4).

Independence and Conditional Probability

In Chapter 2 we defined events A and B to be **independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Note that this definition is symmetric: A independent of B is the same as B independent of A . With the language of conditional probability, we can now see what this really means. If A and B are independent and $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A)\mathbb{P}(B)}{\mathbb{P}(B)} = \mathbb{P}(A). \quad (4.9)$$

Similarly, if $\mathbb{P}(A) > 0$, then $\mathbb{P}(B|A) = \mathbb{P}(B)$. In other words, independence says precisely that conditioning on one event does not change the probability of the other. Knowing that B has occurred gives no new information about whether A occurs, and vice versa. Conversely, if $\mathbb{P}(A|B) = \mathbb{P}(A)$, then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, recovering the definition from Chapter 2.

In the dice example above, the events “sum is 10” and “one 6 is cast” are *not* independent: conditioning on the sum changed the probability from $10/36$ to $2/3$.

Example (Independent Events with Two Dice). Roll two fair dice. Let $A = \{\text{first die shows 6}\}$ and $B = \{\text{second die shows 1}\}$. Then $\mathbb{P}(A) = 1/6$, $\mathbb{P}(B) = 1/6$, and $A \cap B = \{(6, 1)\}$ with $\mathbb{P}(A \cap B) = 1/36 = \mathbb{P}(A)\mathbb{P}(B)$. So A and B are independent, and indeed $\mathbb{P}(A|B) = \mathbb{P}(A) = 1/6$ — knowing the second die shows 1 tells us nothing about the first die.

Conditional Independence

Two events A and B may be independent unconditionally, yet become dependent once we condition on a third event C — or vice versa. This motivates the following definition.

Definition : Conditional Independence — Events A and B are said to be **conditionally independent given C** (with $\mathbb{P}(C) > 0$) if

$$\mathbb{P}(A \cap B|C) = \mathbb{P}(A|C)\mathbb{P}(B|C). \quad (4.10)$$

Equivalently, by the results of the previous subsection, conditional independence means $\mathbb{P}(A|B \cap C) = \mathbb{P}(A|C)$ whenever $\mathbb{P}(B \cap C) > 0$: learning that B occurred provides no additional information about A beyond what C already tells us.

It is important to note that (unconditional) independence does *not* imply conditional independence, nor does conditional independence imply unconditional independence.

Example (Pairwise Independent but Not Conditionally Independent). Recall the two-coin example from Section 2.4. We toss two fair coins with $\Omega = \{HH, HT, TH, TT\}$, each outcome having probability $1/4$, and define

$$A = \{\text{first coin is H}\} = \{HH, HT\}, \quad (4.11)$$

$$B = \{\text{second coin is H}\} = \{HH, TH\}, \quad (4.12)$$

$$C = \{\text{both coins show the same face}\} = \{HH, TT\}. \quad (4.13)$$

We showed there that A , B , and C are pairwise independent. Now consider whether A and B are conditionally independent given C .

We have $\mathbb{P}(C) = 1/2$. The conditional probabilities are:

$$\mathbb{P}(A|C) = \frac{\mathbb{P}(A \cap C)}{\mathbb{P}(C)} = \frac{\mathbb{P}(\{HH\})}{1/2} = \frac{1/4}{1/2} = \frac{1}{2}, \quad \mathbb{P}(B|C) = \frac{1}{2}, \quad (4.14)$$

by symmetry. If A and B were conditionally independent given C , we would need $\mathbb{P}(A \cap B|C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. But

$$\mathbb{P}(A \cap B|C) = \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(C)} = \frac{\mathbb{P}(\{HH\})}{1/2} = \frac{1/4}{1/2} = \frac{1}{2} \neq \frac{1}{4}. \quad (4.15)$$

So A and B are *not* conditionally independent given C . Intuitively, once we know both coins show the same face, learning that the first coin is heads tells us with certainty that the second is also heads.

Product Rule

By the definition of conditional probability we have

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B|A). \quad (4.16)$$

We can generalise this to n intersections $A_1 \cap A_2 \cap \cdots \cap A_n$, which we abbreviate as $A_1 A_2 \cdots A_n$. This gives the **product rule** of probability.

Claim 17: Product Rule — Let $A_1, \dots, A_n$ be a sequence of events with $\mathbb{P}(A_1 \dots A_{n-1}) > 0$. Then,

$$\mathbb{P}(A_1 \cdots A_n) = \mathbb{P}(A_1) \mathbb{P}(A_2|A_1) \mathbb{P}(A_3|A_1 A_2) \cdots \mathbb{P}(A_n|A_1 \cdots A_{n-1}). \quad (4.17)$$

Proof: We show the proof for 4 events A, B, C, D ; the general case follows similarly. By applying (4.1) to each conditional probability on the right-hand side of (4.17), we have

$$\mathbb{P}(A) \mathbb{P}(B|A) \mathbb{P}(C|AB) \mathbb{P}(D|ABC) = \mathbb{P}(A) \frac{\mathbb{P}(AB)}{\mathbb{P}(A)} \frac{\mathbb{P}(ABC)}{\mathbb{P}(AB)} \frac{\mathbb{P}(ABCD)}{\mathbb{P}(ABC)} = \mathbb{P}(ABCD), \quad (4.18)$$

where the last equality follows by cancellation of common factors. The proof for general n follows the same telescoping argument. $\square$

Example (Drawing Balls). We draw consecutively 3 balls from a bowl with 5 white and 5 black balls, without putting them back. What is the probability that all balls will be black?

Solution: Let A_i be the event that the i th ball is black. We wish to find the probability of $A_1 A_2 A_3$, which by the product rule (4.17) is

$$\mathbb{P}(A_1) \mathbb{P}(A_2|A_1) \mathbb{P}(A_3|A_1 A_2) = \frac{5}{10} \frac{4}{9} \frac{3}{8} = \frac{1}{12} \approx 0.083. \quad (4.19)$$

Note that this problem can also be easily solved by counting arguments, as in the previous section.

Example (Birthday Problem). In Section 2.2 we derived by counting arguments that the probability that all people in a group of 40 have different birthdays is

$$\frac{365 \times 364 \times \cdots \times 326}{365 \times 365 \times \cdots \times 365} \approx 0.109. \quad (4.20)$$

We can derive this also via the product rule. Namely, let A_i be the event that the first i people have different birthdays, $i = 1, 2, \dots$. Note that $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$. Therefore $A_n = A_1 \cap A_2 \cap \cdots \cap A_n$, and thus by the product rule

$$\mathbb{P}(A_{40}) = \mathbb{P}(A_1)\mathbb{P}(A_2 | A_1)\mathbb{P}(A_3 | A_1 A_2) \cdots \mathbb{P}(A_{40} | A_1 \cdots A_{39}). \quad (4.21)$$

Since $A_1 \supseteq A_2 \supseteq \cdots$, the intersection $A_1 \cap \cdots \cap A_{k-1}$ is simply A_{k-1} , so each factor simplifies to $\mathbb{P}(A_k | A_{k-1}) = (365 - k + 1)/365$: given that the first $k-1$ people have different birthdays, there are no duplicate birthdays if and only if the birthday of the k -th is chosen from the $365 - (k-1)$ remaining birthdays. Thus, we obtain (4.20). More generally, the probability that n randomly selected people have different birthdays is

$$\mathbb{P}(A_n) = \frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \cdots \times \frac{365 - n + 1}{365}, \quad n \geq 1. \quad (4.22)$$

A graph of $\mathbb{P}(A_n)$ against n is given in Figure 4.3. Note that the probability $\mathbb{P}(A_n)$ rapidly decreases to zero. Indeed, for $n = 23$ the probability of having no duplicate birthdays is already less than $1/2$.

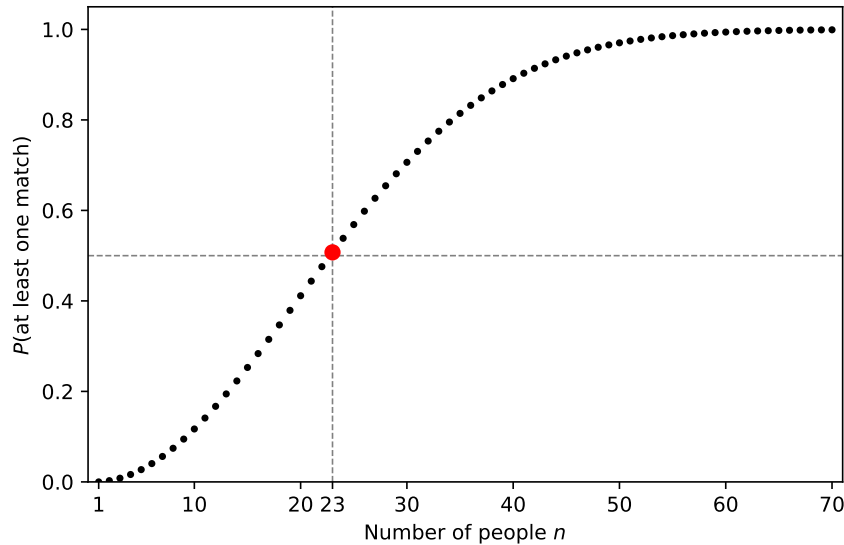


Figure 4.3: The probability of having no duplicate birthday in a group of n people, against n .

4.2 Law of Total Probability and Bayes' Theorem

Conditional probability becomes especially powerful when combined with a decomposition of the sample space into cases. This leads to two fundamental tools: the law of total probability, which computes unconditional probabilities by averaging over cases, and Bayes' theorem, which reverses the direction of conditioning.

Law of Total Probability

Suppose $B_1, B_2, \dots, B_n$ is a **partition** of Ω . That is, $B_1, B_2, \dots, B_n$ are disjoint and their union is Ω , see Figure 4.4.

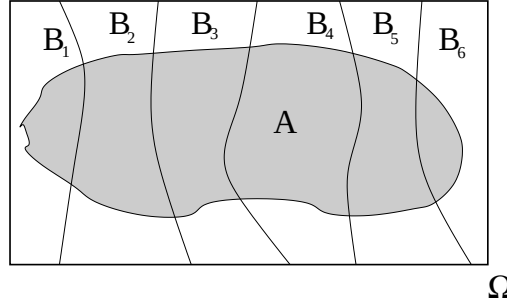


Figure 4.4: A partition of the sample space

Claim 18: Law of Total Probability — Let $B_1, \dots, B_n$ be a partition of Ω with $\mathbb{P}(B_i) > 0$ for each i , and let A be any event. Then,

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A | B_i) \mathbb{P}(B_i). \quad (4.23)$$

Proof: Since $A = A \cap \Omega = A \cap (B_1 \cup \dots \cup B_n) = (A \cap B_1) \cup \dots \cup (A \cap B_n)$ and the sets $A \cap B_i$ are disjoint, the sum rule gives $\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \cap B_i)$. Applying the definition of conditional probability to each term, $\mathbb{P}(A \cap B_i) = \mathbb{P}(A | B_i) \mathbb{P}(B_i)$, gives (4.23). $\square$

Note that if $\mathbb{P}(B_i) = 0$ for some i , the conditional probability $\mathbb{P}(A | B_i)$ is not defined. However, the corresponding term $\mathbb{P}(A | B_i) \mathbb{P}(B_i) = \mathbb{P}(A \cap B_i) = 0$ contributes nothing to the sum (since $A \cap B_i \subseteq B_i$ and $\mathbb{P}(B_i) = 0$). We may therefore simply omit such terms, or equivalently adopt the convention that $0 \cdot \text{undefined} = 0$ in this context.

Example (Three Factories). A company has three factories (1, 2 and 3) that produce the same chip, producing 15%, 35% and 50% of the total production, respectively. The probability of a defective chip at 1, 2, 3 is 0.01, 0.05, 0.02, respectively. What is the overall probability that a randomly selected chip is defective?

Let B_i denote the event that the chip is produced by factory i . The $\{B_i\}$ form a partition of Ω . Let A denote the event that the chip is defective. By the law of total probability,

$$\mathbb{P}(A) = 0.15 \times 0.01 + 0.35 \times 0.05 + 0.5 \times 0.02 = 0.0015 + 0.0175 + 0.01 = 0.029. \quad (4.24)$$

So about 2.9% of all chips are defective.

Bayes' Theorem

Combining the law of total probability with the definition of conditional probability gives **Bayes' theorem**:

Claim 19: Bayes' Theorem — Let $B_1, \dots, B_n$ be a partition of Ω with $\mathbb{P}(B_j) > 0$ for each j , and let A be an event with $\mathbb{P}(A) > 0$. Then,

$$\mathbb{P}(B_j | A) = \frac{\mathbb{P}(A | B_j) \mathbb{P}(B_j)}{\sum_{i=1}^n \mathbb{P}(A | B_i) \mathbb{P}(B_i)}. \quad (4.25)$$

Proof: By the definition of conditional probability, $\mathbb{P}(B_j | A) = \mathbb{P}(A \cap B_j) / \mathbb{P}(A)$. The numerator satisfies $\mathbb{P}(A \cap B_j) = \mathbb{P}(A | B_j) \mathbb{P}(B_j)$, again by the definition of conditional probability. Replacing the denominator $\mathbb{P}(A)$ by the law of total probability gives (4.25). $\square$

The power of Bayes' theorem lies in “reversing the conditioning”: given information about $\mathbb{P}(A | B_j)$ (how likely we are to observe A under each scenario B_j), we can compute $\mathbb{P}(B_j | A)$ (which scenario is most likely, given that we observed A).

Example (Noisy Binary Channel). A transmitter sends a binary digit (0 or 1) over a noisy channel. The digit sent is 1 with probability $\mathbb{P}(B_1) = 0.5$ and 0 with probability $\mathbb{P}(B_0) = 0.5$. The channel occasionally flips a bit: if a 1 is sent, it is received correctly with probability $\mathbb{P}(A_1 | B_1) = 0.9$ (and received as 0 with probability 0.1). If a 0 is sent, it is received correctly with probability $\mathbb{P}(A_0 | B_0) = 0.95$ (and received as 1 with probability 0.05).

Suppose we receive a 1. What is the probability that a 1 was actually sent?

By Bayes' theorem,

$$\mathbb{P}(B_1 | A_1) = \frac{\mathbb{P}(A_1 | B_1) \mathbb{P}(B_1)}{\mathbb{P}(A_1 | B_1) \mathbb{P}(B_1) + \mathbb{P}(A_1 | B_0) \mathbb{P}(B_0)} = \frac{0.9 \times 0.5}{0.9 \times 0.5 + 0.05 \times 0.5} = \frac{0.45}{0.475} \approx 0.947. \quad (4.26)$$

So if we receive a 1, there is about a 94.7% chance that a 1 was indeed sent. The small but nonzero probability of error ($\approx 5.3\%$) comes from the possibility that a 0 was sent but flipped to 1 by the channel noise.

Example (Three Factories (Continued)). Returning to the factory example, suppose someone shows us a defective chip. What is the probability that it came from factory 1? By Bayes' theorem,

$$\mathbb{P}(B_1 | A) = \frac{0.15 \times 0.01}{0.15 \times 0.01 + 0.35 \times 0.05 + 0.5 \times 0.02} = \frac{0.0015}{0.029} \approx 0.052. \quad (4.27)$$

Although factory 1 produces 15% of all chips, it has the lowest defect rate, so only about 5.2% of defective chips come from factory 1.

Example (Monty Hall Problem). The *Monty Hall problem* is a famous example illustrating how probabilistic reasoning may sometimes yield surprising results. Consider a contestant on a game show, with three doors in front of her. One of the doors contains a prize, while the other two are empty. The contestant is asked to guess which door contains the prize, and she makes a random guess. Following this, the game show host (GSH) reveals an empty (losing) door from one of the two remaining doors not chosen. The contestant is then asked if she wishes to stay with her original choice, or if she wishes to switch to the remaining closed door. The question is: should the contestant stay with her original choice, or switch? Or perhaps it doesn't matter.

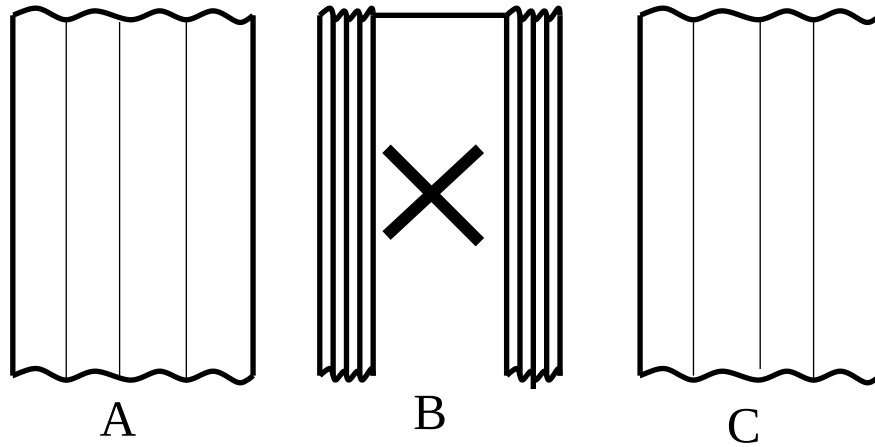


Figure 4.5: The Monty Hall problem: the prize is behind one of three doors.

The two possible policies (or strategies of play) for the contestant are as follows:

Policy I — Stay with the original choice after the door is revealed.

Policy II — Switch after the door is revealed.

If the player adopts Policy I then she always stays with her initial guess regardless of the GSH's action. In this case, her chance of success is $1/3$, that is, she wins if her initial choice is correct.

However, if she adopts Policy II then she always switches after the GSH reveals an empty door. In this case her chance of success is $2/3$, that is, she actually wins if her initial guess is *incorrect*. This is because the GSH must always reveal a losing door. If she originally chose a losing door, then the GSH must reveal the second losing door every time (otherwise he would reveal the prize). That is, if the player chooses an incorrect door at the start, the non-revealed door will always be the winning door. The chance of such an event is $2/3$.

As a further aid for understanding, imagine a case of 100 doors and a single prize behind one of them. The player chooses a door, say door 1, and following this the GSH reveals 98 losing doors. There are now only two doors remaining, her choice door 1, and (say for example) door 38. The intuition of the problem suddenly becomes obvious. The player's original guess was random and hence door 1 had a $1/100$ chance of containing the prize; however, the GSH's actions were constrained — he had to reveal only losing doors, and hence there is a $99/100$ chance that door 38 contains the prize. Hence, Policy II is clearly superior.

We now analyse the case of 3 doors by applying Bayes' theorem. Let A_i be the event that the prize is behind door i . Let B_i be the event that door i is revealed by the GSH. Then, for example, if

the player initially chooses door 1 and then the GSH reveals door 2, we have the following:

$$\mathbb{P}(A_1 | B_2) = \frac{\mathbb{P}(B_2 | A_1) \mathbb{P}(A_1)}{\mathbb{P}(B_2)} = \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3}, \quad (\text{Policy I}) \quad (4.28)$$

$$\mathbb{P}(A_3 | B_2) = \frac{\mathbb{P}(B_2 | A_3) \mathbb{P}(A_3)}{\mathbb{P}(B_2)} = \frac{1 \times \frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}. \quad (\text{Policy II}) \quad (4.29)$$

In the second case note that $\mathbb{P}(B_2 | A_3) = 1$ because the GSH must reveal door 2 if the prize is behind door 3 since door 1 was already picked. Hence, we see that while neither policy guarantees a win, Policy II clearly dominates Policy I.

4.3 Discrete Random Variables Revisited with Conditioning

In Chapter 3 we studied discrete random variables and their distributions. Now that we have the tools of conditional probability, we can revisit some of these distributions.

Conditioning on Events Involving Random Variables

We begin with a simple example. Let X be the outcome of rolling a fair die, so $X \sim \text{DU}(\{1, 2, 3, 4, 5, 6\})$. Suppose we are told that $X \geq 3$. What is the conditional probability that $X = k$?

Using the definition of conditional probability,

$$\mathbb{P}(X = k | X \geq 3) = \frac{\mathbb{P}(X = k \text{ and } X \geq 3)}{\mathbb{P}(X \geq 3)}. \quad (4.30)$$

The denominator is $\mathbb{P}(X \geq 3) = \mathbb{P}(X = 3) + \mathbb{P}(X = 4) + \mathbb{P}(X = 5) + \mathbb{P}(X = 6) = 4/6 = 2/3$. For the numerator, the event $\{X = k\} \cap \{X \geq 3\}$ is empty if $k < 3$, and equals $\{X = k\}$ if $k \geq 3$. Therefore,

$$\mathbb{P}(X = k | X \geq 3) = \begin{cases} 0 & \text{if } k \in \{1, 2\}, \\ \frac{1/6}{2/3} = \frac{1}{4} & \text{if } k \in \{3, 4, 5, 6\}. \end{cases} \quad (4.31)$$

This is exactly the $\text{DU}(\{3, 4, 5, 6\})$ distribution: given that the roll is at least 3, each of the four possible values is equally likely. The conditioning has simply restricted the sample space and renormalised the probabilities.

The Memoryless Property of the Geometric Distribution

Recall from Chapter 3 that if $X \sim \text{Geom}(p)$, then X counts the number of independent $\text{Ber}(p)$ trials until the first success, with pmf $f(k) = (1 - p)^{k-1} p$ for $k = 1, 2, 3, \dots$. We first derive the complementary cdf (ccdf), $\mathbb{P}(X > k)$, and then use it to establish the memoryless property.

The complementary cdf

We compute $\mathbb{P}(X > k)$ in two ways.

Analytic derivation. Summing the pmf,

$$\mathbb{P}(X > k) = \sum_{j=k+1}^{\infty} (1-p)^{j-1} p = p(1-p)^k \sum_{i=0}^{\infty} (1-p)^i = p(1-p)^k \cdot \frac{1}{p} = (1-p)^k. \quad (4.32)$$

Probability argument. The event $\{X > k\}$ means that the first success has not occurred in the first k trials — in other words, all of the first k trials are failures. Since the trials are independent, each with failure probability $1-p$,

$$\mathbb{P}(X > k) = (1-p)^k. \quad (4.33)$$

Both approaches give the same clean result:

$$\boxed{\mathbb{P}(X > k) = (1-p)^k, \quad k = 0, 1, 2, \dots} \quad (4.34)$$

Memoryless property

Suppose we have already performed s trials without a success. What is the probability that we need more than t *additional* trials? In other words, what is $\mathbb{P}(X > s+t | X > s)$?

By the definition of conditional probability,

$$\mathbb{P}(X > s+t | X > s) = \frac{\mathbb{P}(X > s+t \text{ and } X > s)}{\mathbb{P}(X > s)}. \quad (4.35)$$

Since $\{X > s+t\} \subseteq \{X > s\}$, the intersection is simply $\{X > s+t\}$. Using the ccdf,

$$\mathbb{P}(X > s+t | X > s) = \frac{(1-p)^{s+t}}{(1-p)^s} = (1-p)^t = \mathbb{P}(X > t). \quad (4.36)$$

Claim 20: Memoryless Property — If $X \sim \text{Geom}(p)$, then for all $s, t \geq 0$,

$$\boxed{\mathbb{P}(X > s+t | X > s) = \mathbb{P}(X > t).} \quad (4.37)$$

The past — the fact that we have already waited s trials without success — is irrelevant to the future. The probability of needing more than t additional trials is the same as if we were starting from scratch. This is the **memoryless property**: the geometric distribution “forgets” how long it has been waiting.

Intuitively, this makes sense: each Bernoulli trial is independent, so the coin does not “remember” its previous outcomes. If we have had 100 failures in a row, the chance of success on the next trial is still p , exactly as on the first trial.

The geometric distribution is the *only* discrete distribution with the memoryless property. Its continuous counterpart is the exponential distribution (Chapter 5), which is the only continuous memoryless distribution.

4.4 Markov Chains Introduction

Markov chains are very important tools for modelling random phenomena. In this chapter we give a brief introduction to these processes, laying the groundwork for more advanced treatments such as STAT3004.

A First Example: Beyond IID Sequences

In Chapter 3 we studied iid sequences, where each random variable is independent of all the others. Many real-world phenomena, however, exhibit *dependence*: the current state influences what happens next. Markov chains provide the simplest and most important model for such dependent sequences.

Example (Two-State Markov Chain vs. IID Bernoulli). Consider a sequence $X_0, X_1, X_2, \dots$ taking values in $\{0, 1\}$, where the next state depends on the current state via the rule:

$$\mathbb{P}(X_{n+1} = 0 | X_n = 0) = 0.9, \quad \mathbb{P}(X_{n+1} = 1 | X_n = 1) = 0.7. \quad (4.38)$$

In other words, there is a high probability of staying in the current state and a small probability of switching. This is a **two-state Markov chain**. In contrast, an iid $\text{Ber}(0.5)$ sequence has no memory: each value is equally likely to be 0 or 1, regardless of what came before.

The following code simulates both sequences.

Two-state Markov chain vs. iid Bernoulli

LLM Prompt > Write Python code that simulates and plots (a) an iid Bernoulli(0.5) sequence and (b) a two-state Markov chain with staying probabilities $p_{00} = 0.9$ and $p_{11} = 0.7$, both of length 200. Display them as scatter plots stacked vertically to contrast the iid case with the persistent Markov chain.

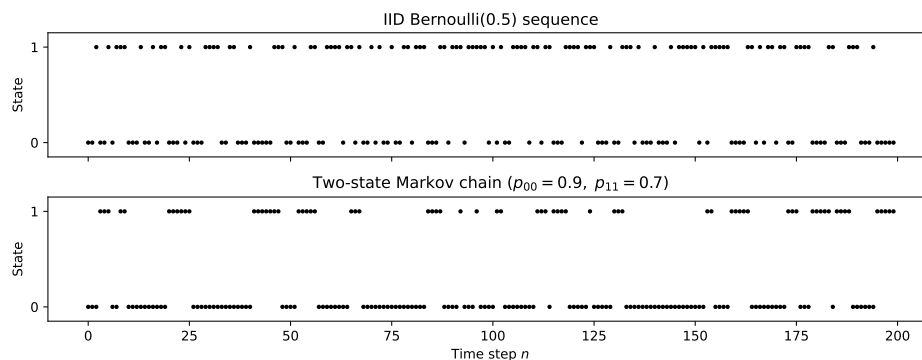


Figure 4.6: An iid Bernoulli sequence (top) versus a two-state Markov chain (bottom). Both take values in $\{0, 1\}$, but the Markov chain exhibits long runs in each state due to the dependence between consecutive values.

The difference is striking. The iid sequence switches between 0 and 1 frequently and erratically. The Markov chain, by contrast, tends to stay in the same state for long stretches before switching — it exhibits *persistence*.

The Markov Property

The key property of the two-state chain in Example 4.4 is that the conditional distribution of X_{n+1} , given the entire history $X_0, X_1, \dots, X_n$, depends only on X_n :

$$\mathbb{P}(X_{n+1} = j | X_0 = x_0, \dots, X_n = i) = \mathbb{P}(X_{n+1} = j | X_n = i). \quad (4.39)$$

This is called the **Markov property**: the future is conditionally independent of the past, given the present. Only the current state matters for predicting the next step.

Computing Distributions via the Product Rule

Knowing the transition probabilities and the starting state, we can compute the distribution of X_n for any n by repeated application of the law of total probability. Let us see how this works for the two-state chain of Example 4.4.

The transition probabilities are:

$$\begin{aligned} \mathbb{P}(X_{n+1} = 0 | X_n = 0) &= 0.9, & \mathbb{P}(X_{n+1} = 1 | X_n = 0) &= 0.1, \\ \mathbb{P}(X_{n+1} = 0 | X_n = 1) &= 0.3, & \mathbb{P}(X_{n+1} = 1 | X_n = 1) &= 0.7. \end{aligned}$$

Suppose the chain starts in state 0 with certainty: $\mathbb{P}(X_0 = 0) = 1$.

Distribution of X_1 . By the law of total probability, conditioning on X_0 :

$$\mathbb{P}(X_1 = 0) = \mathbb{P}(X_1 = 0 | X_0 = 0) \mathbb{P}(X_0 = 0) + \mathbb{P}(X_1 = 0 | X_0 = 1) \mathbb{P}(X_0 = 1) \quad (4.40)$$

$$= 0.9 \times 1 + 0.3 \times 0 = 0.9, \quad (4.41)$$

$$\mathbb{P}(X_1 = 1) = 1 - 0.9 = 0.1. \quad (4.42)$$

After one step the chain is still very likely in state 0, since it started there and has a high probability of staying.

Distribution of X_2 . We apply the same idea, now conditioning on X_1 . By the Markov property, the transition probabilities from X_1 to X_2 are exactly the same as from X_0 to X_1 :

$$\mathbb{P}(X_2 = 0) = \mathbb{P}(X_2 = 0 | X_1 = 0) \mathbb{P}(X_1 = 0) + \mathbb{P}(X_2 = 0 | X_1 = 1) \mathbb{P}(X_1 = 1) \quad (4.43)$$

$$= 0.9 \times 0.9 + 0.3 \times 0.1 = 0.81 + 0.03 = 0.84, \quad (4.44)$$

$$\mathbb{P}(X_2 = 1) = 1 - 0.84 = 0.16. \quad (4.45)$$

Distribution of X_3 . Repeating once more:

$$\mathbb{P}(X_3 = 0) = 0.9 \times 0.84 + 0.3 \times 0.16 = 0.756 + 0.048 = 0.804, \quad (4.46)$$

$$\mathbb{P}(X_3 = 1) = 1 - 0.804 = 0.196. \quad (4.47)$$

We see a clear pattern: the probability of being in state 0 is gradually decreasing from 1, while the probability of being in state 1 is increasing. The chain is “drifting” towards an equilibrium. One can

show (and we will, using matrices) that in the long run $\mathbb{P}(X_n = 0) \rightarrow 0.75$ and $\mathbb{P}(X_n = 1) \rightarrow 0.25$, regardless of where the chain started. The key observation is that at each step, we only needed the *distribution of the current state* and the *transition probabilities* to compute the distribution at the next step. This is a direct consequence of the Markov property.

More Than Two States

The two-state chain above illustrates the key ideas, but Markov chains can have any finite number of states, and even countably infinitely many. The set of all possible states is called the **state space** and we denote it by $\mathcal{S}$. For a finite state space we typically write $\mathcal{S} = \{1, \dots, N\}$ for some finite N ; for a countably infinite state space we write $\mathcal{S} = \{0, 1, 2, \dots\}$. The Markov property and the law of total probability calculations carry over in exactly the same way regardless of the size of $\mathcal{S}$.

Example (Stepping Stones). Consider 6 stones in a pond, labelled $1, \dots, 6$, connected by directed arrows as shown in Figure 4.7. The state space is $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. From the current stone, the next stone is chosen uniformly at random among those reachable by an outgoing arrow. For example, from stone 5 one can step to stone 4 or stone 6, each with probability $1/2$.

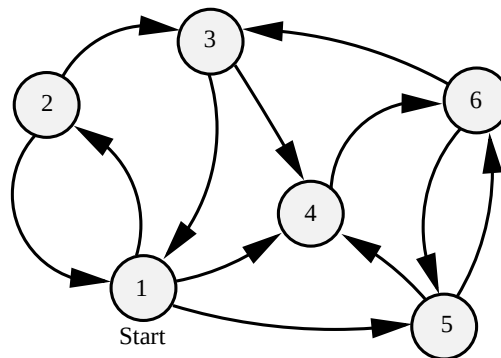


Figure 4.7: Stepping stones: a directed graph on 6 states.

Let X_n denote the position after n steps, with $X_0 = 1$. This is a Markov chain: given the current stone X_n , the next stone X_{n+1} is determined by the outgoing arrows from X_n alone, regardless of the path taken to get there. Where would the walker be after 10 steps? What is the probability of returning to stone 1?

With more than two states, listing all the transition probabilities individually becomes cumbersome. Instead, we arrange them in an array. We can collect all one-step transition probabilities $p_{ij} = \mathbb{P}(X_{n+1} = j | X_n = i)$ into a matrix called the (one-step) *transition matrix* of X . We denote it

usually by P . For example, when $S = \{0, 1, 2, \dots\}$ the transition matrix P has the form

$$\begin{array}{cc} & \text{to} \\ \text{from} & \begin{bmatrix} p_{00} & p_{01} & p_{02} & \dots \\ p_{10} & p_{11} & p_{12} & \dots \\ p_{20} & p_{21} & p_{22} & \dots \\ \vdots & \vdots & & \end{bmatrix} \end{array} \quad (4.48)$$

Note that the elements in every row are non-negative and sum to 1, since from any state the chain must go somewhere. A matrix with non-negative entries whose rows each sum to 1 is called a **stochastic matrix**. Every transition matrix is a stochastic matrix.

Example (Stepping Stones (Continued)). For the stepping stone example the transition matrix is

$$P = \begin{bmatrix} 0 & 1/3 & 0 & 1/3 & 1/3 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1/2 & 0 & 1/2 & 0 \end{bmatrix}. \quad (4.49)$$

For instance, the first row says that from stone 1 we go to stones 2, 4, or 5 each with probability $1/3$.

Another convenient way to describe a Markov chain X is through its *transition graph*. States are indicated by the nodes of the graph, and a strictly positive (> 0) transition probability p_{ij} from state i to j is indicated by an arrow from i to j with weight p_{ij} . For the stepping stone example, the transition graph would give the graph of Figure 4.7 with the addition of weights on the arrows (arcs). For example the arrow from 1 to 2, would have weight $1/3$.

More Examples

We now present several more examples of Markov chains arising from different application areas. These will serve as running examples throughout the rest of the chapter.

Example (Ehrenfest Chain). Consider two containers, A and B , holding a total of N particles between them. At each time step, a particle is chosen uniformly at random (from all N particles) and moved to the other container. Let X_n denote the number of particles in container A after n steps. Then $X_0, X_1, X_2, \dots$ is a Markov chain with state space $S = \{0, 1, \dots, N\}$.

The transition probabilities are:

- From state i , the probability of moving to state $i - 1$ is i/N (choosing one of the i particles in A).
- From state i , the probability of moving to state $i + 1$ is $(N - i)/N$ (choosing one of the $N - i$ particles in B).

All other transitions have probability zero. This model was introduced by Paul and Tatiana Ehrenfest in 1907 to study heat exchange between two bodies.

For $N = 4$, the transition matrix is

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1/4 & 0 & 3/4 & 0 & 0 \\ 0 & 2/4 & 0 & 2/4 & 0 \\ 0 & 0 & 3/4 & 0 & 1/4 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad (4.50)$$

and the transition graph is shown in Figure 4.8.

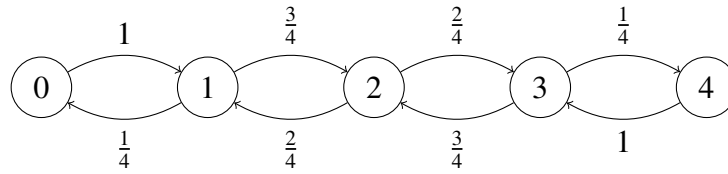


Figure 4.8: Transition graph for the Ehrenfest chain with $N = 4$ particles.

Example (Machine Reliability). A machine can be in one of two states: Working ($W = 0$) or Broken ($B = 1$). If the machine is working, it breaks down with probability p (and continues working with probability $1 - p$). If the machine is broken, it is repaired with probability r (and remains broken with probability $1 - r$). Let X_n denote the state of the machine at time n . Then $X_0, X_1, X_2, \dots$ is a Markov chain with state space $\mathcal{S} = \{0, 1\}$ and transition matrix

$$P = \begin{bmatrix} 1 - p & p \\ r & 1 - r \end{bmatrix}. \quad (4.51)$$

The transition graph is:

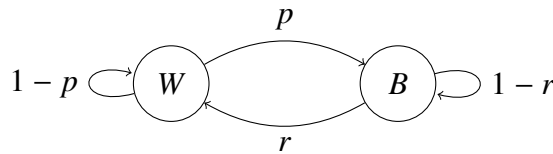


Figure 4.9: Transition graph for the machine reliability model.

This is similar in structure to Example 4.4.

Example (Simple Queueing Model). Consider a service point (e.g., a help desk) with a finite waiting room of capacity K . In each time step, one of three things happens: a new customer arrives (with probability λ), a customer finishes being served and departs (with probability μ), or nothing changes (with probability $1 - \lambda - \mu$), where $\lambda + \mu \leq 1$. Let X_n denote the number of customers present at time n . Then $X_0, X_1, X_2, \dots$ is a Markov chain with state space $\mathcal{S} = \{0, 1, \dots, K\}$.

The transition probabilities for $0 < i < K$ are:

$$p_{i,i+1} = \lambda, \quad p_{i,i-1} = \mu, \quad p_{i,i} = 1 - \lambda - \mu. \quad (4.52)$$

At the boundaries: from state 0 we can only go to state 1 (with probability λ) or stay (with probability $1 - \lambda$); from state K we can only go to $K - 1$ (with probability μ) or stay (with probability $1 - \mu$).

For $K = 3$, the transition matrix is

$$P = \begin{bmatrix} 1 - \lambda & \lambda & 0 & 0 \\ \mu & 1 - \lambda - \mu & \lambda & 0 \\ 0 & \mu & 1 - \lambda - \mu & \lambda \\ 0 & 0 & \mu & 1 - \mu \end{bmatrix}, \quad (4.53)$$

and the transition graph is:

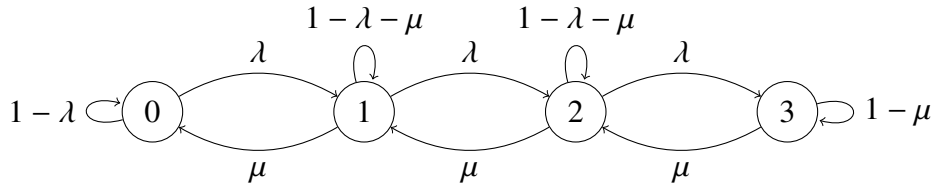


Figure 4.10: Transition graph for the queueing model with capacity $K = 3$.

This is an example of a *birth-death chain*: transitions only occur between neighbouring states. The Ehrenfest chain is also a birth-death chain.

Formal Definitions

Let us now define a Markov chain in a slightly more precise manner.

Definition : Markov Chain — A collection of random variables $X_0, X_1, X_2 \dots$ is called a **Markov chain** with state space $\mathcal{S}$ if

$$\mathbb{P}(X_{n+1} = x_{n+1} | X_0 = x_0, \dots, X_n = x_n) = \mathbb{P}(X_{n+1} = x_{n+1} | X_n = x_n) \quad (4.54)$$

for all $x_0, \dots, x_{n+1} \in \mathcal{S}$ and $n \in \mathbb{N}$. Equation (4.54) is called the **Markov property**.

We restrict ourselves to Markov chains for which the conditional probability

$$\mathbb{P}(X_{n+1} = j | X_n = i), \quad i, j \in \mathcal{S} \quad (4.55)$$

is independent of the time n ; such chains are called **time-homogeneous**. We also restrict ourselves to the case where the state space $\mathcal{S}$ is discrete (countable), like in Example 4.4. The probabilities in (4.55) are called the *(one-step) transition probabilities* of $X := \{X_n, n \geq 0\}$. The distribution of X_0 is called the *initial distribution* of the Markov chain. The one-step transition probabilities and the initial distribution completely specify the distribution of X . Namely, we have by the product rule

(4.17) and the Markov property (4.54)

$$\begin{aligned} \mathbb{P}(X_0 = x_0, \dots, X_n = x_n) \\ &= \mathbb{P}(X_0 = x_0) \mathbb{P}(X_1 = x_1 | X_0 = x_0) \cdots \mathbb{P}(X_n = x_n | X_0 = x_0, \dots, X_{n-1} = x_{n-1}) \\ &= \mathbb{P}(X_0 = x_0) \mathbb{P}(X_1 = x_1 | X_0 = x_0) \cdots \mathbb{P}(X_n = x_n | X_{n-1} = x_{n-1}). \end{aligned}$$

Example (Stepping Stones (Continued)). The probability of simulating the “path” $\{X_0 = 1, X_1 = 2, X_2 = 3, X_3 = 4, X_4 = 6, X_5 = 5\}$ is: $1 \times 1/3 \times 1/2 \times 1/2 \times 1 \times 1/2 = 1/24$.

Countably Infinite State Spaces

We now present examples where the state space is countably infinite.

Example (Queueing with Unlimited Buffer). In Example 4.4 we modelled a service point with a finite waiting room of capacity K . What happens if there is no limit on the number of waiting customers, i.e., $K = \infty$? The state space becomes $\mathcal{S} = \{0, 1, 2, \dots\}$ and the transition probabilities for $i \geq 1$ are

$$p_{i,i+1} = \lambda, \quad p_{i,i-1} = \mu, \quad p_{i,i} = 1 - \lambda - \mu, \quad (4.56)$$

with $p_{0,1} = \lambda$ and $p_{0,0} = 1 - \lambda$ at the boundary. The transition matrix is

$$P = \begin{bmatrix} 1 - \lambda & \lambda & 0 & 0 & \cdots \\ \mu & 1 - \lambda - \mu & \lambda & 0 & \cdots \\ 0 & \mu & 1 - \lambda - \mu & \lambda & \cdots \\ \vdots & \ddots & \ddots & \ddots & \ddots \end{bmatrix}. \quad (4.57)$$

The transition graph extends infinitely to the right:

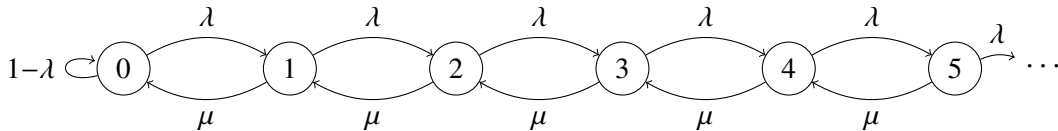


Figure 4.11: Transition graph for the queueing model with unlimited buffer.

Intuitively, if customers arrive “faster” than they are served ($\lambda > \mu$), the queue grows without bound. If the service rate exceeds the arrival rate ($\lambda < \mu$), the queue remains *stable*: it has a limiting distribution and the queue length stays finite on average. This is an example of a *stability condition*: the chain has a proper limiting distribution if and only if $\lambda < \mu$.

Example (Random Walk on the Integers). Let p be a number between 0 and 1. A random walk on $\mathbb{Z}$, with parameter p , is the Markov chain X with state space $\mathbb{Z}$ and transition matrix P , given by

$$p_{i,i+1} = p, \quad p_{i,i-1} = q = 1 - p, \quad \text{for all } i \in \mathbb{Z}. \quad (4.58)$$

Writing out the (doubly infinite) transition matrix with rows and columns indexed by $\dots, -2, -1, 0, 1, 2, \dots$:

$$P = \begin{matrix} & \dots & -2 & -1 & 0 & 1 & 2 & \dots \\ \begin{matrix} \vdots \\ -1 \\ 0 \\ 1 \\ \vdots \end{matrix} & \begin{pmatrix} \ddots & & & & \ddots & & \\ \dots & q & 0 & p & 0 & 0 & \dots \\ \dots & 0 & q & 0 & p & 0 & \dots \\ \dots & 0 & 0 & q & 0 & p & \dots \\ & \ddots & & \ddots & & \ddots & \end{pmatrix} \end{matrix}. \quad (4.59)$$

Each row has exactly two nonzero entries: q on the sub-diagonal and p on the super-diagonal.

Let X start at 0, thus $\mathbb{P}(X_0 = 0) = 1$. The corresponding transition graph is given in Figure 4.12. Starting at 0, we take subsequent steps to the right with probability p and to the left with probability q .

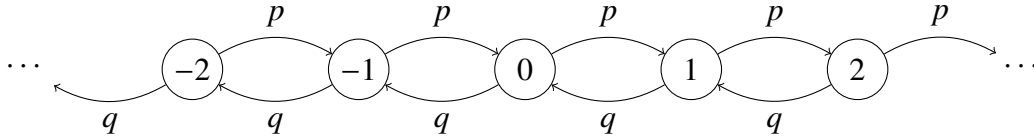


Figure 4.12: Transition graph for the random walk on $\mathbb{Z}$.

Unlike the queueing model, the state space here is all of $\mathbb{Z}$: there is no boundary, and the chain can drift arbitrarily far in either direction.

Example (Random Walks in Higher Dimensions). The random walk on $\mathbb{Z}$ generalises naturally to higher dimensions. A **simple symmetric random walk on $\mathbb{Z}^2$** (the integer lattice in the plane) starts at the origin $(0, 0)$ and at each step moves to one of the four neighbouring lattice points — up, down, left, or right — each with probability $1/4$. Similarly, a **simple symmetric random walk on $\mathbb{Z}^3$** moves at each step to one of the six neighbouring points, each with probability $1/6$.

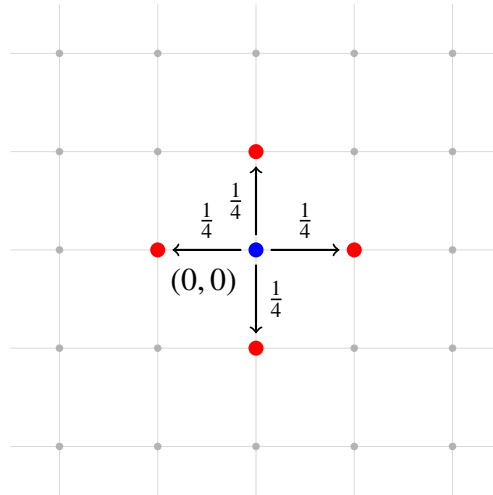


Figure 4.13: The simple symmetric random walk on $\mathbb{Z}^2$: from any lattice point, the walk moves to each of the four neighbours with equal probability.

A natural question is: starting at the origin, will the random walk ever return? A celebrated result of George Pólya (1921) gives a surprising answer that depends on the dimension.

Remark (Pólya's Recurrence Theorem). The simple symmetric random walk starting at the origin returns to the origin with probability 1 in dimensions $d = 1$ and $d = 2$. However, in dimension $d = 3$ (and all higher dimensions), there is a *positive probability of never returning*. In the language of Section 4.7, the walk is *recurrent* in one and two dimensions but *transient* in three or more dimensions.

Concretely, for the three-dimensional walk the probability of ever returning to the origin is approximately 0.34: roughly two times out of three, a random walk in three-dimensional space wanders off to infinity and never comes back. This is sometimes paraphrased as: “a drunk man will find his way home, but a drunk bird may get lost forever.”

Simulating Markov Chains

To simulate a Markov chain with initial distribution $\mathbf{p}^{(0)}$ (a row vector with non-negative entries summing to 1) and transition matrix P , we proceed as follows:

Algorithm 4.4.1: Generating a Markov Chain

1. Draw X_0 from the initial distribution $\mathbf{p}^{(0)}$. Set $n = 0$.
 2. Draw X_{n+1} from the distribution corresponding to the X_n -th row of P .
 3. Set $n = n + 1$ and go to 2.
-

Example (Stepping stones (continued)). Here is how you can simulate the stepping stone Markov chain in Python:

Simulating the stepping stones Markov chain

LLM Prompt > Write Python code that simulates 101 steps of a Markov chain with a given 6×6 transition matrix (the stepping stones example), starting from state 1, and plots the state trajectory.

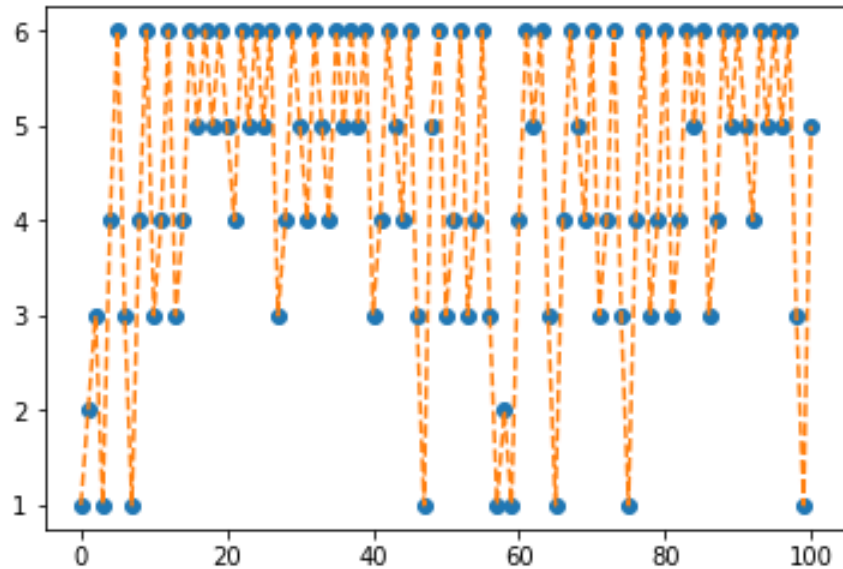


Figure 4.14: A path of the stepping stone Markov chain.

Example (Random Walk on the Integers). Consider the random walk on the integers in Example 4.4. Let $X_0 = 0$ (that is, we start at 0). Suppose the chain is at some time n in state i . Then, in step 2 of Algorithm 4.4.1 we simply need to draw from a 2-point distribution with mass p and q at $i + 1$ and $i - 1$ respectively. In other words, we draw $I_n \sim \text{Ber}(p)$ and set $X_{n+1} = X_n + 2I_n - 1$. Note that when simulating a Markov chain we do not always have to use the transition matrix explicitly. Whenever the transition from X_n to X_{n+1} can be described by a simple algorithmic rule — as here, where we just add $+1$ or -1 — we can use that rule directly. This is especially important when the state space (and hence the matrix) is infinite.

Simulating a symmetric random walk

LLM Prompt > Write Python code that simulates a symmetric random walk on the integers ($p = q = 1/2$) starting at 0 and produces two side-by-side plots: one showing the first 150 steps and one showing 5000 steps, with y-axes symmetric about 0.

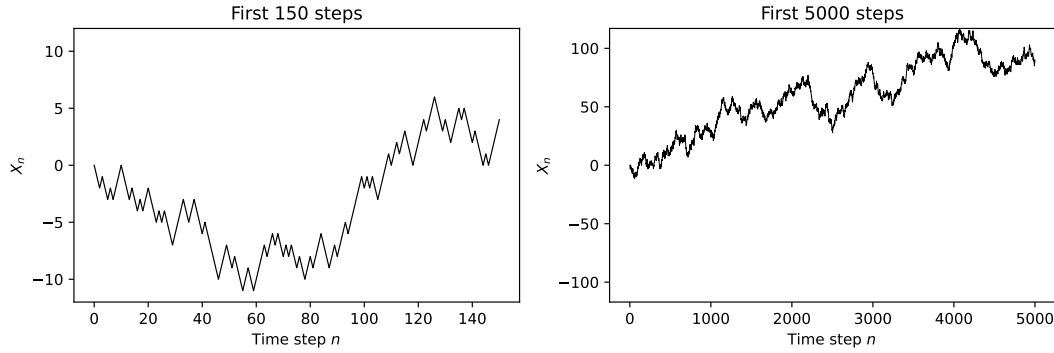


Figure 4.15: Symmetric random walk on $\mathbb{Z}$ with $p = q = 1/2$, starting at the origin. The trajectory resembles Brownian motion at longer time scales.

The sample paths in Figure 4.15 illustrate the symmetric case $p = q = 1/2$. At short time scales the path already looks erratic; at longer time scales the fluctuations grow and the trajectory begins to resemble Brownian motion.

4.5 Markov Chains and Matrix Computations

In the previous section we computed the distribution of X_1, X_2, X_3 for the two-state chain step by step, using the law of total probability. In this section we show that these calculations are really just *matrix multiplications*.

Throughout this section we focus on finite state spaces: $\mathcal{S} = \{1, 2, \dots, r\}$ for some fixed r , so that the transition matrix P is an $r \times r$ matrix.

The n -Step Transition Probability Matrix

The key observation is that P^2 — the ordinary matrix square of P — is the **two-step transition matrix**: its (i, j) -entry is the probability of going from state i to state j in exactly two steps. To see why, apply the law of total probability, summing over the intermediate state ℓ :

$$\mathbb{P}(X_2 = j | X_0 = i) = \sum_{\ell=1}^r \mathbb{P}(X_2 = j | X_1 = \ell) \mathbb{P}(X_1 = \ell | X_0 = i) = \sum_{\ell=1}^r P_{i,\ell} P_{\ell,j}. \quad (4.60)$$

The right-hand side is precisely the (i, j) -entry of the matrix product $P \cdot P = P^2$. (We write $P_{i,j}$ for the (i, j) -entry of a matrix P , and similarly v_j for the j -th component of a vector v .) More generally, the same argument applied one step at a time (i.e. by induction) shows that

$$\mathbb{P}(X_n = j | X_0 = i) = P_{i,j}^n, \quad (4.61)$$

the (i, j) -entry of P^n . Thus P^n is the **n -step transition matrix**: to find the probability of going from state i to state j in n steps, we simply read off the (i, j) -entry of P^n .

Now define the *row* vector

$$\mathbf{p}^{(n)} = (\mathbb{P}(X_n = 1), \dots, \mathbb{P}(X_n = r)). \quad (4.62)$$

We call $\mathbf{p}^{(n)}$ the **distribution vector**, or simply *distribution*, of X at time n , and $\mathbf{p}^{(0)}$ the *initial distribution* of X . Since $\mathbb{P}(X_n = j) = \sum_i \mathbb{P}(X_0 = i) P^n_{i,j} = (\mathbf{p}^{(0)} P^n)_j$, we obtain:

Claim 21: Markov Chain Distribution Vector — The distribution of X at time n is given by

$$\mathbf{p}^{(n)} = \mathbf{p}^{(0)} P^n, \quad (4.63)$$

for all $n = 0, 1, \dots$ (Here P^0 denotes the identity matrix.)

Let us verify this for the two-state chain from Example 4.4, with transition matrix

$$P = \begin{bmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{bmatrix} \quad (4.64)$$

and initial distribution $\mathbf{p}^{(0)} = (1, 0)$ (starting in state 0 with certainty). We showed in Section 4.4 that $\mathbf{p}^{(1)} = (0.9, 0.1)$. Indeed:

$$\mathbf{p}^{(0)} P = (1, 0) \begin{bmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{bmatrix} = (0.9, 0.1) = \mathbf{p}^{(1)}. \quad (4.65)$$

Similarly,

$$\mathbf{p}^{(1)} P = (0.9, 0.1) \begin{bmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{bmatrix} = (0.84, 0.16) = \mathbf{p}^{(2)}, \quad (4.66)$$

and so on — each step of the law of total probability computation is simply a matrix-vector multiplication.

Example (Stepping Stones (Continued)). Below we list some powers of the stepping stones transition matrix P . We see for example that the probability of going from stone 1 to stone 6 in two steps is $1/2$, and the probability of returning to 1 in 10 steps is 0.1049 (approximately).

Matrix powers of the stepping stones chain

LLM Prompt > Write Python code that computes and prints the matrix powers P^n for $n = 1, 2, 5, 10, 20, 100$ of a transition matrix P , to observe convergence to the limiting distribution.

```
[ [0.      0.3333 0.      0.3333 0.3333 0.      ]
  [0.5     0.      0.5     0.      0.      0.      ]
  [0.5     0.      0.      0.5     0.      0.      ]
  [0.      0.      0.      0.      0.      1.      ]
  [0.      0.      0.      0.5     0.      0.5     ]
  [0.      0.      0.5     0.      0.5     0.      ]]

[ [0.1667 0.      0.1667 0.1667 0.      0.5     ]
  [0.25   0.1667 0.      0.4167 0.1667 0.      ]
  [0.      0.1667 0.      0.1667 0.1667 0.5     ]
  [0.      0.      0.5     0.      0.5     0.      ]
  [0.      0.      0.25   0.      0.25   0.5     ]
  [0.25   0.      0.      0.5     0.      0.25   ]]
```

```

[[0.0694 0.0509 0.1597 0.162 0.1968 0.3611]
 [0.1597 0.0278 0.1181 0.3125 0.1319 0.25 ]
 [0.0764 0.0556 0.1181 0.1944 0.1597 0.3958]
 [0.1042 0.      0.2917 0.1667 0.25 0.1875]
 [0.0521 0.0417 0.2083 0.125 0.2292 0.3438]
 [0.1458 0.0347 0.0938 0.3056 0.1285 0.2917]]

[[0.1049 0.0346 0.1581 0.2202 0.1779 0.3044]
 [0.1004 0.0296 0.1848 0.2025 0.1944 0.2882]
 [0.1095 0.0324 0.1598 0.226 0.1773 0.295 ]
 [0.0872 0.0384 0.1693 0.1899 0.1901 0.3249]
 [0.1013 0.0369 0.1543 0.2165 0.177 0.3142]
 [0.1039 0.0291 0.1817 0.2088 0.1916 0.285 ]]

[[0.1006 0.0336 0.1683 0.2097 0.1851 0.3027]
 [0.101 0.0337 0.1675 0.2106 0.1845 0.3026]
 [0.1006 0.0337 0.1681 0.2097 0.185 0.303 ]
 [0.1011 0.0334 0.1685 0.2103 0.185 0.3016]
 [0.1007 0.0336 0.1685 0.2096 0.1852 0.3024]
 [0.1009 0.0337 0.1675 0.2105 0.1845 0.3028]]

[[0.1008 0.0336 0.1681 0.2101 0.1849 0.3025]
 [0.1008 0.0336 0.1681 0.2101 0.1849 0.3025]
 [0.1008 0.0336 0.1681 0.2101 0.1849 0.3025]
 [0.1008 0.0336 0.1681 0.2101 0.1849 0.3025]
 [0.1008 0.0336 0.1681 0.2101 0.1849 0.3025]
 [0.1008 0.0336 0.1681 0.2101 0.1849 0.3025]]

```

Convergence of Matrix Powers

Look carefully at the output in Example 4.5. As n grows, the matrix P^n appears to converge to a matrix P^∞ in which *all rows are equal*:

$$P^{100} \approx \begin{bmatrix} 0.1008 & 0.0336 & 0.1681 & 0.2101 & 0.1849 & 0.3025 \\ 0.1008 & 0.0336 & 0.1681 & 0.2101 & 0.1849 & 0.3025 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0.1008 & 0.0336 & 0.1681 & 0.2101 & 0.1849 & 0.3025 \end{bmatrix}. \quad (4.67)$$

Since $P^n_{i,j}$ is the probability of being in state j after n steps when starting from state i , the fact that all rows are equal means that the long-run distribution does not depend on the starting state. The chain “forgets” where it started.

This is not special to the stepping stones. Let us also check the machine reliability model from Example 4.4, with breakdown probability $p = 0.1$ and repair probability $r = 0.4$:

Matrix powers of the machine reliability chain

LLM Prompt > Write Python code that computes matrix powers P^n for $n = 1, 5, 10, 50$ of the machine reliability transition matrix $[[0.9, 0.1], [0.4, 0.6]]$ to show convergence to the stationary distribution.

```

P^1:
[[0.9 0.1]
 [0.4 0.6]]

```

```

P^5:

```

```
[[0.8063 0.1938]
 [0.775  0.225  ]]
```

```
P^10:
[[0.8002 0.1998]
 [0.7992 0.2008]]
```

```
P^50:
[[0.8 0.2]
 [0.8 0.2]]
```

Again, the rows converge to the same vector.

Computing the Stationary Distribution

Denote the common row of the limiting matrix P^∞ by π ; informally, $\pi = p^{(\infty)}$, the distribution the chain settles into after infinitely many steps. By Claim 21, we have $p^{(n+1)} = p^{(n)}P$. If $p^{(n)} \rightarrow \pi$, then taking the limit gives

$$\pi = \pi P, \quad \text{with } \pi_j \geq 0 \text{ and } \sum_j \pi_j = 1. \quad (4.68)$$

In other words, π is a *fixed point* of the operation “multiply by P ”: once the distribution reaches π , one more step leaves it unchanged. We call π the **stationary distribution**. To find it, we solve a system of linear equations.

Example (Machine Reliability (Continued)). For the machine reliability model with $P = \begin{bmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{bmatrix}$, the equation $\pi = \pi P$ gives

$$\pi_1 = 0.9\pi_1 + 0.4\pi_2, \quad \pi_2 = 0.1\pi_1 + 0.6\pi_2. \quad (4.69)$$

From the first equation, $0.1\pi_1 = 0.4\pi_2$, so $\pi_1 = 4\pi_2$. Together with $\pi_1 + \pi_2 = 1$, we get $\pi_2 = 1/5$ and $\pi_1 = 4/5$, giving $\pi = (0.8, 0.2)$. This matches exactly the rows of P^{50} computed above: in the long run, the machine is working 80% of the time and broken 20% of the time.

Example (Stepping Stones (Continued)). For the 6×6 stepping stones chain, solving $\pi = \pi P$ with $\sum_j \pi_j = 1$ by hand is tedious, but the procedure is the same: write out the six equations, note that one is redundant, replace it with $\sum_j \pi_j = 1$, and solve. In Python, the equation $\pi = \pi P$ can be rewritten as $(P^\top - I)\pi^\top = \mathbf{0}$; we replace one equation with the normalisation constraint and solve:

Solving for the stationary distribution

```
Stationary distribution:
[0.1008 0.0336 0.1681 0.2101 0.1849 0.3025]
```

In exact fractions, $\pi = \frac{1}{119}(12, 4, 20, 25, 22, 36)$, which matches the rows of P^{100} computed earlier.

Example (A Three-State Chain). Let us illustrate the method once more for a three-state chain with transition matrix

$$P = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.2 & 0.4 \end{bmatrix}. \quad (4.70)$$

Writing $\pi = (\pi_1, \pi_2, \pi_3)$, the equation $\pi = \pi P$ gives three equations (one for each state):

$$\pi_1 = 0.5 \pi_1 + 0.1 \pi_2 + 0.4 \pi_3, \quad (4.71)$$

$$\pi_2 = 0.3 \pi_1 + 0.6 \pi_2 + 0.2 \pi_3, \quad (4.72)$$

$$\pi_3 = 0.2 \pi_1 + 0.3 \pi_2 + 0.4 \pi_3. \quad (4.73)$$

Together with the constraint $\pi_1 + \pi_2 + \pi_3 = 1$, this is a system of linear equations. Any one of the three equations above is redundant (since the rows of P sum to 1), so we drop one and solve. For instance, rearranging the first two:

$$0.5 \pi_1 - 0.1 \pi_2 - 0.4 \pi_3 = 0, \quad (4.74)$$

$$-0.3 \pi_1 + 0.4 \pi_2 - 0.2 \pi_3 = 0, \quad (4.75)$$

$$\pi_1 + \pi_2 + \pi_3 = 1. \quad (4.76)$$

Solving gives $\pi \approx (0.316, 0.386, 0.298)$. This means that regardless of where the chain starts, in the long run it spends roughly 31.6% of its time in state 1, 38.6% in state 2, and 29.8% in state 3.

These observations suggest a general pattern. In general, the matrix powers P^n may or may not converge as $n \rightarrow \infty$, and when they do, the limiting matrix need not have all rows equal. However, in the special case where the Markov chain has a finite state space, is *irreducible* (meaning it is possible to get from any state to any other state), and is *aperiodic* (the chain does not cycle with a fixed period), then P^n converges to a matrix P^∞ whose rows are all equal to some probability vector $\mathbf{p}$. This $\mathbf{p}$ is called the **limiting distribution** of the Markov chain. The fact that all rows are equal has a concrete probabilistic meaning: $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = i) = p_i$ regardless of the starting state, so the chain “forgets” where it started. To find $\mathbf{p}$, we can read off any row of P^∞ .

There are, however, many important subcases where this simple picture breaks down: the chain may not be able to reach every state from every other state, or it may exhibit periodic behaviour, or the state space may be infinite. These situations require a more careful analysis and are topics of further study. The next section gives a brief glimpse of some of these issues; the course STAT3004 treats them in much greater depth.

4.6 A Peek at Markov Chain Properties

In the previous section we observed that the matrix powers P^n of several Markov chains converge to a limiting matrix with equal rows. But does this always happen? If so, under what conditions? In this section we develop the vocabulary needed to answer these questions.

Communicating Classes

Definition : Accessibility — State j is **accessible** from state i (written $i \rightarrow j$) if there exists some $n \geq 0$ such that $P_{i,j}^n > 0$. In other words, starting from i , there is a positive probability of reaching j in some number of steps.

Definition : Communication — States i and j **communicate** (written $i \leftrightarrow j$) if $i \rightarrow j$ and $j \rightarrow i$; that is, each state is accessible from the other.

Communication is an *equivalence relation*: it is reflexive ($i \leftrightarrow i$, since $P_{i,i}^0 = 1$), symmetric (by definition), and transitive (if $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$). Therefore, communication partitions the state space into disjoint **communicating classes**.

Example (Stepping Stones). Consider the stepping stones chain from Example 4.4. From the transition matrix, we can verify that every state can be reached from every other state: for instance, to go from state 1 to state 4 we can use the direct transition $1 \rightarrow 4$ (one step), and to go from state 4 to state 1 we can go $4 \rightarrow 6 \rightarrow 3 \rightarrow 1$ (three steps). Since all states communicate, there is a single communicating class: $\{1, 2, 3, 4, 5, 6\}$.

Example (Ehrenfest Chain). The Ehrenfest chain (Example 4.4) with $N = 4$ also has a single communicating class: from any state i we can reach state 0 by moving particles one at a time from A to B , and from state 0 we can reach any state by the reverse process.

Definition : Irreducible Markov Chain — A Markov chain is **irreducible** if all states communicate with each other, i.e., the entire state space forms a single communicating class.

The stepping stones chain, the two-state chain from Example 4.4, the Ehrenfest chain, the machine reliability model, and the queueing model are all irreducible.

Not all Markov chains are irreducible. For instance, consider the transition matrix

$$P = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (4.77)$$

Here states 1 and 2 communicate with each other, but state 3 is not accessible from states 1 or 2, and vice versa. The communicating classes are $\{1, 2\}$ and $\{3\}$. Moreover, state 3 is an *absorbing state*: once the chain enters state 3, it stays there forever.

Recurrent and Transient States

Definition : Recurrence and Transience — A state i is **recurrent** if, starting from i , the chain returns to i with probability 1. A state i is **transient** if, starting from i , there is a positive probability of never returning to i .

Intuitively, the chain visits a recurrent state infinitely often (since each time it returns, it will return again with probability 1), while it visits a transient state only finitely many times.

The following key results connect communicating classes to recurrence:

- All states in the same communicating class are either all recurrent or all transient.
- In a *finite* irreducible Markov chain, all states are recurrent.

The second result makes intuitive sense: in a finite state space where every state can reach every other, the chain must keep revisiting states — it has nowhere to “escape to”.

Example (Random Walk on $\mathbb{Z}$). The random walk on $\mathbb{Z}$ from Example 4.4 is irreducible (every state can reach every other state). When $p = q = 1/2$, all states are recurrent — the symmetric random walk always returns to the origin. However, when $p \neq 1/2$, all states are transient — the walk drifts off to $+\infty$ or $-\infty$ and never returns. This shows that recurrence depends on the transition probabilities, not just the structure of the transition graph.

Limiting Distributions

In Section 4.5 we observed that for several examples, P^n converges to a matrix with equal rows. We now state this more precisely.

The limiting behaviour of Markov chains as $n \rightarrow \infty$, i.e., after the process has been going on for a very long time, is of considerable interest and importance. The results are often much simpler and are quite useful in approximating real situations. It can be shown that for general Markov chains (satisfying some mild conditions)

$$\lim_{n \rightarrow \infty} P^n_{i,j} = \pi_j, \quad (4.78)$$

for some number $0 \leq \pi_j \leq 1$. When the $\{\pi_j\}$ sum up to 1, they form the **limiting distribution** of the Markov chain. The following theorem gives a method for obtaining limiting distributions. Here we assume for simplicity that $\mathcal{S} = \{0, 1, 2, \dots\}$. The limiting distribution is identified with the row vector $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots)$.

Claim 22: Stationary Distribution — The limiting distribution $\boldsymbol{\pi}$, if it exists, is uniquely determined by the solution of

$$\boldsymbol{\pi} = \boldsymbol{\pi}P, \quad (4.79)$$

with $\pi_j \geq 0$ and $\sum_j \pi_j = 1$. Conversely, if there exists a unique positive row vector $\boldsymbol{\pi}$ satisfying (4.79) and summing up to 1, then $\boldsymbol{\pi}$ is the limiting distribution of the Markov chain.

Proof: (Sketch). For the case where $\mathcal{S}$ is finite, the result is simply a consequence of (4.63). Namely, with $\boldsymbol{p}^{(0)}$ being the i -th unit vector, we have

$$P^{n+1}_{i,j} = (\boldsymbol{p}^{(0)} P^n P)_j = \sum_{k \in \mathcal{S}} P^n_{i,k} P_{k,j}. \quad (4.80)$$

Letting $n \rightarrow \infty$, we obtain (4.79) from (4.78), provided that we can change the order of the limit and the summation. To show uniqueness, suppose that another vector $\boldsymbol{y}$, with $y_j \geq 0$ and $\sum_j y_j = 1$, satisfies $\boldsymbol{y} = \boldsymbol{y}P$. Then it is easy to show by induction that also $\boldsymbol{y} = \boldsymbol{y}P^n$, for every n . Hence letting

$n \rightarrow \infty$, we obtain for every j

$$y_j = \sum_i y_i \pi_j = \pi_j, \quad (4.81)$$

since the y_j 's sum up to unity. We omit the proof of the converse statement. $\square$

The equation $\pi = \pi P$ says that π is a *left eigenvector* of P with eigenvalue 1. Equivalently, it says that if the chain's distribution at some time is π , then after one more step the distribution is still π — the chain is in *equilibrium*.

Definition : Period and Aperiodicity — The **period** of a state i is $d_i = \gcd\{n \geq 1 : P_{i,i}^n > 0\}$, the greatest common divisor of all return times. A state with period 1 is called **aperiodic**. If all states of an irreducible chain are aperiodic, the chain itself is called aperiodic.

For example, a chain that can only return to a state at even times has period 2. Self-loops ($P_{i,i} > 0$) guarantee aperiodicity.

For *finite irreducible aperiodic* chains, the limiting distribution always exists and is unique. All the examples we have seen in this chapter (stepping stones, two-state chain, reliability, queueing) satisfy these conditions. (The Ehrenfest chain is irreducible but has period 2, since it alternates between even and odd states.)

Example (Stepping Stones (Continued)). To find the limiting distribution π we need to solve the matrix equation (4.79), or equivalently

$$\pi(P - I) = \mathbf{0}, \quad (4.82)$$

which in turn is equivalent to

$$(P^\top - I)\pi^\top = \mathbf{0}^\top, \quad (4.83)$$

where $^\top$ denotes transposition. In other words, the column vector $\pi^\top$ lies in the null-space of the matrix $P^\top - I$. In Python, we can find such a vector via:

Computing the null space of $P^\top - I$

LLM Prompt > Show how to compute the null space of $P^\top - I$ in Python using *scipy*, to find the stationary distribution vector of a Markov chain.

This vector does not sum up to 1, but this is easily fixed via:

Normalising to a probability vector

LLM Prompt > Show how to normalise a vector so its entries sum to 1 in Python.

Printing the transpose of $\mathbf{v}$ gives the row vector π :

```
[[0.1008 0.0336 0.1681 0.2101 0.1849 0.3025]]
```

Example (Random Walk on the Positive Integers). This is a slightly different random walk than we had before in Example 4.4. Let X be a random walk on $\mathcal{S} = \{0, 1, 2, \dots\}$ with transition matrix

$$P = \begin{bmatrix} q & p & 0 & \dots \\ q & 0 & p & 0 & \dots \\ 0 & q & 0 & p & 0 & \dots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots \end{bmatrix}, \quad (4.84)$$

where $0 < p < 1$ and $q = 1 - p$. X_n could represent for example the number of customers that are waiting in a queue at time n .

Note that all states can be reached from each other. The equation $\pi = \pi P$ becomes

$$\begin{aligned} \pi_0 &= q\pi_0 + q\pi_1, \\ \pi_1 &= p\pi_0 + q\pi_2, \\ \pi_2 &= p\pi_1 + q\pi_3, \\ \pi_3 &= p\pi_2 + q\pi_4, \end{aligned}$$

etc. We can solve this set of equation sequentially. If we let $r = p/q$, then we can express the $\pi_1, \pi_2, \dots$ in terms of π_0 and r as

$$\pi_j = r^j \pi_0, \quad j = 0, 1, 2, \dots \quad (4.85)$$

If $p < q$, then $r < 1$ and $\sum_{j=0}^{\infty} \pi_j = \frac{1}{1-r} \pi_0$, and by choosing $\pi_0 = 1 - r$, we can make the sum $\sum \pi_j = 1$. Hence, for $r < 1$ we have found the limiting distribution $\pi = (1 - r)(1, r, r^2, r^3, \dots)$ for this Markov chain. Notice that $\pi_j = (1 - r)r^j$ for $j = 0, 1, 2, \dots$ is precisely the pmf of a $\text{Geom}_0(1 - r)$ distribution — the geometric distribution starting at 0 that we studied in Chapter 3. So in the long run, the queue length follows a geometric distribution. On the other hand, when $p \geq q$, then $\sum \pi_j$ is either 0 or infinite, and hence the limiting distribution does not exist.

CONTINUOUS RANDOM VARIABLES AND THEIR DISTRIBUTIONS

In this chapter we introduce continuous distributions, their expectation, moments, and properties. We focus on families of continuous distributions including the Normal distribution, exponential, uniform, Gamma, Beta, Pareto, Cauchy, and a few others.

5.1 Continuous Random Variables and Their Distributions

In Chapter 3 we studied discrete random variables, whose possible values form a finite or countably infinite set of isolated points. We now turn to **continuous random variables**, which can take any value in an interval (or union of intervals). Examples include the exact time until a phone call arrives, the height of a randomly chosen person.

Reviewing the Cumulative Distribution Function

Recall from Chapter 3 that the *cumulative distribution function* (cdf) of any random variable X is

$$F(x) = \mathbb{P}(X \leq x), \quad x \in \mathbb{R}. \quad (5.1)$$

Every cdf satisfies the same basic properties, whether X is discrete or continuous:

1. $0 \leq F(x) \leq 1$ for all x ,
2. F is non-decreasing,
3. $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$,
4. $\mathbb{P}(a < X \leq b) = F(b) - F(a)$ for any $a < b$.

For a discrete random variable, the cdf is a step function: it is constant between the support points and has a jump at each value x_k of size $\mathbb{P}(X = x_k)$. The key difference for a continuous random variable is that **the cdf has no jumps** — it is a continuous function.

Definition : Continuous Random Variable — A random variable X is **continuous** if there exists a non-negative function f such that

$$F(x) = \int_{-\infty}^x f(t) dt, \quad \text{for all } x \in \mathbb{R}. \quad (5.2)$$

The function f is called the **probability density function** (pdf) of X .

In simple terms, we can think of a continuous random variable as one whose cdf F is a continuous function (no jumps). There do exist exotic examples of random variables with a continuous cdf that nonetheless have no density, but such cases are beyond the scope of this course.

Since the jump size of F at x equals $\mathbb{P}(X = x)$, a continuous cdf means:

$$\mathbb{P}(X = x) = 0 \quad \text{for every } x \in \mathbb{R}. \quad (5.3)$$

In words: a continuous random variable assigns probability *zero* to every individual value. This may seem paradoxical at first — if every value has probability zero, how can X take any value at all? The resolution is that probability is spread *continuously* over an interval, like mass spread smoothly over a rod. No single point carries any mass, yet the total mass over any interval of positive length is positive.

A useful consequence is that for continuous random variables, *strict and non-strict inequalities do not matter*:

$$\mathbb{P}(a \leq X \leq b) = \mathbb{P}(a < X < b) = \mathbb{P}(a \leq X < b) = \mathbb{P}(a < X \leq b) = F(b) - F(a). \quad (5.4)$$

This is because $\mathbb{P}(X = a) = \mathbb{P}(X = b) = 0$, so including or excluding the endpoints makes no difference. (For discrete random variables this is *not* true — boundary points can carry positive probability.)

The Probability Density Function

For a discrete random variable, we described the distribution via the probability mass function $f(x) = \mathbb{P}(X = x)$. But for a continuous random variable, $\mathbb{P}(X = x) = 0$ for every x , so the pmf is identically zero — it tells us nothing. We need a different tool.

The idea is to look at probabilities over *small intervals* instead of single points. For a small positive number h ,

$$\mathbb{P}(x < X \leq x + h) = F(x + h) - F(x). \quad (5.5)$$

If F is differentiable at x , then by the definition of the derivative,

$$F(x + h) - F(x) = F'(x) \cdot h + o(h) \approx F'(x) \cdot h \quad (5.6)$$

for small h , where $o(h)$ denotes terms that vanish faster than h . In other words,

$$\mathbb{P}(x < X \leq x + h) \approx F'(x) \cdot h. \quad (5.7)$$

The derivative $F'(x)$ measures how fast probability accumulates near x : it is the *probability per unit length* at x .

By the fundamental theorem of calculus, if F is differentiable (except possibly at finitely many points), the pdf can be recovered as

$$f(x) = F'(x) = \frac{dF}{dx}(x). \quad (5.8)$$

Since F is non-decreasing, its derivative is non-negative: $f(x) \geq 0$. And since F rises from 0 to 1, the total area under the density curve is

$$\int_{-\infty}^{\infty} f(x) dx = F(\infty) - F(-\infty) = 1 - 0 = 1. \quad (5.9)$$

Conversely, the Fundamental Theorem of Calculus recovers the cdf from the pdf:

$$F(x) = \int_{-\infty}^x f(t) dt. \quad (5.10)$$

Probabilities are computed as *areas under the density curve*:

$$\mathbb{P}(a < X \leq b) = \int_a^b f(x) dx = F(b) - F(a). \quad (5.11)$$

In summary, the pdf and cdf are related by differentiation and integration:

$$f = F' \quad \text{and} \quad F(x) = \int_{-\infty}^x f(t) dt. \quad (5.12)$$

Remark (The density is not a probability). Unlike the pmf, the value $f(x)$ is *not* the probability that X equals x (that probability is always zero). Rather, $f(x)$ is a *density*: probability per unit length. In particular, $f(x)$ **can exceed** 1. What must equal 1 is the total *area* under f , not the height.

Two Properties that Characterise a PDF

Any function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

1. $f(x) \geq 0$ for all x , and

$$2. \int_{-\infty}^{\infty} f(x) dx = 1,$$

is a valid probability density function. Conversely, every pdf satisfies these two properties. As in the discrete case, the **support** of a continuous distribution is the set of values where $f(x) > 0$. Only the support contributes to the integral in property 2.

Example: Uniform Distribution on $[0, 1]$

Example (Uniform Distribution on $[0, 1]$). The simplest continuous distribution is the *uniform distribution* on $[0, 1]$, denoted $X \sim \mathcal{U}(0, 1)$. Its pdf is

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases} \quad (5.13)$$

This is a valid pdf since $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x) dx = \int_0^1 1 dx = 1$.

Integrating, the cdf is

$$F(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x \leq 1, \\ 1, & x > 1. \end{cases} \quad (5.14)$$

For any sub-interval $[a, b] \subseteq [0, 1]$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b 1 dx = b - a. \quad (5.15)$$

That is, the probability of falling in a sub-interval equals its *length*. Probability is spread uniformly across $[0, 1]$, hence the name.

Example: A Non-Uniform Density

Example (The density $f(x) = 2x$ on $[0, 1]$). Consider the pdf

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases} \quad (5.16)$$

A more compact way to write this is $f(x) = 2x \mathbf{1}_{[0,1]}(x)$, using the **indicator function** $\mathbf{1}_A(x)$, which equals 1 if $x \in A$ and 0 otherwise. We verify the two properties: clearly $f(x) \geq 0$, and

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^1 2x dx = x^2 \Big|_0^1 = 1. \quad (5.17)$$

The cdf is obtained by integration:

$$F(x) = \int_{-\infty}^x f(t) dt = \begin{cases} 0, & x < 0, \\ x^2, & 0 \leq x \leq 1, \\ 1, & x > 1. \end{cases} \quad (5.18)$$

Since f is an increasing function of x , this distribution assigns more probability to values near 1 than near 0.

As an example of computing a probability, suppose we want $\mathbb{P}(0.3 \leq X \leq 0.7)$:

$$\mathbb{P}(0.3 \leq X \leq 0.7) = \int_{0.3}^{0.7} 2x \, dx = x^2 \Big|_{0.3}^{0.7} = 0.49 - 0.09 = 0.40. \quad (5.19)$$

Equivalently, $F(0.7) - F(0.3) = 0.7^2 - 0.3^2 = 0.40$.

PDF and CDF of the $f(x) = 2x$ distribution

LLM Prompt > Write Python code that plots side by side: (a) the PDF $f(x) = 2x$ on $[0, 1]$ with the area $P(0.3 \leq X \leq 0.7)$ shaded, and (b) the corresponding CDF $F(x) = x^2$ with the probability difference marked.

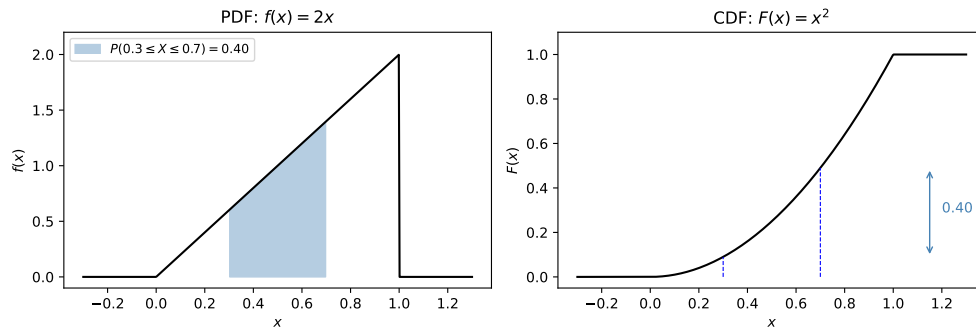


Figure 5.1: Left: the probability density function $f(x) = 2x$ on $[0, 1]$, with the shaded area representing $\mathbb{P}(0.3 \leq X \leq 0.7) = 0.40$. Right: the corresponding cdf $F(x) = x^2$.

Example: A Density Exceeding One

Example (A density with values larger than 1). Let

$$f(x) = \begin{cases} 3, & 0 \leq x \leq 1/3, \\ 0, & \text{otherwise.} \end{cases} \quad (5.20)$$

Here $f(x) = 3 > 1$ on its support, yet this is a valid pdf since $\int_0^{1/3} 3 \, dx = 1$. The probability is concentrated on the short interval $[0, 1/3]$, so the density must be tall to compensate. This reinforces the point that $f(x)$ is a *density* (probability per unit length), not a probability itself.

Comparing the Discrete and Continuous Frameworks

The following table summarises the parallel structures for discrete and continuous random variables.

	Discrete	Continuous
Distribution described by	pmf: $f(x) = \mathbb{P}(X = x) = F(x) - F(x^-)$	pdf: $f(x) = F'(x)$
Non-negativity	$f(x) \geq 0$	$f(x) \geq 0$
Normalisation	$\sum_x f(x) = 1$	$\int_{-\infty}^{\infty} f(x) dx = 1$
$\mathbb{P}(X \in B)$	$\sum_{x \in B} f(x)$	$\int_B f(x) dx$
CDF from pmf/pdf	$F(x) = \sum_{k \leq x} f(k)$	$F(x) = \int_{-\infty}^x f(t) dt$
pmf/pdf from CDF	$f(x) = F(x) - F(x^-)$	$f(x) = F'(x)$
$\mathbb{P}(X = x)$	$f(x)$ (can be > 0)	always 0

The pattern is clear: *sums become integrals*. This correspondence will persist throughout the chapter.

5.2 Expectation, Moments, and Properties

The definitions of expectation, variance, and moments carry over from the discrete case (Section 3.2) with one systematic change: *sums become integrals*. All algebraic properties — linearity of expectation, the computational formula for variance, and variance under linear transformations — are identical, since their proofs rely only on the linearity of the expectation operator.

Expectation (Mean)

Definition : Expectation of a Continuous Random Variable — The **expectation** (or **mean**) of a continuous random variable X with pdf f is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx, \quad (5.21)$$

provided the integral converges absolutely ($\int_{-\infty}^{\infty} |x| f(x) dx < \infty$). We also write $\mu = \mathbb{E}[X]$ or $\mu_X = \mathbb{E}[X]$.

Compare with the discrete case: $\mathbb{E}[X] = \sum_x x f(x)$. The centre-of-mass interpretation from Section 3.2 extends directly: if f is symmetric about a point c (i.e., $f(c + y) = f(c - y)$ for all y), then $\mathbb{E}[X] = c$.

Example (Uniform Distribution on $[0, 1]$). If $X \sim \mathcal{U}(0, 1)$, then $f(x) = 1$ for $0 \leq x \leq 1$. Therefore,

$$\mathbb{E}[X] = \int_0^1 x \cdot 1 \, dx = \left. \frac{x^2}{2} \right|_0^1 = \frac{1}{2}. \quad (5.22)$$

This is the midpoint of $[0, 1]$, as expected by symmetry.

Example (The density $f(x) = 2x$ on $[0, 1]$). For the distribution from Example 5.1, with $f(x) = 2x$ on $[0, 1]$,

$$\mathbb{E}[X] = \int_0^1 x \cdot 2x \, dx = 2 \int_0^1 x^2 \, dx = 2 \cdot \frac{1}{3} = \frac{2}{3}. \quad (5.23)$$

Since the density gives more weight to values near 1 than near 0, the mean is shifted above the midpoint $1/2$.

LOTUS and Moments

The Law of the Unconscious Statistician (LOTUS), introduced for discrete random variables in Section 3.2, extends to the continuous case.

Claim 23: Law of the Unconscious Statistician (Continuous) — If X is a continuous random variable with pdf f and $h : \mathbb{R} \rightarrow \mathbb{R}$ is any function, then

$$\mathbb{E}[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) \, dx, \quad (5.24)$$

provided the integral converges absolutely.

For instance, with $f(x) = 2x$ on $[0, 1]$ and $h(x) = x^2$:

$$\mathbb{E}[X^2] = \int_0^1 x^2 \cdot 2x \, dx = 2 \int_0^1 x^3 \, dx = 2 \cdot \frac{1}{4} = \frac{1}{2}. \quad (5.25)$$

Taking $h(x) = x^k$ gives the **k -th moment** $\mathbb{E}[X^k] = \int_{-\infty}^{\infty} x^k f(x) \, dx$, and taking $h(x) = (x - \mu)^k$ gives the **k -th central moment** $\mathbb{E}[(X - \mu)^k]$, exactly as in the discrete case (Section 3.2). As before, the first moment is the mean, and the second central moment is the variance.

Variance and Standard Deviation

Definition : Variance and Standard Deviation (Continuous) — The **variance** of a continuous random variable X with mean $\mu = \mathbb{E}[X]$ and pdf f is

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) \, dx. \quad (5.26)$$

The **standard deviation** is $\text{SD}(X) = \sqrt{\text{Var}(X)}$.

Example (Uniform on $[0, 1]$: variance). For $X \sim \mathcal{U}(0, 1)$ with $\mu = 1/2$,

$$\mathbb{V}\text{ar}(X) = \int_0^1 \left(x - \frac{1}{2}\right)^2 \cdot 1 \, dx = \int_0^1 \left(x^2 - x + \frac{1}{4}\right) dx = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12}. \quad (5.27)$$

Properties of Expectation and Variance

The key algebraic properties established in Section 3.2 hold identically for continuous random variables. The proofs are the same, with sums replaced by integrals.

Claim 24: Properties of Expectation and Variance (Continuous) — Let X be a continuous random variable with pdf f , and let $a, b \in \mathbb{R}$ be constants. Then:

1. **Linearity of expectation:**

$$\mathbb{E}[aX + b] = a \mathbb{E}[X] + b. \quad (5.28)$$

2. **Additivity of expectation:** For any functions $g, h : \mathbb{R} \rightarrow \mathbb{R}$,

$$\mathbb{E}[g(X) + h(X)] = \mathbb{E}[g(X)] + \mathbb{E}[h(X)]. \quad (5.29)$$

3. **Variance under linear transformation:**

$$\mathbb{V}\text{ar}(aX + b) = a^2 \mathbb{V}\text{ar}(X). \quad (5.30)$$

4. **Computational formula for variance:**

$$\mathbb{V}\text{ar}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \quad (5.31)$$

Proof: We prove property 1; the others follow by similar arguments (see Section 3.2 for the discrete versions).

$$\mathbb{E}[aX + b] = \int_{-\infty}^{\infty} (ax + b) f(x) \, dx = a \int_{-\infty}^{\infty} x f(x) \, dx + b \int_{-\infty}^{\infty} f(x) \, dx = a \mathbb{E}[X] + b. \quad (5.32)$$

□

The computational formula is often easier to use than the definition, since it avoids integrating $(x - \mu)^2$.

Example (Uniform on $[0, 1]$: variance via the computational formula). We already computed $\mathbb{E}[X] = 1/2$. The second moment is

$$\mathbb{E}[X^2] = \int_0^1 x^2 \cdot 1 \, dx = \frac{1}{3}. \quad (5.33)$$

By the computational formula,

$$\mathbb{V}\text{ar}(X) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}, \quad (5.34)$$

which agrees with the direct calculation in Example 5.2.

Example (The density $f(x) = 2x$ on $[0, 1]$: variance). We computed $\mathbb{E}[X] = 2/3$ and $\mathbb{E}[X^2] = 1/2$. Therefore,

$$\mathbb{V}\text{ar}(X) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{9-8}{18} = \frac{1}{18}. \quad (5.35)$$

The standard deviation is $\text{SD}(X) = 1/\sqrt{18} = 1/(3\sqrt{2}) \approx 0.236$.

Example: Continuous Uniform Distribution on $[a, b]$

Example (Continuous Uniform on $[a, b]$). The *continuous uniform distribution* on $[a, b]$, denoted $X \sim \mathcal{U}(a, b)$, has pdf

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases} \quad (5.36)$$

This is a valid pdf: $f(x) \geq 0$ and $\int_a^b \frac{1}{b-a} dx = 1$. This is the continuous counterpart of the discrete uniform distribution $\text{DU}(\{a, \dots, b\})$ introduced in Section 3.2.

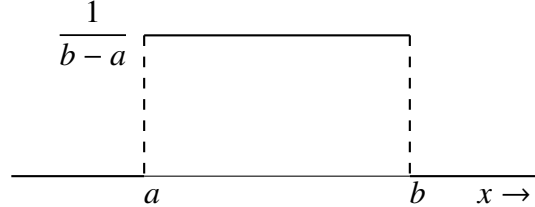


Figure 5.2: The pdf of the uniform distribution on $[a, b]$.

CDF. Integrating the constant density gives

$$F(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases} \quad (5.37)$$

Rather than computing the mean and variance by direct integration, we use a linear transformation, exactly as we did for the discrete uniform in Section 3.2. If $U \sim \mathcal{U}(0, 1)$, then $X = (b-a)U + a$ is uniformly distributed on $[a, b]$. By linearity and the known moments of U ,

Mean. $\mathbb{E}[X] = (b-a)\mathbb{E}[U] + a = (b-a) \cdot \frac{1}{2} + a = \frac{a+b}{2}$.

Variance. $\mathbb{V}\text{ar}(X) = (b-a)^2 \mathbb{V}\text{ar}(U) = (b-a)^2 \cdot \frac{1}{12} = \frac{(b-a)^2}{12}$.

In summary,

$$X \sim \mathcal{U}(a, b) : \quad \mathbb{E}[X] = \frac{a+b}{2}, \quad \mathbb{V}\text{ar}(X) = \frac{(b-a)^2}{12}. \quad (5.38)$$

The mean is the midpoint of $[a, b]$ (by symmetry), and the variance depends only on the length $b-a$ of the interval. Compare with the discrete uniform on $\{a, a+1, \dots, b\}$, which has the same mean $(a+b)/2$ and variance $(b-a)(b-a+2)/12$ — the formulas are strikingly similar.

Other Descriptors

Beyond the mean and variance (which are linked to the first two moments), higher moments and quantile-based quantities provide additional information about the shape of a distribution. These concepts apply to both discrete and continuous random variables; for continuous distributions, the quantile definitions are particularly clean because the cdf is continuous.

Skewness. The **skewness** is based on the third central moment and measures asymmetry of the distribution about its mean:

$$\gamma_1 = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}. \quad (5.39)$$

A symmetric distribution has $\gamma_1 = 0$. If $\gamma_1 > 0$ the distribution is right-skewed (longer right tail); if $\gamma_1 < 0$ it is left-skewed.

Kurtosis. The **kurtosis** is based on the fourth central moment and measures the heaviness of the tails relative to a normal distribution:

$$\kappa = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}. \quad (5.40)$$

The normal distribution has $\kappa = 3$. The **excess kurtosis** $\kappa - 3$ is often reported instead, so that the normal serves as the baseline with excess kurtosis 0.

Median and Quantiles. The **median** is the value m such that $F(m) = \frac{1}{2}$. More generally, the q -**th quantile** (for $0 < q < 1$) is the value x_q satisfying $F(x_q) = q$. Important special cases include the **first quartile** $x_{0.25}$, the **median** $x_{0.5}$, and the **third quartile** $x_{0.75}$.

Example (Skewness, Kurtosis, and Quantiles for $f(x) = 2x$ on $[0, 1]$). From Examples 5.2 and 5.2 we have $\mu = 2/3$, $\mathbb{E}[X^2] = 1/2$, $\sigma^2 = 1/18$. We also need higher moments:

$$\mathbb{E}[X^3] = \int_0^1 x^3 \cdot 2x \, dx = 2 \cdot \frac{1}{5} = \frac{2}{5}, \quad \mathbb{E}[X^4] = \int_0^1 x^4 \cdot 2x \, dx = 2 \cdot \frac{1}{6} = \frac{1}{3}. \quad (5.41)$$

Using $\mathbb{E}[(X - \mu)^3] = \mathbb{E}[X^3] - 3\mu \mathbb{E}[X^2] + 2\mu^3$, we obtain $\mathbb{E}[(X - \mu)^3] = \frac{2}{5} - 1 + \frac{16}{27} = -\frac{1}{135}$. Therefore the skewness is

$$\gamma_1 = \frac{-1/135}{(1/18)^{3/2}} = -\frac{2\sqrt{2}}{5} \approx -0.566. \quad (5.42)$$

The negative value confirms the distribution is left-skewed: the density concentrates mass near $x = 1$ with a tail toward 0.

For the kurtosis, $\mathbb{E}[(X - \mu)^4] = \frac{1}{135}$, giving

$$\kappa = \frac{1/135}{(1/18)^2} = \frac{12}{5} = 2.4, \quad (5.43)$$

so the excess kurtosis is $2.4 - 3 = -0.6$ (lighter tails than a normal distribution).

The cdf is $F(x) = x^2$ on $[0, 1]$, so the q -th quantile satisfies $x_q^2 = q$, giving $x_q = \sqrt{q}$. In particular,

$$\text{first quartile} = \sqrt{0.25} = 0.5, \quad (5.44)$$

$$\text{median} = \sqrt{0.5} \approx 0.707, \quad (5.45)$$

$$\text{third quartile} = \sqrt{0.75} \approx 0.866. \quad (5.46)$$

Note that the median (≈ 0.707) exceeds the mean (≈ 0.667), which is typical for left-skewed distributions.

5.3 The Normal Distribution

The normal (or Gaussian) distribution is the most important distribution in the study of statistics. One of the main reasons for its importance is the *central limit theorem* (Chapter 10), which (roughly speaking) states that the sum of a large number of independent random variables is approximately normally distributed, regardless of the distributions of the individual terms. We will study this remarkable result in Chapter 10; for now, we develop the properties of the normal distribution itself.

We say that a random variable X has a **normal** distribution with parameters μ and σ^2 if its density function f is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad x \in \mathbb{R}. \quad (5.47)$$

We write $X \sim \mathcal{N}(\mu, \sigma^2)$. The parameters μ and σ^2 turn out to be the expectation and variance of the distribution, respectively (as we verify below).

Remark (Parametrisation warning). In the notation $\mathcal{N}(\mu, \sigma^2)$, the second argument is the *variance* σ^2 , not the standard deviation σ . This is the convention used throughout these notes and in most probability textbooks. However, many software implementations parametrise by the standard deviation instead: for example, `scipy.stats.norm(loc= $\mu$ , scale= $\sigma$ )` and `rnorm(n, mean= $\mu$ , sd= $\sigma$ )` in R both take σ , not σ^2 . Always check which convention is in use.

If $\mu = 0$ and $\sigma = 1$ then

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad (5.48)$$

and the distribution is known as the **standard normal** distribution. The cdf of the standard normal is often denoted by Φ and is tabulated in Appendix B. In Figure 5.3 the probability densities for three different normal distributions have been depicted.

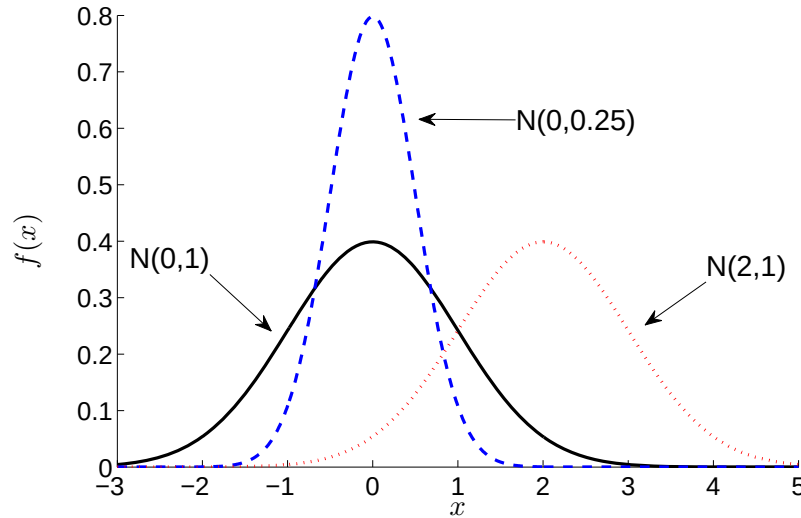


Figure 5.3: Probability density functions for various normal distributions. The labels $\mathcal{N}(\mu, \sigma^2)$ indicate the mean and *variance*: for instance, $\mathcal{N}(0, 0.25)$ has standard deviation $\sigma = 0.5$.

The Gaussian Integral

To confirm that f is a valid pdf we must verify that it integrates to 1. By the substitution $z = (x - \mu)/\sigma$, it suffices to show that

$$I = \int_{-\infty}^{\infty} e^{-z^2/2} dz = \sqrt{2\pi}, \quad (5.49)$$

since then $\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx = \frac{1}{\sigma\sqrt{2\pi}} \cdot \sigma I = 1$.

Proof: The integral I cannot be evaluated by elementary one-variable methods, but a beautiful trick reduces it to a calculation in polar coordinates. Consider I^2 as a double integral:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2/2} dy \right) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy. \quad (5.50)$$

Switching to polar coordinates $x = r \cos \theta$, $y = r \sin \theta$ with $dx dy = r dr d\theta$:

$$I^2 = \int_0^{2\pi} \int_0^{\infty} r e^{-r^2/2} dr d\theta = 2\pi \int_0^{\infty} r e^{-r^2/2} dr = 2\pi \left(-e^{-r^2/2} \right) \Big|_0^{\infty} = 2\pi. \quad (5.51)$$

Since $I > 0$, we conclude $I = \sqrt{2\pi}$. □

This proof will become more natural when we study the bivariate normal distribution in Chapter 7: the double integral $\iint e^{-(x^2+y^2)/2} dx dy$ is precisely the total probability under a standard bivariate normal density with independent components.

No Closed-Form CDF

Unlike the distributions we have seen so far (such as the uniform), the normal cdf

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \quad (5.52)$$

has no elementary closed form — it cannot be expressed in terms of polynomials, exponentials, logarithms, or trigonometric functions. This is not a matter of ingenuity; it is a theorem. Liouville's theorem on integration in finite terms (1835) provides a systematic criterion for deciding when an antiderivative can be expressed in elementary functions, and $e^{-t^2/2}$ provably fails this criterion. Therefore $\Phi(x)$ must be evaluated numerically, either by consulting a table of precomputed values (see Appendix B) or by using software (e.g., `scipy.stats.norm.cdf` in Python or `pnorm` in R). This is the main reason why **normal probability tables** have historically been so important in statistics.

Computing $\Phi(z)$ for $z = -3, -2, \dots, 3$

```
Phi(-3) = 0.0013
Phi(-2) = 0.0228
Phi(-1) = 0.1587
Phi(+0) = 0.5000
Phi(+1) = 0.8413
Phi(+2) = 0.9772
Phi(+3) = 0.9987
```

```
Phi(1) - Phi(-1) = 0.6827
Phi(2) - Phi(-2) = 0.9545
Phi(3) - Phi(-3) = 0.9973
```

The last three lines give the probability of falling within 1, 2, or 3 standard deviations of the mean — the 68–95–99.7 rule. Note the symmetry: $\Phi(-z) = 1 - \Phi(z)$, so each pair $\Phi(-k)$ and $\Phi(k)$ sums to 1.

Standardisation

Claim 25: Standardisation — If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1). \quad (5.53)$$

Thus by subtracting the mean and dividing by the standard deviation we obtain a standard normal distribution. This procedure is called **standardisation**.

Proof: Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Z = (X - \mu)/\sigma$. Then,

$$\begin{aligned}
 \mathbb{P}(Z \leq z) &= \mathbb{P}((X - \mu)/\sigma \leq z) = \mathbb{P}(X \leq \mu + \sigma z) \\
 &= \int_{-\infty}^{\mu + \sigma z} \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\
 &= \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy \quad \left[\text{substitution } y = \frac{x-\mu}{\sigma} \right] \\
 &= \Phi(z).
 \end{aligned} \tag{5.54}$$

Thus Z has a standard normal distribution. $\square$

Standardisation enables us to express the cdf of any normal distribution in terms of the cdf of the standard normal distribution:

$$\mathbb{P}(X \leq x) = \Phi\left(\frac{x - \mu}{\sigma}\right). \tag{5.55}$$

This is the reason why only the table for the standard normal distribution is included in traditional statistics textbooks.

A trivial rewriting of the standardisation formula gives the following important result: if $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$X = \mu + \sigma Z, \quad \text{with } Z \sim \mathcal{N}(0, 1). \tag{5.56}$$

In other words, any Gaussian (normal) random variable can be viewed as an *affine* (linear + constant) transformation of a standard normal random variable.

Mean and Variance

Mean. $\mathbb{E}[X] = \mu$. This follows immediately from the symmetry of the pdf around μ (recall from Section 5.2 that if a pdf is symmetric about c , then $\mathbb{E}[X] = c$).

Variance. $\text{Var}(X) = \sigma^2$. To see this, write $X = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0, 1)$. By the scaling property of variance (Section 5.2), $\text{Var}(X) = \sigma^2 \text{Var}(Z)$. It therefore suffices to show that $\text{Var}(Z) = 1$. Since $\mathbb{E}[Z] = 0$ (by symmetry), we have $\text{Var}(Z) = \mathbb{E}[Z^2]$. Using integration by parts,

$$\mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z^2 \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \int_{-\infty}^{\infty} z \times \frac{z}{\sqrt{2\pi}} e^{-z^2/2} dz. \tag{5.57}$$

Writing the integrand as the product $u = z$ and $v' = \frac{z}{\sqrt{2\pi}} e^{-z^2/2}$ (so that $v = \frac{-1}{\sqrt{2\pi}} e^{-z^2/2}$), integration by parts gives

$$\mathbb{E}[Z^2] = z \cdot \frac{-1}{\sqrt{2\pi}} e^{-z^2/2} \Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 0 + 1 = 1, \tag{5.58}$$

since the boundary term vanishes and the remaining integral is the total area under the standard normal pdf.

In summary,

$$\boxed{X \sim \mathcal{N}(\mu, \sigma^2) : \quad \mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2.} \tag{5.59}$$

Further Properties

The 68–95–99.7 rule. For any normal distribution, the probability of falling within 1, 2, or 3 standard deviations of the mean is approximately:

$$\mathbb{P}(\mu - \sigma < X < \mu + \sigma) \approx 0.683, \quad \mathbb{P}(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.954, \quad (5.60)$$

$$\mathbb{P}(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997. \quad (5.61)$$

These follow from standardisation: $\mathbb{P}(\mu - k\sigma < X < \mu + k\sigma) = \Phi(k) - \Phi(-k)$, and the values $\Phi(k) - \Phi(-k)$ for $k = 1, 2, 3$ were computed in the code output above. In practice, this means that almost all the probability mass of a normal distribution lies within three standard deviations of the mean.

Symmetry. The pdf of $\mathcal{N}(\mu, \sigma^2)$ is symmetric about μ . As a consequence, the *median* and the *mode* (the value where the pdf is maximised) both equal μ , coinciding with the mean. For the standard normal, symmetry gives $\Phi(-z) = 1 - \Phi(z)$ for all z , which is useful for computing tail probabilities.

Linear transformations. If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = aX + b$ for constants $a \neq 0$ and b , then

$$Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2). \quad (5.62)$$

That is, *linear transformations of normal random variables are again normal*. This follows from the standardisation result: $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$, so $Y = a(\mu + \sigma Z) + b = (a\mu + b) + (a\sigma)Z$, which is $\mathcal{N}(a\mu + b, a^2\sigma^2)$.

Computing probabilities. Combining standardisation with the Φ table (or software), we can compute any probability involving a normal random variable. For $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$\mathbb{P}(a < X < b) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right). \quad (5.63)$$

Example (Normal probabilities). Suppose $X \sim \mathcal{N}(100, 15^2)$ (e.g., IQ scores). What is $\mathbb{P}(85 < X < 130)$?

Standardising both endpoints:

$$\mathbb{P}(85 < X < 130) = \Phi\left(\frac{130 - 100}{15}\right) - \Phi\left(\frac{85 - 100}{15}\right) = \Phi(2) - \Phi(-1). \quad (5.64)$$

Using the symmetry property $\Phi(-1) = 1 - \Phi(1)$:

$$\Phi(2) - \Phi(-1) = \Phi(2) - 1 + \Phi(1) \approx 0.9772 - 1 + 0.8413 = 0.8185. \quad (5.65)$$

So about 82% of values fall between 85 and 130.

5.4 The Exponential Distribution

A random variable X with probability density function f given by

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad (5.66)$$

is said to have an **exponential** distribution with parameter $\lambda > 0$. We write $X \sim \text{Exp}(\lambda)$. The exponential distribution can be viewed as a continuous analogue of the geometric distribution: the geometric counts the number of discrete trials until the first success, while the exponential models the continuous waiting time until the first occurrence of an event. Graphs of the pdf for various values of λ are given in Figure 5.4.

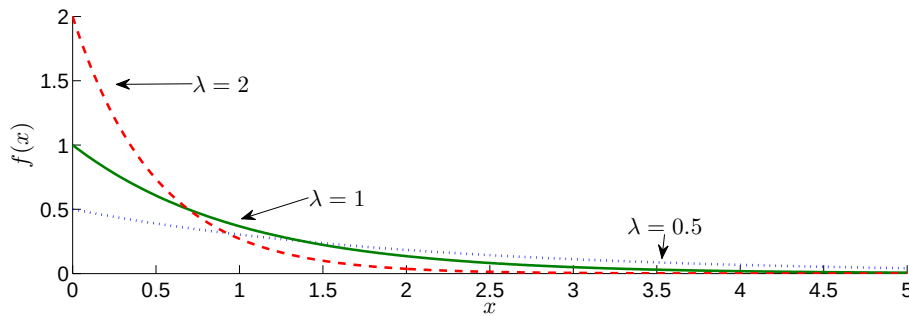


Figure 5.4: The pdf of the $\text{Exp}(\lambda)$ distribution for various λ .

CDF and tail probability

The cdf of $X \sim \text{Exp}(\lambda)$ is

$$F(x) = \mathbb{P}(X \leq x) = \int_0^x \lambda e^{-\lambda u} du = -e^{-\lambda u} \Big|_0^x = 1 - e^{-\lambda x}, \quad x \geq 0. \quad (5.67)$$

As a consequence, the tail probability $\mathbb{P}(X > x)$ decays exponentially:

$$\mathbb{P}(X > x) = e^{-\lambda x}, \quad x \geq 0. \quad (5.68)$$

The function $\bar{F}(x) = 1 - F(x) = \mathbb{P}(X > x)$ is called the **survival function** (or **complementary cdf**, abbreviated CCDF). It gives the probability that X exceeds the value x — in reliability and lifetime modelling, this is the probability of “surviving” past time x .

The survival function provides an elegant alternative formula for the expectation of a non-negative random variable.

Claim 26: Tail-Sum Formula for Expectation — If X is a non-negative continuous random variable with cdf F , then

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > x) dx = \int_0^\infty [1 - F(x)] dx. \quad (5.69)$$

Proof: Starting from the definition of expectation and writing $x = \int_0^x 1 \, dt$:

$$\mathbb{E}[X] = \int_0^\infty x f(x) \, dx = \int_0^\infty \left(\int_0^x 1 \, dt \right) f(x) \, dx. \quad (5.70)$$

Switching the order of integration (the integrand is non-negative, so Tonelli's theorem applies), the region of integration $\{(t, x) : 0 \leq t \leq x < \infty\}$ becomes $\{(t, x) : 0 \leq t < \infty, x \geq t\}$:

$$\mathbb{E}[X] = \int_0^\infty \left(\int_t^\infty f(x) \, dx \right) dt = \int_0^\infty \mathbb{P}(X > t) \, dt. \quad (5.71)$$

□

Geometrically, the mean equals the area *above* the cdf curve and below the line $y = 1$, as illustrated in Figure 5.5 for the exponential distribution.

Tail-sum formula: the area above the CDF equals the mean

LLM Prompt > Plot the CDF of the $\text{Exp}(\lambda)$ distribution for $\lambda = 1$ and $\lambda = 2$ on the same axes. Shade the area between each CDF curve and $y = 1$, illustrating that this area equals the mean $1/\lambda$.

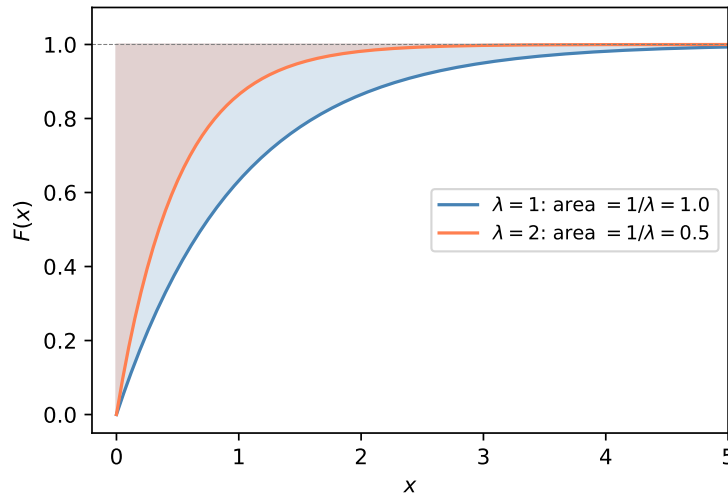


Figure 5.5: CDFs of the $\text{Exp}(\lambda)$ distribution for $\lambda = 1$ and $\lambda = 2$. The shaded area between each curve and the line $y = 1$ equals $\int_0^\infty \mathbb{P}(X > x) \, dx = 1/\lambda$, the mean. A larger rate λ gives a steeper CDF and a smaller shaded area (smaller mean).

Mean and variance

The mean can be computed by integration by parts:

$$\mathbb{E}[X] = \int_0^{\infty} x \lambda e^{-\lambda x} dx = -x e^{-\lambda x} \Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx = 0 + \frac{1}{\lambda} = \frac{1}{\lambda}. \quad (5.72)$$

Alternatively, using the tail-sum formula (Claim 26) with the survival function $\mathbb{P}(X > x) = e^{-\lambda x}$:

$$\mathbb{E}[X] = \int_0^{\infty} e^{-\lambda x} dx = \frac{-1}{\lambda} e^{-\lambda x} \Big|_0^{\infty} = \frac{1}{\lambda}. \quad (5.73)$$

A similar integration-by-parts calculation gives the second moment $\mathbb{E}[X^2] = 2/\lambda^2$, so the variance is

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}. \quad (5.74)$$

In summary,

$$X \sim \text{Exp}(\lambda) : \quad \mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}. \quad (5.75)$$

Note the pattern: both the mean and the standard deviation equal $1/\lambda$, so $\text{SD}(X) = \mathbb{E}[X]$. This is a distinctive feature of the exponential distribution.

The memoryless property

The most important property of the exponential distribution is the following.

Claim 27: Memoryless Property — Let $X \sim \text{Exp}(\lambda)$. Then

$$\mathbb{P}(X > x + y | X > x) = \mathbb{P}(X > y), \quad \text{for all } x, y \geq 0. \quad (5.76)$$

Proof: By the definition of conditional probability,

$$\mathbb{P}(X > x + y | X > x) = \frac{\mathbb{P}(X > x + y, X > x)}{\mathbb{P}(X > x)} = \frac{\mathbb{P}(X > x + y)}{\mathbb{P}(X > x)} \quad (5.77)$$

$$= \frac{e^{-\lambda(x+y)}}{e^{-\lambda x}} = e^{-\lambda y} = \mathbb{P}(X > y), \quad (5.78)$$

where we used that the event $\{X > x + y\}$ is contained in $\{X > x\}$, so their intersection is $\{X > x + y\}$. $\square$

For example, if X denotes the lifetime of a machine, then given that the machine is still alive at time x , the remaining lifetime $X - x$ has the same exponential distribution as a completely new machine. In other words, the machine has no memory of its age and does not “deteriorate” (although it will break down eventually).

This is exactly the continuous-time analogue of the memoryless property of the geometric distribution. Recall from Section 3.4 that $T \sim \text{Geom}(p)$ satisfies $\mathbb{P}(T > m + n | T > m) = \mathbb{P}(T > n)$: given that the first success has not occurred in the first m trials, the remaining waiting time has

the same distribution as starting fresh. The exponential plays the same role for continuous waiting times that the geometric plays for discrete ones.

It is not difficult to prove that the exponential distribution is the *only* continuous distribution on $(0, \infty)$ with the memoryless property, just as the geometric is the only discrete distribution with this property.

5.5 The Gamma and Beta Distributions

Before introducing the gamma and beta families of distributions, we discuss the special functions that give them their names. These functions arise throughout mathematics — in combinatorics, number theory, complex analysis, and physics — and are not specific to probability theory. However, they play a central role in normalising the density functions we study here.

The Gamma and Beta Functions

The Gamma function. The **gamma function** is defined by

$$\Gamma(\alpha) = \int_0^{\infty} u^{\alpha-1} e^{-u} du, \quad \alpha > 0. \quad (5.79)$$

This integral converges for all $\alpha > 0$ and defines a smooth, strictly positive function on $(0, \infty)$. The gamma function extends the factorial to non-integer arguments: as we now show, $\Gamma(n) = (n-1)!$ for every positive integer n .

Key properties.

1. **Recursion:** $\Gamma(\alpha + 1) = \alpha \Gamma(\alpha)$ for all $\alpha > 0$.

Proof: Integration by parts with $u = t^\alpha$ and $dv = e^{-t} dt$ gives

$$\Gamma(\alpha + 1) = \int_0^{\infty} t^\alpha e^{-t} dt = -t^\alpha e^{-t} \Big|_0^{\infty} + \alpha \int_0^{\infty} t^{\alpha-1} e^{-t} dt = \alpha \Gamma(\alpha). \quad (5.80)$$

□

2. **Factorial property:** Since $\Gamma(1) = \int_0^{\infty} e^{-t} dt = 1$, repeated application of the recursion gives

$$\Gamma(n) = (n-1)! \quad \text{for } n = 1, 2, 3, \dots \quad (5.81)$$

For example, $\Gamma(4) = 3! = 6$ and $\Gamma(1) = 0! = 1$.

3. **Half-integer value:** $\Gamma(1/2) = \sqrt{\pi}$.

Proof: By definition,

$$\Gamma(1/2) = \int_0^\infty u^{-1/2} e^{-u} du. \quad (5.82)$$

Substituting $u = z^2/2$ (so $du = z dz$ and $u^{-1/2} = \sqrt{2}/z$) gives

$$\Gamma(1/2) = \int_0^\infty \frac{\sqrt{2}}{z} e^{-z^2/2} z dz = \sqrt{2} \int_0^\infty e^{-z^2/2} dz = \sqrt{2} \cdot \frac{\sqrt{2\pi}}{2} = \sqrt{\pi}, \quad (5.83)$$

where we used the Gaussian integral $\int_{-\infty}^\infty e^{-z^2/2} dz = \sqrt{2\pi}$ (proved in Section 5.3), together with the fact that the integrand is symmetric, so the integral over $(0, \infty)$ is half the total. $\square$

By the recursion, this gives all half-integer values: $\Gamma(3/2) = \frac{1}{2} \Gamma(1/2) = \frac{\sqrt{\pi}}{2}$, $\Gamma(5/2) = \frac{3}{2} \cdot \frac{\sqrt{\pi}}{2} = \frac{3\sqrt{\pi}}{4}$, and so on.

The Beta function. The **beta function** is defined by

$$B(\alpha, \beta) = \int_0^1 u^{\alpha-1} (1-u)^{\beta-1} du, \quad \alpha, \beta > 0. \quad (5.84)$$

The beta function is closely related to the gamma function via the identity

$$B(\alpha, \beta) = \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha + \beta)}. \quad (5.85)$$

This can be proved by writing the product $\Gamma(\alpha) \Gamma(\beta)$ as a double integral and switching to polar-like coordinates; the proof is omitted here but can be found in most analysis textbooks. When α and β are positive integers, this becomes

$$B(m, n) = \frac{(m-1)! (n-1)!}{(m+n-1)!}. \quad (5.86)$$

Note that the beta function is symmetric: $B(\alpha, \beta) = B(\beta, \alpha)$, which follows immediately from the substitution $v = 1 - u$ in the defining integral.

The Gamma Distribution

The **gamma distribution** arises frequently in statistics and applied probability, especially as a model for positive-valued quantities such as waiting times, lifetimes, and accumulated totals.

A random variable X has a **gamma distribution** with shape parameter $\alpha > 0$ and rate parameter $\lambda > 0$ if its density function is given by

$$f(x) = \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, \quad x \geq 0. \quad (5.87)$$

We write $X \sim \text{Gamma}(\alpha, \lambda)$. The parameter $1/\lambda$ is sometimes called the **scale** parameter. One can verify that f integrates to 1 by the substitution $u = \lambda x$, which reduces the integral to $\Gamma(\alpha)/\Gamma(\alpha) = 1$.

Mean and variance. The mean is computed directly using LOTUS and the gamma function recursion:

$$\mathbb{E}[X] = \int_0^\infty x \cdot \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)} dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^\alpha e^{-\lambda x} dx. \quad (5.88)$$

The substitution $u = \lambda x$ gives $\int_0^\infty x^\alpha e^{-\lambda x} dx = \Gamma(\alpha + 1)/\lambda^{\alpha+1}$. Since $\Gamma(\alpha + 1) = \alpha \Gamma(\alpha)$,

$$\mathbb{E}[X] = \frac{\lambda^\alpha}{\Gamma(\alpha)} \cdot \frac{\alpha \Gamma(\alpha)}{\lambda^{\alpha+1}} = \frac{\alpha}{\lambda}. \quad (5.89)$$

A similar calculation gives $\mathbb{E}[X^2] = \alpha(\alpha + 1)/\lambda^2$, so $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \alpha/\lambda^2$. In summary,

$$X \sim \text{Gamma}(\alpha, \lambda) : \quad \mathbb{E}[X] = \frac{\alpha}{\lambda}, \quad \text{Var}(X) = \frac{\alpha}{\lambda^2}. \quad (5.90)$$

Reparametrisation by Mean and Coefficient of Variation

The gamma distribution is often parametrised by the shape α and rate λ , but in applications — particularly in manufacturing and reliability engineering — it is more natural to describe a distribution by its **mean** and its relative spread. The **coefficient of variation** (CV) of a positive random variable X is defined as the ratio of the standard deviation to the mean:

$$c_v = \frac{\text{SD}(X)}{\mathbb{E}[X]} = \frac{\sigma}{\mu}. \quad (5.91)$$

The CV is a dimensionless measure of variability: it tells us how large the spread is *relative to the mean*. A CV of 0.1 indicates low variability (the standard deviation is 10% of the mean), while a CV of 2 indicates high variability.

For the gamma distribution, the CV has a particularly clean form. Since $\mathbb{E}[X] = \alpha/\lambda$ and $\text{SD}(X) = \sqrt{\alpha}/\lambda$,

$$c_v = \frac{\sqrt{\alpha}/\lambda}{\alpha/\lambda} = \frac{1}{\sqrt{\alpha}}. \quad (5.92)$$

Notice that the CV depends *only* on the shape parameter α , not on the rate λ . Solving for the original parameters in terms of the mean μ and CV c_v :

$$\alpha = \frac{1}{c_v^2}, \quad \lambda = \frac{1}{c_v^2 \mu}. \quad (5.93)$$

Example (Gamma distributions with fixed mean). Figure 5.6 shows gamma densities with mean $\mu = 2$ and various coefficients of variation. When $c_v = 1$ we have $\alpha = 1$, which is the exponential distribution. As c_v decreases (equivalently, α increases), the density becomes more concentrated around the mean — approaching a normal shape. As c_v increases beyond 1 ($\alpha < 1$), the density becomes increasingly right-skewed with a singularity at the origin.

This reparametrisation is widely used in manufacturing and queueing theory, where one specifies a process by its mean duration and relative variability rather than by abstract shape and rate parameters.

Gamma densities with fixed mean and varying CV

LLM Prompt > Plot gamma densities with mean $\mu = 2$ for several coefficients of variation $c_v = 0.25, 0.5, 1, 1.5, 2$ on one plot.

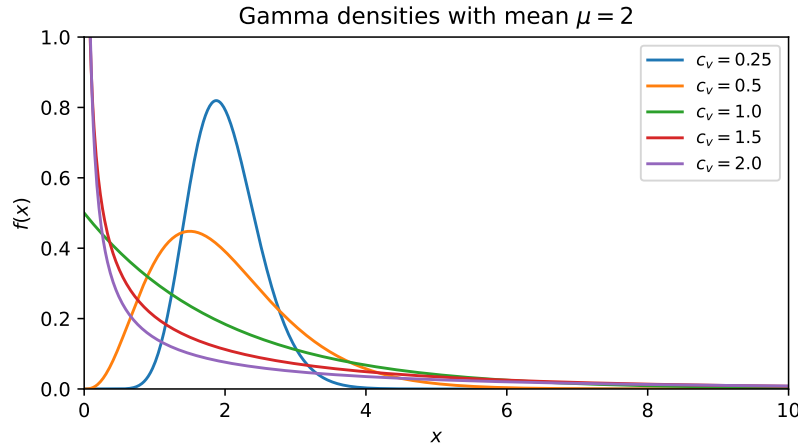


Figure 5.6: Gamma densities with mean $\mu = 2$ and coefficients of variation $c_v = 0.25, 0.5, 1, 1.5, 2$. All distributions have the same mean but differ in spread. The case $c_v = 1$ ($\alpha = 1$) is the exponential distribution.

Special Cases of the Gamma Distribution

The gamma family is remarkably versatile: several important named distributions are special cases.

Exponential distribution. Taking $\alpha = 1$ in the gamma pdf (5.87) gives

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad (5.94)$$

which is precisely the $\text{Exp}(\lambda)$ density from Section 5.4. Thus

$$\boxed{\text{Exp}(\lambda) = \text{Gamma}(1, \lambda).} \quad (5.95)$$

This is consistent with the mean and variance: $\mathbb{E}[X] = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$.

Chi-squared distribution. A random variable X is said to have a **chi-squared distribution** with n **degrees of freedom** (where $n \in \{1, 2, 3, \dots\}$) if

$$\boxed{X \sim \chi_n^2 \iff X \sim \text{Gamma}\left(\frac{n}{2}, \frac{1}{2}\right).} \quad (5.96)$$

The chi-squared distribution plays a central role in statistical inference (hypothesis testing and confidence intervals). Graphs of the pdf for various degrees of freedom are given in Figure 5.7.

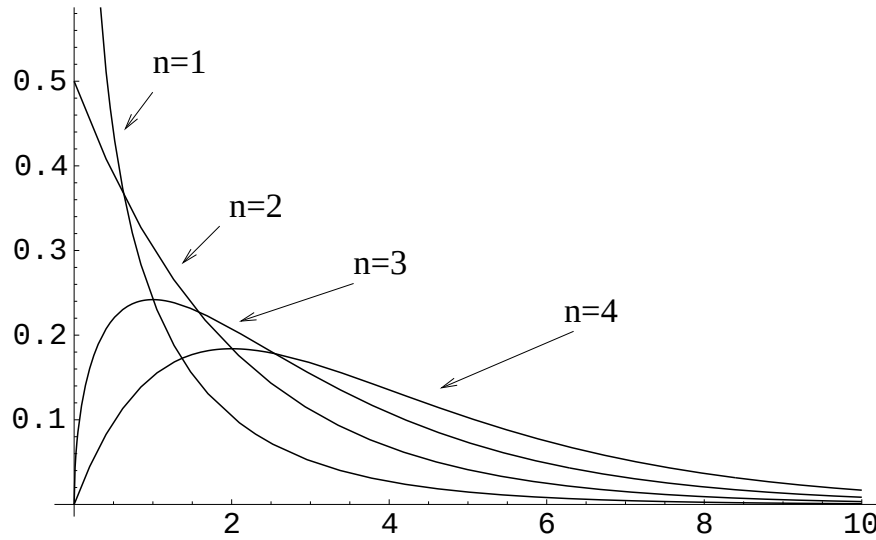


Figure 5.7: Pdfs for the χ_n^2 -distribution, for various degrees of freedom n .

From the gamma mean and variance formulas:

$$X \sim \chi_n^2 : \quad \mathbb{E}[X] = \frac{n/2}{1/2} = n, \quad \mathbb{V}\text{ar}(X) = \frac{n/2}{(1/2)^2} = 2n. \quad (5.97)$$

Erlang distribution. When the shape parameter α is a positive integer, say $\alpha = k$ with $k \in \{1, 2, 3, \dots\}$, the gamma distribution is called the **Erlang distribution**:

$$\boxed{\text{Erl}(k, \lambda) = \text{Gamma}(k, \lambda), \quad k \in \{1, 2, 3, \dots\}.} \quad (5.98)$$

Since $\Gamma(k) = (k-1)!$, the Erlang pdf takes the particularly clean form

$$f(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{(k-1)!}, \quad x \geq 0. \quad (5.99)$$

The Erlang distribution has a natural interpretation: if $T_1, T_2, \dots, T_k$ are independent $\text{Exp}(\lambda)$ random variables (e.g., inter-arrival times in a Poisson process), then their sum $S_k = T_1 + \dots + T_k$ has an $\text{Erl}(k, \lambda)$ distribution. This is the waiting time until the k -th event — compare with the negative binomial distribution, which counts the number of discrete trials until the k -th success.

In summary, the relationships are:

$$\begin{aligned} \text{Exp}(\lambda) &= \text{Gamma}(1, \lambda) = \text{Erl}(1, \lambda), \\ \text{Erl}(k, \lambda) &= \text{Gamma}(k, \lambda) \quad \text{for } k = 1, 2, 3, \dots, \\ \chi_n^2 &= \text{Gamma}(n/2, 1/2). \end{aligned} \quad (5.100)$$

The Beta Distribution

A random variable X has a **beta distribution** with parameters $\alpha > 0$ and $\beta > 0$ if its density function is

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq x \leq 1, \quad (5.101)$$

where $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha + \beta)$ is the beta function defined in (5.84). We write $X \sim \text{Beta}(\alpha, \beta)$. The beta distribution is supported on $[0, 1]$, making it a natural model for proportions, probabilities, and other quantities constrained to lie between 0 and 1.

The shape of the beta density depends strongly on the parameters α and β , as illustrated in Figure 5.8. When both parameters are less than 1, the density is U-shaped; when both are greater than 1, it is unimodal; when one parameter is much larger than the other, the density is skewed.

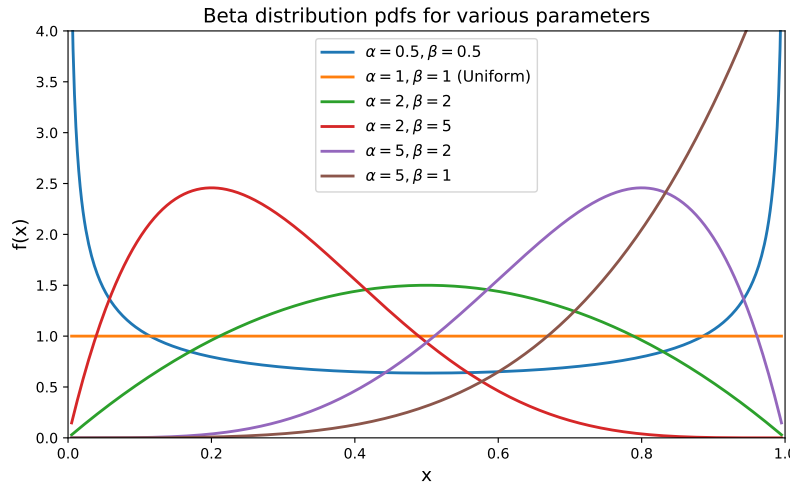


Figure 5.8: Pdfs of the $\text{Beta}(\alpha, \beta)$ distribution for various parameter values.

The uniform as a special case. When $\alpha = \beta = 1$, the beta density becomes

$$f(x) = \frac{x^0(1-x)^0}{B(1, 1)} = \frac{1}{1} = 1, \quad 0 \leq x \leq 1, \quad (5.102)$$

since $B(1, 1) = \Gamma(1)\Gamma(1)/\Gamma(2) = 1$. This is precisely the $\mathcal{U}(0, 1)$ density. Thus

$$\text{Beta}(1, 1) = \mathcal{U}(0, 1). \quad (5.103)$$

Mean and variance. Using the relationship between the beta and gamma functions, one can show that

$$X \sim \text{Beta}(\alpha, \beta) : \quad \mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}. \quad (5.104)$$

Note how the mean $\alpha/(\alpha + \beta)$ is simply the “share” of α in the total $\alpha + \beta$. When $\alpha = \beta$, the distribution is symmetric about $1/2$.

Proof: We compute $\mathbb{E}[X]$ directly:

$$\mathbb{E}[X] = \int_0^1 x \cdot \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)} dx = \frac{1}{B(\alpha, \beta)} \int_0^1 x^{\alpha}(1-x)^{\beta-1} dx = \frac{B(\alpha+1, \beta)}{B(\alpha, \beta)}. \quad (5.105)$$

Using $B(\alpha+1, \beta) = \Gamma(\alpha+1)\Gamma(\beta)/\Gamma(\alpha+\beta+1) = \alpha\Gamma(\alpha)\Gamma(\beta)/[(\alpha+\beta)\Gamma(\alpha+\beta)]$, we get

$$\mathbb{E}[X] = \frac{\alpha\Gamma(\alpha)\Gamma(\beta)/[(\alpha+\beta)\Gamma(\alpha+\beta)]}{\Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)} = \frac{\alpha}{\alpha+\beta}. \quad (5.106)$$

A similar calculation gives $\mathbb{E}[X^2] = \alpha(\alpha+1)/[(\alpha+\beta)(\alpha+\beta+1)]$, from which the variance follows. $\square$

Transformations of the Beta Distribution

The standard beta distribution is supported on $[0, 1]$, but in applications we often need a random variable supported on a general interval $[a, b]$. This is achieved by a simple linear transformation.

If $X \sim \text{Beta}(\alpha, \beta)$, then the transformed random variable

$$Y = a + (b - a)X \quad (5.107)$$

is supported on $[a, b]$. Its density is obtained by the change of variables $x = (y - a)/(b - a)$:

$$f_Y(y) = \frac{1}{b-a} \cdot \frac{\left(\frac{y-a}{b-a}\right)^{\alpha-1} \left(\frac{b-y}{b-a}\right)^{\beta-1}}{B(\alpha, \beta)}, \quad a \leq y \leq b. \quad (5.108)$$

Its mean and variance follow from the linear transformation properties:

$$\mathbb{E}[Y] = a + (b-a) \cdot \frac{\alpha}{\alpha+\beta}, \quad \text{Var}(Y) = (b-a)^2 \cdot \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}. \quad (5.109)$$

When $\alpha = \beta = 1$, this reduces to $Y \sim \mathcal{U}(a, b)$, consistent with the fact that $\text{Beta}(1, 1) = \mathcal{U}(0, 1)$.

MORE ASPECTS OF RANDOM VARIABLES AND DISTRIBUTIONS

In this chapter we explore several additional aspects of random variables and their distributions. We develop techniques for finding the distribution of a transformed random variable, and see how relationships between families of probability distributions arise. We then consider the inverse probability transform, a universal method for simulating random variables. We establish the Markov and Chebyshev inequalities, which bound tail probabilities using only moments. Finally, we encounter random variables whose moments are infinite or even undefined — the Pareto and Cauchy distributions.

6.1 Transformations of Random Variables

Given a random variable X with a known distribution, we often need to determine the distribution of a transformed random variable $Y = g(X)$ for some function g . This is one of the most fundamental operations in probability and statistics.

What We Already Know

Before developing the general theory, let us recall what tools we already have at our disposal.

Computing expectations without finding the distribution. Thanks to LOTUS (Claims 8 and 23), we can compute $\mathbb{E}[g(X)]$ directly from the distribution of X , without ever finding the

distribution of $Y = g(X)$:

$$\mathbb{E}[g(X)] = \sum_x g(x) f_X(x) \quad (\text{discrete}), \quad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx \quad (\text{continuous}). \quad (6.1)$$

This is extremely useful when all we need is the mean (or higher moments) of Y . However, there are many situations where we need the full distribution of Y — its pdf, cdf, or variance — and for that we need the techniques developed in this section.

Linear transformations. We have already handled the special case $Y = aX + b$ (with $a \neq 0$) for several distribution families:

- **Normal:** If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$. In particular, standardisation $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$ is a linear transformation (Claim 25).
- **Uniform:** If $U \sim \mathcal{U}(0, 1)$, then $X = a + (b - a)U \sim \mathcal{U}(a, b)$. The uniform on any interval is obtained by a linear transformation of the standard uniform (Section 5.2).
- **Geometric:** The two conventions for the geometric distribution — counting trials ($X \sim \text{Geom}(p)$ with support $\{1, 2, \dots\}$) and counting failures ($Y = X - 1$ with support $\{0, 1, \dots\}$) — are related by a simple shift $Y = X - 1$ (Section 3.4).
- **General:** For any random variable, $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$ and $\mathbb{V}\text{ar}(aX + b) = a^2 \mathbb{V}\text{ar}(X)$ (see (5.30)).

But what about other distributions and, more generally, *nonlinear* transformations such as $Y = X^2$, $Y = e^X$, or $Y = \sqrt{X}$? For these we need a systematic method.

The CDF Method

The most general and conceptually simple approach is to work through the cdf. Since we now deal with multiple random variables simultaneously, we use subscripts to distinguish their distributions: F_X and f_X denote the cdf and pdf of X , while F_Y and f_Y denote those of Y . (When there is only one random variable in play, we continue to write simply F and f .) If $Y = g(X)$, then

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(\{\omega : g(X(\omega)) \leq y\}). \quad (6.2)$$

We rewrite the event $\{g(X) \leq y\}$ in terms of X and compute the probability using the known distribution of X . This approach always works, regardless of whether g is monotone, one-to-one, or anything else. In the continuous case we can go further and differentiate to obtain the pdf $f_Y(y) = F'_Y(y)$.

One-to-One Transformations: The Change-of-Variable Formula

When g is a strictly monotone (and hence one-to-one) differentiable function, there is a convenient shortcut.

Claim 28: Change-of-variable formula — Let X be a continuous random variable with pdf f_X , and let $Y = g(X)$ where g is a strictly monotone differentiable function. Then Y has pdf

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|. \quad (6.3)$$

Proof: Suppose first that g is strictly increasing. Then g has an inverse g^{-1} , and since $g(X(\omega)) \leq y$ if and only if $X(\omega) \leq g^{-1}(y)$,

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)). \quad (6.4)$$

Differentiating by the chain rule:

$$f_Y(y) = F'_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y). \quad (6.5)$$

Since g is increasing, g^{-1} is also increasing, so $\frac{d}{dy} g^{-1}(y) > 0$ and the absolute value is not needed.

If instead g is strictly decreasing, then $g(X(\omega)) \leq y$ if and only if $X(\omega) \geq g^{-1}(y)$, so

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y)), \quad (6.6)$$

so $f_Y(y) = -f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$. Since g^{-1} is decreasing, the derivative is negative, and the minus sign makes the result positive. In both cases, the formula $f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|$ holds. $\square$

Intuition. The factor $|(d/dy) g^{-1}(y)|$ accounts for the “stretching” or “compression” that g applies: regions where g stretches the x -axis get lower density (probability is spread over a wider range of y), while regions where g compresses get higher density.

Example (Exponential to Rayleigh). Let $X \sim \text{Exp}(1/2)$, so $f_X(x) = \frac{1}{2}e^{-x/2}$ for $x \geq 0$. Define $Y = \sqrt{X}$, i.e., $g(x) = \sqrt{x}$.

Since g is strictly increasing on $[0, \infty)$, we can apply the change-of-variable formula. The inverse is

$$g^{-1}(y) = y^2, \quad \frac{d}{dy} g^{-1}(y) = 2y. \quad (6.7)$$

Therefore,

$$f_Y(y) = f_X(y^2) \cdot |2y| = \frac{1}{2}e^{-y^2/2} \cdot 2y = ye^{-y^2/2}, \quad y \geq 0. \quad (6.8)$$

This is the pdf of the **Rayleigh distribution** $\text{Rayleigh}(1)$. It arises naturally as the distribution of $\sqrt{Z_1^2 + Z_2^2}$, the distance from the origin of a random point (Z_1, Z_2) where Z_1, Z_2 are independent standard normals. The Rayleigh distribution is used in communications engineering and physics to model the magnitude of a two-dimensional random vector.

Figure 6.1 illustrates the transformation: we generate 10^4 samples from $\text{Exp}(1/2)$ and apply $Y = \sqrt{X}$, confirming that the resulting histogram matches the $\text{Rayleigh}(1)$ pdf.

Transformation from exponential to Rayleigh

LLM Prompt > Generate 10^4 samples from $\text{Exp}(1/2)$, apply the transformation $Y = \sqrt{X}$, and plot side-by-side histograms with the theoretical pdfs overlaid.

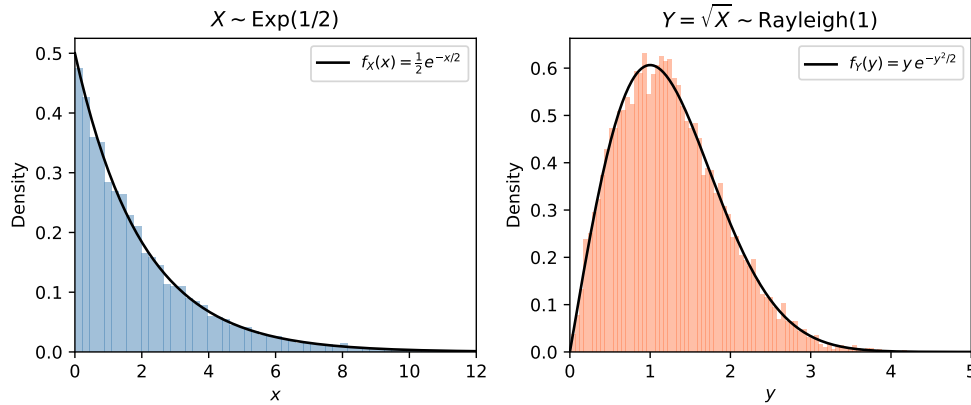


Figure 6.1: Left: histogram of 10^4 samples from $\text{Exp}(1/2)$ with the theoretical pdf. Right: histogram of $Y = \sqrt{X}$ with the Rayleigh(1) pdf, confirming the change-of-variable calculation.

Example (Log-normal distribution). Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and define $Y = e^X$. Since $g(x) = e^x$ is strictly increasing, we apply (6.3). The inverse and its derivative are

$$g^{-1}(y) = \ln y, \quad \frac{d}{dy}g^{-1}(y) = \frac{1}{y}. \quad (6.9)$$

Thus, for $y > 0$,

$$f_Y(y) = f_X(\ln y) \cdot \frac{1}{y} = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right). \quad (6.10)$$

This is the **log-normal** distribution. A random variable is log-normal if and only if its logarithm is normal. The log-normal appears frequently in modelling quantities that are the product of many positive factors (e.g., stock prices, particle sizes, incomes, and pharmacometric analysis).

Note that the parameters μ and σ^2 are the mean and variance of $\ln Y$, *not* of Y itself. Using LOTUS with $Y = e^X$ and $X \sim \mathcal{N}(\mu, \sigma^2)$, one can show (by completing the square in the exponent) that the mean of Y is

$$\mathbb{E}[Y] = e^{\mu + \sigma^2/2}, \quad (6.11)$$

and the variance is

$$\text{Var}(Y) = e^{2\mu + 2\sigma^2}(e^{\sigma^2} - 1). \quad (6.12)$$

The coefficient of variation (Section 5.5) has a particularly clean form:

$$c_v = \frac{\text{SD}(Y)}{\mathbb{E}[Y]} = \sqrt{e^{\sigma^2} - 1}, \quad (6.13)$$

which depends only on σ^2 , not on μ . This makes σ^2 a pure measure of relative variability for log-normal data, just as α plays this role for the gamma distribution.

Conversely, if Y has a log-normal distribution, then $X = \ln Y \sim \mathcal{N}(\mu, \sigma^2)$. This is often used in practice: when data are positively skewed and bounded below by zero, taking logarithms can transform the data to approximate normality, making standard normal-based methods applicable.

Non-Monotone Transformations

When g is not one-to-one, the change-of-variable formula does not directly apply, and we must return to the CDF method (6.2). The key challenge is that the event $\{g(X) \leq y\}$ may correspond to a *union of intervals* for X .

Example ($Z^2 \sim \chi_1^2$ where $Z \sim \mathcal{N}(0, 1)$). Let $Z \sim \mathcal{N}(0, 1)$ and $Y = Z^2$. The function $g(z) = z^2$ is not one-to-one: both z and $-z$ map to the same y . We use the CDF method. For $y > 0$, writing Φ for the standard normal cdf (Section 5.3),

$$F_Y(y) = \mathbb{P}(Z^2 \leq y) = \mathbb{P}(-\sqrt{y} \leq Z \leq \sqrt{y}) = \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) = 2\Phi(\sqrt{y}) - 1, \quad (6.14)$$

where we used the symmetry $\Phi(-z) = 1 - \Phi(z)$. Differentiating with respect to y :

$$f_Y(y) = 2\phi(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = \frac{\phi(\sqrt{y})}{\sqrt{y}} = \frac{1}{\sqrt{y}} \cdot \frac{1}{\sqrt{2\pi}} e^{-y/2} = \frac{1}{\sqrt{2\pi}} y^{-1/2} e^{-y/2}, \quad (6.15)$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the standard normal pdf.

Now recall from Section 5.5 that the $\text{Gamma}(\alpha, \lambda)$ density is $\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$. Setting $\alpha = 1/2$ and $\lambda = 1/2$, and using $\Gamma(1/2) = \sqrt{\pi}$ (also from Section 5.5):

$$\frac{(1/2)^{1/2}}{\Gamma(1/2)} y^{-1/2} e^{-y/2} = \frac{1}{\sqrt{2} \cdot \sqrt{\pi}} y^{-1/2} e^{-y/2} = \frac{1}{\sqrt{2\pi}} y^{-1/2} e^{-y/2}. \quad (6.16)$$

This matches $f_Y(y)$ exactly. Hence

$$\boxed{Z \sim \mathcal{N}(0, 1) \implies Z^2 \sim \text{Gamma}(1/2, 1/2) = \chi_1^2.} \quad (6.17)$$

The square of a standard normal is chi-squared with 1 degree of freedom! This is the fundamental link between the normal and chi-squared families, and explains why the chi-squared distribution appears so prominently in statistical inference.

Example (Absolute value of a standard normal). Let $Z \sim \mathcal{N}(0, 1)$ and $Y = |Z|$. Again, $g(z) = |z|$ is not one-to-one. For $y \geq 0$,

$$F_Y(y) = \mathbb{P}(|Z| \leq y) = \mathbb{P}(-y \leq Z \leq y) = 2\Phi(y) - 1. \quad (6.18)$$

Differentiating:

$$f_Y(y) = 2\phi(y) = \frac{2}{\sqrt{2\pi}} e^{-y^2/2} = \sqrt{\frac{2}{\pi}} e^{-y^2/2}, \quad y \geq 0. \quad (6.19)$$

This is known as the **half-normal** distribution. Its mean and variance are

$$\mathbb{E}[Y] = \mathbb{E}[|Z|] = \sqrt{\frac{2}{\pi}} \approx 0.798, \quad \text{Var}(Y) = 1 - \frac{2}{\pi} \approx 0.363. \quad (6.20)$$

Example ($\mathcal{U}(0, 1)$ squared). Let $U \sim \mathcal{U}(0, 1)$ and $Y = U^2$. Since $U \in [0, 1]$, we have $Y \in [0, 1]$ as well. For $0 < y < 1$,

$$F_Y(y) = \mathbb{P}(U^2 \leq y) = \mathbb{P}(U \leq \sqrt{y}) = \sqrt{y}, \quad (6.21)$$

where the last step uses $F_U(u) = u$ for $u \in [0, 1]$. Differentiating:

$$f_Y(y) = \frac{1}{2\sqrt{y}}, \quad 0 < y < 1. \quad (6.22)$$

This is the $\text{Beta}(1/2, 1)$ distribution. To verify, the $\text{Beta}(1/2, 1)$ pdf is

$$\frac{y^{1/2-1}(1-y)^{1-1}}{B(1/2, 1)} = \frac{y^{-1/2}}{B(1/2, 1)}, \quad (6.23)$$

and $B(1/2, 1) = \Gamma(1/2)\Gamma(1)/\Gamma(3/2) = \sqrt{\pi} \cdot 1/(\sqrt{\pi}/2) = 2$, giving $y^{-1/2}/2 = 1/(2\sqrt{y})$ as required.

Note how the squaring concentrates the density near 0: values of U near 0 are compressed together (many values of U map to small Y), while values near 1 are spread apart.

6.2 The Inverse Probability Transform

In Chapter 3 we developed the *cumulative probability method* (Algorithm 3.6.1) for simulating discrete random variables: partition $[0, 1)$ into sub-intervals of lengths $p_1, p_2, \dots$, draw $U \sim \mathcal{U}(0, 1)$, and return whichever value corresponds to the sub-interval that U falls in. In this section we show that this idea generalises to a beautiful and universal principle — the **inverse probability transform** — which works for *any* distribution, discrete or continuous.

Continuous Distributions Supported on $\mathbb{R}$

We begin with the cleanest case. Suppose X is a continuous random variable whose pdf is positive for every $x \in \mathbb{R}$ (for example, the normal distribution). Then the cdf F is strictly increasing on all of $\mathbb{R}$, rising continuously from 0 to 1. This means F is a bijection from $\mathbb{R}$ onto $(0, 1)$, and it has a well-defined inverse

$$F^{-1} : (0, 1) \rightarrow \mathbb{R}. \quad (6.24)$$

The key observation is: if $U \sim \mathcal{U}(0, 1)$, then the random variable $X = F^{-1}(U)$ has cdf F . We prove this below in the slightly more general setting that also covers distributions with restricted support.

Continuous Distributions Supported on a Subset of $\mathbb{R}$

Many continuous distributions have support on a proper subset of $\mathbb{R}$, such as $[0, \infty)$ (exponential, gamma, Rayleigh) or a bounded interval $[a, b]$ (uniform). In these cases the cdf F is constant outside the support — equal to 0 below the support and 1 above it — but strictly increasing *on* the support. The cdf still maps the support one-to-one onto $(0, 1)$, and the inverse $F^{-1} : (0, 1) \rightarrow \text{support}$ is well-defined. The same principle applies.

Claim 29: Inverse probability transform (continuous case) — Let F be a cdf that is continuous and strictly increasing on the support of the distribution, and let $U \sim \mathcal{U}(0, 1)$. Then

$$X = F^{-1}(U) \quad \text{has cdf } F. \quad (6.25)$$

Proof: Since F is strictly increasing and continuous on the support, F^{-1} is well-defined, and $F^{-1}(u) \leq x$ if and only if $u \leq F(x)$. Therefore,

$$\mathbb{P}(X \leq x) = \mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x), \quad (6.26)$$

where the last equality uses the cdf of the $\mathcal{U}(0, 1)$ distribution:

$$F_U(u) = \mathbb{P}(U \leq u) = \begin{cases} 0, & u < 0, \\ u, & 0 \leq u \leq 1, \\ 1, & u > 1. \end{cases} \quad (6.27)$$

Since $F(x) \in [0, 1]$, we have $\mathbb{P}(U \leq F(x)) = F(x)$. $\square$

Claim 29 immediately gives us a simulation recipe: to generate a random variable with any desired continuous cdf F , simply draw $U \sim \mathcal{U}(0, 1)$ and return $X = F^{-1}(U)$. The method is illustrated graphically in Figure 6.2: given a value u on the vertical axis, we read off $x = F^{-1}(u)$ on the horizontal axis.

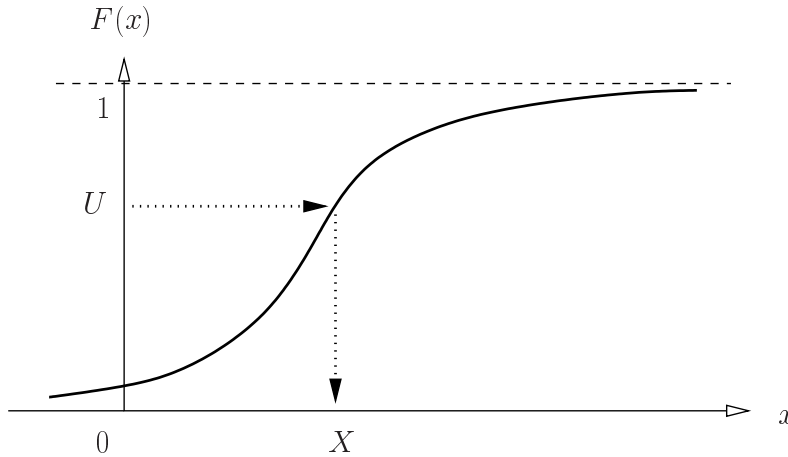


Figure 6.2: The inverse probability transform for a continuous distribution. A uniform draw U on the vertical axis is mapped to $X = F^{-1}(U)$ on the horizontal axis.

The non-uniform shape of F^{-1} transforms the uniform distribution into the desired distribution — regions where F^{-1} is steep (i.e., where F is flat, meaning low density) are stretched out, while regions where F^{-1} is gentle (where F is steep, meaning high density) are compressed.

Example (Generating $\mathcal{U}(a, b)$ from $\mathcal{U}(0, 1)$). Let $X \sim \mathcal{U}(a, b)$, which has support $[a, b]$. The cdf is $F(x) = (x - a)/(b - a)$ for $x \in [a, b]$. Setting $u = (x - a)/(b - a)$ and solving for x gives

$$F^{-1}(u) = a + (b - a)u. \quad (6.28)$$

Therefore,

$$X = a + (b - a)U, \quad U \sim \mathcal{U}(0, 1), \quad (6.29)$$

which is exactly the linear transformation we have used throughout Chapters 3 and 5.

Example (Generating $\text{Exp}(\lambda)$). Let $X \sim \text{Exp}(\lambda)$, which has support $[0, \infty)$. The cdf is $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$. Setting $u = 1 - e^{-\lambda x}$ and solving:

$$e^{-\lambda x} = 1 - u \implies x = -\frac{1}{\lambda} \ln(1 - u). \quad (6.30)$$

So $F^{-1}(u) = -\frac{1}{\lambda} \ln(1 - u)$, and we can generate X via

$$X = -\frac{1}{\lambda} \ln(1 - U), \quad U \sim \mathcal{U}(0, 1). \quad (6.31)$$

Since $1 - U$ has the same distribution as U (both are $\mathcal{U}(0, 1)$), we can simplify to $X = -\frac{1}{\lambda} \ln U$.

Example (The Rayleigh distribution: CDF, moments, and simulation). Let $Y \sim \text{Rayleigh}(1)$, with pdf $f_Y(y) = y e^{-y^2/2}$ for $y \geq 0$ (derived in Example 6.1).

CDF. Integrating the pdf:

$$F_Y(y) = \int_0^y t e^{-t^2/2} dt = \left[-e^{-t^2/2} \right]_0^y = 1 - e^{-y^2/2}, \quad y \geq 0. \quad (6.32)$$

Moments. The k -th moment is computed by the substitution $u = y^2/2$ (so $y = \sqrt{2u}$ and $dy = du/\sqrt{2u}$):

$$\mathbb{E}[Y^k] = \int_0^\infty y^k \cdot y e^{-y^2/2} dy = \int_0^\infty (2u)^{k/2} e^{-u} du = 2^{k/2} \Gamma\left(\frac{k}{2} + 1\right). \quad (6.33)$$

For the first three moments:

$$\mathbb{E}[Y] = \sqrt{2} \Gamma(3/2) = \sqrt{2} \cdot \frac{\sqrt{\pi}}{2} = \sqrt{\frac{\pi}{2}} \approx 1.253, \quad (6.34)$$

$$\mathbb{E}[Y^2] = 2 \Gamma(2) = 2, \quad (6.35)$$

$$\mathbb{E}[Y^3] = 2 \sqrt{2} \Gamma(5/2) = 2 \sqrt{2} \cdot \frac{3 \sqrt{\pi}}{4} = \frac{3 \sqrt{2\pi}}{2} \approx 3.760. \quad (6.36)$$

Inverse probability transform. Setting $u = F_Y(y) = 1 - e^{-y^2/2}$ and solving for y :

$$e^{-y^2/2} = 1 - u \implies y = \sqrt{-2 \ln(1 - u)}. \quad (6.37)$$

So $F_Y^{-1}(u) = \sqrt{-2 \ln(1 - u)}$, and since $1 - U \sim \mathcal{U}(0, 1)$,

$$Y = \sqrt{-2 \ln U}, \quad U \sim \mathcal{U}(0, 1). \quad (6.38)$$

This is the same formula we obtain by first generating $X \sim \text{Exp}(1/2)$ via $X = -2 \ln U$ (Example 6.2) and then applying $Y = \sqrt{X}$ (Example 6.1). The two approaches — direct inversion and transformation through an intermediate exponential — lead to the same result.

The following code generates Rayleigh(1) samples using both approaches and compares the first three sample moments with the theoretical values.

Generating Rayleigh samples: two approaches

LLM Prompt > Generate Rayleigh(1) samples via (1) direct inverse transform $Y = \sqrt{-2 \ln U}$ and (2) via exponential transformation $Y = \sqrt{X}$ with $X \sim \text{Exp}(1/2)$. Print a table comparing the first three sample moments from each approach with the theoretical values.

Moment	Theoretical	Direct IPT	Via Exp
$E[Y^1]$	1.2533	1.2494	1.2557
$E[Y^2]$	2.0000	1.9862	1.9982
$E[Y^3]$	3.7599	3.7137	3.7303

The Quantile Function and the General Case

The inverse cdf F^{-1} is also known as the **quantile function**. Recall from Section 5.2 that the p -th quantile of a distribution is the value x_p satisfying $F(x_p) = p$. In the notation of the inverse cdf, this is simply $x_p = F^{-1}(p)$. For example, $F^{-1}(0.5)$ is the median, $F^{-1}(0.25)$ is the first quartile, and $F^{-1}(0.75)$ is the third quartile. The inverse probability transform can therefore be read as: *a random quantile, evaluated at a uniform random probability level, gives a random variable with the desired distribution.*

Not every cdf is continuous and strictly increasing. A discrete cdf has jumps, and even some continuous cdfs have flat regions. To handle all cases, we define the **generalised inverse** of any cdf F as

$$F^{-1}(u) = \inf\{x : F(x) \geq u\}, \quad 0 < u < 1. \quad (6.39)$$

This is the proper definition of the quantile function: it is well-defined for any cdf, and for a continuous, strictly increasing F it coincides with the ordinary inverse.

Claim 30: Inverse probability transform (general case) — Let F be any cdf and let $U \sim \mathcal{U}(0, 1)$. Then $X = F^{-1}(U)$, with F^{-1} as defined in (6.39), has cdf F .

The proof in the general case uses the key property that $F^{-1}(u) \leq x$ if and only if $u \leq F(x)$ (which holds for any cdf), and then proceeds exactly as before.

Example (Generalised inverse of a step-function cdf). Consider a discrete random variable X taking values 0, 1, 2 with probabilities p_0, p_1, p_2 (as in Example 3.1). The cdf is a step function:

$$F(x) = \begin{cases} 0, & x < 0, \\ p_0, & 0 \leq x < 1, \\ p_0 + p_1, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases} \quad (6.40)$$

Since F has jumps, there is no ordinary inverse. Applying the generalised inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, we ask: for each value of $u \in (0, 1]$, what is the smallest x at which $F(x)$ first reaches or exceeds u ?

- If $0 < u \leq p_0$: the cdf first reaches u at $x = 0$ (where F jumps to p_0), so $F^{-1}(u) = 0$.
- If $p_0 < u \leq p_0 + p_1$: $F(x) \geq u$ is not satisfied until $x = 1$ (where F jumps to $p_0 + p_1$), so $F^{-1}(u) = 1$.
- If $p_0 + p_1 < u \leq 1$: $F(x) \geq u$ is not satisfied until $x = 2$ (where F jumps to 1), so $F^{-1}(u) = 2$.

In summary:

$$F^{-1}(u) = \begin{cases} 0, & 0 < u \leq p_0, \\ 1, & p_0 < u \leq p_0 + p_1, \\ 2, & p_0 + p_1 < u \leq 1. \end{cases} \quad (6.41)$$

Drawing $U \sim \mathcal{U}(0, 1)$ and returning $X = F^{-1}(U)$ therefore assigns $X = k$ with probability equal to the length of the corresponding sub-interval, which is exactly p_k . This is the cumulative probability method from Chapter 3 (Algorithm 3.6.1), now derived from the general theory.

The pattern in the example above extends to any discrete distribution. For a random variable taking values $x_1 < x_2 < x_3 < \dots$ with probabilities $p_1, p_2, p_3, \dots$, the generalised inverse $F^{-1}(u)$ returns x_k whenever u falls in the interval $(F(x_{k-1}), F(x_k)]$, i.e., when

$$p_1 + \dots + p_{k-1} < u \leq p_1 + \dots + p_k. \quad (6.42)$$

The “partition $[0, 1)$ into sub-intervals” idea that we developed intuitively for discrete distributions is, in fact, a special case of the inverse probability transform. The general theory unifies the discrete and continuous cases under a single principle. Figure 6.3 illustrates the construction: the step function cdf maps the uniform draw U to the value x_k at which the cdf first reaches or exceeds U .

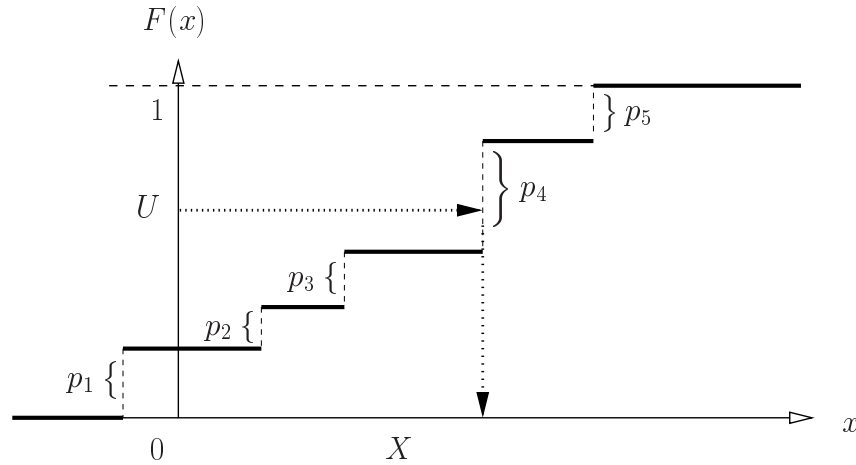


Figure 6.3: The inverse probability transform for a discrete distribution. The generalised inverse of the step-function cdf maps U to the value x_k where the cdf jumps past U .

Limitations: When the Inverse CDF Has No Closed Form

The inverse probability transform is a universal theoretical result: it works for *every* distribution. However, as a practical simulation method, it requires computing $F^{-1}(u)$ — and for some important distributions, no closed-form expression for F^{-1} exists.

The most prominent example is the **normal distribution**. The cdf of $\mathcal{N}(0, 1)$ is

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt, \quad (6.43)$$

and this integral cannot be expressed in terms of elementary functions. Consequently, the inverse $\Phi^{-1}(u)$ — also called the **probit function** — has no closed-form expression either. One can still compute $\Phi^{-1}(u)$ numerically (and software like Python's `scipy.stats.norm.ppf` does so efficiently), but this is slower than methods that avoid the inverse altogether. In practice, normal random variables are usually generated by other techniques such as the *Box–Muller transform*, which cleverly uses two uniform random variables and trigonometric functions to produce two independent standard normals.

Similarly, the **gamma** and **beta** distributions do not have closed-form inverse cdfs (except in special cases like the exponential). For these, specialised algorithms based on acceptance–rejection methods (a Monte Carlo technique not covered in these notes) are typically used. It is important to appreciate that while the inverse probability transform provides the *conceptual foundation* for simulation, it is not always the most practical *computational tool*.

On the other hand, many distributions *do* have tractable inverse cdfs, including: the exponential ($F^{-1}(u) = -\ln(1-u)/\lambda$), the Rayleigh ($F^{-1}(u) = \sigma \sqrt{-2 \ln(1-u)}$), the Cauchy ($F^{-1}(u) = \tan(\pi(u - 1/2))$), and the Weibull. For these, the inverse probability transform provides the simplest and most efficient simulation method.

6.3 The Markov and Chebyshev Inequalities

In many situations we do not know the full distribution of a random variable, but we do know its mean or variance. Remarkably, even this limited information allows us to bound tail probabilities. The two key tools are the **Markov inequality** and its consequence, the **Chebyshev inequality**.

The Markov Inequality

The Markov inequality gives an upper bound on the probability that a nonnegative random variable exceeds a given threshold, using only its mean.

Claim 31: Markov's inequality — Let X be a nonnegative random variable with finite mean. Then for any $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}. \quad (6.44)$$

We give two proofs. The first is a direct calculation for continuous random variables; the second uses indicator functions and works in complete generality.

Proof 1 (continuous case). Suppose X has pdf f . Since $X \geq 0$,

$$\mathbb{E}[X] = \int_0^\infty x f(x) dx \geq \int_a^\infty x f(x) dx \geq \int_a^\infty a f(x) dx = a \mathbb{P}(X \geq a). \quad (6.45)$$

The first inequality drops the non-negative contribution from $[0, a)$; the second replaces x by a on $[a, \infty)$ (where $x \geq a$). Dividing both sides by a gives the result. $\square$

Proof 2 (general, using indicators). Define the indicator random variable

$$\mathbf{1}(X \geq a) = \begin{cases} 1 & \text{if } X \geq a, \\ 0 & \text{if } X < a. \end{cases} \quad (6.46)$$

Since $X \geq 0$ and $X \geq a$ whenever $\mathbf{1}(X \geq a) = 1$, we have the pointwise inequality

$$X \geq a \cdot \mathbf{1}(X \geq a). \quad (6.47)$$

(If $X \geq a$, both sides equal a or more; if $X < a$, the right side is 0 and the left side is ≥ 0 .) Taking expectations of both sides:

$$\mathbb{E}[X] \geq a \cdot \mathbb{E}[\mathbf{1}(X \geq a)] = a \cdot \mathbb{P}(X \geq a). \quad (6.48)$$

Dividing by $a > 0$ gives $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$. $\square$

The second proof is elegant because it makes no assumption about whether X is discrete, continuous, or mixed — it works for *any* nonnegative random variable.

Example (Bounding a tail probability). Suppose the average commute time in a city is 30 minutes. Without knowing anything else about the distribution, what can we say about the fraction of commuters with a commute of at least 90 minutes?

By Markov's inequality with $a = 90$:

$$\mathbb{P}(X \geq 90) \leq \frac{30}{90} = \frac{1}{3} \approx 33\%. \quad (6.49)$$

At most one-third of commuters have a commute of 90 minutes or more. This bound holds regardless of the shape of the distribution.

Remark. The Markov bound is often quite loose — the true probability is typically much smaller. Its power lies in its generality: it requires only $\mathbb{E}[X]$ and the nonnegativity of X .

The Chebyshev Inequality

By applying the Markov inequality to the nonnegative random variable $(X - \mu)^2$, we obtain a much more useful bound that involves both the mean and the variance.

Claim 32: Chebyshev's inequality — Let X be a random variable with mean $\mu = \mathbb{E}[X]$ and finite variance $\sigma^2 = \mathbb{V}\text{ar}(X)$. Then for any $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}. \quad (6.50)$$

Proof: The event $\{|X - \mu| \geq k\sigma\}$ is the same as $\{(X - \mu)^2 \geq k^2\sigma^2\}$. Since $(X - \mu)^2$ is nonnegative, we may apply Markov's inequality (6.44) with $a = k^2\sigma^2$:

$$\mathbb{P}(|X - \mu| \geq k\sigma) = \mathbb{P}((X - \mu)^2 \geq k^2\sigma^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{k^2\sigma^2} = \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}. \quad (6.51)$$

□

Equivalently, by taking the complement:

$$\mathbb{P}(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}. \quad (6.52)$$

This says that the probability of being within k standard deviations of the mean is at least $1 - 1/k^2$, for *any* distribution with a finite variance.

Standard Deviations and Universal Bounds

The Chebyshev inequality gives concrete meaning to “being k standard deviations from the mean” without specifying the distribution. The following table shows the Chebyshev bound alongside the exact normal probability (recall the 68–95–99.7 rule from Section 5.3):

k	Chebyshev: $\mathbb{P}(X - \mu < k\sigma) \geq$	Normal: $\mathbb{P}(X - \mu < k\sigma) \approx$
1	0%	68.3%
2	75%	95.4%
3	88.9%	99.7%
4	93.75%	99.994%

For $k = 1$, the Chebyshev bound is vacuous ($1 - 1/1 = 0$), telling us nothing. But from $k = 2$ onward it becomes informative: for *any* distribution, at least 75% of the probability lies within 2 standard deviations of the mean, at least 88.9% within 3, and at least 93.75% within 4.

The normal distribution does much better than these bounds, which is to be expected: its tails decay extremely fast (as $e^{-x^2/2}$). The Chebyshev inequality must account for the worst case — distributions with much heavier tails — and so its bounds are necessarily conservative. Nevertheless, the guarantee that *no distribution* can place more than $1/k^2$ of its mass beyond k standard deviations from the mean is a powerful and universal statement.

Example (Exam scores). Suppose exam scores have mean $\mu = 72$ and standard deviation $\sigma = 8$, but the shape of the distribution is unknown. What fraction of students scored between 56 and 88?

The interval $[56, 88]$ is $\mu \pm 2\sigma = 72 \pm 16$. By Chebyshev with $k = 2$:

$$\mathbb{P}(56 \leq X \leq 88) = \mathbb{P}(|X - 72| \leq 16) \geq 1 - \frac{1}{4} = 75\%. \quad (6.53)$$

At least 75% of students scored in this range, regardless of the distribution.

Further Inequalities

The Markov and Chebyshev inequalities are the simplest members of a large family of probability inequalities. More advanced courses cover many others, including:

- **Chernoff bounds**, which use the moment generating function to obtain exponentially tight tail bounds;
- **Jensen's inequality**, which relates $\mathbb{E}[g(X)]$ to $g(\mathbb{E}[X])$ for convex functions g ;
- **Hoeffding's inequality**, which gives sharp concentration bounds for sums of bounded independent random variables;
- **Cauchy–Schwarz** and **Hölder inequalities**, fundamental tools in analysis and probability alike.

These inequalities are central to fields such as information theory, machine learning, and mathematical finance. The Markov and Chebyshev inequalities are the starting point for all of them.

6.4 Heavy Tails, Infinite Moments, and Undefined Means

All the continuous distributions we met in Chapter 5 — the normal, uniform, exponential, gamma, and beta — possess finite means and finite variances. In this section we encounter two important families for which this is not always the case: the **Pareto** and **Cauchy** distributions, both examples of so-called *heavy-tailed* distributions.

The Pareto Distribution

The Pareto distribution is named after the Italian economist Vilfredo Pareto, who used it to model the distribution of wealth: a small number of individuals hold a disproportionately large share of the total wealth. It has since found applications in many other areas — city sizes, file sizes on the internet, earthquake magnitudes, and insurance claims — wherever extreme values are much more likely than a normal distribution would predict.

A random variable X has a **Pareto distribution** with shape parameter $\alpha > 0$ if its density function is

$$f(x) = \frac{\alpha}{x^{\alpha+1}}, \quad x \geq 1. \quad (6.54)$$

We write $X \sim \text{Pareto}(\alpha)$. The support is $[1, \infty)$: the smallest possible value is 1. Graphs of the pdf for various values of α are given in Figure 6.4.

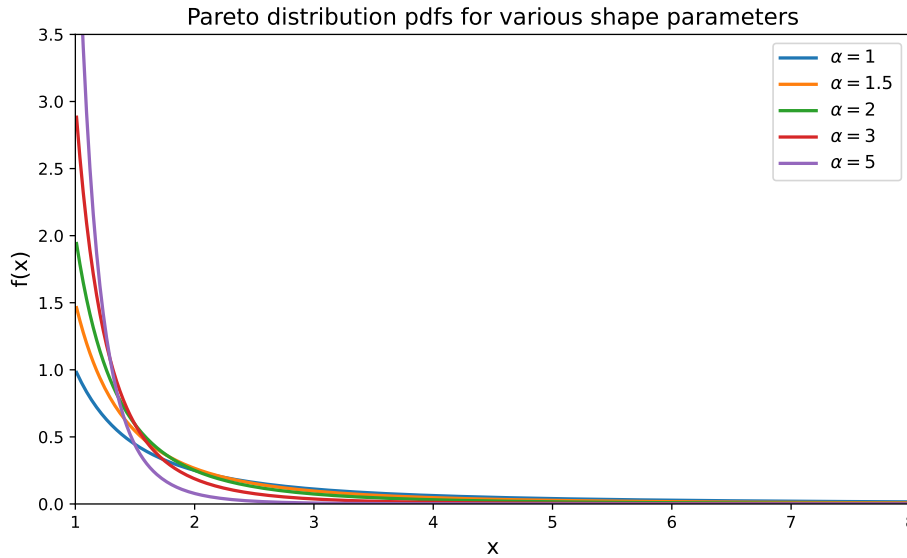


Figure 6.4: Pdfs of the $\text{Pareto}(\alpha)$ distribution for various shape parameters α .

One can verify that f integrates to 1:

$$\int_1^{\infty} \frac{\alpha}{x^{\alpha+1}} dx = \alpha \left[\frac{x^{-\alpha}}{-\alpha} \right]_1^{\infty} = \alpha \cdot \frac{1}{\alpha} = 1. \quad (6.55)$$

CDF and tail probability. Integrating the pdf:

$$F(x) = \int_1^x \frac{\alpha}{t^{\alpha+1}} dt = \left[-\frac{1}{t^{\alpha}} \right]_1^x = 1 - \frac{1}{x^{\alpha}}, \quad x \geq 1. \quad (6.56)$$

so the survival function (recall Section 5.4) is $\mathbb{P}(X > x) = 1/x^{\alpha}$. This is a *power-law* decay, much slower than the exponential decay $e^{-\lambda x}$ of the exponential distribution. It is this slow, polynomial decay of the tail that makes the Pareto distribution “heavy-tailed.”

Remark. There is also a more general version of the Pareto distribution with a scale parameter $x_0 > 0$ (the minimum value), where $f(x) = \alpha x_0^\alpha / x^{\alpha+1}$ for $x \geq x_0$. The version above corresponds to $x_0 = 1$. For simplicity, we focus on this case; results for general x_0 follow by the substitution $X \rightarrow X/x_0$.

Heavy Tails and Finite Moments

The most striking feature of the Pareto distribution is that its moments are not always finite. Whether $\mathbb{E}[X^k]$ exists depends on the relationship between k and α .

Which moments are finite? For $X \sim \text{Pareto}(\alpha)$ and any positive integer k ,

$$\mathbb{E}[X^k] = \int_1^\infty x^k \cdot \frac{\alpha}{x^{\alpha+1}} dx = \alpha \int_1^\infty x^{k-\alpha-1} dx. \quad (6.57)$$

The integrand is $x^{k-\alpha-1}$, which is integrable on $[1, \infty)$ if and only if $k - \alpha - 1 < -1$, i.e., $k < \alpha$. Therefore:

$$\boxed{\mathbb{E}[X^k] < \infty \iff k < \alpha.} \quad (6.58)$$

When $k < \alpha$, the integral evaluates to $\mathbb{E}[X^k] = \alpha/(\alpha - k)$.

This has dramatic consequences:

- If $\alpha > 2$: both $\mathbb{E}[X]$ and $\text{Var}(X)$ are finite.
- If $1 < \alpha \leq 2$: $\mathbb{E}[X]$ is finite but $\text{Var}(X) = \infty$ (since $\mathbb{E}[X^2] = \infty$).
- If $0 < \alpha \leq 1$: even $\mathbb{E}[X] = \infty$ — the distribution has *infinite mean*.

When $\alpha > 1$, the mean and variance (when they exist) are:

$$\mathbb{E}[X] = \frac{\alpha}{\alpha - 1} \quad (\alpha > 1), \quad \text{Var}(X) = \frac{\alpha}{(\alpha - 1)^2(\alpha - 2)} \quad (\alpha > 2). \quad (6.59)$$

Example (Pareto with $\alpha = 3/2$: finite mean, infinite variance). Let $X \sim \text{Pareto}(3/2)$. Since $\alpha = 3/2 > 1$, the mean is finite:

$$\mathbb{E}[X] = \frac{3/2}{3/2 - 1} = \frac{3/2}{1/2} = 3. \quad (6.60)$$

However, since $\alpha = 3/2 \leq 2$, the second moment $\mathbb{E}[X^2] = \int_1^\infty (3/2) x^{-1/2} dx = \infty$ diverges, so $\text{Var}(X) = \infty$. The distribution has a well-defined centre ($\mathbb{E}[X] = 3$) but its spread is “infinite” — extreme values are so likely that the variance does not exist.

Example (Pareto with $\alpha = 1/2$: infinite mean). Let $X \sim \text{Pareto}(1/2)$. The mean is

$$\mathbb{E}[X] = \int_1^\infty x \cdot \frac{1/2}{x^{3/2}} dx = \frac{1}{2} \int_1^\infty x^{-1/2} dx = \frac{1}{2} [2\sqrt{x}]_1^\infty = \infty. \quad (6.61)$$

The tail decays so slowly that the distribution places enough probability on extremely large values to make the mean infinite.

A discrete analogue. This is not the first time in the course that we encounter a distribution without a finite mean. Recall from Chapter 3 the distribution on $\{1, 2, 3, \dots\}$ with $\mathbb{P}(X = i) = c/i^2$ where $c = 6/\pi^2$ (Example 3.1). Its mean is

$$\mathbb{E}[X] = \sum_{i=1}^{\infty} i \cdot \frac{c}{i^2} = c \sum_{i=1}^{\infty} \frac{1}{i} = \infty, \quad (6.62)$$

since the harmonic series diverges. So even though the probabilities c/i^2 sum to 1, the values i grow fast enough that the weighted average diverges. The Pareto distribution with $\alpha \leq 1$ is the continuous counterpart of this phenomenon.

Remark on heavy tails. The study of heavy-tailed distributions is a rich topic in its own right, with connections to extreme value theory, risk management, and network science. Besides the Pareto, other notable heavy-tailed distributions include the log-normal, the stable distributions, and the Cauchy distribution which we discuss next. What these distributions share is that rare, extreme events are far more probable than they would be under a normal (or exponential) distribution.

Higher Moments Imply Lower Moments

The Pareto distribution illustrates a striking “staircase” pattern: if $\alpha = 3$, the first and second moments are finite but the third is not; if $\alpha = 3/2$, the first moment is finite but the second is not. This is not a peculiarity of the Pareto — it is a general fact about all distributions.

Claim 33: Moment hierarchy — Let X be a continuous random variable. If $\mathbb{E}[|X|^s] < \infty$ for some $s > 0$, then

$$\mathbb{E}[|X|^r] < \infty \quad \text{for all } 0 < r < s. \quad (6.63)$$

In other words, the set of orders for which the absolute moment is finite is always a “downward closed” interval: if a higher moment exists, all lower moments exist too.

Proof: We split the integral at $|x| = 1$:

$$\mathbb{E}[|X|^r] = \int_{-\infty}^{\infty} |x|^r f(x) dx = \underbrace{\int_{|x| \leq 1} |x|^r f(x) dx}_{I_1} + \underbrace{\int_{|x| > 1} |x|^r f(x) dx}_{I_2}. \quad (6.64)$$

For I_1 : when $|x| \leq 1$, we have $|x|^r \leq 1$, so

$$I_1 \leq \int_{|x| \leq 1} f(x) dx \leq 1. \quad (6.65)$$

For I_2 : when $|x| > 1$ and $r < s$, we have $|x|^r < |x|^s$ (since raising a number greater than 1 to a higher power makes it larger). Therefore,

$$I_2 \leq \int_{|x| > 1} |x|^s f(x) dx \leq \mathbb{E}[|X|^s] < \infty. \quad (6.66)$$

Combining: $\mathbb{E}[|X|^r] = I_1 + I_2 \leq 1 + \mathbb{E}[|X|^s] < \infty$. $\square$

Central moments. The same hierarchy holds for central moments. If $\mathbb{E}[|X - c|^s] < \infty$ for some constant c and some $s > 0$, then $\mathbb{E}[|X - c|^r] < \infty$ for all $0 < r < s$. This follows immediately by applying the claim to the random variable $Y = X - c$.

In particular, if $\text{Var}(X) = \mathbb{E}[(X - \mu)^2] < \infty$ (where $\mu = \mathbb{E}[X]$), then $\mathbb{E}[|X - \mu|^r] < \infty$ for all $0 < r < 2$. And if the fourth central moment $\mathbb{E}[(X - \mu)^4]$ is finite (as it is for the normal distribution), then all central moments of order less than 4 are finite.

Remark. The converse is false: finiteness of $\mathbb{E}[|X|^r]$ does not imply finiteness of $\mathbb{E}[|X|^s]$ for $s > r$. The Pareto distribution provides a clean example: $X \sim \text{Pareto}(3/2)$ has $\mathbb{E}[X] < \infty$ but $\mathbb{E}[X^2] = \infty$.

The Cauchy Distribution

The Cauchy distribution is the most extreme example of a heavy-tailed distribution we will encounter: not only is its variance infinite, but its mean does not even exist in any meaningful sense.

A random variable X has a **standard Cauchy distribution** if its density function is

$$f(x) = \frac{1}{\pi(1 + x^2)}, \quad x \in \mathbb{R}. \quad (6.67)$$

We write $X \sim \text{Cauchy}(0, 1)$. More generally, $X \sim \text{Cauchy}(\mu, \gamma)$ has pdf $f(x) = 1/[\pi\gamma(1 + ((x - \mu)/\gamma)^2)]$ with location parameter μ and scale parameter $\gamma > 0$.

The standard Cauchy density looks superficially similar to the standard normal density — both are symmetric, bell-shaped curves centred at the origin. However, the Cauchy has much heavier tails, as shown in Figure 6.5. At $x = 5$, the Cauchy density is about $1/(\pi \cdot 26) \approx 0.012$, while the normal density is approximately 1.5×10^{-6} — four orders of magnitude smaller.

Comparing the Cauchy and normal pdfs

LLM Prompt > Write Python code that plots the standard Cauchy and standard normal PDFs on the same axes to illustrate the heavier tails of the Cauchy distribution.

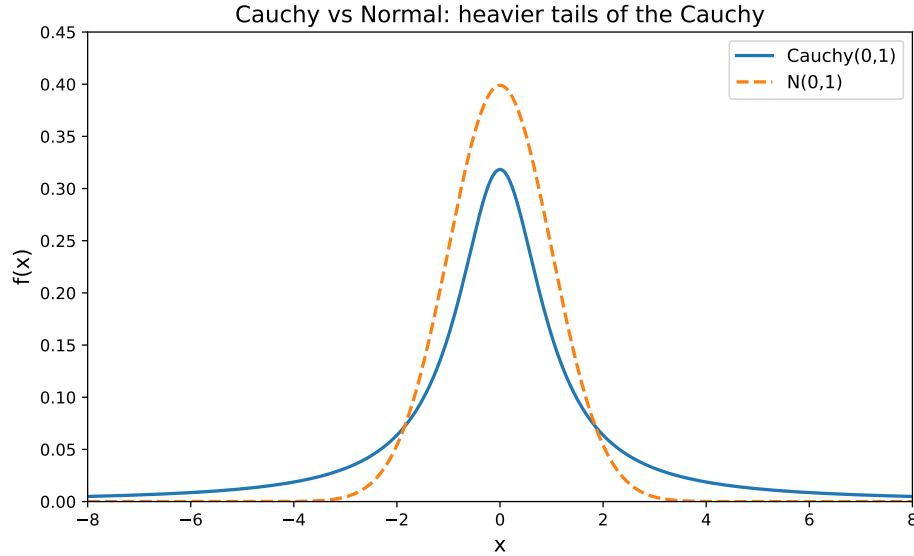


Figure 6.5: The standard Cauchy and standard normal pdfs. The Cauchy has much heavier tails.

One can verify that the Cauchy pdf integrates to 1 using the substitution $x = \tan \theta$:

$$\int_{-\infty}^{\infty} \frac{1}{\pi(1+x^2)} dx = \frac{1}{\pi} [\arctan(x)]_{-\infty}^{\infty} = \frac{1}{\pi} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) = 1. \quad (6.68)$$

CDF. The cdf of the standard Cauchy is

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan(x). \quad (6.69)$$

The Cauchy Has No Mean

It is natural to ask: what is $\mathbb{E}[X]$ when $X \sim \text{Cauchy}(0, 1)$? Since the density is symmetric about 0, one might guess that $\mathbb{E}[X] = 0$. However, the expectation is not defined at all. Let us see why.

For the expectation to be well-defined, we need $\mathbb{E}[|X|] < \infty$ (recall from Section 5.2 that $\mathbb{E}[X]$ is only defined when $\int |x|f(x) dx < \infty$). But

$$\mathbb{E}[|X|] = \int_{-\infty}^{\infty} \frac{|x|}{\pi(1+x^2)} dx = \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx = \frac{1}{\pi} [\ln(1+x^2)]_0^{\infty} = \infty. \quad (6.70)$$

Since $\mathbb{E}[|X|] = \infty$, the expectation $\mathbb{E}[X]$ does not exist.

Infinite mean vs. undefined mean. It is important to distinguish between two types of pathology:

- **Infinite mean:** $\mathbb{E}[X] = +\infty$ (or $-\infty$). This occurs for the Pareto distribution with $\alpha \leq 1$, where $\int_0^{\infty} x f(x) dx = +\infty$. The integral “exists” in the extended sense — it simply diverges to $+\infty$. We can meaningfully say that the average is “infinitely large.”

- **Undefined mean:** $\mathbb{E}[|X|] = \infty$ but the positive and negative parts both contribute infinitely. This is the Cauchy case. The integral $\int_{-\infty}^{\infty} x f(x) dx$ is not well-defined, not even as $+\infty$ or $-\infty$ — it depends on the order in which we sum the positive and negative contributions. Indeed, one can show that $\lim_{a \rightarrow \infty} \int_{-a}^a x f(x) dx = 0$ (by symmetry), but $\lim_{a \rightarrow \infty} \int_{-2a}^a x f(x) dx \neq 0$. The “mean” changes depending on how we take the limit, so it is genuinely undefined.

Failure of the Law of Large Numbers

For “well-behaved” distributions (those with a finite mean), the *law of large numbers* guarantees that the sample mean $\bar{X}_n = (X_1 + \dots + X_n)/n$ converges to $\mathbb{E}[X]$ as $n \rightarrow \infty$. We will study this fundamental result in detail later in the course. The Cauchy distribution provides a striking example of what can go wrong when the mean does not exist.

Figure 6.6 shows the running sample mean $\bar{X}_n$ for $n = 1, 2, \dots, 10,000$ independent standard Cauchy random variables. Unlike the normal or exponential case — where the running mean quickly stabilises near the true mean — the Cauchy sample mean continues to jump erratically, even after thousands of observations. Occasional extreme values (which the heavy tails make far more likely than one might expect) cause sudden jumps in the average, and these jumps never die out no matter how many samples are collected.

Running sample mean of Cauchy random variables

LLM Prompt > Write Python code that generates 10,000 standard Cauchy random variables, plots the running sample mean, and observe that it does not converge (unlike distributions with a finite mean).

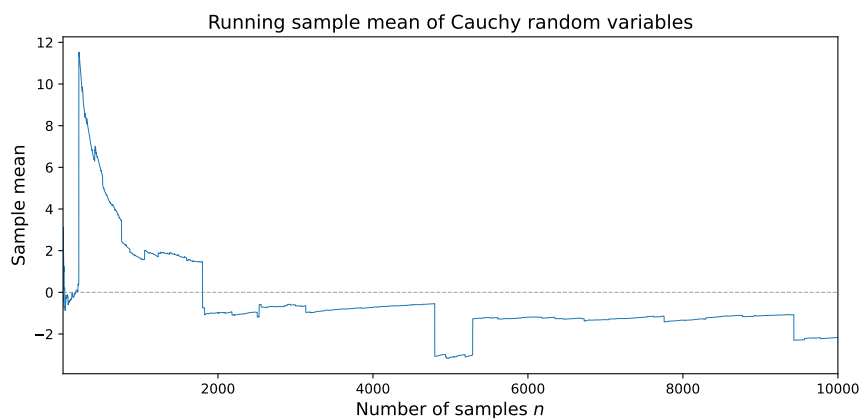


Figure 6.6: Running sample mean of n independent standard Cauchy random variables. The sample mean does not converge — it continues to fluctuate wildly, in contrast to distributions with a finite mean.

In fact, one can prove that for independent $X_1, X_2, \dots \sim \text{Cauchy}(0, 1)$, the sample mean $\bar{X}_n = (X_1 + \dots + X_n)/n$ is itself $\text{Cauchy}(0, 1)$ for every n . The average of n Cauchy random variables

has the same distribution as a single observation! Collecting more data does nothing to reduce the uncertainty — the law of large numbers simply does not apply.

JOINT DISTRIBUTIONS

In previous chapters we studied individual random variables — their distributions, moments, and transformations. In this chapter we consider situations where two or more random variables are observed simultaneously. We develop the theory of joint distributions, marginal and conditional distributions, covariance and correlation, and the multivariate normal and multinomial distributions. We then introduce conditional expectation and the tower property (law of total expectation), and apply these tools to random sums and the law of total variance.

7.1 Joint Distributions of Multiple Random Variables

Often a random experiment is described via more than one random variable. Examples are:

1. We select a random sample of $n = 10$ people and observe their lengths. Let $X_1, \dots, X_n$ be the individual lengths.
2. We flip a coin repeatedly. Let $X_i = 1$ if the i th flip is “heads” and 0 else. The experiment is described by the sequence $X_1, X_2, \dots$ of coin flips.
3. We randomly select a person from a large population and measure his/her weight X and height Y .

How can we specify the behaviour of the random variables above? We should not just specify the pdf or pmf of the individual random variables, but also say something about the “interaction” (or lack thereof) between the random variables. For example, in the third experiment above if the height Y is large, we expect that X is large as well. On the other hand, for the first and second experiment it is reasonable to assume that information about one of the random variables does not give extra information about the others. What we need to specify is the **joint distribution** of the random variables.

The theory for multiple random variables is quite similar to that of a single random variable. The most important extra feature is perhaps the concept of *independence* of random variables. Independent random variables play a crucial role in stochastic modelling.

Random Vectors and the Joint CDF

Let $X_1, \dots, X_n$ be random variables describing some random experiment. We can accumulate the X_i 's into a vector $\mathbf{X} = (X_1, \dots, X_n)$, which we always interpret as a column vector. $\mathbf{X}$ is called a **random vector**.

Recall from Chapters 3 and 5 that the distribution of a *single* random variable X is completely specified by its cumulative distribution function. Analogously, the joint distribution of $X_1, \dots, X_n$ is specified by the **joint cumulative distribution function** F , defined by

$$F(x_1, \dots, x_n) = \mathbb{P}(\{X_1 \leq x_1\} \cap \dots \cap \{X_n \leq x_n\}) = \mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n). \quad (7.1)$$

If we know F then we can in principle derive any probability involving the X_i 's. Note the abbreviation on the right-hand side. We will henceforth use this kind of abbreviation throughout the notes.

Similar to the 1-dimensional case we distinguish between the case where the X_i are discrete and continuous. The corresponding joint distributions are again called discrete and continuous, respectively.

7.2 Joint Density and Probability Mass Functions

The joint distribution of a random vector is described by its joint pmf (in the discrete case) or joint pdf (in the continuous case). We treat the two cases in turn, beginning with the discrete setting.

Discrete Joint Distributions

To see how things work in the discrete case, let's start with an example.

Example (Box With Different Dice). In a box are three dice. Die 1 is a normal die; die 2 has no 6 face, but instead two 5 faces; die 3 has no 5 face, but instead two 6 faces. The experiment consists of selecting a die at random, followed by a toss with that die. Let X be the die number that is selected, and let Y be the face value of that die. The probabilities $\mathbb{P}(X = x, Y = y)$ are specified below.

x	y						Σ
	1	2	3	4	5	6	
1	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{3}$
2	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{9}$	0	$\frac{1}{3}$
3	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	$\frac{1}{18}$	0	$\frac{1}{9}$	$\frac{1}{3}$
Σ	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	1

The function $f : (x, y) \mapsto \mathbb{P}(X = x, Y = y)$ is called the *joint pmf* of X and Y . The following definition is just a generalization of this.

Definition : Joint Probability Mass Function — Let $X_1, \dots, X_n$ be *discrete* random variables. The function f defined by $f(x_1, \dots, x_n) = \mathbb{P}(X_1 = x_1, \dots, X_n = x_n)$ is called the **joint probability mass function** (pmf) of $X_1, \dots, X_n$.

We sometimes write $f_{X_1, \dots, X_n}$ instead of f to show that this is the pmf of the random variables $X_1, \dots, X_n$. Or, if $\mathbf{X}$ is the corresponding random vector, we could write $f_{\mathbf{X}}$ instead.

Note that, by the sum rule, if we are given the joint pmf of $X_1, \dots, X_n$ we can in principle calculate *all possible probabilities* involving these random variables. For example, in the 2-dimensional case

$$\mathbb{P}((X, Y) \in B) = \sum_{(x,y) \in B} \mathbb{P}(X = x, Y = y), \quad (7.2)$$

for any subset B of possible values for (X, Y) . In particular, we can find the pmf of X by summing the joint pmf over all possible values of y :

$$\mathbb{P}(X = x) = \sum_y \mathbb{P}(X = x, Y = y). \quad (7.3)$$

The converse is *not* true: from the individual distributions (so-called **marginal** distributions) of X and Y we cannot in general reconstruct the joint distribution of X and Y . We are missing the “dependency” information. E.g., in Example 7.2 we cannot reconstruct the inside of the two-dimensional table if only given the column and row totals.

However, there is one *important exception* to this, namely when we are dealing with *independent* random variables. We have so far only defined what independence is for *events*. The following definition says that random variables $X_1, \dots, X_n$ are independent if the events $\{X_1 \in A_1\}, \dots, \{X_n \in A_n\}$ are independent for any subsets $A_1, \dots, A_n$ of $\mathbb{R}$. Intuitively, this means that any information about one of them does not affect our knowledge about the others.

Definition : Independent Random Variables — The random variables $X_1, \dots, X_n$ are called **independent** if for all $A_1, \dots, A_n$, with $A_i \subseteq \mathbb{R}$, $i = 1, \dots, n$

$$\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \mathbb{P}(X_1 \in A_1) \cdots \mathbb{P}(X_n \in A_n). \quad (7.4)$$

The following claim is a direct consequence of the definition above.

Claim 34: Independent Discrete Random Variables — Discrete random variables $X_1, \dots, X_n$, are independent if and only if

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_1 = x_1) \cdots \mathbb{P}(X_n = x_n), \quad (7.5)$$

for all $x_1, x_2, \dots, x_n$.

Proof: The necessary condition is obvious: if $X_1, \dots, X_n$ are independent random variables, then $\{X_1 = x_1\}, \dots, \{X_n = x_n\}$ are (mutually) independent events. To prove the sufficient condition, write

$$\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \sum_{x_1 \in A_1} \cdots \sum_{x_n \in A_n} \mathbb{P}(X_1 = x_1, \dots, X_n = x_n). \quad (7.6)$$

Then, if (7.5) holds, the multiple sum can be written as

$$\sum_{x_1 \in A_1} \mathbb{P}(X_1 = x_1) \cdots \sum_{x_n \in A_n} \mathbb{P}(X_n = x_n) = \mathbb{P}(X_1 \in A_1) \cdots \mathbb{P}(X_n \in A_n), \quad (7.7)$$

which implies that $X_1, \dots, X_n$ are independent random variables. $\square$

Example (Box with Same Dice). We repeat the experiment in Example 7.2 with three ordinary fair dice. What are now the joint probabilities in the table? Since the events $\{X = x\}$ and $\{Y = y\}$ are now independent, each entry in the pmf table is $\frac{1}{3} \times \frac{1}{6}$. Clearly in the first experiment not *all* events $\{X = x\}$ and $\{Y = y\}$ are independent (why not?).

Example (Coin Flip Experiment). Consider the experiment where we flip a coin n times. We can model this experiments in the following way. For $i = 1, \dots, n$ let X_i be the result of the i th toss: $\{X_i = 1\}$ means Heads, $\{X_i = 0\}$ means Tails. Also, let

$$\mathbb{P}(X_i = 1) = p = 1 - \mathbb{P}(X_i = 0), \quad i = 1, 2, \dots, n. \quad (7.8)$$

Thus, p can be interpreted as the probability of Heads, which may be known or unknown. Finally, assume that $X_1, \dots, X_n$ are *independent*.

This completely specifies our model. In particular we can find any probability related to the X_i 's. For example, let $X = X_1 + \cdots + X_n$ be the total number of Heads in n tosses. Obviously X is a random variable that takes values between 0 and n . Denote by A the set of all binary vectors $\mathbf{x} = (x_1, \dots, x_n)$ such that $\sum_{i=1}^n x_i = k$. Note that A has $\binom{n}{k}$ elements. We now have

$$\begin{aligned} \mathbb{P}(X = k) &= \sum_{\mathbf{x} \in A} \mathbb{P}(X_1 = x_1, \dots, X_n = x_n) \\ &= \sum_{\mathbf{x} \in A} \mathbb{P}(X_1 = x_1) \cdots \mathbb{P}(X_n = x_n) = \sum_{\mathbf{x} \in A} p^k (1-p)^{n-k} \\ &= \binom{n}{k} p^k (1-p)^{n-k}. \end{aligned}$$

In other words, $X \sim \text{Bin}(n, p)$.

Remark. If $f_{X_1, \dots, X_n}$ denotes the joint pmf of $X_1, \dots, X_n$ and f_{X_i} the marginal pmf of X_i , $i = 1, \dots, n$, then the theorem above states that independence of the X_i 's is equivalent to

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n) \quad (7.9)$$

for all possible $x_1, \dots, x_n$.

Remark. An *infinite* sequence $X_1, X_2, \dots$ of random variables is called independent if for any finite choice of parameters $i_1, i_2, \dots, i_n$ (none of them the same) the random variables $X_{i_1}, \dots, X_{i_n}$ are independent.

Continuous Joint Distributions

Joint distributions for continuous random variables are usually defined via the joint pdf. The results are very similar to the discrete case discussed above. Compare this with the one-dimensional case from Chapter 5.

Definition : Joint Probability Density Function — We say that the continuous random variables $X_1, \dots, X_n$ have a **joint probability density function** (pdf) f if

$$\mathbb{P}(a_1 \leq X_1 \leq b_1, \dots, a_n \leq X_n \leq b_n) = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_n \cdots dx_1 \quad (7.10)$$

for all $a_1, \dots, b_n$.

We sometimes write $f_{X_1, \dots, X_n}$ instead of f to show that this is the pdf of the random variables $X_1, \dots, X_n$. Or, if $\mathbf{X}$ is the corresponding random vector, we could write $f_{\mathbf{X}}$ instead.

We can interpret $f(x_1, \dots, x_n)$ as a continuous analogue of a pmf, or as the “density” that $X_1 = x_1$, $X_2 = x_2$, ..., and $X_n = x_n$. For example in the 2-dimensional case:

$$\begin{aligned} \mathbb{P}(x \leq X \leq x+h, y \leq Y \leq y+h) \\ = \int_x^{x+h} \int_y^{y+h} f(u, v) dv du \approx h^2 f(x, y) . \end{aligned}$$

Note that if the joint pdf is given, then in principle we can calculate *all probabilities*. Specifically, in the 2-dimensional case we have

$$\mathbb{P}((X, Y) \in B) = \iint_{(x, y) \in B} f(x, y) dx dy , \quad (7.11)$$

for any subset B of possible values for $\mathbb{R}^2$. Thus, the calculation of probabilities is reduced to *integration*.

Similarly to the discrete case, if $X_1, \dots, X_n$ have joint pdf f , then the (individual, or marginal) pdf of each X_i can be found by integrating f over all other variables. For example, in the two-

dimensional case

$$f_X(x) = \int_{y=-\infty}^{\infty} f(x, y) dy. \quad (7.12)$$

However, we usually cannot reconstruct the joint pdf from the marginal pdf's unless we assume that the random variables are *independent*. The definition of independence is exactly the same as for discrete random variables, see Definition 7.2. More importantly, we have the following analogue of Claim 34.

Claim 35: Independent Continuous Random Variables — Let $X_1, \dots, X_n$ be continuous random variables with joint pdf f and marginal pdf's $f_{X_1}, \dots, f_{X_n}$. The random variables $X_1, \dots, X_n$ are independent if and only if, for all $x_1, \dots, x_n$,

$$f(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n). \quad (7.13)$$

Example (Uniform Numbers). Consider the experiment where we select randomly and independently n points from the interval $[0,1]$. We can carry this experiment out using a calculator or computer, using the *random generator*. On your calculator this means pushing the RAN# or Rand button. Here is a possible outcome, or **realisation**, of the experiment, for $n = 12$.

0.9451226800 0.2920864820 0.0019900900 0.8842189383 0.8096459523
 0.3503489150 0.9660027079 0.1024852543 0.7511286891 0.9528386400
 0.2923353821 0.0837952423

A model for this experiment is: Let $X_1, \dots, X_n$ be independent random variables, each with a uniform distribution on $[0,1]$. The joint pdf of $X_1, \dots, X_n$ is very simple, namely

$$f(x_1, \dots, x_n) = 1, \quad 0 \leq x_1 \leq 1, \dots, \quad 0 \leq x_n \leq 1, \quad (7.14)$$

(and 0 else). In principle we can now calculate any probability involving the X_i 's. For example for the case $n = 2$ what is the probability

$$\mathbb{P} \left(\frac{X_1 + X_2^2}{X_1 X_2} > \sin(X_1^2 - X_2) \right) ? \quad (7.15)$$

The answer, by (7.11), is

$$\iint_A 1 \, dx_1 \, dx_2 = \text{Area}(A), \quad (7.16)$$

where

$$A = \left\{ (x_1, x_2) \in [0, 1]^2 : \frac{x_1 + x_2^2}{x_1 x_2} > \sin(x_1^2 - x_2) \right\}. \quad (7.17)$$

(Here $[0, 1]^2$ is the unit square in $\mathbb{R}^2$).

Random samples. The type of model used in the previous example, i.e., $X_1, \dots, X_n$ are independent and all have the same distribution, is the most widely used model in statistics. We say that $X_1, \dots, X_n$ is a **random sample** of **size n** , from some given distribution. In Example 7.2 $X_1, \dots, X_n$ is a random sample from a $\mathcal{U}[0, 1]$ -distribution. In Example 7.2 we also had a random sample, this time from a $\text{Ber}(p)$ -distribution. The common distribution of a random sample is sometimes called the **sampling distribution**.

Using the computer we can generate the outcomes of random samples from many (sampling) distributions. In Figure 7.1 the outcomes of two random samples, both of size 1000, are depicted in a **histogram**. Here the x-axis is divided into 20 intervals, and the number of points in each interval is counted. The first sample is from the $\mathcal{U}[0, 1]$ -distribution, and the second sample is from the $\mathcal{N}(1/2, 1/12)$ -distribution.

Histograms of random samples

LLM Prompt > Write Python code that generates 1000 samples from a $U[0, 1]$ distribution and 1000 samples from a $N(1/2, 1/12)$ distribution, and plots a histogram for each.

Note that the true expectation and variance of the distributions are the same. However, the “density” of points in the two samples is clearly different, and follows the shape of the corresponding pdf’s.

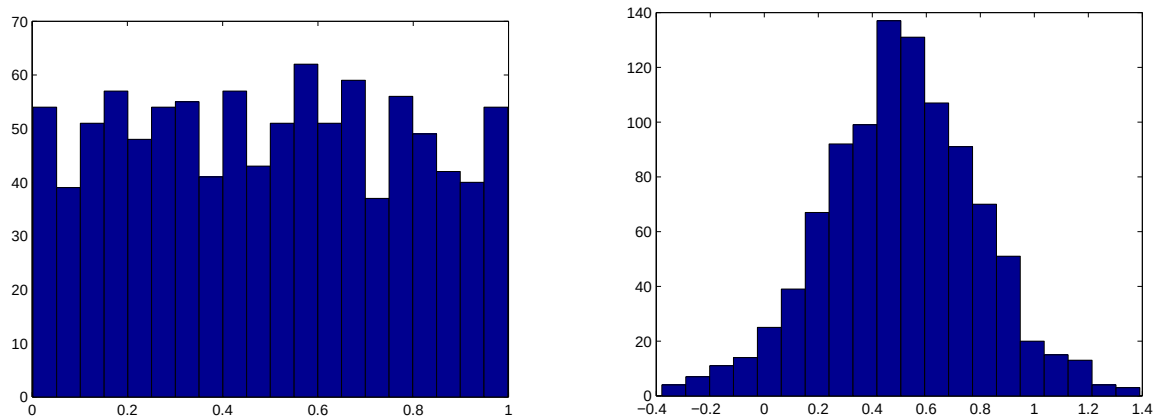


Figure 7.1: A histogram of a random sample of size 1000 from the $\mathcal{U}[0, 1]$ -distribution (left) and the $\mathcal{N}(1/2, 1/12)$ -distribution (right).

7.3 Marginal and Conditional Distributions

Given the joint distribution of (X, Y) , we often want to know how Y behaves when we have information about X . This leads to the notion of a conditional distribution, which extends the idea of conditional probability from Chapter 4 to the setting of random variables.

Conditional Distributions

Suppose X and Y are both discrete or both continuous, with joint pmf/pdf $f_{X,Y}$, and suppose $f_X(x) > 0$. The *conditional pdf/pmf* of Y given $X = x$ is defined as

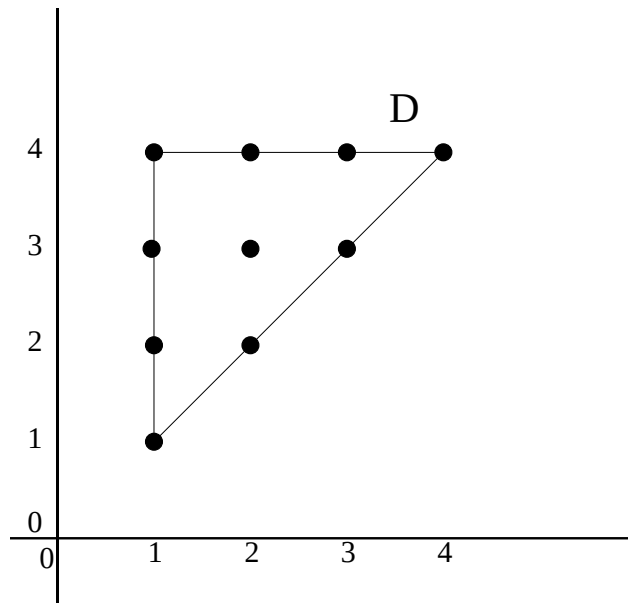
$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}, \text{ for all } y. \quad (7.18)$$

We can interpret $f_{Y|X}(\cdot|x)$ as the pmf/pdf of Y given the information that X takes the value x . For discrete random variables the definition is simply a consequence of the conditioning formula for events (recall Chapter 4). Namely,

$$f_{Y|X}(y|x) = \mathbb{P}(Y = y | X = x) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(X = x)} = \frac{f_{X,Y}(x,y)}{f_X(x)}. \quad (7.19)$$

Example (Uniformly Drawing from a Discrete Triangle). We draw “uniformly” a point (X, Y) from the 10 points on the triangle D below. Thus, each point is equally likely to be drawn. That is, the joint pmf is

$$f_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y) = \frac{1}{10}, (x,y) \in D, \quad (7.20)$$



The marginal pmf's of X and Y are

$$f_X(x) = \mathbb{P}(X = x) = \frac{5-x}{10}, \quad x \in \{1, 2, 3, 4\}, \quad (7.21)$$

and

$$f_Y(y) = \mathbb{P}(Y = y) = \frac{y}{10}, \quad y \in \{1, 2, 3, 4\}. \quad (7.22)$$

Clearly X and Y are not independent. In fact, if we know that $X = 2$, then Y can only take the values $j = 2, 3$ or 4 . The corresponding probabilities are

$$\mathbb{P}(Y = y | X = 2) = \frac{\mathbb{P}(Y = y, X = 2)}{\mathbb{P}(X = 2)} = \frac{1/10}{3/10} = \frac{1}{3}. \quad (7.23)$$

In other words, the conditional pmf of Y given $X = 2$ is

$$f_{Y|X}(y|2) = \frac{f_{X,Y}(2, y)}{f_X(2)} = \frac{1}{3}, \quad y = 2, 3, 4. \quad (7.24)$$

Thus, given $X = 2$, Y takes the values 2, 3 and 4 with equal probability.

Conditional Distributions in the Continuous Case

When X is *continuous*, we cannot directly apply (4.1) to define the conditional density. Instead, we define first the **conditional cdf** of Y given $X = x$ as the limit

$$F_{Y|X}(y|x) := \lim_{h \rightarrow 0} F_Y(y | x < X \leq x + h). \quad (7.25)$$

Now, (4.1) can be applied to $F_Y(y | x < X \leq x + h)$ to yield

$$F_Y(y | x < X \leq x + h) = \frac{\int_{-\infty}^y \int_x^{x+h} f_{X,Y}(u, v) \, du \, dv}{\int_x^{x+h} f_X(u) \, du}. \quad (7.26)$$

Now, for small h the integral $\int_x^{x+h} f_{X,Y}(u, v) \, du$ is approximately equal to $h f_{X,Y}(x, v)$ plus some small term which goes to zero faster than h . Similarly, $\int_x^{x+h} f_X(u) \, du \approx h f_X(x)$ (plus smaller order terms). Hence, for $h \rightarrow 0$, the limit of $F_Y(y | x < X \leq x + h)$ is

$$F_{Y|X}(y|x) = \frac{\int_{-\infty}^y f_{X,Y}(x, v) \, dv}{f_X(x)}. \quad (7.27)$$

Note that $F_{Y|X}(y|x)$ as a function of y has all the properties of a cdf. By differentiating this cdf with respect to y we obtain the conditional pdf of Y given $X = x$ in the continuous case, which gives the same formula (7.18) as for the discrete case.

Product Rule and the Law of Total Probability

Rewriting (7.18), we find the “product rule” for densities:

$$f_{X,Y}(x, y) = f_X(x) f_{Y|X}(y | x), \quad (7.28)$$

and by integrating out x , we have (in the continuous case) the following version of the law of total probability for probability densities:

$$f_Y(y) = \int f_{Y|X}(y | x) f_X(x) dx. \quad (7.29)$$

Example (Conditional Distribution for a Continuous Pair). Let (X, Y) have joint pdf

$$f_{X,Y}(x, y) = 2, \quad 0 \leq x \leq y \leq 1. \quad (7.30)$$

The joint density is constant (uniform) on the triangle $\{(x, y) : 0 \leq x \leq y \leq 1\}$.

Marginal of X . Integrating out y :

$$f_X(x) = \int_x^1 2 dy = 2(1 - x), \quad 0 \leq x \leq 1. \quad (7.31)$$

Conditional pdf of Y given $X = x$. For $0 < x < 1$,

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \frac{2}{2(1 - x)} = \frac{1}{1 - x}, \quad x \leq y \leq 1. \quad (7.32)$$

This is the $\mathcal{U}(x, 1)$ density: given that $X = x$, the variable Y is uniformly distributed on $[x, 1]$. The conditional mean is

$$\mathbb{E}[Y | X = x] = \frac{x + 1}{2}. \quad (7.33)$$

Conditional pmf’s and pdf’s for more than two random variables are defined analogously. For example, the conditional pmf of X_n given $X_1, \dots, X_{n-1}$ is given by

$$f_{X_n | X_1, \dots, X_{n-1}}(x_n | x_1, \dots, x_{n-1}) = \frac{\mathbb{P}(X_1 = x_1, \dots, X_n = x_n)}{\mathbb{P}(X_1 = x_1, \dots, X_{n-1} = x_{n-1})}. \quad (7.34)$$

Mixed Distributions: One Continuous, One Discrete

Not every pair (X, Y) falls neatly into the all-discrete or all-continuous setting. It is perfectly possible for X to be continuous and Y to be discrete. Such a **mixed** pair has no single joint pdf or joint pmf, but its distribution is fully specified by a marginal distribution for one variable together with a conditional distribution for the other.

Structure of a mixed pair. Suppose X is continuous with pdf f_X and, given $X = x$, the variable Y is discrete with conditional pmf

$$f_{Y|X}(y|x) = \mathbb{P}(Y = y | X = x). \quad (7.35)$$

Applying the law of total probability (7.29) with integration over the continuous variable X gives the **marginal pmf** of Y :

$$\mathbb{P}(Y = y) = \int f_{Y|X}(y|x) f_X(x) dx. \quad (7.36)$$

Provided $\mathbb{P}(Y = y) > 0$, the **conditional pdf** of X given $Y = y$ is

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x) f_X(x)}{\mathbb{P}(Y = y)}, \quad (7.37)$$

which is Bayes' formula applied at the level of densities and probabilities.

Example (Beta–Bernoulli Model). Suppose a coin has an unknown probability of Heads. We model this unknown bias as a random variable $X \sim \text{Beta}(\alpha, \beta)$, the beta distribution introduced in Section 5.5 of Chapter 5. Recall that X is supported on $[0, 1]$ with pdf

$$f_X(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq x \leq 1, \quad (7.38)$$

and mean $\mathbb{E}[X] = \alpha/(\alpha + \beta)$. Given $X = x$, we toss the coin once and let $Y \in \{0, 1\}$ record the outcome ($Y = 1$ for Heads). Since the coin's bias is x ,

$$Y | X = x \sim \text{Ber}(x), \quad (7.39)$$

so $\mathbb{P}(Y = 1 | X = x) = x$ and $\mathbb{P}(Y = 0 | X = x) = 1 - x$.

Marginal distribution of Y . By (7.36),

$$\mathbb{P}(Y = 1) = \int_0^1 x f_X(x) dx = \mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}, \quad (7.40)$$

and $\mathbb{P}(Y = 0) = \beta/(\alpha + \beta)$. The marginal probability of Heads equals the prior mean of the bias—a clean consequence of the law of total probability.

Conditional distribution of X given Y . By (7.37), the conditional pdf of X given $Y = 1$ is

$$f_{X|Y}(x|1) = \frac{x \cdot f_X(x)}{\mathbb{P}(Y = 1)} = \frac{x^\alpha(1-x)^{\beta-1}}{B(\alpha, \beta) \cdot \alpha/(\alpha + \beta)}. \quad (7.41)$$

Using the identity $B(\alpha + 1, \beta) = \alpha/(\alpha + \beta) \cdot B(\alpha, \beta)$, this simplifies to

$$f_{X|Y}(x|1) = \frac{x^\alpha(1-x)^{\beta-1}}{B(\alpha + 1, \beta)}, \quad (7.42)$$

which is the $\text{Beta}(\alpha + 1, \beta)$ density. Observing Heads therefore shifts the distribution of X from $\text{Beta}(\alpha, \beta)$ to $\text{Beta}(\alpha + 1, \beta)$, effectively incrementing the Heads count by one. By a symmetric calculation, $f_{X|Y}(\cdot|0)$ is the $\text{Beta}(\alpha, \beta + 1)$ density, shifting the Tails count instead.

7.4 Expectation, Covariance, and Correlation

With the machinery of joint and conditional distributions in place, we now study expectations of functions of several random variables, and introduce two quantities — covariance and correlation — that measure the strength of the linear relationship between them.

Expectation of Functions of Several Random Variables

Similar to the one-dimensional case (recall LOTUS from Claims 8 and 23), the expected value of any real-valued function of $X_1, \dots, X_n$ is the weighted average of all values that this function can take. Specifically, if $Z = g(X_1, \dots, X_n)$ then in the discrete case

$$\mathbb{E}Z = \sum_{x_1} \cdots \sum_{x_n} g(x_1, \dots, x_n) f(x_1, \dots, x_n), \quad (7.43)$$

where f is the joint pmf; and in the continuous case

$$\mathbb{E}Z = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} g(x_1, \dots, x_n) f(x_1, \dots, x_n) dx_1 \cdots dx_n, \quad (7.44)$$

where f is the joint pdf.

Example (Expectation of the Sum of Two Random Variables). Let X and Y be continuous, possibly *dependent*, random variables with joint pdf f . Then,

$$\begin{aligned} \mathbb{E}(X + Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y) f(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy \\ &= \int_{-\infty}^{\infty} x f_X(x) dx + \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= \mathbb{E}X + \mathbb{E}Y. \end{aligned}$$

The previous example is easily generalized to the following result:

Claim 36: Linearity of Expectation — Suppose $X_1, X_2, \dots, X_n$ are discrete or continuous random variables with means $\mu_1, \mu_2, \dots, \mu_n$. Let

$$Y = a + b_1 X_1 + b_2 X_2 + \cdots + b_n X_n \quad (7.45)$$

where $a, b_1, b_2, \dots, b_n$ are constants. Then

$$\begin{aligned} \mathbb{E}Y &= a + b_1 \mathbb{E}X_1 + \cdots + b_n \mathbb{E}X_n \\ &= a + b_1 \mu_1 + \cdots + b_n \mu_n. \end{aligned} \quad (7.46)$$

Example (Expectation of $\text{Bin}(n, p)$ and $\text{Hyp}(n, r, N)$). We can now prove that the expectation of a $\text{Bin}(n, p)$ random variable is np , without having to resort to difficult arguments. Namely, if $X \sim$

$\text{Bin}(n, p)$, then X can be written as the sum $X_1 + \cdots + X_n$ of iid $\text{Ber}(p)$ random variables, see Example 7.2. Thus,

$$\mathbb{E}X = \mathbb{E}(X_1 + \cdots + X_n) = \mathbb{E}X_1 + \cdots + \mathbb{E}X_n = \underbrace{p + \cdots + p}_{n \text{ times}} = np, \quad (7.47)$$

because the expectation of each X_i is p .

Notice that we do not use anywhere the independence of the $\{X_i\}$. Now, let $X \sim \text{Hyp}(n, r, N)$, and let $p = r/N$. We can think of X as the total number of red balls when n balls are drawn from an urn with r red balls and $N - r$ other balls. Without loss of generality we may assume the balls are drawn one-by-one. Let $X_i = 1$ if the i -th ball is red, and 0 otherwise. Then, again $X = X_1 + \cdots + X_n$, and each $X_i \sim \text{Ber}(p)$, but now the $\{X_i\}$ are *dependent*. However, this does not affect the result (7.47), so that the expectation of X is $np = nr/N$.

Expectation of Products of Independent Random Variables

Claim 37: Products of Independent Variables — If $X_1, \dots, X_n$ are *independent*, then

$$\mathbb{E}X_1 X_2 \cdots X_n = \mathbb{E}X_1 \mathbb{E}X_2 \cdots \mathbb{E}X_n. \quad (7.48)$$

Proof: We prove it only for the 2-dimensional continuous case. Let f denote the joint pdf of X and Y , and f_X and f_Y the marginals. Then, $f(x, y) = f_X(x)f_Y(y)$ for all x, y . Thus

$$\begin{aligned} \mathbb{E}XY &= \iint xy f(x, y) dx dy = \iint xy f_X(x) f_Y(y) dx dy \\ &= \int x f_X(x) dx \int y f_Y(y) dy = \mathbb{E}X \mathbb{E}Y. \end{aligned} \quad (7.49)$$

The generalization to the n -dimensional continuous/discrete case is obvious. $\square$

Claim 37 is particularly handy in combination with transform techniques. We give two examples.

Example (Sum of Poisson Random Variables). Let $X \sim \text{Poi}(\lambda)$, then its PGF (see Chapter 9) is given by

$$G(z) = e^{-\lambda(1-z)}. \quad (7.50)$$

Now let $Y \sim \text{Poi}(\mu)$ be independent of X . Then, the PGF of $X + Y$ is given by

$$\mathbb{E}z^{X+Y} = \mathbb{E}z^X \mathbb{E}z^Y = e^{-\lambda(1-z)} e^{-\mu(1-z)} = e^{-(\lambda+\mu)(1-z)}. \quad (7.51)$$

Thus, by the uniqueness property of the PGF, $X + Y \sim \text{Poi}(\lambda + \mu)$.

Example (Sum of Gamma Random Variables). The MGF of $X \sim \text{Gamma}(\alpha, \lambda)$ is (see Chapter 9)

$$\mathbb{E}e^{sX} = \left(\frac{\lambda}{\lambda - s} \right)^\alpha. \quad (7.52)$$

As a special case, the moment generating function of the $\text{Exp}(\lambda)$ distribution is given by $\lambda/(\lambda - s)$. Now let $X_1, \dots, X_n$ be iid $\text{Exp}(\lambda)$ random variables. The MGF of $S_n = X_1 + \dots + X_n$ is

$$\mathbb{E}e^{sS_n} = \mathbb{E}[e^{sX_1} \dots e^{sX_n}] = \mathbb{E}e^{sX_1} \dots \mathbb{E}e^{sX_n} = \left(\frac{\lambda}{\lambda - s} \right)^n, \quad (7.53)$$

which shows that $S_n \sim \text{Gamma}(n, \lambda)$.

Covariance

Definition : Covariance — The **covariance** of two random variables X and Y is defined as the number

$$\mathbb{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]. \quad (7.54)$$

The covariance is a measure for the amount of linear dependency between the variables. If small values of X (smaller than the expected value of X) go together with small values of Y , and at the same time large values of X go together with large values of Y , then $\mathbb{Cov}(X, Y)$ will be *positive*. If on the other hand small values of X go together with large values of Y , and large values of X go together with small values of Y , then $\mathbb{Cov}(X, Y)$ will be *negative*.

For easy reference we list some important properties of the variance and covariance in Table 7.1. The proofs follow directly from the definitions of covariance and variance and the properties of the expectation.

Table 7.1: Properties of variance and covariance

1.	$\text{Var}(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2.$
2.	$\text{Var}(aX + b) = a^2 \text{Var}(X).$
3.	$\mathbb{Cov}(X, Y) = \mathbb{E}XY - \mathbb{E}X\mathbb{E}Y.$
4.	$\mathbb{Cov}(X, Y) = \mathbb{Cov}(Y, X).$
5.	$\mathbb{Cov}(aX + bY, Z) = a \mathbb{Cov}(X, Z) + b \mathbb{Cov}(Y, Z).$
6.	$\mathbb{Cov}(X, X) = \text{Var}(X).$
7.	$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \mathbb{Cov}(X, Y).$
8.	$X \text{ and } Y \text{ indep.} \implies \mathbb{Cov}(X, Y) = 0.$

Proof: Properties 1. and 2. were already proved in Chapters 3 and 5 (see (5.30)). Properties 4. and 6. follow directly from the definitions of covariance and variance. Denote for convenience $\mu_X = \mathbb{E}X$ and $\mu_Y = \mathbb{E}Y$.

$$3. \text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY - Y\mu_X - X\mu_Y + \mu_X\mu_Y] = \mathbb{E}XY - \mu_Y\mu_X - \mu_X\mu_Y + \mu_X\mu_Y = \mathbb{E}XY - \mu_X\mu_Y.$$

5. Using property 3. we get

$$\begin{aligned} \text{Cov}(aX + bY, Z) &= \mathbb{E}[(aX + bY)Z] - \mathbb{E}[aX + bY] \mathbb{E}[Z] \\ &= a \mathbb{E}[XZ] + b \mathbb{E}[YZ] - a \mathbb{E}[X] \mathbb{E}[Z] - b \mathbb{E}[Y] \mathbb{E}[Z] \\ &= a\{\mathbb{E}[XZ] - \mathbb{E}[X]\mathbb{E}[Z]\} + b\{\mathbb{E}[YZ] - \mathbb{E}[Y]\mathbb{E}[Z]\} \\ &= a \text{Cov}(X, Z) + b \text{Cov}(Y, Z). \end{aligned} \quad (7.55)$$

7. By property 6. we have

$$\text{Var}(X + Y) = \text{Cov}(X + Y, X + Y). \quad (7.56)$$

By property 5. we can expand this to

$$\begin{aligned} \text{Cov}(X + Y, X + Y) &= \text{Cov}(X, X + Y) + \text{Cov}(Y, X + Y) \\ &= \text{Cov}(X, X) + \text{Cov}(X, Y) + \text{Cov}(Y, X) + \text{Cov}(Y, Y), \end{aligned} \quad (7.57)$$

where we have also used the symmetry property 4. to expand the second argument of the covariance. Now, by properties 4. and 6. we can simplify the above sum to $\text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$, which had to be shown.

8. If X and Y are independent, then $\mathbb{E}XY = \mathbb{E}X\mathbb{E}Y$, so that $\text{Cov}(X, Y) = \mathbb{E}XY - \mathbb{E}X\mathbb{E}Y = 0$.

□

Correlation Coefficient

A scaled version of the covariance is given by the **correlation coefficient**.

$$\rho(X, Y) := \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}. \quad (7.58)$$

Claim 38: Correlation Coefficient Bounds — The correlation coefficient always lies between -1 and 1 .

Proof: Let a be an arbitrary real number, and denote the standard deviations of X and Y by σ_X and σ_Y . Obviously the variance of $\pm aX + Y$ is always non-negative. Thus, using the properties of

covariance and variance

$$\mathbb{V}\text{ar}(-aX + Y) = a^2\sigma_X^2 + \sigma_Y^2 - 2a\mathbb{C}\text{ov}(X, Y) \geq 0. \quad (7.59)$$

So that after rearranging and dividing by $\sigma_X\sigma_Y$, we obtain

$$\varrho(X, Y) \leq \frac{1}{2} \left(\frac{a\sigma_X}{\sigma_Y} + \frac{\sigma_Y}{a\sigma_X} \right). \quad (7.60)$$

Similarly,

$$\mathbb{V}\text{ar}(aX + Y) = a^2\sigma_X^2 + \sigma_Y^2 + 2a\mathbb{C}\text{ov}(X, Y) \geq 0, \quad (7.61)$$

so that

$$\varrho(X, Y) \geq -\frac{1}{2} \left(\frac{a\sigma_X}{\sigma_Y} + \frac{\sigma_Y}{a\sigma_X} \right). \quad (7.62)$$

By choosing $a = \sigma_Y/\sigma_X$, we see that $-1 \leq \varrho(X, Y) \leq 1$. $\square$

In Figure 7.2 an illustration of the correlation coefficient is given. Each figure corresponds to samples of size 40 from a different 2-dimensional distribution. In each case $\mathbb{E}X = \mathbb{E}Y = 0$ and $\mathbb{V}\text{ar}(X) = \mathbb{V}\text{ar}(Y) = 1$.

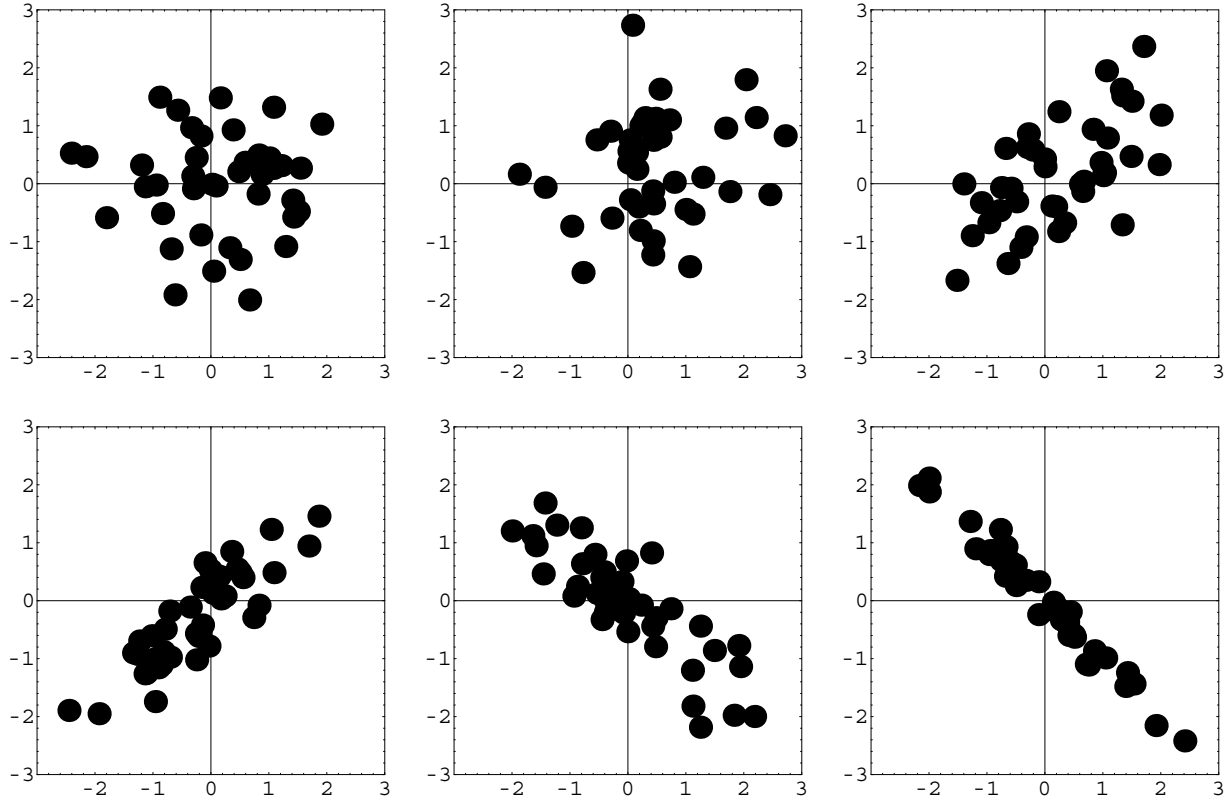


Figure 7.2: *Illustration of correlation coefficient. Above: $\varrho = 0$, $\varrho = 0.4$, $\varrho = 0.7$. Below: $\varrho = 0.9$, $\varrho = -0.8$, $\varrho = -0.98$.*

Variance of Linear Combinations

As a consequence of properties 2. and 7. in Table 7.1, we have

Claim 39: Variance of Independent Sums — Suppose $X_1, X_2, \dots, X_n$ are discrete or continuous *independent* random variables with variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$. Let

$$Y = a + b_1X_1 + b_2X_2 + \dots + b_nX_n \quad (7.63)$$

where $a, b_1, b_2, \dots, b_n$ are constants. Then

$$\begin{aligned} \text{Var}(Y) &= b_1^2 \text{Var}(X_1) + \dots + b_n^2 \text{Var}(X_n) \\ &= b_1^2 \sigma_1^2 + \dots + b_n^2 \sigma_n^2. \end{aligned} \quad (7.64)$$

Proof: By virtue of property 6., and repetitive application of property 5., we have (note that the constant a does not play a role in the variance):

$$\begin{aligned} \text{Var}(Y) &= \mathbb{Cov}(b_1X_1 + b_2X_2 + \dots + b_nX_n, b_1X_1 + b_2X_2 + \dots + b_nX_n) \\ &= \sum_{i=1}^n \mathbb{Cov}(b_iX_i, b_iX_i) + 2 \sum_{i<j} \mathbb{Cov}(b_iX_i, b_jX_j). \end{aligned} \quad (7.65)$$

Since $\mathbb{Cov}(b_iX_i, b_iX_i) = b_i^2 \text{Var}(X_i)$ and all covariance terms are zero because of the independence of X_i and X_j ($i \neq j$), the result follows. $\square$

Example (Variance of $\text{Bin}(n, p)$ and $\text{Hyp}(n, r, N)$). Consider again Example 7.4 where we derived the expectation of $X \sim \text{Bin}(n, p)$ and $X \sim \text{Hyp}(n, r, N)$ by writing X as

$$X = X_1 + \dots + X_n \quad (7.66)$$

of independent (in the binomial case) or dependent (in the hypergeometric case) $\text{Ber}(p)$ random variables, where $p = r/N$ in the hypergeometric case. Using Claim 39, the variance of the binomial distribution follows directly from

$$\text{Var}(X) = \text{Var}(X_1) + \dots + \text{Var}(X_n) = n \text{Var}(X_1) = np(1 - p). \quad (7.67)$$

For the hypergeometric case we must include the covariance terms as well:

$$\text{Var}(X) = \text{Var}(X_1) + \dots + \text{Var}(X_n) + 2 \sum_{i<j} \mathbb{Cov}(X_i, X_j). \quad (7.68)$$

By symmetry all the $\mathbb{Cov}(X_i, X_j)$ are the same ($i \neq j$). Hence,

$$\text{Var}(X) = n \text{Var}(X_1) + n(n - 1) \mathbb{Cov}(X_1, X_2). \quad (7.69)$$

Since $\text{Var}(X_1) = p(1 - p)$, and $\mathbb{Cov}(X_1, X_2) = \mathbb{E}X_1X_2 - p^2$, it remains to find $\mathbb{E}X_1X_2 = \mathbb{P}(X_1 = 1, X_2 = 1)$, which is

$$\mathbb{P}(X_1 = 1, X_2 = 1) = \mathbb{P}(X_1 = 1) \mathbb{P}(X_2 = 1 | X_1 = 1) = p \frac{r - 1}{N - 1}. \quad (7.70)$$

Simplifying gives,

$$\text{Var}(X) = np(1 - p) \frac{N - n}{N - 1}. \quad (7.71)$$

7.5 The Mean Vector and Covariance Matrix

Let $\mathbf{X} = (X_1, \dots, X_n)$ be a random vector. Sometimes it is convenient to write the expectations and covariances in vector notation.

Definition : Expectation Vector and Covariance Matrix — For any random vector $\mathbf{X}$ we define the **expectation vector** as the vector of expectations

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix} := \begin{bmatrix} \mathbb{E}X_1 \\ \vdots \\ \mathbb{E}X_n \end{bmatrix}. \quad (7.72)$$

The **covariance matrix** Σ is defined as the matrix whose (i, j) th element is

$$\mathbb{Cov}(X_i, X_j) = \mathbb{E}(X_i - \mu_i)(X_j - \mu_j). \quad (7.73)$$

If we define the expectation of a vector (matrix) to be the vector (matrix) of expectations, then we can write:

$$\boldsymbol{\mu} = \mathbb{E}\mathbf{X} \quad (7.74)$$

and

$$\Sigma = \mathbb{E}(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top. \quad (7.75)$$

Note that $\boldsymbol{\mu}$ and Σ take the same role as μ and σ^2 in the 1-dimensional case. We sometimes write $\boldsymbol{\mu}_X$ and Σ_X if we wish to emphasise that $\boldsymbol{\mu}$ and Σ belong to the vector $\mathbf{X}$.

Positive semidefiniteness. Note that any covariance matrix Σ is a *symmetric* matrix. In fact, it is *positive semidefinite*, i.e., for any (column) vector $\mathbf{u}$, we have

$$\mathbf{u}^\top \Sigma \mathbf{u} \geq 0. \quad (7.76)$$

To see this, suppose Σ is the covariance matrix of some random vector $\mathbf{X}$ with expectation vector $\boldsymbol{\mu}$. Write $\mathbf{Y} = \mathbf{X} - \boldsymbol{\mu}$. Then

$$\begin{aligned} \mathbf{u}^\top \Sigma \mathbf{u} &= \mathbf{u}^\top \mathbb{E}\mathbf{Y}\mathbf{Y}^\top \mathbf{u} = \mathbb{E}\mathbf{u}^\top \mathbf{Y}\mathbf{Y}^\top \mathbf{u} \\ &= \mathbb{E}(\mathbf{Y}^\top \mathbf{u})^\top \mathbf{Y}^\top \mathbf{u} = \mathbb{E}(\mathbf{Y}^\top \mathbf{u})^2 \geq 0. \end{aligned}$$

Note that $\mathbf{Y}^\top \mathbf{u}$ is a random variable.

Example (Mean Vector and Covariance Matrix of a Bivariate Distribution). Let $X_1 \sim \text{Exp}(\lambda)$ and $X_2 \sim \text{Exp}(\mu)$ be independent exponential random variables. We have $\mathbb{E}[X_1] = 1/\lambda$, $\mathbb{E}[X_2] = 1/\mu$, $\text{Var}(X_1) = 1/\lambda^2$, and $\text{Var}(X_2) = 1/\mu^2$. Since X_1 and X_2 are independent, $\mathbb{Cov}(X_1, X_2) = 0$. Thus, the expectation vector and covariance matrix of $\mathbf{X} = (X_1, X_2)$ are

$$\boldsymbol{\mu} = \begin{bmatrix} 1/\lambda \\ 1/\mu \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 1/\lambda^2 & 0 \\ 0 & 1/\mu^2 \end{bmatrix}. \quad (7.77)$$

The covariance matrix is diagonal, reflecting the independence. For instance, with $\lambda = 2$ and $\mu = 3$,

$$\boldsymbol{\mu} = \begin{bmatrix} 0.5 \\ 1/3 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 0.25 & 0 \\ 0 & 1/9 \end{bmatrix}. \quad (7.78)$$

7.6 The Multivariate Normal Distribution

In this section we have a closer look at normally distributed random variables and their properties, and introduce normally distributed random *vectors*.

It is helpful to view normally distributed random variables as simple transformations of standard normal random variables. For example, let $X \sim \mathcal{N}(0, 1)$. Then, X has density f_X given by

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}. \quad (7.79)$$

Now consider the transformation $Z = \mu + \sigma X$. Then, by the affine transformation formula, Z has density

$$f_Z(z) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(z-\mu)^2}{2\sigma^2}}. \quad (7.80)$$

In other words, $Z \sim \mathcal{N}(\mu, \sigma^2)$. We could also write this as follows: if $Z \sim \mathcal{N}(\mu, \sigma^2)$, then $(Z - \mu)/\sigma \sim \mathcal{N}(0, 1)$. This **standardisation** procedure was already mentioned in Section 5.3.

Let's generalise this to n dimensions. Let $X_1, \dots, X_n$ be independent and standard normal random variables. The joint pdf of $\mathbf{X} = (X_1, \dots, X_n)$ is given by

$$f_{\mathbf{X}}(\mathbf{x}) = (2\pi)^{-n/2} e^{-\frac{1}{2} \mathbf{x}^\top \mathbf{x}}, \quad \mathbf{x} \in \mathbb{R}^n. \quad (7.81)$$

Consider the transformation

$$\mathbf{Z} = \boldsymbol{\mu} + B\mathbf{X}, \quad (7.82)$$

for some $(m \times n)$ matrix B . Note that $\mathbf{Z}$ has expectation vector $\boldsymbol{\mu}$ and covariance matrix $\Sigma = BB^\top$. Indeed, since $X_1, \dots, X_n$ are independent standard normals, $\mathbb{E}\mathbf{X} = \mathbf{0}$ and $\mathbb{E}\mathbf{X}\mathbf{X}^\top = I$, so by linearity of expectation $\mathbb{E}\mathbf{Z} = \boldsymbol{\mu} + B\mathbb{E}\mathbf{X} = \boldsymbol{\mu}$, and by the definition of the covariance matrix,

$$\text{Cov}(\mathbf{Z}) = \mathbb{E}(\mathbf{Z} - \boldsymbol{\mu})(\mathbf{Z} - \boldsymbol{\mu})^\top = \mathbb{E}(B\mathbf{X})(B\mathbf{X})^\top = B\mathbb{E}(\mathbf{X}\mathbf{X}^\top)B^\top = BIB^\top = BB^\top = \Sigma. \quad (7.83)$$

Any random vector of the form (7.82) is said to have a **jointly normal** (or multivariate normal) distribution. We write $\mathbf{Z} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$.

Remark (Standard Definition of the Multivariate Normal). The standard definition in probability theory is more general: a random vector $\mathbf{Z}$ has a multivariate normal distribution if and only if $\mathbf{t}^\top \mathbf{Z}$ has a (univariate) normal distribution for every $\mathbf{t} \in \mathbb{R}^n$. This definition is equivalent to the constructive one above when Σ is invertible, but it also covers cases where Σ is singular and no density exists — for example, the joint distribution of residuals from a normal linear regression. For our purposes the constructive definition suffices.

Suppose B is an invertible $(n \times n)$ -matrix. Set $\mathbf{Y} = \mathbf{Z} - \boldsymbol{\mu} = B\mathbf{X}$. The **linear transformation rule for densities** (Section 8.4) states that for $\mathbf{Y} = B\mathbf{X}$ with B invertible,

$$f_Y(\mathbf{y}) = \frac{f_X(B^{-1}\mathbf{y})}{|B|}, \quad |B| := |\det B|. \quad (7.84)$$

Substituting the standard normal density (7.81), the density of $\mathbf{Y}$ is

$$f_Y(\mathbf{y}) = \frac{1}{|B| \sqrt{(2\pi)^n}} e^{-\frac{1}{2} (B^{-1}\mathbf{y})^\top B^{-1}\mathbf{y}} = \frac{1}{|B| \sqrt{(2\pi)^n}} e^{-\frac{1}{2} \mathbf{y}^\top (B^{-1})^\top B^{-1} \mathbf{y}}. \quad (7.85)$$

We have $|B| = \sqrt{|\Sigma|}$ and $(B^{-1})^\top B^{-1} = (B^\top)^{-1} B^{-1} = (BB^\top)^{-1} = \Sigma^{-1}$, so that

$$f_Y(\mathbf{y}) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2} \mathbf{y}^\top \Sigma^{-1} \mathbf{y}}. \quad (7.86)$$

Because $\mathbf{Z}$ is obtained from $\mathbf{Y}$ by simply adding a constant vector $\boldsymbol{\mu}$, we have $f_Z(\mathbf{z}) = f_Y(\mathbf{z} - \boldsymbol{\mu})$, and therefore

$$f_Z(\mathbf{z}) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} e^{-\frac{1}{2} (\mathbf{z} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{z} - \boldsymbol{\mu})}, \quad \mathbf{z} \in \mathbb{R}^n. \quad (7.87)$$

Note that this formula is very similar to the 1-dimensional case.

Example (Bivariate Normal Distribution). Consider the 2-dimensional case with $\boldsymbol{\mu} = (\mu_1, \mu_2)$, and

$$B = \begin{bmatrix} \sigma_1 & 0 \\ \sigma_2 \varrho & \sigma_2 \sqrt{1 - \varrho^2} \end{bmatrix}. \quad (7.88)$$

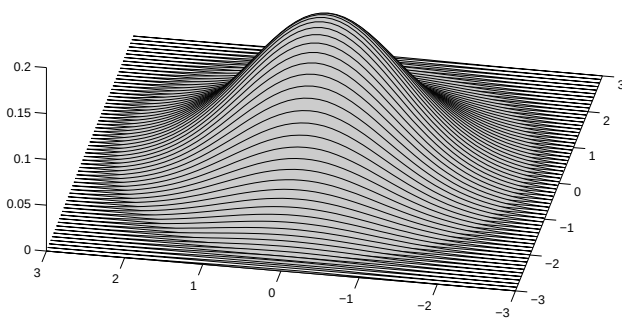
The covariance matrix is now

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \varrho \sigma_1 \sigma_2 \\ \varrho \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix}. \quad (7.89)$$

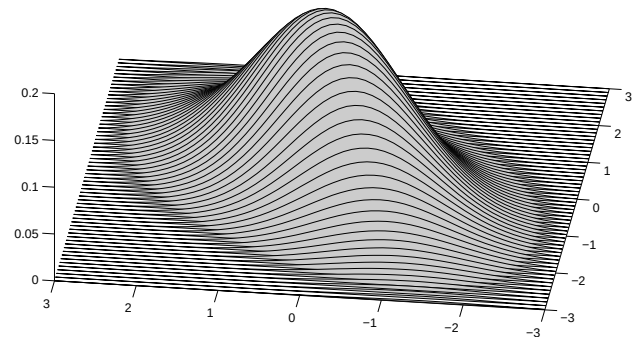
Therefore, the density is

$$f_Z(\mathbf{z}) = \frac{1}{2\pi \sigma_1 \sigma_2 \sqrt{1 - \varrho^2}} \exp \left\{ -\frac{1}{2(1 - \varrho^2)} \left(\frac{(z_1 - \mu_1)^2}{\sigma_1^2} - 2\varrho \frac{(z_1 - \mu_1)(z_2 - \mu_2)}{\sigma_1 \sigma_2} + \frac{(z_2 - \mu_2)^2}{\sigma_2^2} \right) \right\}. \quad (7.90)$$

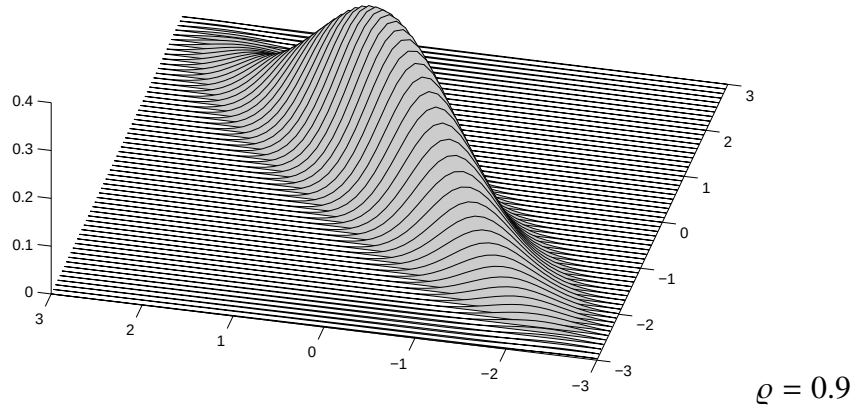
Here are some pictures of the density, for $\mu_1 = \mu_2 = 0$ and $\sigma_1 = \sigma_2 = 1$, and for various ϱ .



$\varrho = 0$



$\varrho = 0.5$



We say that (Z_1, Z_2) has a **bivariate normal** distribution. Note that in this example $\mathbb{E}Z_i = \mu_i, i = 1, 2$. Moreover, since we have chosen B such that the covariance matrix has the form (7.89), we have $\text{Var}(Z_i) = \sigma_i^2, i = 1, 2$, and $\rho(Z_1, Z_2) = \rho$.

Uncorrelated Jointly Normal Variables are Independent

Compare the following with property 8 of Table 7.1.

Claim 40: Uncorrelated Normal Independence — If Z_1 and Z_2 have a jointly normal distribution then

$$\text{Cov}(Z_1, Z_2) = 0 \implies Z_1 \text{ and } Z_2 \text{ are independent.} \quad (7.91)$$

Proof: If $\text{Cov}(Z_1, Z_2) = 0$, then B in (7.88) is a diagonal matrix. Thus, trivially $Z_1 = \sigma_1 X_1$ and $Z_2 = \sigma_2 X_2$ are independent. $\square$

Affine Combinations of Normals

One of the most (if not the most) important properties of the normal distribution is that affine combinations (that is, linear combinations plus a constant) of independent normal random variables are normally distributed. Here is a more precise formulation.

Claim 41: Affine Combinations of Normals — If $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, independently, for $i = 1, 2, \dots, n$, then

$$Y = a + \sum_{i=1}^n b_i X_i \sim \mathcal{N}\left(a + \sum_{i=1}^n b_i \mu_i, \sum_{i=1}^n b_i^2 \sigma_i^2\right). \quad (7.92)$$

Proof: The easiest way to prove this is by using moment generating functions (MGFs), which are formally introduced in Chapter 9. The reader may wish to revisit this proof after studying that

chapter. Recall that the MGF of a $\mathcal{N}(\mu, \sigma^2)$ -distributed random variable X is given by

$$M_X(s) = e^{\mu s + \frac{1}{2} \sigma^2 s^2}. \quad (7.93)$$

Let M_Y be the moment generating function of Y . Since $X_1, \dots, X_n$ are independent, we have

$$\begin{aligned} M_Y(s) &= \mathbb{E} \exp\{as + \sum_{i=1}^n b_i X_i s\} \\ &= e^{as} \prod_{i=1}^n M_{X_i}(b_i s) \\ &= e^{as} \prod_{i=1}^n \exp\{\mu_i(b_i s) + \frac{1}{2} \sigma_i^2 (b_i s)^2\} \\ &= \exp\{sa + s \sum_{i=1}^n b_i \mu_i + \frac{1}{2} \sum_{i=1}^n b_i^2 \sigma_i^2 s^2\}, \end{aligned}$$

which is the MGF of a normal distribution of the form (7.92). $\square$

Remark (Distribution of Normal Affine Combinations). Note that from Claims 36 and 39 we had already established the expectation and variance of Y in (7.92). But we have now found that the *distribution* is normal.

Example (Ball Bearings). A machine produces ball bearings with a $\mathcal{N}(1, 0.01)$ diameter (cm). The balls are placed on a sieve with a $\mathcal{N}(1.1, 0.04)$ diameter. The diameter of the balls and the sieve are assumed to be independent of each other.

Question: What is the probability that a ball will fall through?

Answer: Let $X \sim \mathcal{N}(1, 0.01)$ and $Y \sim \mathcal{N}(1.1, 0.04)$. We need to calculate $\mathbb{P}(Y > X) = \mathbb{P}(Y - X > 0)$. But, $Z := Y - X \sim \mathcal{N}(0.1, 0.05)$. Hence

$$\mathbb{P}(Z > 0) = \mathbb{P}\left(\frac{Z - 0.1}{\sqrt{0.05}} > \frac{-0.1}{\sqrt{0.05}}\right) = \Phi(0.447) \approx 0.67, \quad (7.94)$$

where Φ is the cdf of the $\mathcal{N}(0, 1)$ -distribution.

7.7 The Multinomial Distribution

The multinomial distribution can be viewed as a generalization of the binomial distribution to experiments with more than two outcomes.

Definition : Multinomial Distribution — We say that $(X_1, X_2, \dots, X_k)$ has a **multinomial** distribution, with parameters n and $p_1, p_2, \dots, p_k$, if

$$\mathbb{P}(X_1 = x_1, \dots, X_k = x_k) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}, \quad (7.95)$$

for all $x_1, \dots, x_k \in \{0, 1, \dots, n\}$ such that $x_1 + x_2 + \dots + x_k = n$. We write $(X_1, \dots, X_k) \sim \text{Mnom}(n, p_1, \dots, p_k)$.

Example (Multinomial Distribution). We independently throw n balls into k urns, such that each ball is thrown in urn i with probability p_i , $i = 1, \dots, k$; see Figure 7.3.

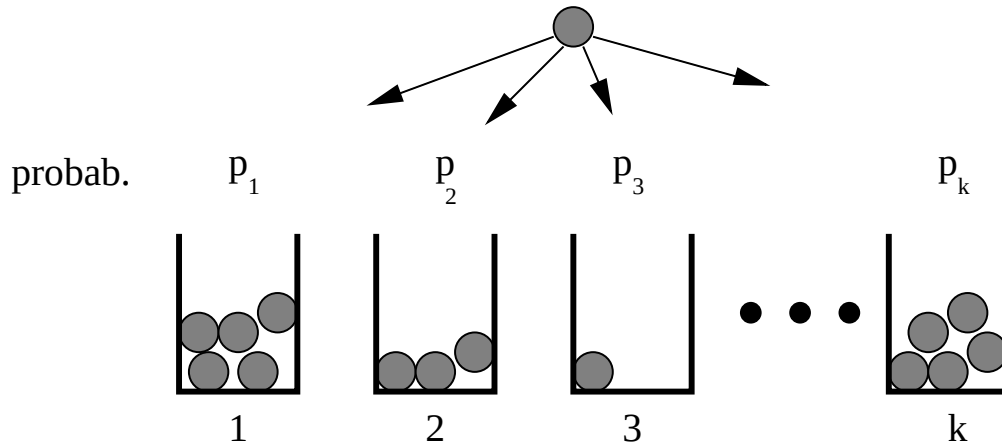


Figure 7.3: Throwing n balls into k urns.

Let X_i be the total number of balls in urn i , $i = 1, \dots, k$. We show that $(X_1, \dots, X_k) \sim \text{Mnom}(n, p_1, \dots, p_k)$. Let $x_1, \dots, x_k$ be integers between 0 and n that sum up to n . The probability that the *first* x_1 balls fall in the first urn, the *next* x_2 balls fall in the second urn, etcetera, is

$$p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}. \quad (7.96)$$

To find the probability that there are x_1 balls in the first urn, x_2 in the second, etcetera, we have to multiply the probability above with the number of ways in which we can fill the urns with $x_1, x_2, \dots, x_k$ balls, i.e., $n!/(x_1! x_2! \dots x_k!)$. This gives (7.95).

Remark. Note that for the *binomial* distribution there are only *two* possible urns. Also, note that for each $i = 1, \dots, k$, $X_i \sim \text{Bin}(n, p_i)$.

Covariance Structure

The components of a multinomial random vector are negatively correlated: if more balls fall in one urn, fewer are available for the others. Specifically,

$$\text{Cov}(X_i, X_j) = -n p_i p_j, \quad i \neq j. \quad (7.97)$$

To see this, consider two independent draws from the same experiment. For a single trial, let I_i be the indicator that the ball lands in urn i . Then $\mathbb{E}[I_i I_j] = 0$ for $i \neq j$ (a ball cannot be in two urns

simultaneously), and $\mathbb{E}[I_i] = p_i$. Hence $\mathbb{Cov}(I_i, I_j) = 0 - p_i p_j = -p_i p_j$. Since $X_i = \sum_{t=1}^n I_i^{(t)}$ and $X_j = \sum_{t=1}^n I_j^{(t)}$ with independent trials, the covariance scales by n .

Example (Rolling a Fair Die). Roll a fair die $n = 60$ times and let X_i count the number of times face i appears, $i = 1, \dots, 6$. Then $(X_1, \dots, X_6) \sim \text{Mnom}(60, 1/6, \dots, 1/6)$.

Each marginal has $X_i \sim \text{Bin}(60, 1/6)$, so $\mathbb{E}[X_i] = 10$ and $\mathbb{Var}(X_i) = 60 \cdot \frac{1}{6} \cdot \frac{5}{6} = \frac{25}{3} \approx 8.33$. The covariance between any two faces is

$$\mathbb{Cov}(X_i, X_j) = -60 \cdot \frac{1}{6} \cdot \frac{1}{6} = -\frac{5}{3} \approx -1.67, \quad i \neq j, \quad (7.98)$$

and the corresponding correlation coefficient is

$$\rho(X_i, X_j) = \frac{-5/3}{25/3} = -\frac{1}{5}. \quad (7.99)$$

7.8 Conditional Expectation and the Law of Total Expectation

Conditional distributions allow us to “update” our beliefs about one random variable given information about another. Conditional expectation extends this idea to expectations: it answers the question “what is the expected value of Y if we know the value of X ?”

Since a conditional pmf (pdf) has all the properties of a probability mass (density) function, it makes sense to define the corresponding *conditional expectation* (in the continuous case) as

$$\mathbb{E}[Y | X = x] = \int y f_{Y|X}(y | x) dy. \quad (7.100)$$

For discrete X and Y we have instead:

$$\mathbb{E}[Y | X = x] = \sum_y y \mathbb{P}(Y = y | X = x). \quad (7.101)$$

Note that $y \mapsto \mathbb{P}(Y = y | X = x)$ is a genuine pmf for each fixed x .

In both the discrete and continuous case, $\mathbb{E}[Y | X = x]$ is a function of x , say $h(x)$. The corresponding random variable $h(X)$ is written as $\mathbb{E}[Y | X]$. One of the most useful results in probability is that its expectation is simply the expectation of Y . It simplifies the law of total probability calculations for random variables as in (7.29).

Claim 42: Tower Property — For any pair of random variables X, Y , it holds that

$$\mathbb{E}[\mathbb{E}[Y | X]] = \mathbb{E}[Y]. \quad (7.102)$$

Proof: We prove it for the case where X and Y are jointly continuous. Then, for x such that $f_X(x) > 0$,

$$\mathbb{E}[Y | X = x] = h(x) = \int_{-\infty}^{\infty} y f_{Y|X}(y | x) dy, \quad (7.103)$$

where $f_{Y|X}(\cdot | x)$ is the pdf of Y given X . It follows that

$$\begin{aligned} \mathbb{E}h(X) &= \int_{-\infty}^{\infty} h(x) f_X(x) dx = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} y f_{Y|X}(y | x) dy \right) f_X(x) dx \\ &= \int_{-\infty}^{\infty} y \left(\underbrace{\int_{-\infty}^{\infty} f_{Y|X}(y | x) f_X(x) dx}_{\text{by (7.29)}} \right) dy = \int_{-\infty}^{\infty} y f_Y(y) dy = \mathbb{E}Y. \end{aligned} \quad (7.104)$$

□

The tower property is also called the **law of iterated expectations** or the **law of total expectation**. It is an indispensable tool: to compute $\mathbb{E}[Y]$, one can first compute $\mathbb{E}[Y | X = x]$ as a function of x , and then average over X .

Example (Computing $\mathbb{E}[Y]$ via Conditioning). A fair coin is tossed. If heads, a fair die is rolled and Y is the outcome. If tails, $Y = 0$. Let $X = 1$ for heads and $X = 0$ for tails. Then

$$\mathbb{E}[Y | X = 1] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5, \quad \mathbb{E}[Y | X = 0] = 0. \quad (7.105)$$

By the tower property,

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y | X]] = \mathbb{E}[Y | X = 1] \mathbb{P}(X = 1) + \mathbb{E}[Y | X = 0] \mathbb{P}(X = 0) = 3.5 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = 1.75. \quad (7.106)$$

Example (Conditional Expectation for a Continuous Pair). Let (X, Y) have joint pdf $f_{X,Y}(x, y) = 2$ on the triangle $\{0 \leq x \leq y \leq 1\}$ (as in Example 7.3). We showed that $\mathbb{E}[Y | X = x] = (x + 1)/2$. Hence

$$\mathbb{E}[Y] = \mathbb{E}\left[\frac{X + 1}{2}\right] = \frac{\mathbb{E}[X] + 1}{2}. \quad (7.107)$$

Since $f_X(x) = 2(1 - x)$ for $x \in [0, 1]$, we compute $\mathbb{E}[X] = \int_0^1 2x(1 - x) dx = 1/3$, giving $\mathbb{E}[Y] = (1/3 + 1)/2 = 2/3$. One can verify this directly: $\mathbb{E}[Y] = \int_0^1 \int_0^y 2y dx dy = \int_0^1 2y^2 dy = 2/3$.

7.9 Random Sums and the Law of Total Variance

A powerful application of conditional expectation is to problems involving a *random* number of random variables. In such problems, the tower property and a companion result — the *law of total variance* — allow us to compute both the mean and the variance of the resulting sum.

Random Sums

Example (Random Sums). Let $X_1, X_2, \dots$ be a sequence of iid random variables with common mean μ and variance σ^2 , and let

$$S_n = \begin{cases} 0 & \text{if } n = 0 \\ \sum_{i=1}^n X_i & \text{if } n \geq 1. \end{cases} \quad (7.108)$$

Then, clearly $\mathbb{E}S_n = n\mu$.

Now let N be a random variable (independent of the X_i) which takes values in the non-negative integers. By the tower property,

$$\mathbb{E}S_N = \mathbb{E}(\mathbb{E}[S_N | N]) = \mathbb{E}(N\mu) = \mu \mathbb{E}N. \quad (7.109)$$

This is called **Wald's equation** for the mean.

To find the variance of S_N , we need a new tool.

The Law of Total Variance

Claim 43: Law of Total Variance — For any pair of random variables X, Y with $\mathbb{V}\text{ar}(Y) < \infty$,

$$\mathbb{V}\text{ar}(Y) = \mathbb{E}[\mathbb{V}\text{ar}(Y | X)] + \mathbb{V}\text{ar}(\mathbb{E}[Y | X]). \quad (7.110)$$

Proof: By the computational formula for variance, $\mathbb{V}\text{ar}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2$. Applying the tower property to Y^2 and Y :

$$\mathbb{E}[Y^2] = \mathbb{E}[\mathbb{E}[Y^2 | X]], \quad \mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y | X]]. \quad (7.111)$$

Now, the conditional variance satisfies $\mathbb{V}\text{ar}(Y | X) = \mathbb{E}[Y^2 | X] - (\mathbb{E}[Y | X])^2$, so $\mathbb{E}[Y^2 | X] = \mathbb{V}\text{ar}(Y | X) + (\mathbb{E}[Y | X])^2$. Taking expectations:

$$\mathbb{E}[Y^2] = \mathbb{E}[\mathbb{V}\text{ar}(Y | X)] + \mathbb{E}[(\mathbb{E}[Y | X])^2]. \quad (7.112)$$

Subtracting $(\mathbb{E}[Y])^2 = (\mathbb{E}[\mathbb{E}[Y | X]])^2$:

$$\mathbb{V}\text{ar}(Y) = \mathbb{E}[\mathbb{V}\text{ar}(Y | X)] + \underbrace{\mathbb{E}[(\mathbb{E}[Y | X])^2] - (\mathbb{E}[\mathbb{E}[Y | X]])^2}_{\mathbb{V}\text{ar}(\mathbb{E}[Y | X])}. \quad (7.113)$$

□

The two terms have intuitive names. The first, $\mathbb{E}[\mathbb{V}\text{ar}(Y | X)]$, is the *average unexplained variance* — the variability in Y that remains even after knowing X . The second, $\mathbb{V}\text{ar}(\mathbb{E}[Y | X])$, is the *explained variance* — how much the conditional mean of Y varies as X changes.

Variance of a Random Sum

Example (Variance of S_N). Returning to Example 7.9, we apply the law of total variance to find $\mathbb{V}\text{ar}(S_N)$. Since the X_i are iid with mean μ and variance σ^2 , and N is independent of the X_i :

$$\mathbb{E}[S_N | N] = N\mu, \quad \mathbb{V}\text{ar}(S_N | N) = N\sigma^2. \quad (7.114)$$

Applying (7.110):

$$\mathbb{V}\text{ar}(S_N) = \mathbb{E}[N\sigma^2] + \mathbb{V}\text{ar}(N\mu) = \sigma^2 \mathbb{E}[N] + \mu^2 \mathbb{V}\text{ar}(N). \quad (7.115)$$

In summary,

$$\boxed{\mathbb{V}\text{ar}(S_N) = \sigma^2 \mathbb{E}[N] + \mu^2 \mathbb{V}\text{ar}(N).} \quad (7.116)$$

This is **Wald's equation for the variance**.

Compound Poisson Distribution

An important special case arises when $N \sim \text{Poi}(\lambda)$.

Example (Compound Poisson). An insurance company receives claims at a rate of λ per month, modelled as $N \sim \text{Poi}(\lambda)$. Each claim amount X_i is iid with mean μ and variance σ^2 , independent of N . The total monthly payout is $S_N = X_1 + \cdots + X_N$.

Since $\mathbb{E}[N] = \mathbb{V}\text{ar}(N) = \lambda$ for a Poisson variable,

$$\mathbb{E}[S_N] = \mu\lambda, \quad \mathbb{V}\text{ar}(S_N) = \sigma^2\lambda + \mu^2\lambda = \lambda(\sigma^2 + \mu^2) = \lambda \mathbb{E}[X_1^2]. \quad (7.117)$$

For instance, with $\lambda = 100$ claims per month and claim amounts having mean $\mu = 500$ and standard deviation $\sigma = 200$:

$$\mathbb{E}[S_N] = 500 \cdot 100 = 50,000, \quad \mathbb{V}\text{ar}(S_N) = 100 \cdot (200^2 + 500^2) = 29,000,000, \quad (7.118)$$

giving a standard deviation of approximately 5,385.

MORE ASPECTS OF JOINT DISTRIBUTIONS

In this chapter we study transformations of vector random variables — their densities via the Jacobian, linear maps and their effects on mean and covariance, and the Cholesky decomposition for generating multivariate normal samples. We then derive distributions of sums, products, ratios, and order statistics of independent random variables, with convolutions as a central tool.

8.1 Transformations of Vector Random Variables

In Chapter 7 we studied joint distributions and their properties. A natural next question is: if we know the joint distribution of a random vector $\mathbf{X}$, what can we say about the distribution of a *transformed* vector $\mathbf{Z} = \mathbf{g}(\mathbf{X})$?

This question arises frequently. For example, given two independent standard normals (X_1, X_2) , what is the distribution of their polar coordinates (R, Θ) ? Or given a random vector with covariance Σ , what happens to the distribution after a linear transformation $\mathbf{Z} = \mathbf{A}\mathbf{X}$?

In one dimension, we studied the change-of-variable formula and the CDF method (Section 6.1). In this chapter, we extend these tools to multiple dimensions. The key ingredients are:

- The **Jacobian matrix** of partial derivatives, which describes how volumes are distorted by the transformation.
- The **transformation rule**, which gives the density of $\mathbf{Z} = \mathbf{g}(\mathbf{X})$ in terms of the density of $\mathbf{X}$ and the Jacobian determinant.

We begin with the general transformation rule, then specialise to the important case of linear transformations.

8.2 Densities of Transformed Distributions

Consider a general (possibly nonlinear) transformation $\mathbf{x} \mapsto \mathbf{g}(\mathbf{x})$, written out:

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \mapsto \begin{bmatrix} g_1(\mathbf{x}) \\ g_2(\mathbf{x}) \\ \vdots \\ g_n(\mathbf{x}) \end{bmatrix}. \quad (8.1)$$

For a fixed $\mathbf{x}$, let $\mathbf{z} = \mathbf{g}(\mathbf{x})$. Suppose $\mathbf{g}$ is invertible, hence, $\mathbf{x} = \mathbf{g}^{-1}(\mathbf{z})$. Any infinitesimal n -dimensional rectangle at $\mathbf{x}$ with volume V is transformed into a n -dimensional parallelepiped at $\mathbf{z}$ with volume $V |\det(J_{\mathbf{x}}(\mathbf{g}))|$, where $J_{\mathbf{x}}(\mathbf{g})$ is the **matrix of Jacobi** at $\mathbf{x}$ of the transformation $\mathbf{g}$:

$$J_{\mathbf{x}}(\mathbf{g}) = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_1}{\partial x_n} \\ \vdots & \cdots & \vdots \\ \frac{\partial g_n}{\partial x_1} & \cdots & \frac{\partial g_n}{\partial x_n} \end{bmatrix}. \quad (8.2)$$

Now consider a random column vector $\mathbf{Z} = \mathbf{g}(\mathbf{X})$. Let C be a small cube around $\mathbf{z}$ with volume h^n . Let D be image of C under $\mathbf{g}^{-1}$. Then, as in the linear case,

$$\mathbb{P}(\mathbf{Z} \in C) \approx h^n f_{\mathbf{Z}}(\mathbf{z}) \approx h^n |\det(J_{\mathbf{z}}(\mathbf{g}^{-1}))| f_{\mathbf{X}}(\mathbf{x}). \quad (8.3)$$

Hence, we have derived the following transformation rule.

Claim 44: Transformation Rule — Let $\mathbf{X}$ be a continuous random vector with joint pdf $f_{\mathbf{X}}$, and let $\mathbf{g}$ be an invertible function with Jacobi matrix $J_{\mathbf{x}}(\mathbf{g})$ at $\mathbf{x}$. Then the random vector $\mathbf{Z} = \mathbf{g}(\mathbf{X})$ has joint pdf $f_{\mathbf{Z}}$ given by

$$f_{\mathbf{Z}}(\mathbf{z}) = \frac{f_{\mathbf{X}}(\mathbf{x})}{|\det(J_{\mathbf{x}}(\mathbf{g}))|} = f_{\mathbf{X}}(\mathbf{x}) |\det(J_{\mathbf{z}}(\mathbf{g}^{-1}))|, \quad \mathbf{z} \in \mathbb{R}^n, \quad (8.4)$$

where $\mathbf{z} = \mathbf{g}(\mathbf{x})$ and $\mathbf{x} = \mathbf{g}^{-1}(\mathbf{z})$. The second equality holds because $|\det(J_{\mathbf{z}}(\mathbf{g}^{-1}))| = 1/|\det(J_{\mathbf{x}}(\mathbf{g}))|$, which follows from applying the chain rule to $\mathbf{g}^{-1} \circ \mathbf{g} = \text{id}$.

Remark (One-dimensional Transformation Rule). For the one-dimensional case, $Z = g(X)$, we have that $J_x(g)$ is simply the derivative of g at x ; that is, $g'(x)$. This recovers the change-of-variable formula from Section 6.1 (Claim 28).

Example (Square of a Standard Normal). Let $X \sim \mathcal{N}(0, 1)$ and consider the quadratic mapping $g : x \mapsto x^2$ on $\mathbb{R}$. We cannot directly apply the one-dimensional transformation rule, as g is not invertible. However, this problem can be circumvented by taking the square of the random variable $|X|$ instead. The latter has the pdf

$$f_{|X|}(x) = \sqrt{\frac{2}{\pi}} e^{-x^2/2}, \quad x \geq 0. \quad (8.5)$$

Since X^2 and $|X|^2$ mean the same thing, and the mapping $g : x \mapsto x^2$ is invertible on the interval $[0, \infty)$, we can apply the transformation rule to find the pdf of $Z = g(|X|)$. Taking a $z \geq 0$ and the

corresponding $x = \sqrt{z}$, we have

$$f_Z(z) = \frac{f_{|X|}(x)}{g'(x)} = \frac{\sqrt{\frac{2}{\pi}} e^{-x^2/2}}{2x} = \frac{\sqrt{\frac{2}{\pi}} e^{-z/2}}{2\sqrt{z}} = \frac{1}{\sqrt{2\pi z}} e^{-z/2}, \quad z \geq 0. \quad (8.6)$$

This is the pdf of a χ_1^2 -distribution. Thus $Z \sim \chi_1^2$ (recall Example 6.1, where we derived the same result using the one-dimensional change-of-variable formula).

Remark (Coordinate Transformation). In most coordinate transformations it is $\mathbf{g}^{-1}$ that is given — that is, an expression for $\mathbf{x}$ as a function of $\mathbf{z}$, — rather than $\mathbf{g}$.

8.3 Bivariate Normal in Polar Coordinates and the Box-Muller Transform

The transformation rule has a beautiful application to the bivariate standard normal distribution. By switching to polar coordinates, we obtain a surprisingly simple factorisation that leads to the **Box-Muller method** — one of the most elegant algorithms for generating normal random variables.

Example (Box-Muller). Let X and Y be two independent standard normal random variables. (X, Y) is a random point in the plane. Let (R, Θ) be the corresponding polar coordinates. The joint pdf $f_{R,\Theta}$ of R and Θ is given by

$$f_{R,\Theta}(r, \theta) = \frac{1}{2\pi} e^{-r^2/2} r, \quad \text{for } r \geq 0 \text{ and } \theta \in [0, 2\pi). \quad (8.7)$$

Namely, specifying x and y in terms of r and θ gives

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta. \quad (8.8)$$

The Jacobian of this coordinate transformation is

$$\det \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r. \quad (8.9)$$

The result now follows from the transformation rule (8.4), noting that the joint pdf of X and Y is $f_{X,Y}(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2}$. It is not difficult to verify that R and Θ are independent, that $\Theta \sim \mathcal{U}[0, 2\pi)$ and that $\mathbb{P}(R > r) = e^{-r^2/2}$. This means that R has the same distribution as $\sqrt{V}$, with $V \sim \text{Exp}(1/2)$ — that is, R has a Rayleigh distribution (see Example 6.1). Namely, $\mathbb{P}(\sqrt{V} > v) = \mathbb{P}(V > v^2) = e^{-v^2/2}$. Both Θ and R are easy to generate, and are transformed via (8.8) to independent standard normal random variables.

The Box-Muller Algorithm

Since the inverse probability transform gives $V = -2 \ln U_1$ when $V \sim \text{Exp}(1/2)$ and $U_1 \sim \mathcal{U}(0, 1)$, and $\Theta = 2\pi U_2$ when $U_2 \sim \mathcal{U}(0, 1)$, the polar decomposition above yields a simple algorithm for generating standard normal variables.

Algorithm 8.3.1: Generation of a pair of $\mathcal{N}(0, 1)$ variables (Box–Muller)

1. Generate two independent random variables, U_1 and U_2 , from $\mathcal{U}(0, 1)$.
2. Return two independent standard normal variables, X and Y , via

$$\begin{aligned} X &= (-2 \ln U_1)^{1/2} \cos(2\pi U_2) , \\ Y &= (-2 \ln U_1)^{1/2} \sin(2\pi U_2) . \end{aligned} \tag{8.10}$$

To generate $Z \sim \mathcal{N}(\mu, \sigma^2)$, simply compute $Z = \mu + \sigma X$, where X is a standard normal obtained from the algorithm above. The Box–Muller method is efficient — it transforms two uniform random variables into two independent normal random variables with no rejection step.

8.4 Linear Transformations: Effects on Mean and Covariance

Let $\mathbf{x} = (x_1, \dots, x_n)$ be a (column) vector in $\mathbb{R}^n$ and A an $(m \times n)$ -matrix. The mapping $\mathbf{x} \mapsto \mathbf{z}$, with

$$\mathbf{z} = A\mathbf{x} \tag{8.11}$$

is called a **linear transformation**. Now consider a *random* vector $\mathbf{X} = (X_1, \dots, X_n)$, and let

$$\mathbf{Z} = A\mathbf{X} . \tag{8.12}$$

Then $\mathbf{Z}$ is a random vector in $\mathbb{R}^m$. Again, in principle, if we know the joint distribution of $\mathbf{X}$ then we can derive the joint distribution of $\mathbf{Z}$. Let us first see how the expectation vector and covariance matrix are transformed.

Claim 45: Mean and Covariance Under Linear Maps — If $\mathbf{X}$ has expectation vector $\boldsymbol{\mu}_X$ and covariance matrix Σ_X , then the expectation vector and covariance matrix of $\mathbf{Z} = A\mathbf{X}$ are respectively given by

$$\boldsymbol{\mu}_Z = A\boldsymbol{\mu}_X \tag{8.13}$$

and

$$\Sigma_Z = A \Sigma_X A^\top . \tag{8.14}$$

Proof: We have $\boldsymbol{\mu}_Z = \mathbb{E}\mathbf{Z} = \mathbb{E}A\mathbf{X} = A\mathbb{E}\mathbf{X} = A\boldsymbol{\mu}_X$ and

$$\begin{aligned} \Sigma_Z &= \mathbb{E}(\mathbf{Z} - \boldsymbol{\mu}_Z)(\mathbf{Z} - \boldsymbol{\mu}_Z)^\top = \mathbb{E}A(\mathbf{X} - \boldsymbol{\mu}_X)(A(\mathbf{X} - \boldsymbol{\mu}_X))^\top \\ &= A\mathbb{E}(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^\top A^\top \\ &= A \Sigma_X A^\top \end{aligned}$$

which completes the proof. □

Density of a Linear Transformation

From now on assume A is an *invertible* $(n \times n)$ -matrix. If $\mathbf{X}$ has joint density f_X , what is the joint density f_Z of $\mathbf{Z}$?

Consider Figure 8.1. For any fixed $\mathbf{x}$, let $\mathbf{z} = A\mathbf{x}$. Hence, $\mathbf{x} = A^{-1}\mathbf{z}$. Consider the n -dimensional cube $C = [z_1, z_1 + h] \times \cdots \times [z_n, z_n + h]$. Let D be the *image* of C under A^{-1} , i.e., the parallelepiped of all points $\mathbf{x}$ such that $A\mathbf{x} \in C$. Then,

$$\mathbb{P}(\mathbf{Z} \in C) \approx h^n f_Z(\mathbf{z}). \quad (8.15)$$

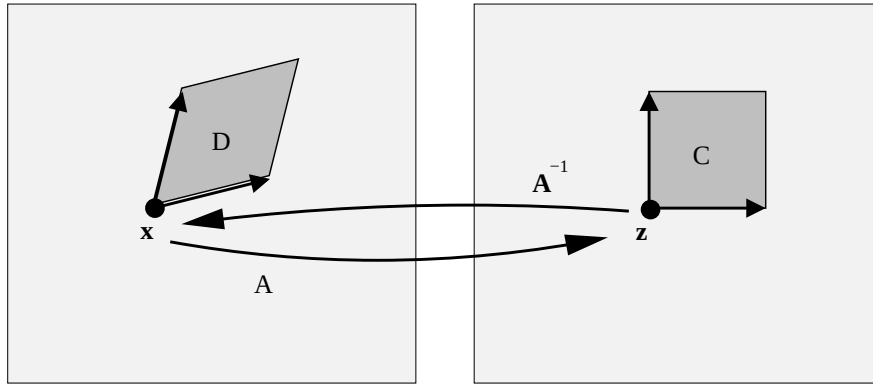


Figure 8.1: Linear transformation

Now recall from linear algebra that any n -dimensional rectangle with “volume” V is transformed into a n -dimensional parallelepiped with volume $V|A|$, where $|A| := |\det(A)|$. Thus,

$$\mathbb{P}(\mathbf{Z} \in C) = \mathbb{P}(\mathbf{X} \in D) \approx h^n |A^{-1}| f_X(\mathbf{x}) = h^n |A|^{-1} f_X(\mathbf{x}) \quad (8.16)$$

Letting h go to 0 we conclude that

$$f_Z(\mathbf{z}) = \frac{f_X(A^{-1}\mathbf{z})}{|A|}, \quad \mathbf{z} \in \mathbb{R}^n. \quad (8.17)$$

Remark (Intuition for the Division by the Determinant). The determinant $|A|$ is the factor by which A scales n -dimensional volumes. If A stretches space by $|A|$, then a small region of volume V in $\mathbf{X}$ -space is mapped to a region of volume $|A|V$ in $\mathbf{Z}$ -space. Since the total probability is unchanged, the same probability mass is now spread over a larger region, so the density must be divided by $|A|$ to compensate. Conversely, if A compresses space ($|A| < 1$), the density increases.

For example, take

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad |A| = 6. \quad (8.18)$$

This stretches the x_1 -axis by 2 and the x_2 -axis by 3, so a unit square in $\mathbf{X}$ -space becomes a 2×3 rectangle in $\mathbf{Z}$ -space with area 6. The density f_Z is therefore $f_X(A^{-1}\mathbf{z})/6$: the same probability mass spread over 6 times the area.

8.5 Generating Multivariate Normal Distributions

In Chapter 7 we showed that the multivariate normal distribution $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ arises by applying an affine transformation to independent standard normals. This provides a practical recipe for simulation: given any target mean vector $\boldsymbol{\mu}$ and positive definite covariance matrix Σ , we can generate samples from $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ in two steps.

The Cholesky Decomposition Method

A matrix Σ admits a **Cholesky decomposition**

$$\Sigma = L L^\top, \quad (8.19)$$

where L is a lower triangular matrix with positive diagonal entries, if and only if Σ is **symmetric positive definite**. Moreover, this decomposition is then unique.

Remark (Positive Semidefiniteness of Covariance Matrices). A covariance matrix is always symmetric positive *semidefinite*—that is, $\mathbf{a}^\top \Sigma \mathbf{a} \geq 0$ for all $\mathbf{a}$ —but need not be positive definite. It is positive definite exactly when no component of $\mathbf{X}$ is an exact affine function of the others. For symmetric positive semidefinite matrices, positive definiteness is equivalent to invertibility: a zero eigenvalue makes Σ both singular and merely semidefinite. When Σ is only positive semidefinite (and hence singular), the Cholesky decomposition does not apply. For example, if $X_1 \sim \mathcal{N}(0, 1)$ and $X_2 = X_1$, then

$$\Sigma = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad (8.20)$$

which is singular ($\det \Sigma = 0$) and admits no Cholesky decomposition.

Now let $\mathbf{X} = (X_1, \dots, X_n)$ be a vector of independent standard normal random variables. Define

$$\mathbf{Z} = \boldsymbol{\mu} + L \mathbf{X}. \quad (8.21)$$

By Claim 45, the expectation vector and covariance matrix of $\mathbf{Z}$ are

$$\mathbb{E}[\mathbf{Z}] = \boldsymbol{\mu} + L \mathbb{E}[\mathbf{X}] = \boldsymbol{\mu}, \quad \Sigma_{\mathbf{Z}} = L I L^\top = L L^\top = \Sigma. \quad (8.22)$$

Since $\mathbf{Z}$ is an affine transformation of jointly normal random variables, $\mathbf{Z} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$.

Example (Generating Bivariate Normal Samples). Suppose we wish to generate $(Z_1, Z_2) \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ with

$$\boldsymbol{\mu} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 4 & 3 \\ 3 & 9 \end{bmatrix}. \quad (8.23)$$

The Cholesky factor is

$$L = \begin{bmatrix} 2 & 0 \\ 3/2 & 3\sqrt{3}/2 \end{bmatrix}, \quad (8.24)$$

which one can verify satisfies $LL^\top = \Sigma$. To generate a sample, draw $X_1, X_2 \sim \mathcal{N}(0, 1)$ independently (e.g. via the Box–Muller method), then compute

$$Z_1 = 1 + 2X_1, \quad Z_2 = 3 + \frac{3}{2}X_1 + \frac{3\sqrt{3}}{2}X_2. \quad (8.25)$$

The correlation between Z_1 and Z_2 is $\rho = 3/\sqrt{4 \cdot 9} = 0.5$, as encoded in the covariance matrix.

Remark. In practice, the Cholesky decomposition is computed numerically. In Python, `numpy.linalg.cholesky` provides L directly from Σ .

8.6 Distributions of Sums, Products, and Ratios of Independent Random Variables

When X and Y are independent continuous random variables with known densities, we can derive the density of various combinations — sums, products, and ratios — using the transformation rule from Section 8.2.

Distribution of a Product

Let X and Y be independent with joint pdf $f_{X,Y}(x, y) = f_X(x)f_Y(y)$. To find the pdf of $Z = XY$, we introduce the auxiliary variable $W = Y$ and consider the transformation $(X, Y) \mapsto (Z, W) = (XY, Y)$. The inverse is $X = Z/W$, $Y = W$, and the Jacobian determinant is $|\partial(x, y)/\partial(z, w)| = 1/|w|$. By the transformation rule,

$$f_{Z,W}(z, w) = f_X(z/w) f_Y(w) \cdot \frac{1}{|w|}. \quad (8.26)$$

Integrating out w gives the **product density**:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z/w) f_Y(w) \frac{1}{|w|} dw. \quad (8.27)$$

Example (Product of Independent Uniforms). Let $X, Y \sim \mathcal{U}(0, 1)$ independently. The density of $Z = XY$ is, for $0 < z < 1$,

$$f_Z(z) = \int_z^1 \frac{1}{w} dw = -\ln z. \quad (8.28)$$

The integrand is $1/w$ only when $0 < z/w < 1$ and $0 < w < 1$, i.e. when $z < w < 1$.

Distribution of a Ratio

Similarly, to find the pdf of $Z = X/Y$, we use $(Z, W) = (X/Y, Y)$, giving $X = ZW$, $Y = W$, with Jacobian $|w|$. The **ratio density** is

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(zw) f_Y(w) |w| dw. \quad (8.29)$$

Example (Ratio of Independent Standard Normals and the Cauchy Distribution). Let $X, Y \sim \mathcal{N}(0, 1)$ independently. The density of $Z = X/Y$ is

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-(z^2 w^2 + w^2)/2} |w| dw = \frac{1}{\pi} \int_0^{\infty} w e^{-w^2(1+z^2)/2} dw = \frac{1}{\pi(1+z^2)}, \quad (8.30)$$

where the last integral evaluates to $1/(1+z^2)$. This is the **Cauchy(0, 1)** density. The ratio of two independent standard normals is Cauchy-distributed — providing an alternative derivation of the result mentioned in Section 6.4.

8.7 Convolutions

In the previous section we saw formulas for the density of products and ratios. The most important case — the density of a *sum* — deserves its own treatment, and leads to the concept of *convolution*.

Sum of Two Random Variables

We can find the density of the sum of two random variables with a general joint pdf or pmf $f_{X,Y}$. Let us consider the jointly continuous case only, so X and Y have joint pdf $f_{X,Y}$ and we are interested in finding the pdf of $Z = X + Y$. By conditioning on X , we have (see (7.29)):

$$f_Z(z) = \int_{-\infty}^{\infty} f_{Z|X}(z|x) f_X(x) dx. \quad (8.31)$$

Since $Z = X + Y$, we have

$$\begin{aligned} f_{Z|X}(z|x) &= \lim_{\delta \rightarrow 0} \frac{\mathbb{P}(z \leq Z \leq z + \delta | X = x)}{\delta} \\ &= \lim_{\delta \rightarrow 0} \frac{\mathbb{P}(z - x \leq Y \leq z - x + \delta | X = x)}{\delta} = f_{Y|X}(z - x | x). \end{aligned} \quad (8.32)$$

Hence, we have found

$$f_Z(z) = \int_{-\infty}^{\infty} f_{Y|X}(z - x | x) f_X(x) dx. \quad (8.33)$$

In particular, when X and Y are *independent*, we have

$$f_Z(z) = \int_{-\infty}^{\infty} f_Y(z - x) f_X(x) dx = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y) dy, \quad (8.34)$$

where second equation follows from swapping the roles of X and Y . We say that f_Z is the **convolution** of f_X and f_Y and write this as

$$f_Z = f_X \star f_Y \quad (= f_Y \star f_X). \quad (8.35)$$

Example: Sum of Two Uniforms

Let X and Y be independent and both $\mathcal{U}(0, 1)$ distributed. What does the pdf of $Z = X + Y$ look like? Note that Z takes values in $(0, 2)$. The following Python code draws 10,000 times from the distribution of Z and plots a histogram of the data (Figure 8.2):

LLM Prompt > Write Python code that generates 10,000 samples of $Z = U_1 + U_2$ where U_1, U_2 are independent $U(0, 1)$ and plots a histogram, revealing the triangular shape.

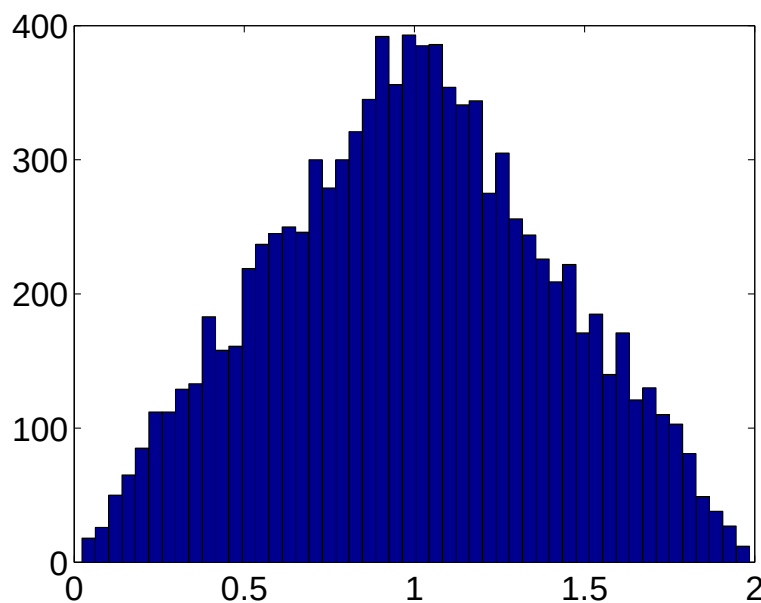
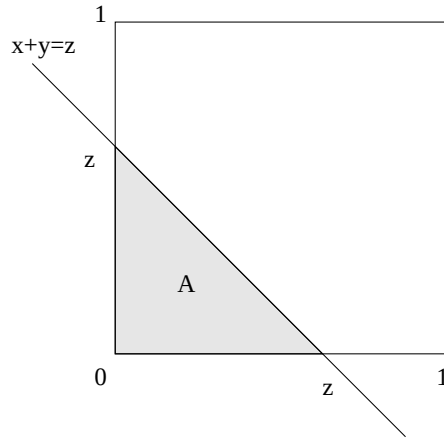


Figure 8.2: Histogram for adding two uniform random variables

This looks remarkably like a triangle. Perhaps the true pdf of $Z = X + Y$ has a triangular shape? This is indeed easily proved. Namely, first observe that the pdf of Z must be symmetrical around 1. Thus to find the pdf, it suffices to find its form for $z \in [0, 1]$. Take such a z and consider the event $\{Z \leq z\}$, which corresponds to (X, Y) lying in the shaded region A of Figure 8.3.

Figure 8.3: The random point (X, Y) must lie in set A

We thus have:

$$F_Z(z) = \mathbb{P}(Z \leq z) = \mathbb{P}((X, Y) \in A) = \iint_A f(x, y) \, dx \, dy = \text{area}(A) = \frac{1}{2} z^2. \quad (8.36)$$

where we have used the fact that the joint density $f(x, y)$ is equal to 1 on the square $[0, 1] \times [0, 1]$. By differentiating the cdf F_Z we get the pdf f_Z

$$f_Z(z) = z, \quad z \in [0, 1], \quad (8.37)$$

and by symmetry

$$f_Z(z) = 2 - z, \quad z \in [1, 2], \quad (8.38)$$

which is indeed a triangular density. If we rescaled the histogram such that the total area under the bars would be 1, the fit with the true distribution would be very good.

Sum of Uniforms via Convolution

Let us derive the pdf of Z for the sum of two independent $\mathcal{U}(0, 1)$ variables via convolution. For $z \in [0, 1]$, we have

$$f_Y(z - x) = \begin{cases} 1 & \text{if } 0 \leq x \leq z \\ 0 & \text{otherwise.} \end{cases} \quad (8.39)$$

Hence,

$$f_Z(z) = \int_{-\infty}^{\infty} f_Y(z - x) f_X(x) \, dx = \int_0^z f_Y(z - x) \, dx = \int_0^z 1 \, dx = z. \quad (8.40)$$

Similarly, for $z \in [1, 2]$, we have

$$f_Y(z - x) = \begin{cases} 1 & \text{if } z - 1 \leq x \leq 1 \\ 0 & \text{otherwise,} \end{cases} \quad (8.41)$$

so that in this case

$$f_Z(z) = \int_{-\infty}^{\infty} f_Y(z - x) f_X(x) \, dx = \int_0^1 f_Y(z - x) \, dx = \int_{z-1}^1 1 \, dx = 2 - z. \quad (8.42)$$

Sum of Two Independent Exponentials

Example (Sum of Two Independent Exponentials). Let $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(\lambda)$ be independent, both with rate λ . The convolution gives, for $z > 0$:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda(z-x)} \cdot \lambda e^{-\lambda x} dx = \lambda^2 e^{-\lambda z} \int_0^z dx = \lambda^2 z e^{-\lambda z}. \quad (8.43)$$

This is the $\text{Gamma}(2, \lambda)$ density, confirming the result from Chapter 7 that the sum of n independent $\text{Exp}(\lambda)$ random variables has a $\text{Gamma}(n, \lambda)$ distribution.

Remark. While convolution provides a direct way to derive the density of a sum, it often involves tedious integral calculations. In Chapter 9 we will see that moment generating functions offer a much simpler approach: the MGF of a sum of independent random variables is the *product* of the individual MGFs, and the density can then be identified by recognising the resulting MGF.

8.8 Distributions of Maxima, Minima, and Order Statistics

Given a random sample $X_1, \dots, X_n$ from some distribution, we often want to know the distribution of the largest or smallest observation, or more generally the k -th smallest. These are called **order statistics**.

Maximum and Minimum

Example (Maximum and Minimum). Suppose $X_1, \dots, X_n$ are independent and have cdf F . Let $Y = \min(X_1, \dots, X_n)$ and $Z = \max(X_1, \dots, X_n)$. The cdf of Y and Z are easily obtained. First, note that the maximum of the $\{X_i\}$ is less than some number z if and only if *all* X_i are less than z . Thus,

$$\mathbb{P}(Z \leq z) = \mathbb{P}(X_1 \leq z, X_2 \leq z, \dots, X_n \leq z) = \mathbb{P}(X_1 \leq z) \mathbb{P}(X_2 \leq z) \cdots \mathbb{P}(X_n \leq z), \quad (8.44)$$

where the second equation follows from the independence assumption. It follows that

$$F_Z(z) = (F(z))^n. \quad (8.45)$$

Similarly,

$$\mathbb{P}(Y > y) = \mathbb{P}(X_1 > y, X_2 > y, \dots, X_n > y) = \mathbb{P}(X_1 > y) \mathbb{P}(X_2 > y) \cdots \mathbb{P}(X_n > y), \quad (8.46)$$

so that

$$F_Y(y) = 1 - (1 - F(y))^n. \quad (8.47)$$

If the X_i are continuous with common pdf f , we can differentiate to obtain the densities:

$$f_Z(z) = n (F(z))^{n-1} f(z), \quad f_Y(y) = n (1 - F(y))^{n-1} f(y). \quad (8.48)$$

Example (Maximum of Uniform Random Variables). Let $X_1, \dots, X_n \sim \mathcal{U}(0, 1)$ independently. Then $F(x) = x$ for $x \in [0, 1]$, and the maximum $Z = \max(X_1, \dots, X_n)$ has cdf $F_Z(z) = z^n$ and pdf

$$f_Z(z) = n z^{n-1}, \quad 0 \leq z \leq 1. \quad (8.49)$$

This is the $\text{Beta}(n, 1)$ density (recall Section 5.5). Its mean is $\mathbb{E}[Z] = n/(n+1)$, which approaches 1 as $n \rightarrow \infty$ — the maximum of many uniform samples concentrates near the upper boundary.

Example (Minimum of Exponential Random Variables). Let $X_1, \dots, X_n \sim \text{Exp}(\lambda)$ independently. Then $1 - F(y) = e^{-\lambda y}$, and the minimum $Y = \min(X_1, \dots, X_n)$ has

$$\mathbb{P}(Y > y) = (e^{-\lambda y})^n = e^{-n\lambda y}. \quad (8.50)$$

Thus $Y \sim \text{Exp}(n\lambda)$: the minimum of n independent exponentials with rate λ is again exponential, but with rate $n\lambda$. This has a natural interpretation: if n machines each fail independently at rate λ , the time until the first failure has rate $n\lambda$.

General Order Statistics

Let $X_1, \dots, X_n$ be iid with cdf F and pdf f . The **order statistics** are the values arranged in increasing order:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}, \quad (8.51)$$

so that $X_{(1)} = \min(X_1, \dots, X_n)$ and $X_{(n)} = \max(X_1, \dots, X_n)$.

Claim 46: Density of the k -th Order Statistic — The pdf of the k -th order statistic $X_{(k)}$ is

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} (F(x))^{k-1} (1 - F(x))^{n-k} f(x). \quad (8.52)$$

Proof: For $X_{(k)}$ to lie in a small interval $[x, x+h)$, we need exactly $k-1$ of the n observations below x , exactly one in $[x, x+h)$, and $n-k$ above $x+h$. By the multinomial distribution with probabilities $F(x)$, $f(x)h$, and $1 - F(x) - f(x)h$:

$$\mathbb{P}(x \leq X_{(k)} < x+h) \approx \frac{n!}{(k-1)! 1! (n-k)!} (F(x))^{k-1} f(x) h (1 - F(x))^{n-k}. \quad (8.53)$$

Dividing by h and letting $h \rightarrow 0$ gives the result. $\square$

Setting $k = 1$ recovers the minimum density and $k = n$ recovers the maximum density derived above.

Example (Median of Five Uniform Observations). Let $X_1, \dots, X_5 \sim \mathcal{U}(0, 1)$ independently. The median is $X_{(3)}$. Its density is

$$f_{X_{(3)}}(x) = \frac{5!}{2! 2!} x^2 (1-x)^2 = 30 x^2 (1-x)^2, \quad 0 \leq x \leq 1. \quad (8.54)$$

This is the **Beta(3, 3)** density (see Section 5.5). Its mean is $\mathbb{E}[X_{(3)}] = 1/2$ and its variance is $\text{Var}(X_{(3)}) = 1/28$. The sample median is thus centred at the population median, with a standard deviation of about 0.189.

MOMENT GENERATING FUNCTIONS, RELATED TRANSFORMS, AND EXAMPLES

In this chapter we introduce probability generating functions and moment generating functions — transforms that encode a distribution and turn convolutions into simple algebra. We develop their properties, show how to extract probabilities and moments, and apply them to sums of independent random variables and a branching process model.

9.1 Probability and Moment Generating Functions

Many calculations and manipulations involving probability distributions are facilitated by the use of *transforms*. A transform encodes the entire distribution of a random variable into a single function, in such a way that many operations — computing moments, summing independent random variables, identifying distributions — become algebraically simple. In this section we define the two most important transforms: the probability generating function and the moment generating function.

The Probability Generating Function

Definition : Probability Generating Function — Let X be a *non-negative* and *integer-valued* random variable. The **probability generating function** (PGF) of X is the function $G : [-1, 1] \rightarrow \mathbb{R}$ defined by

$$G(z) := \mathbb{E} z^X = \sum_{x=0}^{\infty} z^x \mathbb{P}(X = x) . \quad (9.1)$$

The series converges absolutely for all $|z| \leq 1$, so G is well-defined on $[-1, 1]$ and infinitely differentiable on $(-1, 1)$. We illustrate the PGF for several of the discrete distributions introduced in Chapter 3. At $z = 1$, derivatives are interpreted as left-hand limits (see [7]). Note that $G(1) = 1$

always (since the probabilities sum to 1), and $G(0) = \mathbb{P}(X = 0)$. The PGF is a power series whose coefficients are exactly the probabilities $\mathbb{P}(X = x)$.

Example (PGF of the Bernoulli distribution). Let $X \sim \text{Ber}(p)$. Then $\mathbb{P}(X = 0) = 1 - p$ and $\mathbb{P}(X = 1) = p$, so

$$G(z) = (1 - p)z^0 + pz^1 = 1 - p + pz. \quad (9.2)$$

Example (PGF of the Binomial distribution). Let $X \sim \text{Bin}(n, p)$. Since X can be written as the sum of n independent $\text{Ber}(p)$ random variables, and the PGF of a sum of independent random variables is the product of their PGFs (a fact we will establish in Section 9.5), we get

$$G(z) = (1 - p + pz)^n. \quad (9.3)$$

One can also verify this directly:

$$G(z) = \sum_{x=0}^n \binom{n}{x} p^x (1 - p)^{n-x} z^x = \sum_{x=0}^n \binom{n}{x} (pz)^x (1 - p)^{n-x} = (1 - p + pz)^n, \quad (9.4)$$

by the binomial theorem.

Example (PGF of the Poisson distribution). Let $X \sim \text{Poi}(\lambda)$, so $\mathbb{P}(X = x) = e^{-\lambda} \lambda^x / x!$ for $x = 0, 1, 2, \dots$. The PGF is

$$G(z) = \sum_{x=0}^{\infty} z^x e^{-\lambda} \frac{\lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(z\lambda)^x}{x!} = e^{-\lambda} e^{z\lambda} = e^{-\lambda(1-z)}.$$

Example (PGF of the Geometric distribution). Let $X \sim \text{Geom}(p)$ with support $\{1, 2, 3, \dots\}$, so $\mathbb{P}(X = x) = (1 - p)^{x-1} p$. Then

$$G(z) = \sum_{x=1}^{\infty} (1 - p)^{x-1} p z^x = pz \sum_{k=0}^{\infty} ((1 - p)z)^k = \frac{pz}{1 - (1 - p)z}, \quad (9.5)$$

where the geometric series converges for $|z| \leq 1$ since $0 < 1 - p < 1$.

The Moment Generating Function

Definition : Moment Generating Function — The **moment generating function** (MGF) of a random variable X is the function $M : I \rightarrow [0, \infty)$ given by

$$M(s) = \mathbb{E} e^{sX} . \quad (9.6)$$

Here I is an open interval containing 0 for which the expectation is finite for all $s \in I$. In particular, for a discrete random variable with pmf f ,

$$M(s) = \sum_x e^{sx} f(x), \quad (9.7)$$

and for a continuous random variable with pdf f ,

$$M(s) = \int_{-\infty}^{\infty} e^{sx} f(x) dx . \quad (9.8)$$

Note that $M(0) = 1$ always. The MGF is defined for both discrete and continuous random variables, whereas the PGF is only defined for non-negative integer-valued random variables.

Example (MGF of the Exponential distribution). Let $X \sim \text{Exp}(\lambda)$, so $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Then, for $s < \lambda$,

$$M(s) = \int_0^{\infty} e^{sx} \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} e^{-(\lambda-s)x} dx = \frac{\lambda}{\lambda - s} . \quad (9.9)$$

Example (MGF of the Normal distribution). Let $Z \sim \mathcal{N}(0, 1)$. Then

$$M(s) = \int_{-\infty}^{\infty} e^{sz} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z^2-2sz)/2} dz . \quad (9.10)$$

Completing the square: $z^2 - 2sz = (z - s)^2 - s^2$, so

$$M(s) = e^{s^2/2} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z-s)^2/2} dz = e^{s^2/2} , \quad (9.11)$$

since the remaining integral is 1 (it is the integral of a $\mathcal{N}(s, 1)$ density). More generally, if $X \sim \mathcal{N}(\mu, \sigma^2)$, then $X = \mu + \sigma Z$, so

$$M_X(s) = \mathbb{E}[e^{s(\mu + \sigma Z)}] = e^{s\mu} M_Z(s\sigma) = e^{s\mu + s^2\sigma^2/2} . \quad (9.12)$$

Example (MGF of the Gamma distribution). Let $X \sim \text{Gamma}(\alpha, \lambda)$. Then, for $s < \lambda$,

$$M(s) = \mathbb{E} e^{sX} = \int_0^{\infty} e^{sx} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx = \left(\frac{\lambda}{\lambda - s} \right)^\alpha . \quad (9.13)$$

This is obtained by writing $e^{sx} e^{-\lambda x} = e^{-(\lambda-s)x}$ and recognising the remaining integral as a $\text{Gamma}(\alpha, \lambda - s)$ density integrated over its support, giving $(\lambda - s)^\alpha / \lambda^\alpha$ in the denominator. As a check, setting $\alpha = 1$ recovers the MGF of the $\text{Exp}(\lambda)$ distribution.

Uniqueness

The most important theoretical property of both transforms is that they *uniquely determine* the distribution.

Claim 47: Uniqueness of the PGF — Two non-negative integer-valued random variables have the same distribution if and only if they have the same PGF.

Claim 48: Uniqueness of the MGF — If the MGF $M(s)$ exists (i.e., is finite) for all s in some open interval containing 0, then M uniquely determines the distribution of X . That is, two random variables with the same MGF have the same distribution.

These uniqueness results are powerful: to show that a random variable has a particular distribution, it suffices to compute its transform and recognise it as the transform of that distribution. We will use this strategy repeatedly.

9.2 Other Transforms: Laplace-Stieltjes and Characteristic Functions

The PGF and MGF are the most commonly used transforms in this course, but there are several related transforms that appear in more advanced work. We briefly survey them here.

The Laplace-Stieltjes Transform

For a *non-negative* random variable X (not necessarily integer-valued), the **Laplace-Stieltjes transform** is defined as

$$\widehat{F}(s) := \mathbb{E}[e^{-sX}], \quad s \geq 0. \quad (9.14)$$

Note that this is simply the MGF evaluated at $-s$: $\widehat{F}(s) = M(-s)$. The Laplace-Stieltjes transform is preferred in some areas (e.g., queueing theory and reliability) because it is always well-defined for $s \geq 0$ when $X \geq 0$, whereas the MGF $M(s) = \mathbb{E}[e^{sX}]$ may be infinite for $s > 0$ if X has a heavy right tail.

The Characteristic Function

For a random variable X that can take *any* real value (positive or negative), the MGF may not exist at all (e.g., for the Cauchy distribution). The **characteristic function** avoids this problem by using

complex exponentials:

$$\varphi(\omega) := \mathbb{E}[e^{i\omega X}], \quad \omega \in \mathbb{R}, \quad (9.15)$$

where $i = \sqrt{-1}$. Since $|e^{i\omega x}| = 1$ for all real x and ω , the expectation always exists. When X has a pdf f , the characteristic function is the *Fourier transform* of f :

$$\varphi(\omega) = \int_{-\infty}^{\infty} e^{i\omega x} f(x) dx. \quad (9.16)$$

The characteristic function shares the key properties of the MGF: it uniquely determines the distribution, and moments can be recovered via

$$\mathbb{E}[X^n] = \frac{1}{i^n} \varphi^{(n)}(0). \quad (9.17)$$

The characteristic function is the transform of choice in advanced probability theory (e.g., in the proof of the central limit theorem), but we will not use it further in this course.

Summary. The following table summarises the transforms and their domains of applicability:

Transform	Definition	Applies to
PGF	$G(z) = \mathbb{E}[z^X]$	Non-negative integer X
MGF	$M(s) = \mathbb{E}[e^{sX}]$	Any X (if it exists)
Laplace-Stieltjes	$\widehat{F}(s) = \mathbb{E}[e^{-sX}]$	Non-negative X
Characteristic fn.	$\varphi(\omega) = \mathbb{E}[e^{i\omega X}]$	Any X (always exists)

9.3 Basic Properties of Generating Functions

The PGF encodes the entire pmf of a non-negative integer-valued random variable. In this section we show how to extract probabilities and factorial moments from it.

Retrieving Probabilities from the PGF

Knowing only the PGF of X , we can recover the pmf:

$$\mathbb{P}(X = x) = \left. \frac{1}{x!} \frac{d^x}{dz^x} G(z) \right|_{z=0}, \quad x = 0, 1, 2, \dots \quad (9.18)$$

Proof: By definition

$$G(z) = \mathbb{P}(X = 0) + z\mathbb{P}(X = 1) + z^2\mathbb{P}(X = 2) + \dots \quad (9.19)$$

Substituting $z = 0$ gives $G(0) = \mathbb{P}(X = 0)$. Differentiating once:

$$G'(z) = \mathbb{P}(X = 1) + 2z\mathbb{P}(X = 2) + 3z^2\mathbb{P}(X = 3) + \dots \quad (9.20)$$

Thus, $G'(0) = \mathbb{P}(X = 1)$. Differentiating again, $G''(0) = 2\mathbb{P}(X = 2)$, and in general $G^{(x)}(0) = x! \mathbb{P}(X = x)$. $\square$

Example (Recovering the Poisson pmf from its PGF). The PGF of $X \sim \text{Poi}(\lambda)$ is $G(z) = e^{-\lambda(1-z)}$. The x -th derivative is

$$G^{(x)}(z) = \lambda^x e^{-\lambda(1-z)}. \quad (9.21)$$

Setting $z = 0$: $G^{(x)}(0) = \lambda^x e^{-\lambda}$. Dividing by $x!$:

$$\mathbb{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad (9.22)$$

which is indeed the Poisson pmf.

Factorial Moments from the PGF

The PGF also provides a convenient route to the moments of X , via the so-called **factorial moments**. Differentiating $G(z) = \mathbb{E}[z^X]$ with respect to z :

$$\begin{aligned} G'(z) &= \mathbb{E}[Xz^{X-1}], \\ G''(z) &= \mathbb{E}[X(X-1)z^{X-2}], \\ G'''(z) &= \mathbb{E}[X(X-1)(X-2)z^{X-3}], \end{aligned}$$

et cetera. Evaluating at $z = 1$ gives the **factorial moments**:

$$G^{(k)}(1) = \mathbb{E}[X(X-1)(X-2)\cdots(X-k+1)], \quad k = 1, 2, 3, \dots \quad (9.23)$$

In particular, the first two factorial moments yield the mean and variance:

$$\boxed{\mathbb{E}X = G'(1)}, \quad (9.24)$$

and, since $\mathbb{E}[X(X-1)] = \mathbb{E}[X^2] - \mathbb{E}[X]$, we have $\mathbb{E}[X^2] = G''(1) + G'(1)$, so

$$\boxed{\mathbb{V}\text{ar}(X) = G''(1) + G'(1) - (G'(1))^2}. \quad (9.25)$$

Example (Mean and variance of $\text{Poi}(\lambda)$ via the PGF). From $G(z) = e^{-\lambda(1-z)}$ we compute

$$G'(z) = \lambda e^{-\lambda(1-z)}, \quad G''(z) = \lambda^2 e^{-\lambda(1-z)}. \quad (9.26)$$

Evaluating at $z = 1$: $G'(1) = \lambda$ and $G''(1) = \lambda^2$. Therefore,

$$\mathbb{E}[X] = \lambda, \quad \mathbb{V}\text{ar}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda. \quad (9.27)$$

Example (Mean and variance of $\text{Bin}(n, p)$ via the PGF). From $G(z) = (1 - p + pz)^n$ we compute

$$G'(z) = np(1 - p + pz)^{n-1}, \quad G''(z) = n(n-1)p^2(1 - p + pz)^{n-2}. \quad (9.28)$$

Evaluating at $z = 1$: $G'(1) = np$ and $G''(1) = n(n-1)p^2$. Therefore,

$$\mathbb{E}[X] = np, \quad \mathbb{V}\text{ar}(X) = n(n-1)p^2 + np - (np)^2 = np(1 - p). \quad (9.29)$$

Example (Mean and variance of $\text{Geom}(p)$ via the PGF). From $G(z) = pz/(1 - (1 - p)z)$ we compute, using the quotient rule,

$$G'(z) = \frac{p}{(1 - (1 - p)z)^2}. \quad (9.30)$$

Evaluating at $z = 1$: $G'(1) = p/p^2 = 1/p$. Similarly, $G''(1) = 2(1 - p)/p^2$. Therefore,

$$\mathbb{E}[X] = \frac{1}{p}, \quad \mathbb{V}\text{ar}(X) = \frac{2(1 - p)}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1 - p}{p^2}. \quad (9.31)$$

9.4 Moments from the Moment Generating Function

The moment generating function earns its name: the moments of X are obtained simply by differentiating M and evaluating at $s = 0$.

Moments from the MGF

If $\mathbb{E}[|X|^n] < \infty$, then M is n times differentiable at $s = 0$, and

$$\boxed{\mathbb{E}[X^n] = M^{(n)}(0).} \quad (9.32)$$

To see why, note that $e^{sX} = \sum_{k=0}^{\infty} (sX)^k/k!$, so (formally) $M(s) = \sum_{k=0}^{\infty} s^k \mathbb{E}[X^k]/k!$. Differentiating n times and setting $s = 0$ picks out the coefficient $\mathbb{E}[X^n]$.

As a consequence, the mean and variance of X are

$$\mathbb{E}[X] = M'(0), \quad \mathbb{V}\text{ar}(X) = M''(0) - (M'(0))^2. \quad (9.33)$$

Example (Mean and variance of $\text{Exp}(\lambda)$ via the MGF). From Example 9.1, $M(s) = \lambda/(\lambda - s)$ for $s < \lambda$. Then

$$M'(s) = \frac{\lambda}{(\lambda - s)^2}, \quad M''(s) = \frac{2\lambda}{(\lambda - s)^3}. \quad (9.34)$$

Evaluating at $s = 0$: $M'(0) = 1/\lambda$ and $M''(0) = 2/\lambda^2$. Therefore,

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \mathbb{V}\text{ar}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}. \quad (9.35)$$

Example (Mean and variance of $\mathcal{N}(\mu, \sigma^2)$ via the MGF). From Example 9.1, $M(s) = e^{s\mu + s^2\sigma^2/2}$. Then

$$M'(s) = (\mu + s\sigma^2) M(s), \quad M''(s) = [\sigma^2 + (\mu + s\sigma^2)^2] M(s). \quad (9.36)$$

Evaluating at $s = 0$ (where $M(0) = 1$): $M'(0) = \mu$ and $M''(0) = \sigma^2 + \mu^2$. Therefore,

$$\mathbb{E}[X] = \mu, \quad \mathbb{V}\text{ar}(X) = \sigma^2 + \mu^2 - \mu^2 = \sigma^2. \quad (9.37)$$

Example (Mean and variance of $\text{Gamma}(\alpha, \lambda)$ via the MGF). From Example 9.1, $M(s) = (\lambda/(\lambda-s))^\alpha$ for $s < \lambda$. Then

$$M'(s) = \frac{\alpha\lambda^\alpha}{(\lambda-s)^{\alpha+1}}, \quad M''(s) = \frac{\alpha(\alpha+1)\lambda^\alpha}{(\lambda-s)^{\alpha+2}}. \quad (9.38)$$

Evaluating at $s = 0$: $M'(0) = \alpha/\lambda$ and $M''(0) = \alpha(\alpha+1)/\lambda^2$. Therefore,

$$\mathbb{E}[X] = \frac{\alpha}{\lambda}, \quad \mathbb{V}\text{ar}(X) = \frac{\alpha(\alpha+1)}{\lambda^2} - \frac{\alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}. \quad (9.39)$$

As a check, setting $\alpha = 1$ recovers $\mathbb{E}[X] = 1/\lambda$ and $\mathbb{V}\text{ar}(X) = 1/\lambda^2$ for the $\text{Exp}(\lambda)$ distribution.

9.5 Transforms and Convolution Simplification

In Chapter 8 we saw that the distribution of a sum $X + Y$ of independent random variables can be found by *convolving* their distributions — an operation that involves summing (or integrating) over all ways the individual values can add up to a given total. Convolutions are conceptually simple but computationally tedious: they require nested sums or integrals that quickly become unwieldy.

Transforms turn convolutions into *multiplications*. This is their single most powerful application.

The Product Rule for PGFs

Claim 49: PGF of a sum of independent random variables — Let $X_1, X_2, \dots, X_n$ be *independent* non-negative integer-valued random variables with PGFs $G_{X_1}, G_{X_2}, \dots, G_{X_n}$. Then the PGF of $S_n = X_1 + X_2 + \dots + X_n$ is

$$G_{S_n}(z) = G_{X_1}(z) G_{X_2}(z) \cdots G_{X_n}(z). \quad (9.40)$$

Proof: By definition, $G_{S_n}(z) = \mathbb{E}[z^{S_n}] = \mathbb{E}[z^{X_1+X_2+\dots+X_n}] = \mathbb{E}[z^{X_1} z^{X_2} \cdots z^{X_n}]$. Since $X_1, \dots, X_n$ are independent, the expectation of a product equals the product of expectations (Claim 37):

$$G_{S_n}(z) = \mathbb{E}[z^{X_1}] \mathbb{E}[z^{X_2}] \cdots \mathbb{E}[z^{X_n}] = G_{X_1}(z) G_{X_2}(z) \cdots G_{X_n}(z). \quad (9.41)$$

□

The Product Rule for MGFs

Claim 50: MGF of a sum of independent random variables — Let $X_1, X_2, \dots, X_n$ be *independent* random variables (not necessarily integer-valued) with MGFs $M_{X_1}, M_{X_2}, \dots, M_{X_n}$. Then the MGF of $S_n = X_1 + X_2 + \dots + X_n$ is

$$M_{S_n}(s) = M_{X_1}(s) M_{X_2}(s) \cdots M_{X_n}(s). \quad (9.42)$$

Proof: By definition, $M_{S_n}(s) = \mathbb{E}[e^{sS_n}] = \mathbb{E}[e^{s(X_1 + \dots + X_n)}] = \mathbb{E}[e^{sX_1} e^{sX_2} \dots e^{sX_n}]$. Since $X_1, \dots, X_n$ are independent:

$$M_{S_n}(s) = \mathbb{E}[e^{sX_1}] \mathbb{E}[e^{sX_2}] \dots \mathbb{E}[e^{sX_n}] = M_{X_1}(s) M_{X_2}(s) \dots M_{X_n}(s). \quad (9.43)$$

□

Remark. For iid random variables $X_1, \dots, X_n$ (all with the same PGF G or MGF M), the results simplify to $G_{S_n}(z) = [G(z)]^n$ and $M_{S_n}(s) = [M(s)]^n$.

The strategy for finding the distribution of S_n is now:

1. Compute the transform of S_n using the product rule.
2. Recognise the result as the transform of a known distribution.
3. Conclude (by the uniqueness of the transform) that S_n has that distribution.

Continuous Examples

Example (Sum of iid exponentials is Gamma). Let $X_1, \dots, X_n$ be iid $\text{Exp}(\lambda)$ random variables. From Example 9.1, each has MGF $M(s) = \lambda/(\lambda - s)$. By the product rule,

$$M_{S_n}(s) = \left(\frac{\lambda}{\lambda - s} \right)^n. \quad (9.44)$$

From Example 9.1, this is exactly the MGF of the $\text{Gamma}(n, \lambda)$ distribution. By uniqueness,

$$\boxed{X_1 + \dots + X_n \sim \text{Gamma}(n, \lambda).} \quad (9.45)$$

This recovers, in one line, the result that the sum of n independent exponential waiting times has a Gamma distribution (recall Section 5.5) — a fact that required considerably more effort to establish via convolution. More generally, the same argument shows that if $X_1 \sim \text{Gamma}(\alpha_1, \lambda)$ and $X_2 \sim \text{Gamma}(\alpha_2, \lambda)$ are independent (same rate), then $X_1 + X_2 \sim \text{Gamma}(\alpha_1 + \alpha_2, \lambda)$: the shape parameters add.

Example (Sum of independent normals is normal). Let $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ be independent. From Example 9.1, their MGFs are $e^{s\mu_1 + s^2\sigma_1^2/2}$ and $e^{s\mu_2 + s^2\sigma_2^2/2}$, so

$$M_{X_1+X_2}(s) = e^{s\mu_1 + s^2\sigma_1^2/2} \cdot e^{s\mu_2 + s^2\sigma_2^2/2} = e^{s(\mu_1 + \mu_2) + s^2(\sigma_1^2 + \sigma_2^2)/2}, \quad (9.46)$$

which is the MGF of $\mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. Hence

$$\boxed{X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).} \quad (9.47)$$

The normal family is *closed under addition*: the sum of independent normals is again normal, with means and variances that simply add.

Discrete Examples

Example (Sum of independent Poisson random variables). Let $X \sim \text{Poi}(\lambda)$ and $Y \sim \text{Poi}(\mu)$ be independent. Their PGFs are $e^{-\lambda(1-z)}$ and $e^{-\mu(1-z)}$, so

$$G_{X+Y}(z) = e^{-\lambda(1-z)} \cdot e^{-\mu(1-z)} = e^{-(\lambda+\mu)(1-z)}, \quad (9.48)$$

which is the PGF of $\text{Poi}(\lambda + \mu)$. Hence

$$\boxed{X + Y \sim \text{Poi}(\lambda + \mu)}. \quad (9.49)$$

The Poisson family is closed under addition: the rates simply add.

Example (Sum of iid Bernoullis is Binomial). Let $X_1, \dots, X_n$ be iid $\text{Ber}(p)$ random variables. Each has PGF $G(z) = 1 - p + pz$. By the product rule,

$$G_{S_n}(z) = (1 - p + pz)^n, \quad (9.50)$$

which is the PGF of $\text{Bin}(n, p)$. Hence $S_n = X_1 + \dots + X_n \sim \text{Bin}(n, p)$. This gives a clean proof that the binomial distribution counts the number of successes in n independent Bernoulli trials. Similarly, if $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ are independent, then $X + Y \sim \text{Bin}(n + m, p)$: the number-of-trials parameters add.

Example (Sum of independent Negative Binomials). Let $X \sim \text{NegBin}(r_1, p)$ and $Y \sim \text{NegBin}(r_2, p)$ be independent, where $\text{NegBin}(r, p)$ counts the number of failures before the r -th success. The PGF of $\text{NegBin}(r, p)$ is

$$G(z) = \left(\frac{p}{1 - (1-p)z} \right)^r. \quad (9.51)$$

By the product rule,

$$G_{X+Y}(z) = \left(\frac{p}{1 - (1-p)z} \right)^{r_1+r_2}, \quad (9.52)$$

which is the PGF of $\text{NegBin}(r_1 + r_2, p)$. Hence $X + Y \sim \text{NegBin}(r_1 + r_2, p)$: the number-of-successes parameters add. This makes intuitive sense — if you wait for r_1 successes and then wait for r_2 more, the total number of failures has a $\text{NegBin}(r_1 + r_2, p)$ distribution.

9.6 Branching Process Example Using Probability Generating Functions

The **Galton–Watson branching process** is a simple model of population growth that provides a beautiful application of probability generating functions and conditional expectation.

The Model

A population evolves in discrete generations. Generation 0 consists of a single individual (the “ancestor”). Each individual in generation n independently produces a random number of offspring

according to a common distribution with pmf

$$p_k = \mathbb{P}(\text{an individual has } k \text{ offspring}), \quad k = 0, 1, 2, \dots \quad (9.53)$$

These offspring form generation $n + 1$. Let Z_n denote the population size in generation n , so $Z_0 = 1$.

The offspring distribution has PGF

$$G(z) = \sum_{k=0}^{\infty} p_k z^k = \mathbb{E}[z^X], \quad (9.54)$$

where X is the number of offspring of a single individual. We write $\mu = \mathbb{E}[X] = G'(1)$ for the mean number of offspring per individual.

Mean Population Size via Conditional Expectation

We can find $\mathbb{E}[Z_{n+1}]$ using the tower property (Claim 42). Given $Z_n = m$, generation $n + 1$ consists of the offspring of m independent individuals, each producing a random number of offspring with mean μ . Therefore,

$$\mathbb{E}[Z_{n+1} | Z_n = m] = m\mu, \quad (9.55)$$

so $\mathbb{E}[Z_{n+1} | Z_n] = \mu Z_n$. Applying the tower property:

$$\mathbb{E}[Z_{n+1}] = \mathbb{E}[\mathbb{E}[Z_{n+1} | Z_n]] = \mathbb{E}[\mu Z_n] = \mu \mathbb{E}[Z_n]. \quad (9.56)$$

Since $Z_0 = 1$, iterating gives

$$\boxed{\mathbb{E}[Z_n] = \mu^n}. \quad (9.57)$$

The mean population grows geometrically if $\mu > 1$, stays constant if $\mu = 1$, and decays to 0 if $\mu < 1$.

Figure 9.1 illustrates the difference between the *supercritical* ($\mu > 1$) and *subcritical* ($\mu < 1$) regimes via simulation of a binary splitting process (where each individual either dies or produces two offspring). In the supercritical case some realisations explode while others go extinct; in the subcritical case all realisations quickly die out.

Simulating a binary splitting branching process

LLM Prompt > Write Python code that simulates 20 realisations of a binary splitting branching process over 15 generations for both a supercritical ($p = 0.7$) and subcritical ($p = 0.4$) case. Plot them side by side with the theoretical mean μ^n as a dashed line.

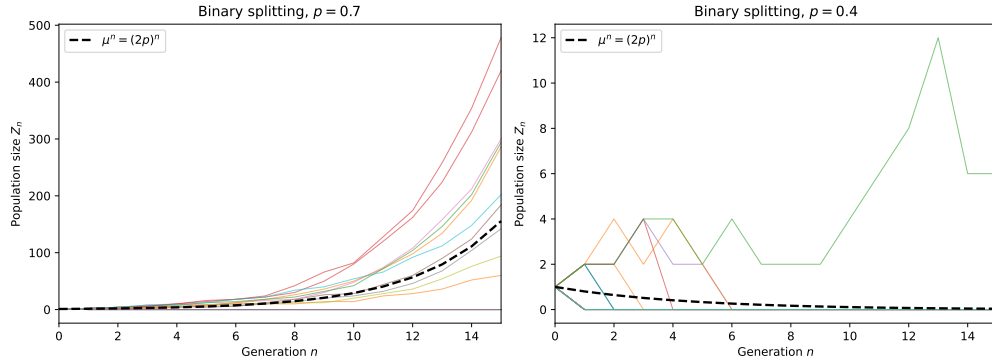


Figure 9.1: Twenty simulated realisations of a binary splitting branching process. Left: supercritical case ($p = 0.7$, $\mu = 1.4$) — some paths survive and grow, others go extinct. The dashed line shows the mean $\mathbb{E}[Z_n] = \mu^n$. Right: subcritical case ($p = 0.4$, $\mu = 0.8$) — all paths quickly die out.

The PGF of Z_n

Let $G_n(z) = \mathbb{E}[z^{Z_n}]$ be the PGF of Z_n . We claim that G_n is obtained by *iterating* G :

$$G_n(z) = \underbrace{G(G(\cdots G(z) \cdots))}_{n \text{ times}}. \quad (9.58)$$

We write $G_n = G \circ G \circ \cdots \circ G$ (n -fold composition).

Proof: We proceed by induction. The base case $n = 1$ is clear: $Z_1 = X$ (the offspring of the single ancestor), so $G_1(z) = G(z)$.

For the inductive step, suppose $G_n(z)$ is the n -fold composition. In generation $n + 1$, each of the Z_n individuals independently produces offspring. If $Z_n = m$, then $Z_{n+1} = X_1 + X_2 + \cdots + X_m$, where $X_1, \dots, X_m$ are iid with PGF G . By the product rule (Claim 49),

$$\mathbb{E}[z^{Z_{n+1}} | Z_n = m] = [G(z)]^m. \quad (9.59)$$

Therefore,

$$G_{n+1}(z) = \mathbb{E}[z^{Z_{n+1}}] = \mathbb{E}[\mathbb{E}[z^{Z_{n+1}} | Z_n]] = \mathbb{E}[G(z)^{Z_n}] = G_n(G(z)), \quad (9.60)$$

where the last step uses the definition of G_n . By the inductive hypothesis, $G_n(G(z))$ is the $(n + 1)$ -fold composition of G . $\square$

Extinction Probability

A key question is: what is the probability that the population eventually dies out? Define the **extinction probability**

$$q = \mathbb{P}(Z_n = 0 \text{ for some } n \geq 1). \quad (9.61)$$

Since $\{Z_n = 0\}$ implies $\{Z_{n+1} = 0\}$ (an extinct population stays extinct), the events are nested and $\mathbb{P}(Z_n = 0) \uparrow q$ as $n \rightarrow \infty$.

Now, $\mathbb{P}(Z_n = 0) = G_n(0)$. Taking the limit:

$$q = \lim_{n \rightarrow \infty} G_n(0). \quad (9.62)$$

One can show that q is the *smallest non-negative solution* of the fixed-point equation

$$G(z) = z. \quad (9.63)$$

The key result is:

- If $\mu \leq 1$: extinction is certain ($q = 1$), unless $p_1 = 1$ (every individual has exactly one offspring).
- If $\mu > 1$: there is a positive probability of survival ($q < 1$). The extinction probability q is the unique solution of $G(z) = z$ in $[0, 1)$.

The graphical interpretation is simple: plot $G(z)$ and the diagonal $y = z$ on $[0, 1]$. The extinction probability is the *leftmost* intersection point. When $\mu \leq 1$, the curve $G(z)$ lies above the diagonal for all $z < 1$ and meets it only at $z = 1$. When $\mu > 1$, the curve crosses the diagonal at some $q < 1$. Figure 9.2 illustrates this for the binary splitting and Poisson offspring distributions.

Graphical solution of the extinction equation $G(z) = z$

LLM Prompt > Write Python code that plots the PGF $G(z)$ and the diagonal $y = z$ on $[0, 1]$ for (a) a binary splitting process with several values of p , and (b) a Poisson offspring distribution with several values of λ , to illustrate the graphical solution of the extinction equation $G(z) = z$.

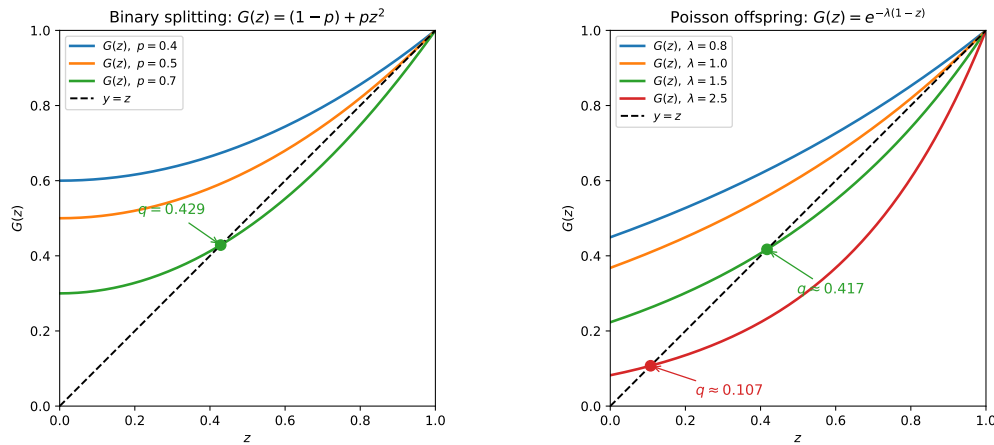


Figure 9.2: Graphical solution of the fixed-point equation $G(z) = z$. The extinction probability q is the smallest intersection of $G(z)$ with the diagonal. Left: binary splitting — for $p \leq 0.5$ ($\mu \leq 1$) the only intersection is $z = 1$; for $p = 0.7$ ($\mu = 1.4$) the curve crosses at $q = 3/7 \approx 0.429$. Right: Poisson offspring — for $\lambda \leq 1$ the only intersection is $z = 1$; for $\lambda > 1$ a second intersection appears at $q < 1$.

Examples

Example (Binary splitting). Each individual either dies (with probability $1 - p$) or splits into two offspring (with probability p). Thus $p_0 = 1 - p$, $p_2 = p$, and all other $p_k = 0$. The offspring PGF is

$$G(z) = (1 - p) + pz^2, \quad (9.64)$$

with mean offspring $\mu = G'(1) = 2p$, so $\mathbb{E}[Z_n] = (2p)^n$. We solve $G(z) = z$ for the extinction probability:

$$(1 - p) + pz^2 = z \implies pz^2 - z + (1 - p) = 0. \quad (9.65)$$

By the quadratic formula,

$$z = \frac{1 \pm \sqrt{1 - 4p(1 - p)}}{2p} = \frac{1 \pm \sqrt{(2p - 1)^2}}{2p} = \frac{1 \pm |2p - 1|}{2p}. \quad (9.66)$$

The two roots are $z = 1$ and $z = (1 - p)/p$.

If $p \leq 1/2$: $(1 - p)/p \geq 1$, so the smallest non-negative root in $[0, 1]$ is $q = 1$. Extinction is certain.

If $p > 1/2$: $(1 - p)/p < 1$, so $q = (1 - p)/p < 1$. There is a positive probability of survival: $1 - q = (2p - 1)/p$.

For example, with $p = 0.7$: $q = 0.3/0.7 = 3/7 \approx 0.429$, and the population survives with probability $4/7 \approx 0.571$.

Example (Poisson offspring). Each individual produces $\text{Poi}(\lambda)$ offspring. The PGF is $G(z) = e^{-\lambda(1-z)}$, and $\mu = \lambda$, so $\mathbb{E}[Z_n] = \lambda^n$.

PGF iteration. $G_2(z) = G(G(z)) = e^{-\lambda(1-e^{-\lambda(1-z)})}$. Even for $n = 2$ the explicit form is complicated, illustrating the power of the PGF framework: we can reason about G_n without needing a closed form.

Extinction probability. We solve $e^{-\lambda(1-z)} = z$. For $\lambda \leq 1$ the only solution in $[0, 1]$ is $z = 1$, so $q = 1$. For $\lambda > 1$ there is a unique solution $q \in (0, 1)$, which must be found numerically. The following code computes q by iterating G :

Extinction probability for Poisson branching process

LLM Prompt > Write Python code that numerically computes the extinction probability of a Poisson branching process by iterating $z \mapsto \exp(-\lambda(1 - z))$ from $z = 0$ until convergence, for several values of λ .

Example (Geometric offspring). Each individual produces k offspring with probability $p_k = (1 - p)p^k$ for $k = 0, 1, 2, \dots$ (a geometric distribution on $\{0, 1, 2, \dots\}$). The PGF is

$$G(z) = \frac{1 - p}{1 - pz}, \quad (9.67)$$

with $\mu = p/(1 - p)$. Solving $G(z) = z$ gives $pz^2 - z + (1 - p) = 0$ — the same quadratic as the binary splitting example! The roots are $z = 1$ and $z = (1 - p)/p$, so $q = 1$ if $p \leq 1/2$ and $q = (1 - p)/p$ if $p > 1/2$.

RANDOM SAMPLES AND LIMIT THEOREMS

In this chapter we study statistics computed from random samples — the sample mean and sample variance, their distributions, and their unbiasedness. We then establish the laws of large numbers and the central limit theorem, which together describe the behaviour of averages and sums for large samples.

10.1 Random Samples and Statistics

In follow-up courses on statistics, the focus shifts to **statistical inference**: given observed data, what can we learn about the underlying process that generated it? In the most basic setup one has an iid sample $X_1, X_2, \dots, X_n$ from some density f (or probability mass function in the discrete case). If f depends on unknown parameters θ , the goal of inference is to estimate or test hypotheses about θ . That is not our concern here. Instead, we study the *distributional properties* of summary functions of the sample — these properties form the probabilistic foundation on which all inference methods rest.

Definition : Random Sample — We call $X_1, X_2, \dots, X_n$ a **random sample** of size n from a distribution with density (or pmf) f if the X_i are independent and identically distributed (iid), each with density f .

A **statistic** is any function $T: \mathbb{R}^n \rightarrow \mathbb{R}$ that maps the sample $(X_1, \dots, X_n)$ to a real number. Since a statistic is a function of random variables, it is itself a random variable whose distribution is determined by the joint distribution of the sample. The two most important statistics are the

sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad (10.1)$$

and the **sample variance**

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2. \quad (10.2)$$

Throughout this chapter we assume that each X_i has mean $\mu = \mathbb{E}[X_i]$ and variance $\sigma^2 = \mathbb{V}\text{ar}(X_i)$ (recall Claim 8 and Claim 23 for the definitions of expectation in the discrete and continuous cases), both finite. The distribution does *not* have to be normal.

Claim 51: Mean and Variance of the Sample Mean — Let $X_1, \dots, X_n$ be iid with mean μ and variance σ^2 . Then

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \mathbb{V}\text{ar}(\bar{X}_n) = \frac{\sigma^2}{n}. \quad (10.3)$$

Proof: By linearity of expectation,

$$\mathbb{E}[\bar{X}_n] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mu. \quad (10.4)$$

For the variance, since the X_i are independent,

$$\mathbb{V}\text{ar}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}\text{ar}(X_i) = \frac{\sigma^2}{n}. \quad (10.5)$$

□

So the sample mean has the same expectation as each individual observation, but its variance shrinks with n — the estimate becomes more precise as the sample grows.

Claim 52: Expectation of the Sample Variance — Let $X_1, \dots, X_n$ be iid with mean μ and variance σ^2 . Then

$$\mathbb{E}[S^2] = \sigma^2. \quad (10.6)$$

Proof: Expand the sum of squares:

$$\sum_{i=1}^n (X_i - \bar{X}_n)^2 = \sum_{i=1}^n X_i^2 - n\bar{X}_n^2. \quad (10.7)$$

Taking expectations, using $\mathbb{E}[X_i^2] = \sigma^2 + \mu^2$ and $\mathbb{E}[\bar{X}_n^2] = \sigma^2/n + \mu^2$:

$$\mathbb{E}\left[\sum_{i=1}^n X_i^2 - n\bar{X}_n^2\right] = n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) = (n-1)\sigma^2. \quad (10.8)$$

Dividing by $n-1$ gives $\mathbb{E}[S^2] = \sigma^2$.

□

In mathematical statistics, an estimator whose expectation equals the parameter it targets is called **unbiased**. The results above show that $\bar{X}_n$ is unbiased for μ and S^2 is unbiased for σ^2 . The divisor $n - 1$ (rather than n) in S^2 is precisely what achieves this.

The following code verifies these results via Monte Carlo simulation using an $\text{Exp}(2)$ distribution, which has $\mu = 1/2$ and $\sigma^2 = 1/4$.

Monte Carlo: properties of sample mean and variance

LLM Prompt > Write Python code that draws 50,000 samples of size $n = 20$ from an $\text{Exp}(2)$ distribution, computes the sample mean $\bar{X}$ and sample variance S^2 for each, and compares their Monte Carlo averages with the theoretical values. Plot histograms of both statistics.

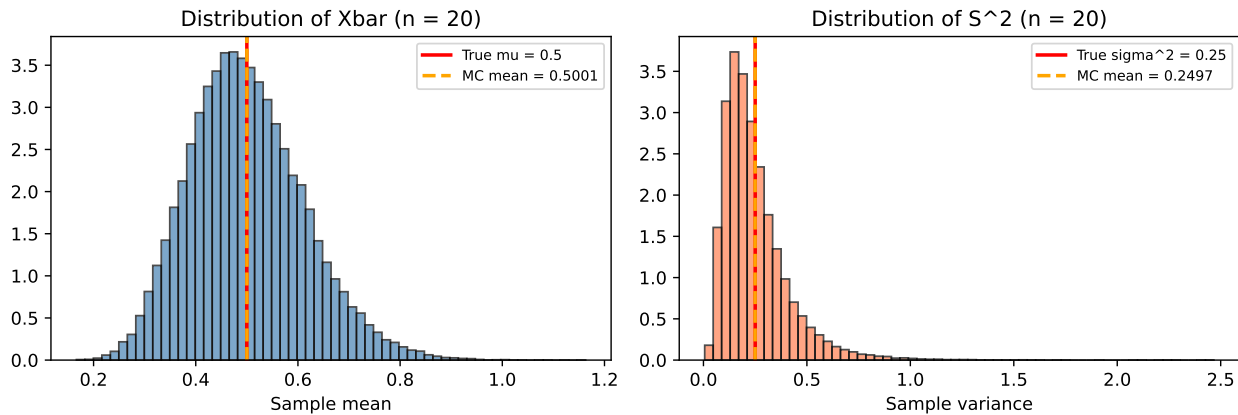


Figure 10.1: Monte Carlo verification of sample mean and variance properties for $N = 50,000$ samples of size $n = 20$ from an $\text{Exp}(2)$ distribution.

10.2 Properties of Normal Samples

When the underlying distribution is normal, we can determine the *exact* distributions of the sample mean and sample variance. These results form the backbone of classical methods such as t -tests and confidence intervals.

Distribution of the Sample Mean

Claim 53: Distribution of the Sample Mean for Normal Populations — If $X_1, \dots, X_n$ are iid $\mathcal{N}(\mu, \sigma^2)$, then

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right). \quad (10.9)$$

Proof: Write $\bar{X}_n = \frac{1}{n}(X_1 + \cdots + X_n)$. Since the X_i are independent normals, their sum $S_n = X_1 + \cdots + X_n$ is normal by Claim 41. A scalar multiple of a normal random variable is again normal, so $\bar{X}_n = S_n/n$ is normal. Its mean and variance are $\mathbb{E}[\bar{X}_n] = \mu$ and $\text{Var}(\bar{X}_n) = \sigma^2/n$ by Claim 51. $\square$

The Chi-Squared Distribution

To describe the distribution of the sample variance from a normal sample, we first introduce the chi-squared distribution.

Definition : Chi-Squared Distribution — If $Z_1, \dots, Z_k$ are iid $\mathcal{N}(0, 1)$, then

$$Q = Z_1^2 + Z_2^2 + \cdots + Z_k^2 \quad (10.10)$$

has a **chi-squared distribution with k degrees of freedom**, written $Q \sim \chi^2(k)$. In Example 6.1 we showed that a single $Z^2 \sim \chi^2_1$; the definition here generalises this to a sum of k squared normals. Its density is

$$f_Q(x) = \frac{1}{2^{k/2} \Gamma(k/2)} x^{k/2-1} e^{-x/2}, \quad x > 0. \quad (10.11)$$

This is a $\text{Gamma}(k/2, 1/2)$ distribution (rate parametrisation; see Section 5.5).

The chi-squared distribution has $\mathbb{E}[Q] = k$ and $\text{Var}(Q) = 2k$.

Distribution of the Sample Variance

Claim 54: Sample Variance Distribution — Let $X_1, \dots, X_n$ be iid $\mathcal{N}(\mu, \sigma^2)$. Then:

1. $\bar{X}_n$ and S^2 are **independent**.
2. $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$.

The proof uses orthogonal projections in $\mathbb{R}^n$ and is beyond the scope of this course. The loss of one degree of freedom reflects the constraint $\sum_{i=1}^n (X_i - \bar{X}_n) = 0$: only $n-1$ of the deviations are free.

Part (2) together with the properties of the χ^2 distribution gives $\mathbb{E}[S^2] = \sigma^2$ (consistent with Claim 52) and $\text{Var}(S^2) = 2\sigma^4/(n-1)$.

The t -Distribution and the t -Statistic

Definition : Student's t -Distribution — If $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi^2(k)$ are independent, then

$$T = \frac{Z}{\sqrt{V/k}} \quad (10.12)$$

has a **Student's t -distribution with k degrees of freedom**, written $T \sim t(k)$. Its density is

$$f_T(t) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi} \Gamma\left(\frac{k}{2}\right)} \left(1 + \frac{t^2}{k}\right)^{-(k+1)/2}, \quad t \in \mathbb{R}. \quad (10.13)$$

The t -distribution is symmetric about 0 and bell-shaped, but with heavier tails than the normal. The special case $k = 1$ gives the Cauchy(0, 1) distribution (see Section 6.4), which has such heavy tails that its mean does not even exist. As k increases, the tails become lighter and the $t(k)$ distribution converges to $\mathcal{N}(0, 1)$. In this sense, the t -distribution interpolates between two extremes: the very heavy-tailed Cauchy at $k = 1$ and the light-tailed normal at $k = \infty$.

Figure 10.2 illustrates this by plotting the 0.975 quantile of $t(k)$ as a function of k . At $k = 1$ (Cauchy) the quantile is approximately 12.7; it drops rapidly and approaches the standard normal value of 1.96 from above.

The 0.975 quantile of Student's t -distribution vs. degrees of freedom

LLM Prompt > Write Python code that estimates the 0.975 quantile of Student's t -distribution for degrees of freedom $k = 1, \dots, 60$ via Monte Carlo, and plots it alongside the Cauchy ($k = 1$) and standard normal reference values to show convergence.

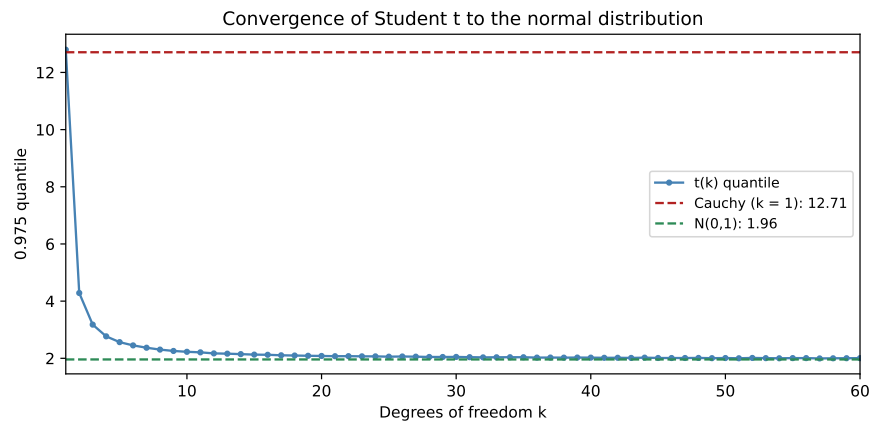


Figure 10.2: The 0.975 quantile of $t(k)$ as a function of the degrees of freedom k , interpolating between the Cauchy ($k = 1$) and normal ($k \rightarrow \infty$) extremes.

Now consider the *t*-statistic formed from a normal random sample:

$$T = \frac{\bar{X}_n - \mu}{S / \sqrt{n}}. \quad (10.14)$$

We can rewrite this as

$$T = \frac{(\bar{X}_n - \mu) / (\sigma / \sqrt{n})}{S / \sigma} = \frac{Z}{\sqrt{V/(n-1)}}, \quad (10.15)$$

where $Z = \frac{\bar{X}_n - \mu}{\sigma / \sqrt{n}} \sim \mathcal{N}(0, 1)$ and $V = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$. By Claim 54, $\bar{X}_n$ and S^2 are independent, so Z and V are independent. Therefore,

$$\boxed{\frac{\bar{X}_n - \mu}{S / \sqrt{n}} \sim t(n-1)}. \quad (10.16)$$

Remark. The *t*-statistic is the basis for the well-known *t*-test and *t*-based confidence intervals used throughout applied statistics. These methods exploit the fact that the *t*-statistic has a known distribution even when σ is unknown (since σ cancels in the ratio). We do not pursue inference further in this course, but the distributional result above is the key ingredient.

The following code verifies the chi-squared and *t*-distribution results via Monte Carlo for a $\mathcal{N}(5, 4)$ sample of size $n = 10$.

Monte Carlo: chi-squared and t-distribution for normal samples

LLM Prompt > Write Python code that draws 50,000 normal samples of size $n = 10$ from $\mathcal{N}(5, 4)$, computes the chi-squared statistic $(n-1)S^2/\sigma^2$ and the *t*-statistic, and compares their histograms with the theoretical $\chi^2(9)$ and $t(9)$ densities (implemented manually using the gamma function).

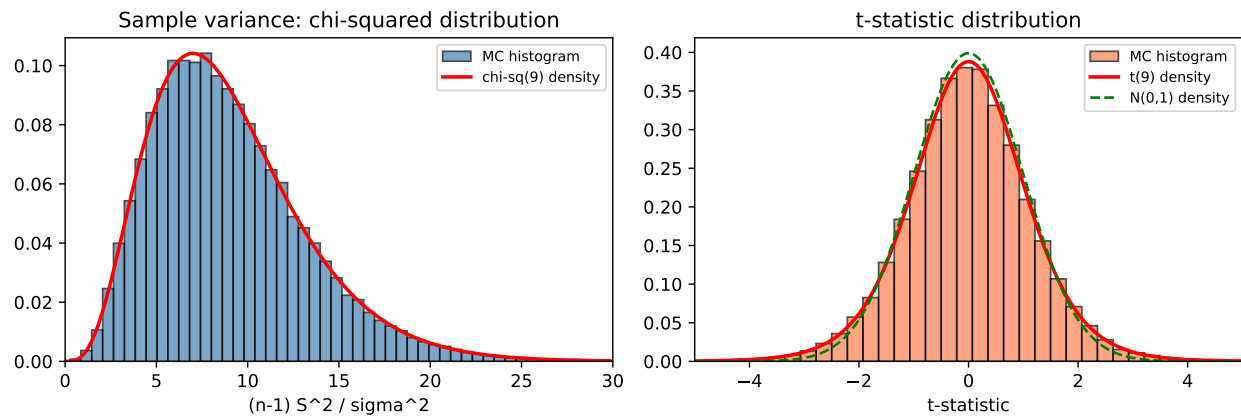


Figure 10.3: Monte Carlo verification of the χ^2 and *t*-distribution results for a $\mathcal{N}(5, 4)$ sample of size $n = 10$.

The F-Distribution

Definition : The F-Distribution — If $U \sim \chi^2(d_1)$ and $V \sim \chi^2(d_2)$ are independent, then

$$F = \frac{U/d_1}{V/d_2} \quad (10.17)$$

has an **F-distribution with d_1 and d_2 degrees of freedom**, written $F \sim F(d_1, d_2)$. Its density is

$$f(x) = \frac{\Gamma\left(\frac{d_1+d_2}{2}\right)}{\Gamma\left(\frac{d_1}{2}\right)\Gamma\left(\frac{d_2}{2}\right)} \left(\frac{d_1}{d_2}\right)^{d_1/2} \frac{x^{d_1/2-1}}{\left(1 + \frac{d_1}{d_2}x\right)^{(d_1+d_2)/2}}, \quad x > 0. \quad (10.18)$$

The F-distribution has $\mathbb{E}[F] = d_2/(d_2 - 2)$ for $d_2 > 2$. If $X_1, \dots, X_m$ are iid $\mathcal{N}(\mu_X, \sigma_X^2)$ and $Y_1, \dots, Y_n$ are iid $\mathcal{N}(\mu_Y, \sigma_Y^2)$, independent of the X_i 's, then the ratio of sample variances satisfies

$$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F(m-1, n-1). \quad (10.19)$$

This result is central to *analysis of variance* (ANOVA) and hypothesis tests in linear models, where one tests whether two populations share the same variance, or more generally whether a model explains a significant portion of the variability in the data. These methods are covered in detail in other courses.

F-distribution densities for several choices of d_1 and d_2

LLM Prompt > Plot the F-distribution density for $(d_1, d_2) \in \{(5, 20), (10, 10), (20, 50), (50, 50)\}$ on a single figure, implementing the density manually using the gamma function.

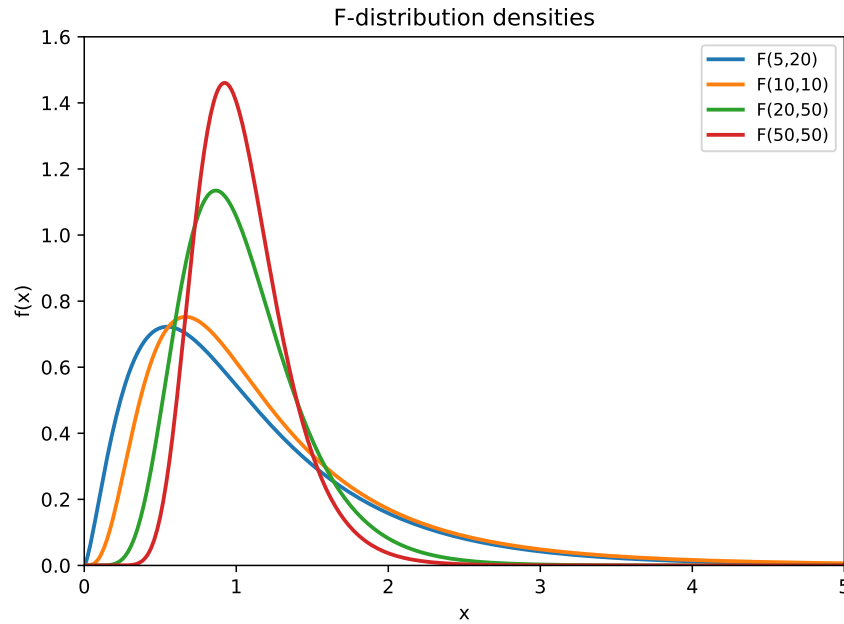


Figure 10.4: The $F(d_1, d_2)$ density for several choices of degrees of freedom.

10.3 The Laws of Large Numbers

Let $X_1, X_2, \dots$ be an iid sequence with mean μ and let $S_n = X_1 + \dots + X_n$. The law of large numbers roughly states that $\bar{X}_n = S_n/n$ is close to μ for large n . There are two versions: the *weak* law and the *strong* law. Both guarantee convergence of the sample mean to the population mean, but they differ in the *mode of convergence* — a distinction explored in more advanced probability courses.

An important special case arises when the X_i are indicator random variables of an event A , so that $X_i \sim \text{Ber}(p)$ with $p = \mathbb{P}(A)$. Then S_n counts the number of times A occurs in n independent trials, and the sample mean

$$\bar{X}_n = \frac{S_n}{n} \quad (10.20)$$

is the **sample proportion** — the fraction of trials in which A occurred. Since $\mu = \mathbb{E}[X_i] = p$, the law of large numbers tells us that this proportion converges to the true probability p as $n \rightarrow \infty$. This is precisely the frequentist justification for interpreting probability as a long-run relative frequency.

The Weak Law

Claim 55: Weak Law of Large Numbers — If $X_1, \dots, X_n$ are independent and identically distributed with expectation μ and finite variance σ^2 , then for all $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{S_n}{n} - \mu \right| \geq \varepsilon \right) = 0. \quad (10.21)$$

Proof: Apply the Markov inequality (Claim 31) to the nonnegative random variable $Z = (S_n/n - \mu)^2$ with $a = \varepsilon^2$:

$$\mathbb{P}((S_n/n - \mu)^2 \geq \varepsilon^2) \leq \frac{\mathbb{E}(S_n/n - \mu)^2}{\varepsilon^2} \quad (10.22)$$

The left-hand side of the above equation can also be written as $\mathbb{P}(|S_n/n - \mu| \geq \varepsilon)$, and the right-hand side is equal to the variance of S_n/n , which is σ^2/n . Combining gives,

$$\mathbb{P}(|S_n/n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}, \quad (10.23)$$

which is precisely Chebyshev's inequality (Claim 32) applied to $\bar{X}_n$, for any $\varepsilon > 0$. As $n \rightarrow \infty$ the quotient $\frac{\sigma^2}{n\varepsilon^2}$ tends to zero, and therefore $\mathbb{P}(|S_n/n - \mu| \geq \varepsilon)$ goes to zero as well. $\square$

The weak law is a statement about **convergence in probability**: for any fixed tolerance $\varepsilon > 0$, the probability that $\bar{X}_n$ deviates from μ by more than ε vanishes as $n \rightarrow \infty$. However, this does not by itself guarantee that the *entire sequence* $\bar{X}_1, \bar{X}_2, \dots$ settles down to μ .

The Strong Law

The **strong law of large numbers** makes a stronger assertion.

Claim 56: Strong Law of Large Numbers — If $X_1, X_2, \dots$ are iid with $\mathbb{E}[X_i] = \mu$, then

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mu\right) = 1. \quad (10.24)$$

This means that the set of outcomes ω for which $\frac{S_n(\omega)}{n} \rightarrow \mu$ has probability one. Intuitively, if we were to run a computer simulation, then every simulated path of $\bar{X}_1, \bar{X}_2, \dots$ would converge to μ . The proof is considerably harder than that of the weak law and is omitted.

Convergence in Probability vs. Almost Sure Convergence

The two laws illustrate two fundamental *modes of convergence* in probability theory:

- The weak law says $\bar{X}_n \rightarrow \mu$ **in probability**: for every $\varepsilon > 0$, $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \rightarrow 0$.
- The strong law says $\bar{X}_n \rightarrow \mu$ **almost surely** (a.s.): with probability 1, the entire sequence converges.

Almost sure convergence implies convergence in probability, but not conversely. More advanced probability courses study these and other modes of convergence (such as convergence in distribution and convergence in L^p) systematically.

10.4 The Central Limit Theorem and Its Applications

The Central Limit Theorem says something about the approximate *distribution* of S_n (or $\bar{X}_n$). Roughly it says this:

*The sum of a large number of iid random variables has approximately a **normal** distribution.*

Here is a more precise statement.

Claim 57: Central Limit Theorem — If $X_1, \dots, X_n$ are independent and identically distributed with expectation μ and variance $\sigma^2 < \infty$, then for all $x \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{S_n - n\mu}{\sigma \sqrt{n}} \leq x \right) = \Phi(x), \quad (10.25)$$

where Φ is the cdf of the standard normal distribution.

The CLT from Two Perspectives

The CLT can be stated equivalently in terms of the sum S_n or the sample mean $\bar{X}_n$.

Sum perspective. The standardised sum $(S_n - n\mu)/(\sigma \sqrt{n})$ (recall the idea of standardisation from Claim 25) converges in distribution to $\mathcal{N}(0, 1)$. This means that for large n ,

$$S_n \approx \mathcal{N}(n\mu, n\sigma^2). \quad (10.26)$$

This form is natural when we are interested in the *total* of n iid contributions.

Sample mean perspective. Dividing by n , the standardised sample mean $(\bar{X}_n - \mu)/(\sigma/\sqrt{n})$ converges in distribution to $\mathcal{N}(0, 1)$. Equivalently, for large n ,

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right). \quad (10.27)$$

This form is natural when we are interested in the *average* of n iid observations.

Note that the distribution of X_i does not have to be normal, or even continuous — the CLT applies to *any* iid sequence with finite variance. The requirement $\sigma^2 < \infty$ is essential: for the Cauchy distribution (Section 6.4), which has no finite variance, the sample mean does not converge to a normal distribution no matter how large n is.

Visualising the CLT: Sums

To see the CLT in action consider Figure 10.5. The first picture shows the pdf's of $S_1, \dots, S_4$ for the case where the X_i have a $\mathcal{U}[0, 1]$ distribution. The second shows the same, but now for an $\text{Exp}(1)$ distribution. We clearly see convergence to a bell shaped curve.

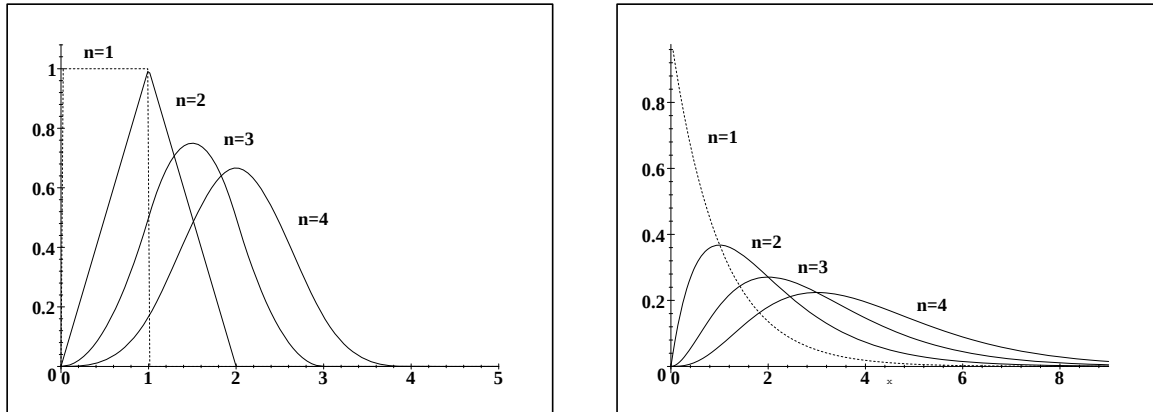


Figure 10.5: Illustration of the CLT for the uniform and exponential distribution

Visualising the CLT: Sample Means

Figure 10.6 shows the same phenomenon from the sample mean perspective. As n increases, the distribution of $\bar{X}_n$ concentrates around μ and becomes increasingly well approximated by $\mathcal{N}(\mu, \sigma^2/n)$ (dashed curve).

CLT for the sample mean: uniform and exponential distributions

LLM Prompt > Write Python code that illustrates the Central Limit Theorem by plotting overlaid histograms of $\bar{X}_n$ for $n = 1, 2, 5, 30$ for both $U[0, 1]$ and $\text{Exp}(1)$ distributions, with the CLT normal approximation for $n = 30$ overlaid as a dashed curve.

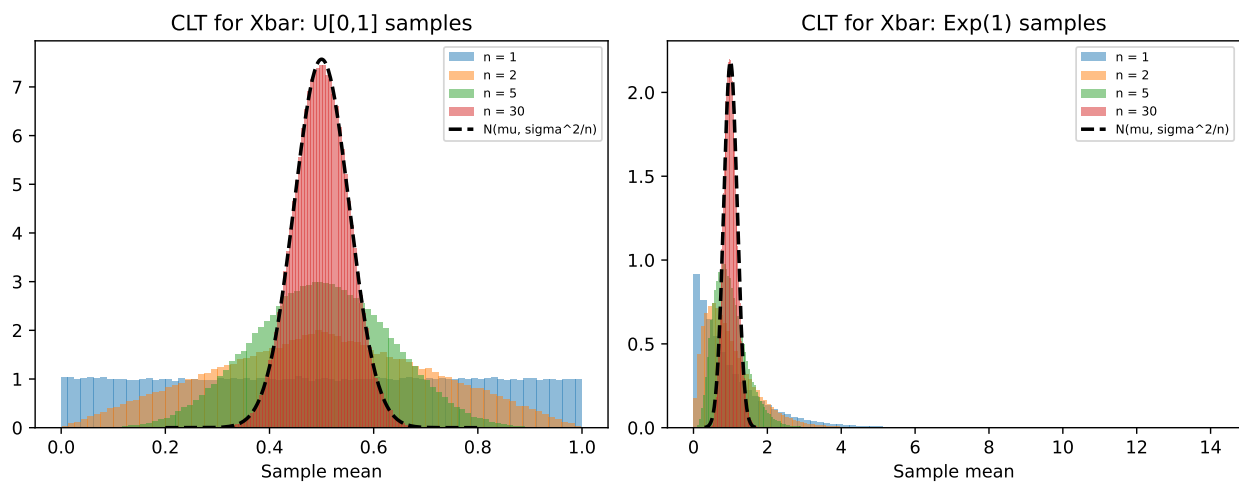


Figure 10.6: Distribution of the sample mean $\bar{X}_n$ for $n = 1, 2, 5, 30$, with the CLT normal approximation (dashed) for $n = 30$. Left: $\mathcal{U}[0, 1]$. Right: $\text{Exp}(1)$.

Normal Approximation to the Binomial

An important application of the CLT is to the binomial distribution. If $X \sim \text{Bin}(n, p)$, then $X = X_1 + \cdots + X_n$ where $X_i \sim \text{Ber}(p)$ are iid. Applying the CLT to this sum:

Claim 58: Normal Approximation to the Binomial — Let $X \sim \text{Bin}(n, p)$. For large n ,

$$\mathbb{P}(X \leq k) \approx \Phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right), \quad (10.28)$$

or equivalently, $X \approx \mathcal{N}(np, np(1-p))$. As a rule of thumb, the approximation is accurate if both np and $n(1-p)$ are larger than 5.

Figure 10.7 compares the $\text{Bin}(30, 0.4)$ pmf with its normal approximation $\mathcal{N}(12, 7.2)$.

Normal approximation to the binomial

LLM Prompt > Write Python code that plots the $\text{Bin}(30, 0.4)$ PMF as a bar chart alongside the normal approximation $\mathcal{N}(12, 7.2)$ density curve to illustrate the CLT approximation to the binomial.

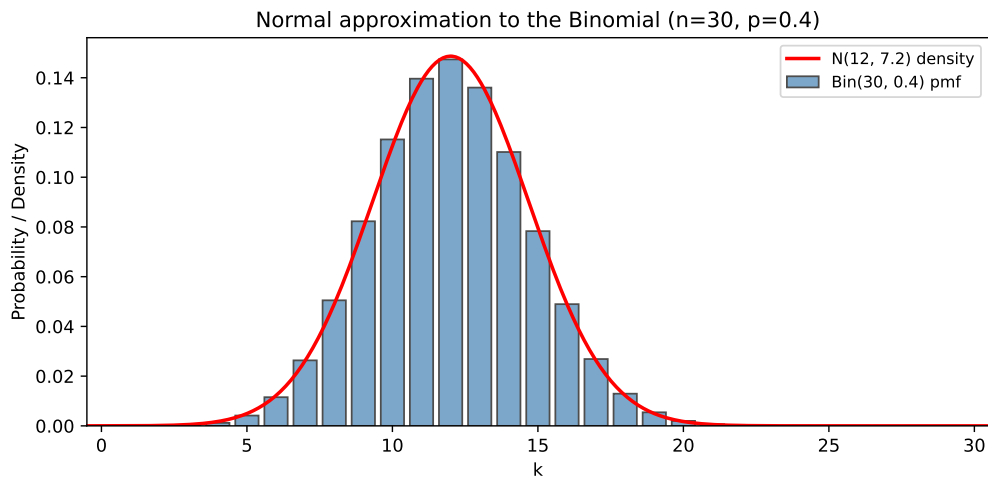


Figure 10.7: The $\text{Bin}(30, 0.4)$ pmf and its normal approximation $\mathcal{N}(12, 7.2)$.

Example (Normal approximation to a binomial probability). A factory produces light bulbs, each of which is defective independently with probability $p = 0.05$. In a batch of $n = 200$ bulbs, what is the probability that more than 15 are defective?

Let $X \sim \text{Bin}(200, 0.05)$, so $\mu = np = 10$ and $\sigma^2 = np(1-p) = 9.5$. We check: $np = 10 > 5$ and $n(1-p) = 190 > 5$, so the normal approximation applies.

$$\mathbb{P}(X > 15) = 1 - \mathbb{P}(X \leq 15) \approx 1 - \Phi\left(\frac{15 - 10}{\sqrt{9.5}}\right) = 1 - \Phi(1.622) \approx 1 - 0.9476 = 0.0524. \quad (10.29)$$

So there is roughly a 5.2% chance that more than 15 bulbs are defective.

Using the CLT for Sums: An Engineering Example

The CLT is especially useful when we need to find the distribution of a total formed from many independent contributions, none of which need be normally distributed.

Example (Load on a bridge). A bridge is designed to carry a maximum load of $W = 200$ tonnes. Vehicles cross the bridge one at a time, and the weight of each vehicle is a random variable with mean $\mu = 1.8$ tonnes and standard deviation $\sigma = 0.6$ tonnes. Suppose $n = 100$ vehicles are on the bridge simultaneously.

Let $S_{100} = X_1 + \cdots + X_{100}$ be the total weight. By the CLT,

$$S_{100} \approx \mathcal{N}(n\mu, n\sigma^2) = \mathcal{N}(180, 36), \quad (10.30)$$

so S_{100} is approximately normal with mean 180 tonnes and standard deviation $\sqrt{36} = 6$ tonnes. The probability that the load exceeds the design limit is

$$\mathbb{P}(S_{100} > 200) \approx 1 - \Phi\left(\frac{200 - 180}{6}\right) = 1 - \Phi(3.33) \approx 0.0004. \quad (10.31)$$

Although we have no idea what the distribution of individual vehicle weights looks like, the CLT tells us the *total* is approximately normal, and the probability of exceeding the limit is very small.

Monte Carlo verification of the bridge load example

LLM Prompt > Write Python code that uses Monte Carlo simulation to estimate the probability that the total weight of 100 vehicles exceeds 200 tonnes, where each vehicle has mean weight 1.8 and standard deviation 0.6. Compare the Monte Carlo estimate with the CLT approximation.

CLT Proof Sketch

We outline the main ideas behind the proof of the CLT using moment generating functions.

Proof: (Sketch) Without loss of generality, assume $\mu = 0$ and $\sigma = 1$ (this amounts to replacing each X_i by $(X_i - \mu)/\sigma$, which does not change the standardised sum). The goal is to show that the MGF of $S_n/\sqrt{n}$ converges to the MGF of the standard normal.

Step 1: Taylor expand the MGF. Let $M(s) = \mathbb{E}[e^{sX_1}]$ be the common MGF. Since $\mathbb{E}[X_1] = 0$ and $\mathbb{E}[X_1^2] = 1$, a Taylor expansion around $s = 0$ gives

$$M(s) = 1 + s \mathbb{E}[X_1] + \frac{s^2}{2} \mathbb{E}[X_1^2] + o(s^2) = 1 + \frac{s^2}{2} + o(s^2), \quad (10.32)$$

where $o(s^2)$ denotes terms that vanish faster than s^2 as $s \rightarrow 0$.

Step 2: Compute the MGF of $S_n/\sqrt{n}$. By independence,

$$\mathbb{E}\left[\exp\left(s \frac{S_n}{\sqrt{n}}\right)\right] = \prod_{i=1}^n \mathbb{E}\left[\exp\left(\frac{s}{\sqrt{n}} X_i\right)\right] = \left[M\left(\frac{s}{\sqrt{n}}\right)\right]^n. \quad (10.33)$$

Substituting the Taylor expansion with s replaced by $s/\sqrt{n}$:

$$\left[M\left(\frac{s}{\sqrt{n}}\right)\right]^n = \left[1 + \frac{s^2}{2n} + o\left(\frac{s^2}{n}\right)\right]^n. \quad (10.34)$$

Step 3: Take $n \rightarrow \infty$. Recall the classical limit $\lim_{n \rightarrow \infty} (1 + a/n)^n = e^a$. Applying this with $a = s^2/2$:

$$\left[1 + \frac{s^2}{2n} + o\left(\frac{s^2}{n}\right)\right]^n \rightarrow e^{s^2/2}, \quad (10.35)$$

which is the MGF of the $\mathcal{N}(0, 1)$ distribution.

Step 4: Conclude. If the MGFs converge, so do the corresponding cdfs (this is a known result called the *continuity theorem* for MGFs). Therefore $S_n/\sqrt{n} \xrightarrow{d} \mathcal{N}(0, 1)$. $\square$

Limitations of the MGF approach. The proof above assumes that the MGF $M(s) = \mathbb{E}[e^{sX}]$ exists in a neighbourhood of $s = 0$. This fails for some distributions that have finite variance but extremely heavy tails. For example, the lognormal distribution has finite moments of all orders, yet its MGF is infinite for every $s > 0$. Since the CLT applies to *any* iid sequence with finite variance, the MGF-based argument does not cover all cases.

A fully rigorous proof replaces the MGF with the **characteristic function** $\varphi(t) = \mathbb{E}[e^{itX}]$, $t \in \mathbb{R}$, which exists for *every* random variable (not just those with light tails). The Taylor expansion and limiting argument proceed in essentially the same way, but now the continuity theorem (Lévy's continuity theorem) guarantees that pointwise convergence of characteristic functions implies convergence in distribution. This approach is covered in more advanced probability courses.

TABLE OF DISTRIBUTIONS

Discrete Distributions $(q = 1 - p)$

Distribution	PMF $p(x)$	Support	$\mathbb{E}X$	$\mathbb{V}\text{ar}(X)$	PGF $G_X(z)$
DU($\{1, \dots, N\}$)	$\frac{1}{N}$	$\{1, \dots, N\}$	$\frac{N+1}{2}$	$\frac{N^2-1}{12}$	$\frac{z(1-z^N)}{N(1-z)}$
Ber(p)	$p^x q^{1-x}$	$\{0, 1\}$	p	pq	$q + pz$
Bin(n, p)	$\binom{n}{x} p^x q^{n-x}$	$\{0, \dots, n\}$	np	npq	$(q + pz)^n$
Geom(p)	$p q^{x-1}$	$\{1, 2, \dots\}$	$\frac{1}{p}$	$\frac{q}{p^2}$	$\frac{pz}{1-qz}$
NegBin(r, p)	$\binom{x-1}{r-1} p^r q^{x-r}$	$\{r, r+1, \dots\}$	$\frac{r}{p}$	$\frac{rq}{p^2}$	$\left(\frac{pz}{1-qz}\right)^r$
Poi(λ)	$\frac{e^{-\lambda} \lambda^x}{x!}$	$\{0, 1, 2, \dots\}$	λ	λ	$e^{\lambda(z-1)}$
Hyp(N, K, n)	$\frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$	$\{\max(0, n+K-N), \dots, \min(n, K)\}$	$\frac{nK}{N}$	$\frac{nK(N-K)(N-n)}{N^2(N-1)}$	—

Continuous Distributions

Distribution	PDF $f(x)$	Support	$\mathbb{E}X$	$\mathbb{V}\text{ar}(X)$	MGF $M_X(t)$
$\mathcal{U}(a, b)$	$\frac{1}{b-a}$	$[a, b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb} - e^{ta}}{t(b-a)}$
Exp(λ)	$\lambda e^{-\lambda x}$	$[0, \infty)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda - t}$
$\mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	$\mathbb{R}$	μ	σ^2	$e^{t\mu + \sigma^2 t^2/2}$
Gamma(α, λ)	$\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$	$(0, \infty)$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$\left(\frac{\lambda}{\lambda - t}\right)^\alpha$
$\chi^2(n)$	$\frac{x^{n/2-1} e^{-x/2}}{2^{n/2} \Gamma(n/2)}$	$(0, \infty)$	n	$2n$	$(1-2t)^{-n/2}$
Beta(α, β)	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$	$(0, 1)$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	—
Pareto(α)	$\alpha x^{-(\alpha+1)}$	$[1, \infty)$	$\frac{\alpha}{\alpha-1} \quad (\alpha > 1)$	$\frac{\alpha}{(\alpha-1)^2(\alpha-2)} \quad (\alpha > 2)$	—
Cauchy	$\frac{1}{\pi(1+x^2)}$	$\mathbb{R}$	—	—	—
t(k)	$\frac{\Gamma(\frac{k+1}{2})}{\sqrt{k\pi} \Gamma(\frac{k}{2})} \left(1 + \frac{x^2}{k}\right)^{-(k+1)/2}$	$\mathbb{R}$	$0 \quad (k > 1)$	$\frac{k}{k-2} \quad (k > 2)$	—
F(d_1, d_2)	$\frac{\Gamma(\frac{d_1+d_2}{2})}{\Gamma(\frac{d_1}{2})\Gamma(\frac{d_2}{2})} \left(\frac{d_1}{d_2}\right)^{d_1/2} \frac{x^{d_1/2-1}}{(1 + \frac{d_1}{d_2}x)^{(d_1+d_2)/2}}$	$(0, \infty)$	$\frac{d_2}{d_2-2} \quad (d_2 > 2)$	$\frac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)} \quad (d_2 > 4)$	—

BIBLIOGRAPHY

- [1] W. Feller, *An Introduction to Probability Theory and Its Applications*, Volume I, 2nd edition, John Wiley & Sons, 1970.
- [2] H. Hsu, *Probability, Random Variables & Random Processes*, Schaum's Outline Series, McGraw-Hill, 1997.
- [3] D.P. Kroese, Z.I. Botev, T. Taimre, and R. Vaisman, *Data Science and Machine Learning: Mathematical and Statistical Methods*, Chapman & Hall/CRC Press, 2019.
- [4] Y. Nazarathy and H. Klok, *Statistics with Julia: Fundamentals for Data Science, Machine Learning and Artificial Intelligence*, Springer, 2021.
- [5] A. Leon-Garcia, *Probability, Statistics, and Random Processes for Electrical Engineering*, 3rd edition, Pearson, 2008.
- [6] S.M. Ross, *A First Course in Probability*, 7th edition, Prentice-Hall, 2005.
- [7] D. Stirzaker, *Elementary Probability*, 2nd edition, Cambridge University Press, 2003.

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